



TEAMCENTER

Change Management on Active Workspace — Usage

Software Version 2606

Beta Release

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1. Overview of change management in Active Workspace

Managing Changes in Active Workspace

Change Management enables your organization to ensure the quality of every change made to a product. This is done through mechanisms for problem identification, change authorization, coordination and planning, cost and benefit analysis, and record keeping.

By automating your change process, you can minimize change-related rework and coordinate tasks to be performed by individuals across your organization. You can evaluate the impact of changes, track the status and completion of tasks, and maintain a comprehensive history of product changes throughout the lifecycle. Change Management on Active Workspace — Usage is also tightly integrated with Schedule Management on Active Workspace — Usage and Workflows and Tasks on Active Workspaces so you can schedule implementation activities and guide a change through its phases.



You can manage the work required to:

- Implement a change.
- Assess its impact on any managed business items, such as parts or documents.
- Notify participants about proposed and authorized changes.
- Track progress and completion of work.
- Compare before-and-after product configurations.

Where do I go from here?

 Change Administrator/Change Specialist	Change administrators and specialists facilitate and manage movement of a change through the appropriate processes.
Configure specific Change features.	Configure Change Management
Learn about managing the change process.	• Create workflows with Workflow Designer. • Create work breakdown structures with Schedule Manager to plan and schedule changes. • Assign or replace participants.

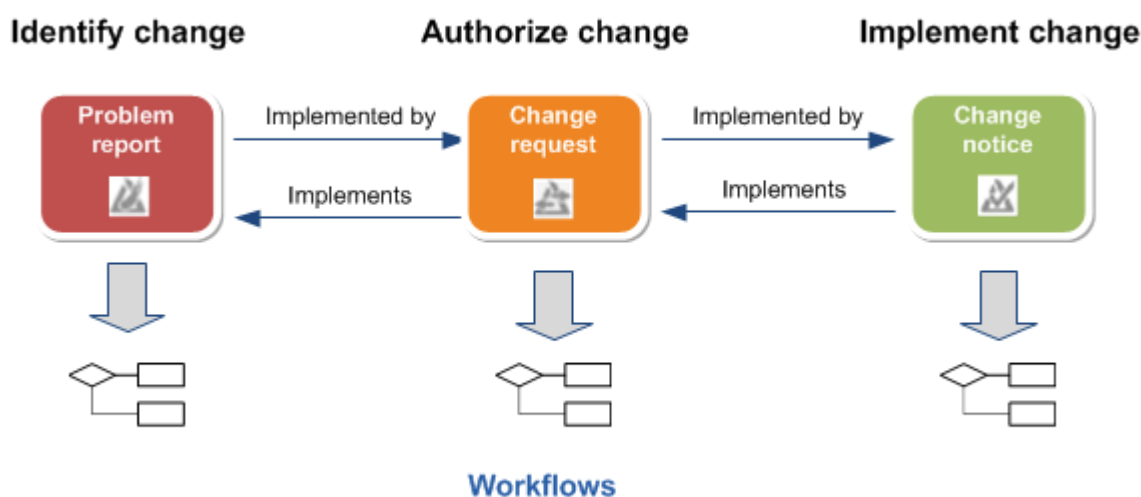
 Change Analyst/Change Contributor	A change analyst elaborates the details of a change by providing a technical recommendation, performing an impact analysis, and planning the implementation. A change contributor can create solutions for the change notice just like an analyst and functions as a secondary analyst.
Show me the basics of using Active Workspace.	Refer to Active Workspace Fundamentals
Learn more about team member participation in Change Management on Active Workspace — Usage.	See the section on: Who are the participants in a change?
Identify a problem and create a change.	See: • Create a change and send it for resolution • View your changes • Create a change in the context of an object and attach it to the change • Edit a change • Derive a change from another change
Other features	
Show me other important features that work with Change Management.	See: • Workflows and Tasks • Schedule Management on Active Workspace — Usage • Structure Management on Active Workspace — Usage • Design and BOM Alignment — Usage

2. Exploring Change Management

What is a change?

Using Teamcenter, you can manage changes to support the continued evolution of released product data. Typically, the change process includes identifying the goal (problems or enhancements), assigning them to users who analyze and identify the impacts, develop solutions to address the goal, and the approval steps to authorize and validate the change.

Three main types of changes shown in the figure implement the key elements of Change Management:



These objects are Item types in the business model, and are therefore revisionable in Teamcenter. Each object type encapsulates the data for a phase in the change process. For example, a problem report contains the data that defines the problem, the change request contains the analysis of the impact of the problem, and the change notice contains the data for the implementation of the solution.

The **Implemented By** and **Implements** relationships associate the change objects with each other. Each object type can be optionally associated with a workflow that defines the sequence of tasks and moves each object through a set of states to bring the problem to a resolution.

Using Teamcenter, you can create a problem report to identify and formally track an issue with your product information. Others who are involved in the change process can review and confirm the problem and provide input into what business objects must be updated to resolve the issue. Members of a change review or change implementation board can review and approve or disapprove the changes.

What are the types of changes?

The following change objects are used to manage a change:

Change object	Description
Problem report	Initiates a change. A problem report (PR) captures information about a problem or an enhancement. It includes the information necessary to confirm and reproduce any problems observed or to document the specifics of a request for an enhancement. Additional attributes document the perceived severity of the problem and set the priority for addressing the issue relative to other PRs. The processing of a PR sometimes leads to the creation of a change request. Creating a PR is an optional step in the change process. Depending on the conventions at your site, you may first identify a problem or enhancement with a change request or change notice, not a PR. A PR may be implemented by one or more change requests.
Change request	Initiates a proposal that recommends a change and captures business decisions associated with the change. A change request (CR) proposes a solution to the problem, populates impacted items, and performs impact analysis, with cost estimates and benefits of making the change. The actual solution (for example, a new item revision) is implemented in the change notice. A CR is typically a response to a problem report unless the problem report stage is skipped. A single CR may logically group and address issues identified in multiple PRs. A CR may be implemented by one or more change notices.
Deviation request	A deviation request is a type of change request. Seeks consent to deviate from a solution in production to resolve a set of problems to initiate improvements. Typically, there are two types of deviation requests:

Change object	Description
	- Request for deviation Seeks consent to deviate from a part solution. - Request for waiver Seeks consent to accept a non-conforming part.
Change notice	Implements a change. It provides authorization to generate solution items and a detailed work plan to resolve one or more change requests or a portion of a single change request. A change notice (CN) identifies all items and documents affected by a change and authorizes the actions that address a change.
Simple change	Identifies a problem and implements a change. A simple change (SC) is designed for minor changes and a reduced number of change participants. A simple change cannot be derived from another change type.
Non-functional change	Non-functional changes (NFCs) allow modifications to parts in an assembly without affecting their form, fit, or function, thus avoiding major revisions and complex reviews. Form refers to the physical characteristics of an object, fit describes how well a part connects with others, and function pertains to the object's capabilities. NFCs are useful for updating specific attributes, such as vendor parts or custom descriptions, without altering the core aspects of the product.

What are the states of a change?

A change has three key change states that capture where in the change process it is and what decisions about the change have been made.

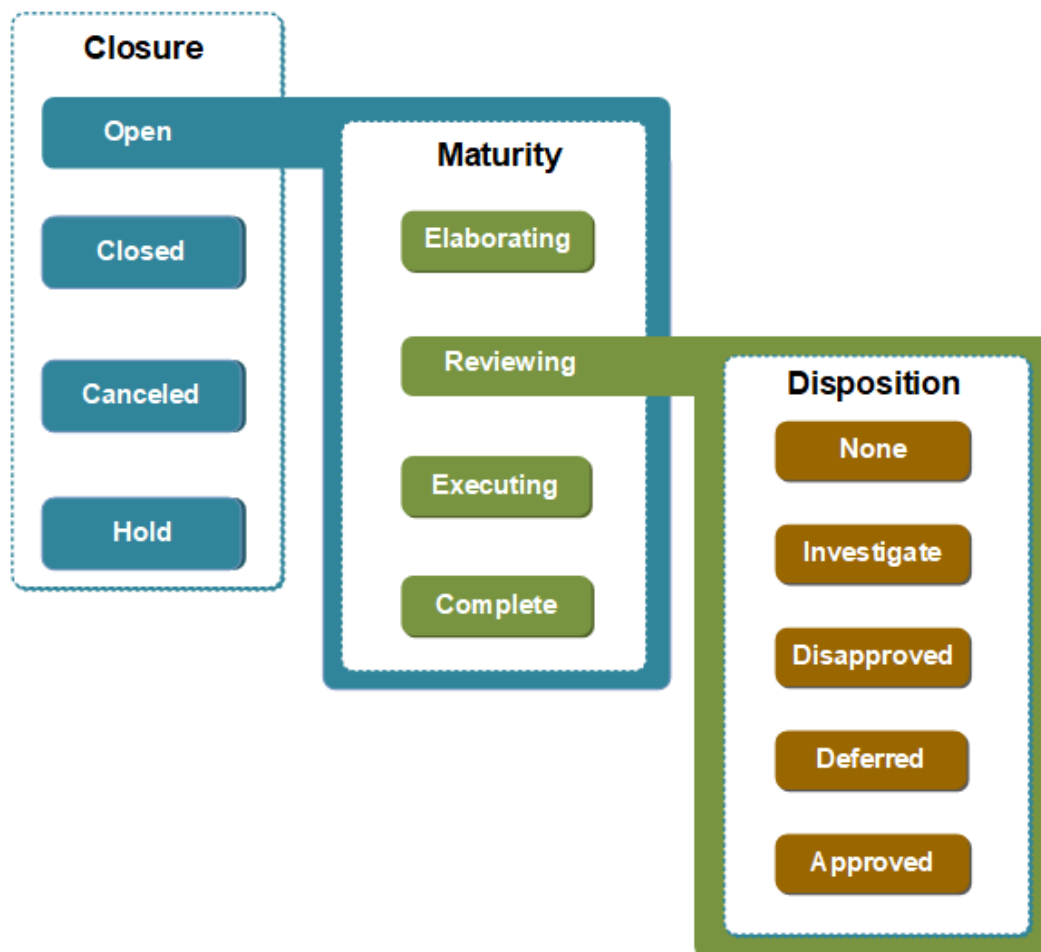
- Its status or *Closure* (for example, Open or Canceled).
- The degree of completion of the overall change process, its *Maturity*.

- The technical, business, or implementation decision by a person or review board about a change object's approval, its *Disposition*.

The change states interconnect and are dependent on the other change states. For example, Maturity is a substate of the *Closure* state and *Disposition* is a substate of *Maturity*. The states of a change are set during the workflow process. To move a change to the next phase of the change process, for example, to move a problem report to a change request so a solution can be determined, the states of a change must be appropriate.

You can view the states of a change in the **Overview** tab.

The following graphic illustrates the change states. For detailed information on change states, see *Change Management — Deployment and Rich Client Usage* in the Teamcenter help.



What are the contents of a change?

The **Affected Items** tab displays and organizes the contents of the change.

The change types, the relationships they define, and the permissions and change states required to add an object to a change are as follows:

Contents of a Change	Information adding the objects				
Problem Items					
Contains item revisions with the problems that the change is to address. This could be the parent assembly.	To create a relationship between an item revision and a change object (problem report, change request, change notice), add the item revision to the change object's Problem Items table. You must be an assigned participant and the change object settings must be as follows. <table> <tr> <th>Assigned participant</th><th>Closure property settings</th></tr> <tr> <td>All participants</td><td>Open</td></tr> </table>	Assigned participant	Closure property settings	All participants	Open
Assigned participant	Closure property settings				
All participants	Open				
Impacted Items					
Contains item revisions that are being changed as a result of the change process.	Creates a relationship between an item revision that a change request or change notice affects and the change request or change notice. To add the item revision to the change request or change notice's Impacted Items table, you must be an assigned participant and the change request or change notice property settings must be as follows. <table> <tr> <th>Assigned participant</th><th>Closure property settings</th></tr> <tr> <td>Must have an Author license level and be the Requestor, Analyst, or Change Contributor</td><td>Open</td></tr> </table>	Assigned participant	Closure property settings	Must have an Author license level and be the Requestor, Analyst, or Change Contributor	Open
Assigned participant	Closure property settings				
Must have an Author license level and be the Requestor, Analyst, or Change Contributor	Open				
Solution Items					

Contents of a Change	Information adding the objects					
Contains item revisions that are generated as a result of the change (for example, the new piece parts and the new revision of the parent assembly to contain them).	To create a relationship between an item revision and the change notice that implements the change, add the item revision to the change notice's Solution Items table.					
	You must be an assigned participant and the change notice property settings must be as follows.					
	<table><tr><th>Assigned participant</th><th>Closure property settings</th></tr><tr><td>Must have an Author license level and be the Analyst or Change Contributor</td><td> Open Note The Maturity must not be equal to Reviewing. </td></tr></table>	Assigned participant	Closure property settings	Must have an Author license level and be the Analyst or Change Contributor	Open Note The Maturity must not be equal to Reviewing.	
Assigned participant	Closure property settings					
Must have an Author license level and be the Analyst or Change Contributor	Open Note The Maturity must not be equal to Reviewing.					
Reference Items						
Contains any object that references related information (such as analysis documents and system logs).	To associate related information (such as analysis documents and system logs) with a change, add any object to the Reference Items table.					
	You must be an assigned participant.					

The **Reference Items** tab displays and organizes the following contents.

Plan Items				
Contains schedules that define tasks in a work breakdown structure.	To associate related work breakdowns (schedules) with a change request or change notice, add any schedules to the Plan Items table.			
	You must be an assigned participant and the property settings for the change request or change notice must be as follows.			
	<table><tr><th>Assigned participant</th><th>Closure property settings</th></tr><tr><td>Must have an Author license level</td><td>Open</td></tr></table>	Assigned participant	Closure property settings	Must have an Author license level
Assigned participant	Closure property settings			
Must have an Author license level	Open			
Implements				
Contains change object revisions that reference this	Note A problem report does not have an Implements table because it only identifies problems.			

change object. A change request implements problem reports. A change notice implements change requests.

Usually, the **Implements** table is automatically populated as part of the change process, but you can add items to it if you are an assigned participant and the property settings are set as follows.

Assigned participant	Closure property settings
Must have an Author license level	Open

Implemented By

Contains change object revisions that are referenced by this change object. For example, a problem report is *implemented by* a change request and a change request is *implemented by* a change notice.

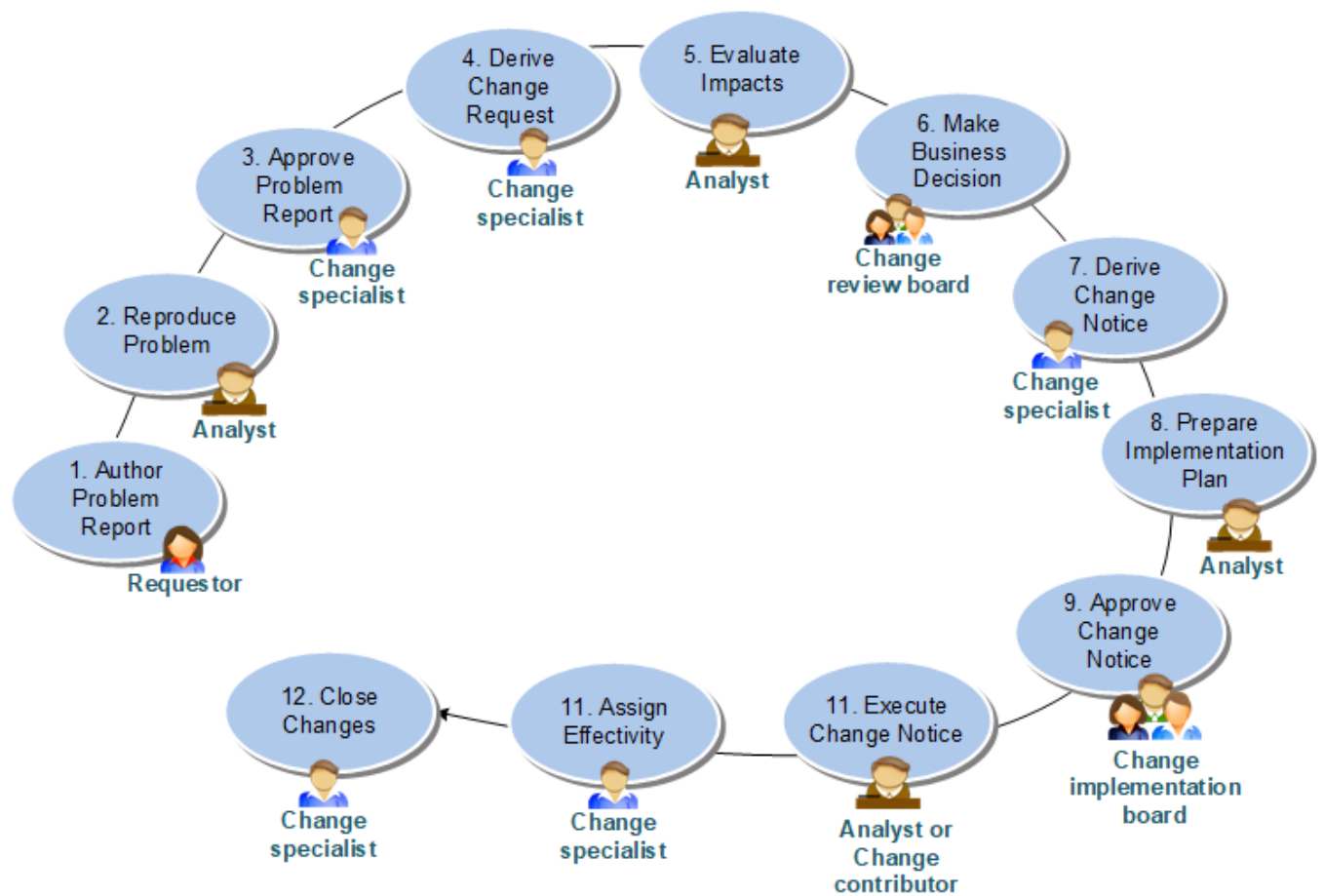
Note

The **Implemented By** table is automatically populated as part of the change process by objects in the **Implements** table.

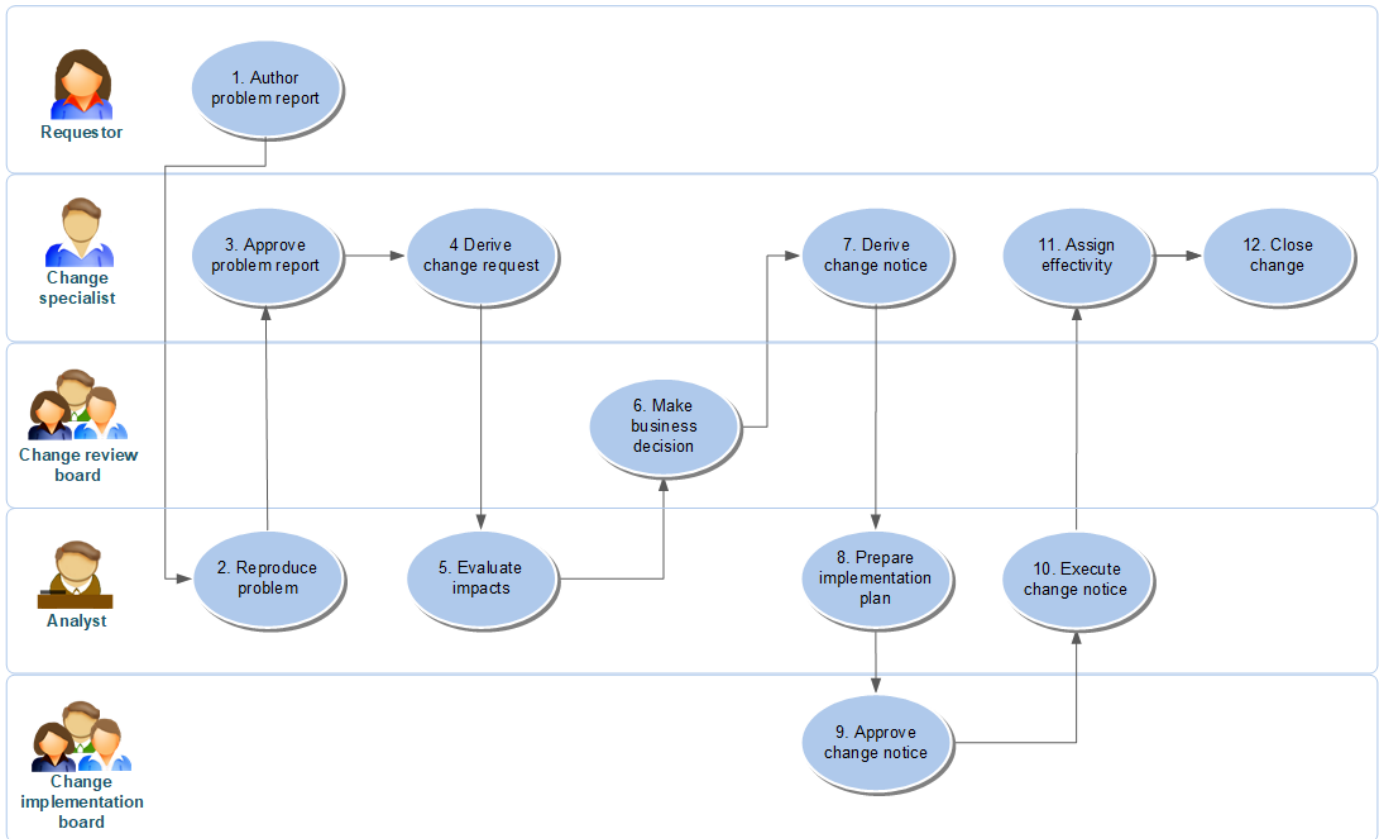
Defining your change management process

You can manage your changes in the way that works best for your company's processes. You should define a change process that is flexible enough to impose the appropriate level of rigor and control based on the level of risk, cost, and the business items impacted by the change. You can classify a change as fast or standard track. In a fast track process, the change does not go through a formal review process, while a standard track follows a more rigorous process, and may include a schedule to manage the required tasks.

The following graphic shows a typical standard track process along with the associated roles.



The following graphic shows the tasks performed by each role in the Change Management process.



1. Author a problem report (PR).

A requestor creates a PR to identify a problem or enhancement, provide a preliminary assessment, and show the steps necessary to reproduce the problem.

2. Reproduce problem.

A change analyst performs the steps to reproduce the problem as detailed by the requestor.

3. Approve a PR.

A change specialist assigns a priority to the PR and assigns it to an analyst for technical review. The specialist or analyst recommends a disposition, such as **Approved**.

4. Derive a change request (CR).

A change specialist creates a CR to address the PR.

At this stage, the analyst develops a solution or several alternative solutions. The analyst does this by creating markups on documents, Word documents, presentations, and so on. No decision has been made at this stage about whether to proceed or what new items or item revisions may be required.

Sometimes the PR may propose a solution if the problem is simple to fix. However, the solution would still be formally documented in the CR. Typically, though, the requestor is unlikely to be in a position to know what the solution should be and may likely have no idea at all.

Note

The CR can address more than one PR.

5. Evaluate the impacts.

The analyst identifies the items impacted by the change, prepares supporting documentation, and prepares a high-level proposal for the actions required to implement the change.

6. Make a business decision.

A change specialist submits the change request to a change review board who decides if the change will be made. The change review board can approve the change request, reject it, or require additional investigation. If this is a fast track change, the review board is the owner of the change and the process moves to the execute change step.

At this stage, a decision is made about whether to revise or create new items, according to the form, fit and function and interoperability.

7. Derive a change notice (CN).

The change specialist either derives a new CN to address the approved CR or associates the CR with an existing CN. The CN addresses the implementation details of the change. It may address multiple CRs. The requestor can delegate responsibility for elaborating the details of the implementation plan.

Note

A CN is always derived to implement a solution, even for a CN that went through a fast track process. However, the workflow for the fast track CN is very short, with a minimal number of steps. It is necessary to create an CN so the analyst can add solution items, which is not possible in the CR, whose purpose is only to define a proposed solution, and perform impact analysis.

8. Prepare an implementation plan.

The analyst develops a detailed plan to address the set of approved CRs addressed by the CN.

At this stage the agreed solution is implemented in the new/revised items.

9. Approve the CN.

For a standard track process, the change implementation board reviews and approves the plan to address the change. For a fast track process, the approval is informal and may just require the change specialist managing the change.

10. Execute the CN.

The analyst implements and tracks the detailed plan for addressing the change. A change specialist tracks the implementation progress at a high level. The analyst and/or change contributors will generate solutions to implement the change.

11. Assign an effectivity.

A change specialist can assign effectivities to the CN. The effectivities specify the timing of when the change takes effect.

12. Close changes.

The change specialist closes the change after all the actions associated with each level of the implementation plan are complete.

About managing the change process through a workflow

Workflows guide a change through the different phases of a change process: authoring, review and approval, execution, and closure. These phases are modeled as states of the change.

For example, a workflow process moves the **Disposition** attribute of a change from **Investigate** to **Approve**. Change states cannot be manually updated. Using Change Management with Workflow Designer, therefore, tracks the evolution of changes through your organization according to a controlled, repeatable process. In addition, you can:

- Tie a workflow to a scheduled task so as tasks are worked on and updated in Teamcenter workflow, the information is sent to Schedule Manager where the corresponding information is updated. You can also configure a Teamcenter scheduled task so the associated workflow can be initiated when certain conditions are met.
- Set up a workflow to automatically assign review members from assigned participants by configuring the workflow for dynamic participants.

To do this, you need to use the following workflow action handlers:

- The **EPM-set-property** handler to set the change state properties when they change during the workflow.
- The **EPM-auto-assign** handler to assign a single, appropriate participant on the change object as the responsible party for the current task.
- The **EPM-adhoc-signoffs** handler to assign review tasks.

Note

Use the **EPM-assign-responsible-party-dynamic-participant** and **EPM-assign-signoff-dynamic-participant** handlers to automate the assignment of participants.

These Change Management preferences determine which workflows start when a change revision is submitted:

Preference	Change Object
ChangeNoticeRevision_default_workflow_template	Change Notice Revision
ChangeRequestRevision_default_workflow_template	Change Request Revision

Preference	Change Object
Cm0SimpleChangeRevision_default_workflow_template	Simple Change Revision
ProblemReportRevision_default_workflow_template	Problem Report Revision

Note

A default workflow process template is used when **Submit** is selected on the **Create** panel. For a change revision submitted *after* creation, the default template is pre-selected for the matching change type, making it easier to submit to the default workflow on pre-existing change revisions.

Examples of the workflow process for changes

The following examples are default workflow process templates for changes. A default workflow process template is used when **Submit** is selected on the **Create** panel.

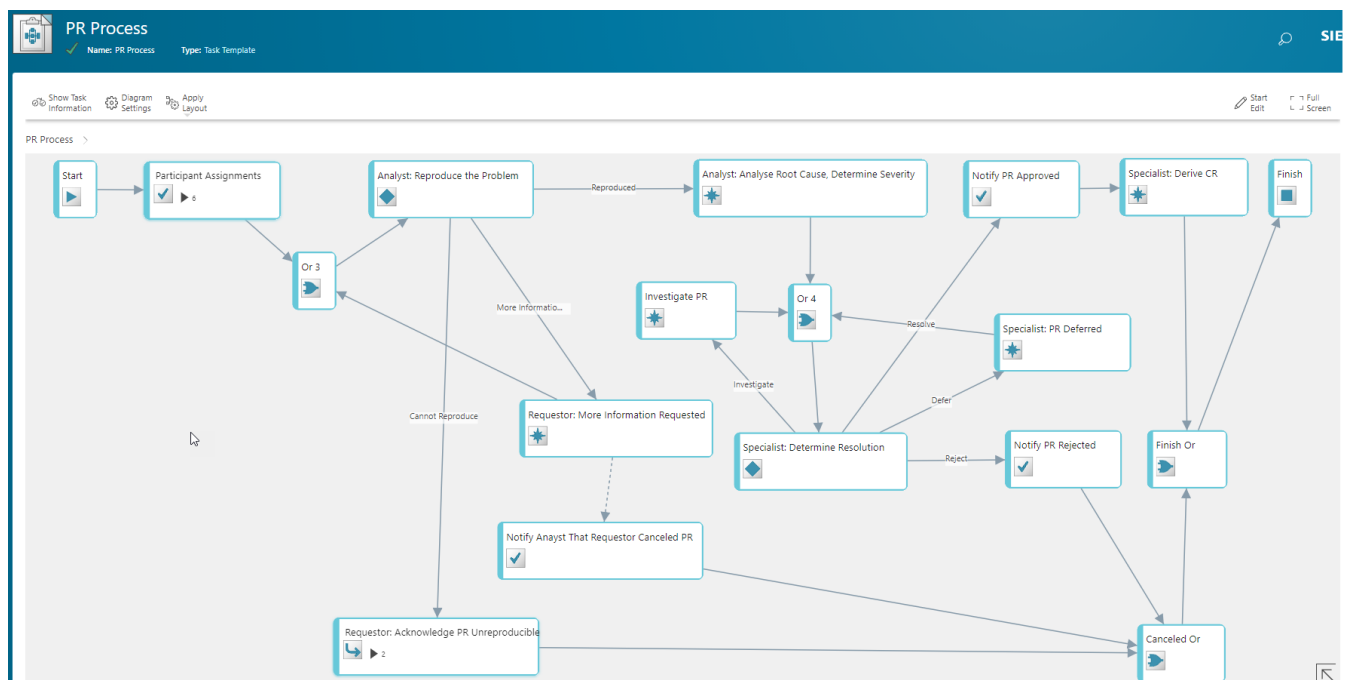
For a change revision submitted *after* creation, the default template is pre-selected for the matching change type, making it easier to submit to the default workflow on pre-existing change revisions.

The following workflow examples for a problem report, change request, simple change, and change notice are the default workflow process templates in Workflow Designer. These examples illustrate the automated and interactive steps, along with the associated participant (if available). In Workflow Designer, you can select an individual task to view the specific participant instructions.

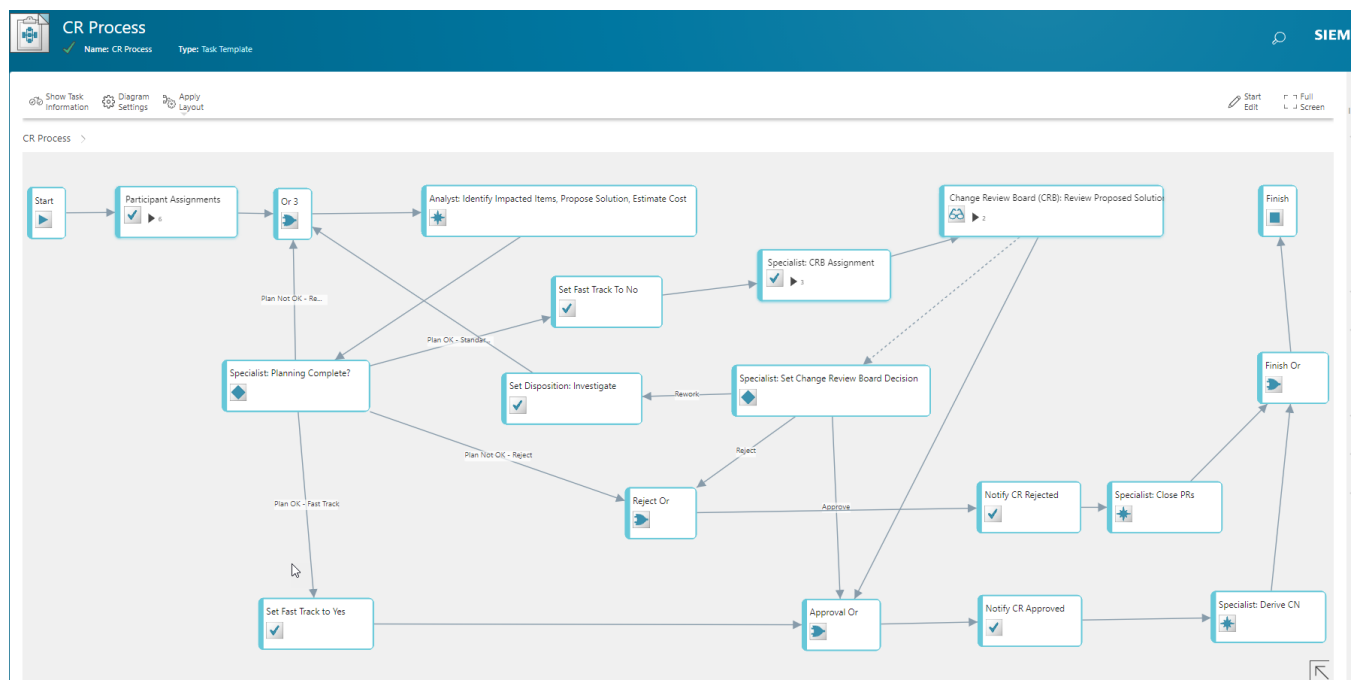
Note

For more information on workflow templates refer to the Workflow Designer documentation.

Problem report default workflow process template



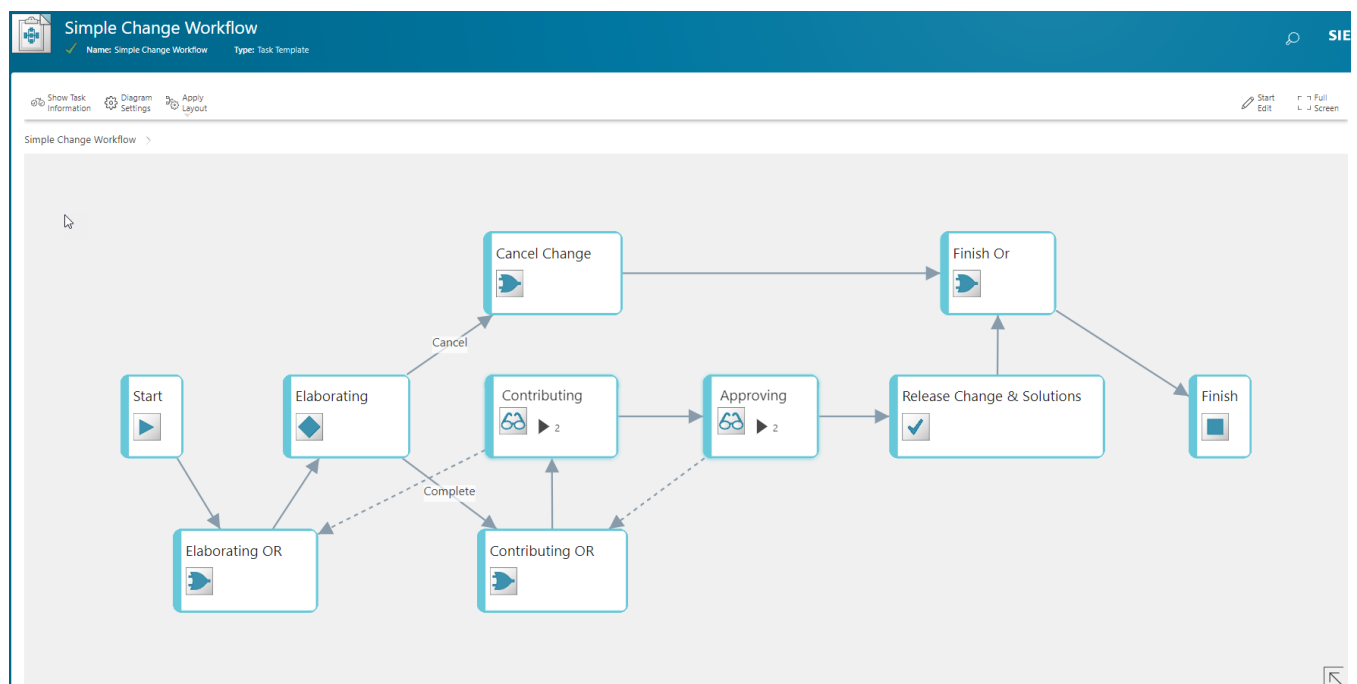
Change request default workflow process template



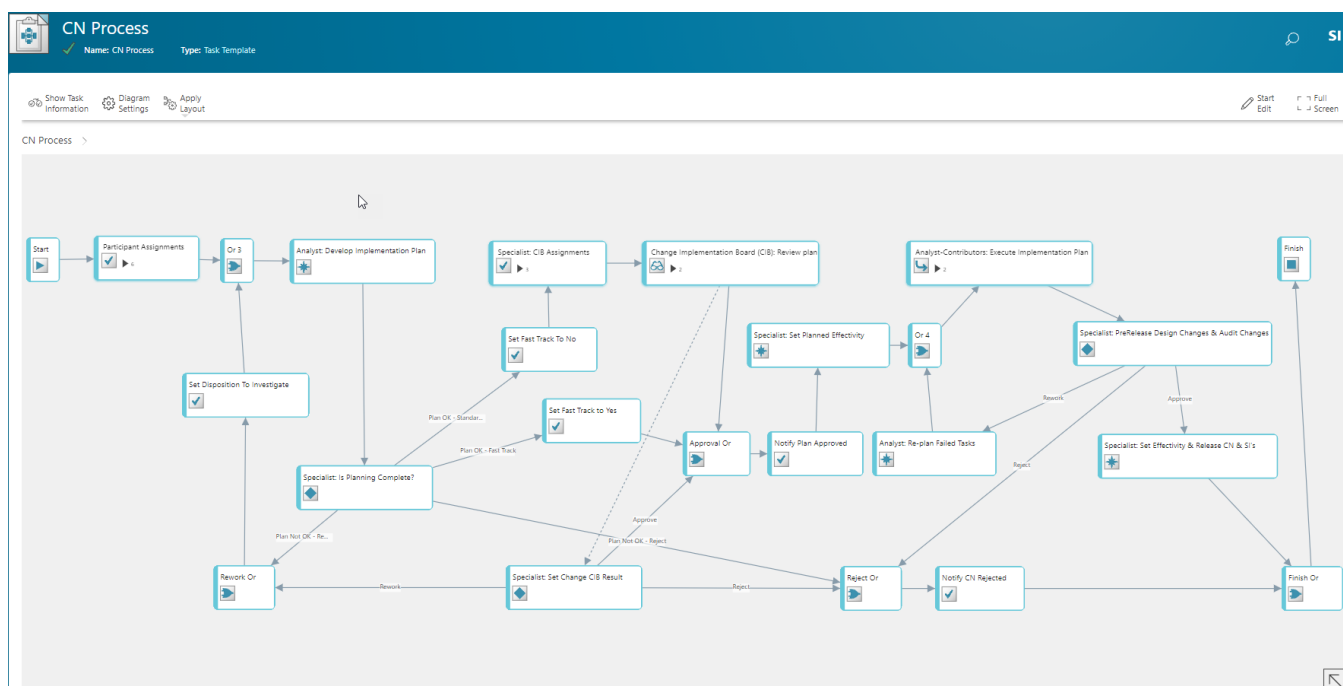
Simple change default workflow process template

Note

Simple change automates the assignment of contributors and approvers. The requestor may assign change contributors and approvers, or a single participant may perform all of the tasks.



Change notice default workflow process template



Change participants

Who are the participants in a change?

When a change workflow begins, often participants need to be assigned, which generally includes the following roles. Other roles may be included based on the type of change and your company's change process. The participant type of *requestor* is assigned automatically to the user who created the change. You can change this, for example, if you created a change on behalf of someone else.

Note

If you created a custom participant or a custom user profile, you must add the new attribute to the **Search Criteria** section for the **My Change Notices** query.

Some workflows automatically assign participants. For example, a workflow can automatically assign the analyst to implement the change. Users who are automatically assigned are referred to as *dynamic participants*.

Participant	Does the following
 Requestor	Creates a change object or is delegated a change object by another requestor. The requestor is responsible for elaborating the definition of a change and for providing as much detail as possible to define the problem or request at hand. A requestor may perform the elaboration, or a change specialist may delegate responsibility for elaborating a change to another user (an analyst

Participant	Does the following
	or requestor). When you create a change object, you are its requestor. If responsibility is delegated to another user, that user becomes the current requestor and you are no longer the requestor.
 Specialist	Facilitates and manages movement of a change through the appropriate processes. The specialist is responsible for reviewing and approving the change object and determines if more planning is required. The specialist assigns the analyst and the change implementation board and audits all changes. The specialist may set the effectivity, as well.
 Analyst	Assesses the technical feasibility of a change or the technical feasibility of implementing the low-level details of a change. An analyst elaborates the details of a change by providing a technical recommendation, performing an impact analysis, and planning the implementation. The analyst may perform the implementation or delegate the work to others. During the implementation phase, the analyst monitors the change execution and is ultimately responsible for ensuring the implementation is performed correctly and according to the plan. This person is generally a senior technical person with knowledge of technical issues and a site's products and business goals.
 Change review board	Reviews, approves, and authorizes engineering change requests and deviation requests. The review board makes a business decision about whether a change or deviation request should proceed. Review board members are generally senior individuals from various functional areas within a site or individuals who have expertise in some aspect of a change or deviation.
 Change implementation board	Reviews whether a change notice is to be executed or canceled. The change implementation board makes a business decision about how the change notice should proceed. The implementation board members are generally senior individuals from various functional areas within a site or individuals who have expertise in the context of a change or deviation.

Use the **Participants** tab in **Changes** to view the participants in a particular change.

Note

Assigning participants is not the same as selecting a signoff team in a workflow. It is a separate task for a change only. The change does not have to be in a workflow to assign participants.

Reassign participants

You can select multiple changes from the **Changes** tab or the search results and reassign a selected participant, such as a Change Specialist, from one user to a different user. The assignable participants

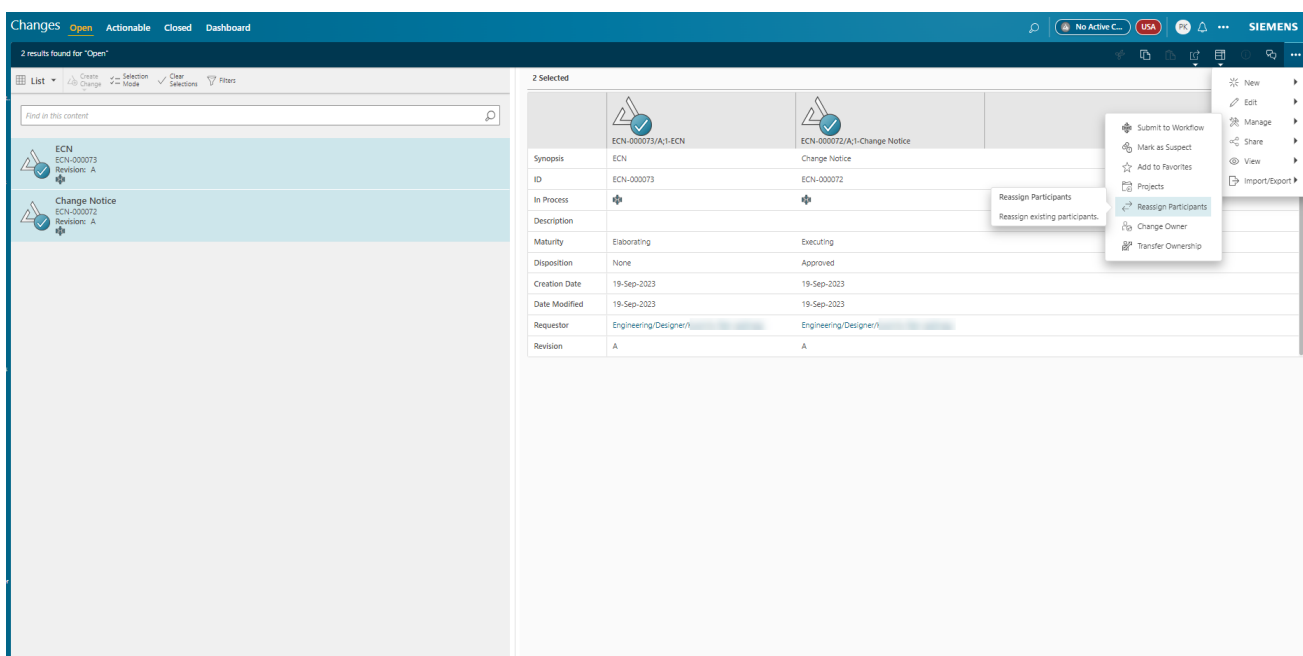
depend on the types of changes you select and the common set of participants available for these changes. For example, if you select a mix of change notices and change requests, the **Participant** list only displays the participants common to the selected change requests and change notices.

You can modify the participants for a large number of changes assigned to a user that has left your organization, moved to a different department, or needs the workload balanced across other members of the team.

The participants you can reassign are dependent on your privileges and your participant type in the change process.

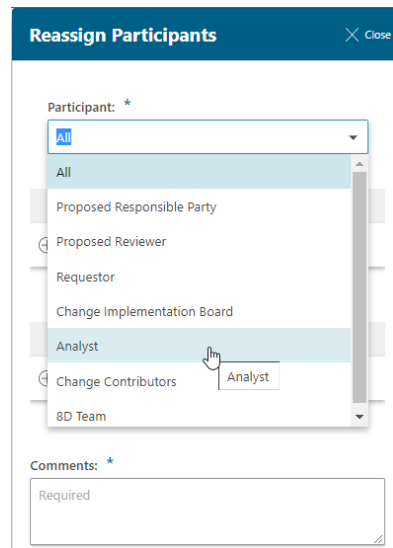
Procedure

1. Use **Ctrl+click** or **Shift+click** to select two or more changes from the **Changes** tab or the search results.
2. Go to **More Commands *** > Manage  > Reassign Participants ** from the **Page level** toolbar.



The **Reassign Participants** panel appears on the right.

3. Select the participant type. If there is more than one common type, you can select **All** to change the participants for all.



4. In the **From** section, click **Add** ⊕.
5. Select a participant to reassign. Or, type the name of the participant to whom you want to reassign the change in the top search box and press **Enter** or click **Search** 🔍 to see results. You can enter a partial name or wildcard characters.

Filters 🏷️ are defaulted to **Type**, **Group**, or **Role**. A search box narrows your search to a specific property.

You can increase the size of the panel by dragging the two bars on the left side of the panel to the left.
6. Once selected, click **Add**.
7. Repeat for the **To** section and select the new participant.

Reassign Participants × Close

★ Participant:
Analyst

▼ From

⊕ From

EE Edward Engineer
edward.engineer
Group: Systems Engineering
Role: Systems Engineer

▼ To

⊕ Assign To

C ClassificationUser
clsuser_16
Group: Engineering
Role: BOM Analyst

★ Comments:
Reassigned

Reassign

8. Comments are required for reassignment. Enter any comments, such as the reason for reassigning the participants, and then click **Reassign** to finish.
A message appears at the bottom of the browser informing you how many of your selections were reassigned. For example, if you selected five changes and Alex was the change specialist on four of them, the message states that four of the five selected changes were reassigned. The change that Alex was not the specialist on had no actions taken on it.

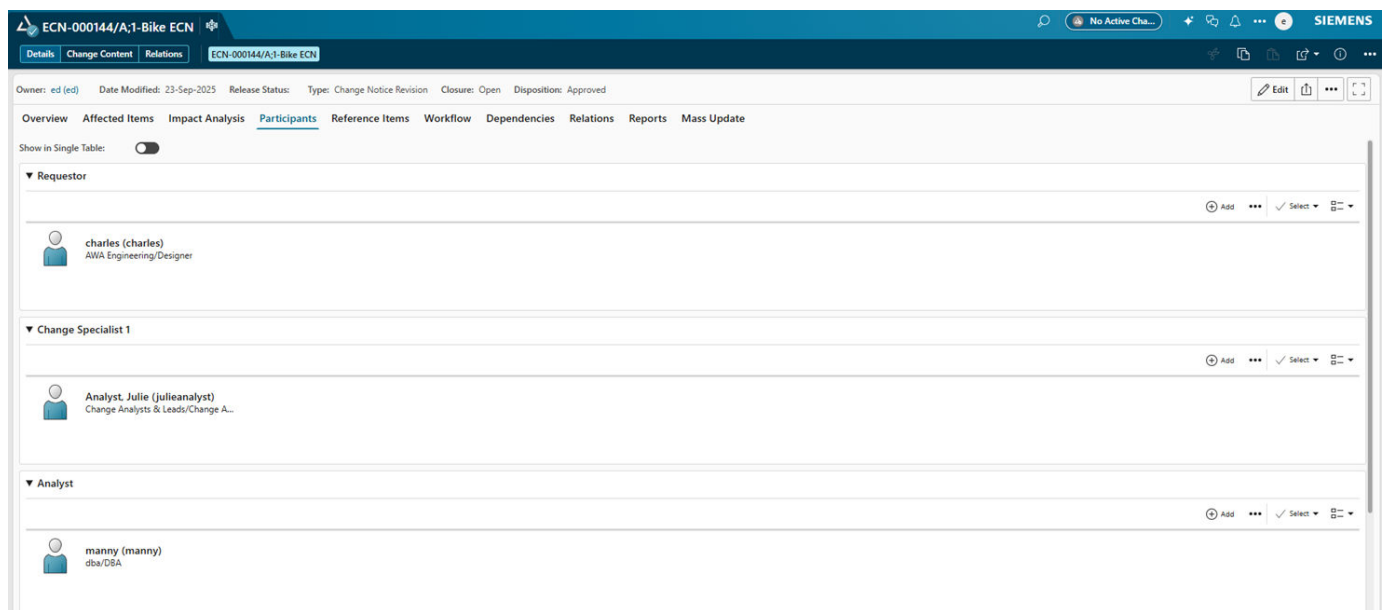
Assign or replace participants

The **Participants** tab contains all assigned participants for a change object. You can change, assign, or replace various participants.

You can select more than one user when assigning a review board.

Select a change and click the **Participants** tab to view or manage assignments.

Add or replace participants in a change object



Note

Turn on **Show in Single Table** to consolidate all tables on the **Participants** tab into one, offering an alternative way to track participants without navigating through multiple sections. You can add additional participants to the change notice via the **Add** ⊕ button.

Add or replace participants in a change object

Use the **Participants** tab to add or replace participants in a change object.

Procedure

1. To add a participant, select **Add** ⊕.



2. Select the participant type.
3. Select one or more users to assign. You can find detailed instructions for assigning users in the People Picker.
4. Once selected, click **Add**.
The change object displays the assigned participants in the table.

Remove a participant from a change

Use the **Participants** tab to remove a participant from a change object.

Procedure

1. Select the change object from the **Participant** tab.
2. Click **Remove**  in the toolbar, or go to **More Commands**  > **Edit**  > **Remove**  from the **Page level** toolbar. You can also right-click and select **Remove** .
The participant is removed from the changed (but not added to the clipboard).

Multiple change contributors

Change contributors can create solutions for the change notice just like the analyst. In effect, they function as secondary analysts. The **Cm0ChangeContributors** and **fnd0IsParticipant** operators handle this operation, verifying that the user is a participant of the change notice. The change analyst can add other users as change contributors, who can then add solutions to the change notice.

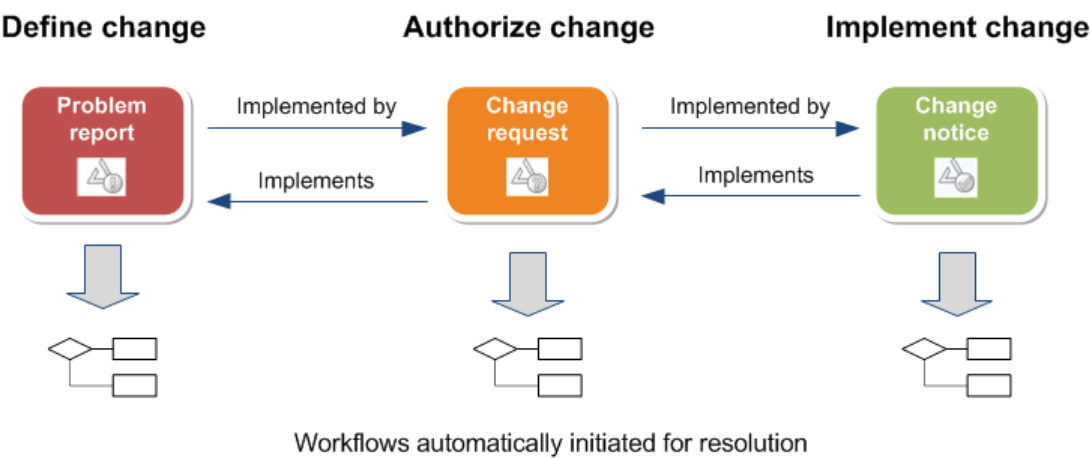
3. Creating a change

How to identify problems as changes

When you identify a potential problem that needs investigation, capture it as a change. Four change object types are available by default to represent the different stages of a change. Your company can also create its own types of changes. When you create a change, you are the *requestor*.

Create the change	To
 Problem report	Identify a problem and initiate a change process.
 Change request	Initiate a proposal to recommend a change and capture business decisions associated with the change.
 Change notice	Implement a change.
 Deviation request	Seek consent to deviate from a solution in production so you can resolve a set of problems and initiate improvements.

The following graphic shows how the three main change object types work together to solve a change.



You create a change in **Changes** or create it in the context of a reference object. The types of changes you can create depend on your permissions and the current state of the change. **Changes** can be configured to show you only the types of changes that you have permission to create.

Once you capture a problem as a change, you can:

- Automatically submit the change to a workflow process that routes it through your company's business process to resolve the change. Siemens Digital Industries Software provides a default workflow for each change object type. The default workflow automatically assigns you as the requestor of the change. Your organization can create its own workflow process. Check with your administrator for the default workflow set up for each type of change object in your organization.
- Save your change for later editing and resolution.
- Derive a change from another change to move it along in the process. For example, you can derive a change request to plan the solution from the problem report that identified the problem. The earlier figure shows how derived objects are related using the *Implements* and *Implemented by* relationships. An administrator sets up how the changes are derived and by whom.

Tip

If you only want to create a problem report, use **Report a Problem**.

Create a change and send it for resolution

Create a change and send it for resolution

A change identifies the issues and components that require resolution. Submitting the change to the appropriate workflow for its specific change type provides the mechanism to resolve the change.

Procedure

1. You can create a change using any of these methods.
 - Create a change from an object in a folder or in a list.

Select an object in a list or folder, go to **More Commands** *** > **New** ✨ > **Create Change** △⊕.

This includes items in any folders, including custom folders.
 - Create a change from within an object.

Open an object and go to **More Commands** *** > **New** ✨ > **Create Change** △⊕.
 - Click the **CHANGES** tile on the **Home** page or click the **Changes** △ icon to view all changes.
 - Click **New** ✨ > **Create Change**.
 - Click **Create Change** △⊕ on the top toolbar, then select the change type. The available subtypes of the change display, where you can further your selection. Skip to step 3.
2. Select the type of change you want to create.
 - **Change Notice**
 - **Change Request**

- Deviation Request
- Problem Report
- Simple Change
- Non-Functional Change

3. On the **Create Change** panel, enter the properties for the object:

- Specify the item ID.
- Enter the **Revision**.
- Enter the **Synopsis**.
- Enter a **Description**.

Note

In Active Workspace 6.2/Teamcenter 14.2, the **Description** is optional. In all other versions, it is required.

4. (Optional) To submit the new change to a workflow, select the workflow template from the **Workflow** drop-down list. Templates display based on the type of the object selected.

If a default workflow exists for the content type, it is automatically specified as the workflow template. You may also choose to leave this blank and not select a workflow.

If an administrator disables the drop-down menu, it will not offer any selectable options.

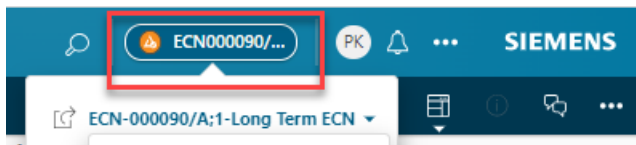
5. (Optional) Click **Add** ⊕ under **Attachments** to add supporting objects and reference documents to the change.
6. If applicable, the project the object is owned by is listed under **Owning Project**.
To assign a new project to an object, click **Add Project** to search for and select a project.
7. Select **Open New Change** to open the change once it has been created.
Open New Change is checked if the preference **CM_open_change_on_create** is set to true.
8. Select **Set as Active Change** to set the newly-created change notice as the active change. The current user is added as an **Analyst**.

Set as Active Change is checked if the preference **CM_set_active_change_on_create** is set to true.

Note

Non-functional changes cannot be set as an active change.

Once the change is created, the **Active Change** indicator is highlighted in the global navigation toolbar 🏠 and the newly created change is listed as the active change.



9. Click the **Participants** tab to include any members who participate in the review and execution of the change. The participant types are based on type of the change object selected in the first panel.

Note

This tab is not available for a simple change.

10. Expand the participant section, then click **Add Participants** ⊕ under each section to add the required participants. You can find detailed instructions for assigning users in the People Picker.
11. Once selected, click **Add**.
12. Do one of the following on the **Create** tab:
 - If you want to continue editing later, click **Create**.
The change is created and displayed in edit mode.
You can send it for resolution later.
 - If you want to send it for resolution immediately and a workflow has been selected, click **Create and Submit**.

The change is sent through the workflow. Participants to review and approve the change may be automatically assigned depending on how your organization's change process workflow is configured.

Tip

If a task is assigned to you as part of the change workflow, **Perform Task**  appears on the tools and information bar. You can perform the task as you do from your Inbox.

Results

Note

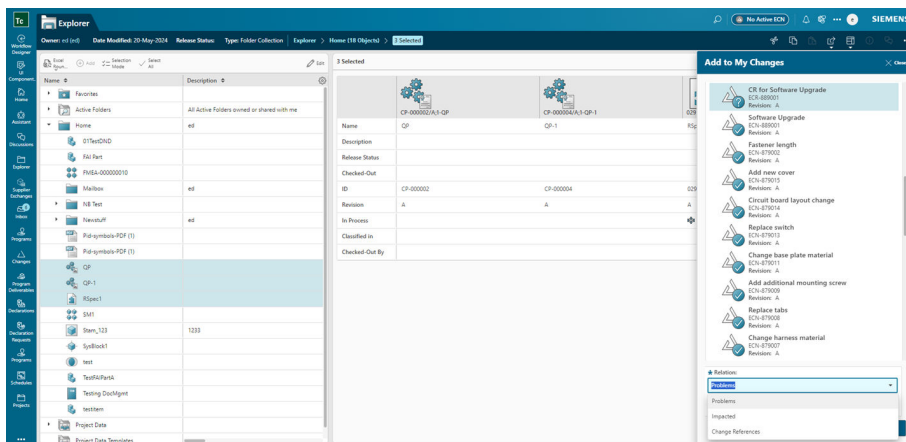
Depending on how your organization is configured, the **Create and Submit** button may be the only button available. The business object constant **Awp0EnableCreateForCreatePanel** controls display of the **Create** button. The default value is **true**.

Add an object to a change

You can add an object to an existing change using **Add to My Changes**.

Procedure

1. Select the object you want to associate with a change.
You can select multiple items at one time. Selected items are highlighted.
2. Go to **More Commands** *** > **Manage**  > **Add to My Changes** from the **Page level** toolbar.



You can select an existing change from the list, or use the filter to find a specific change.

3. Select the type of **Relation** for the object.
Relation options include **Problems**, **Impacted**, **Solutions**, or **Reference Items**.
4. Click **Add** to add the object to the selected change and apply its appropriate relation.

Objects added to a change appear in the **Affected Items** tab of the selected change.

Depending on the type of change and the relation, the added object also appears in the **Overview** tab.

Copy and paste objects in a change

You can copy and paste objects from the clipboard into the **Affected Items** or **Reference Items** sections of a change. Adding change objects depends on your role and access.

You can also copy and paste an object from one section to another. For example, you can copy an object from **Problem Items** into **Impacted Items**, or select an object from any of your folders and paste it into the appropriate section.

Note

You cannot paste an object into the **Solution Items** table.

Procedure

1. Select the change object and click **Copy**  in the toolbar.
2. To include the copied change object, select a change from the list of **Changes**  and click the **Affected Items** or **Reference Items** tab.
3. Click **Paste**  on the appropriate change object section.

Note

If you are copying an object from **Problem Items** into **Impacted Items**, you must first clear the checkbox against the problem item before pasting it into the **Impacted Items** table.

The change object is included in the table. You can associate multiple change objects in the same manner.

Remove an object from a change

Removing an object extracts it from the **Solution Items**, **Impacted Items**, or **Problem Items** tables.

Procedure

1. Select the change object from either the **Solution Items**, **Impacted Items**, or **Problem Items** tables.
2. Click **Remove**  in the toolbar, or go to **More Commands** ******* > **Edit**  > **Remove**  from the **Page level** toolbar. You can also right-click and select **Remove** .

The object is removed from the table (but not added to the clipboard).

Make a copy of a change

If you have authoring privileges, you can make a copy of a change. Copying a change saves time by copying attributes and supporting information to ensure consistency across changes.

Depending on how Active Workspace is set up, you can choose to copy the supporting reference documents and problem items or participants.

Unlike changes you create new, a copied change is not automatically submitted to workflow.

Procedure

1. If necessary, search for the change you want to copy.

Tip

Be sure to select the correct revision of the change prior to copying.

2. Click **New** ✱ > **Save As** .
3. In the **Save As** panel, select **New**.

Save As ✕

Revision New

Synopsis:

Description:

COPY OPTIONS

☒ Impacted

☒ Problems

Save

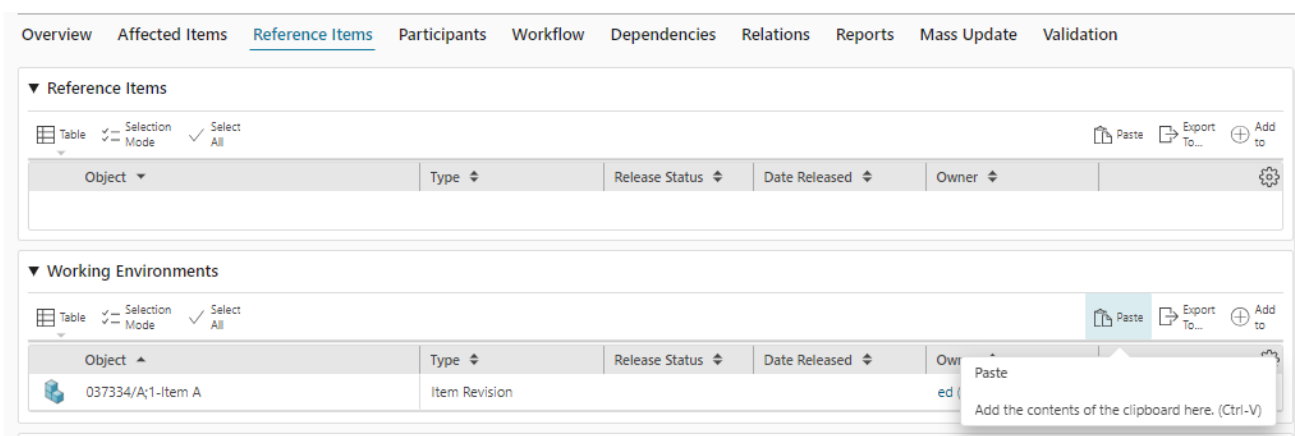
4. Enter the synopsis and description.
5. Choose **Copy Options** to copy into the new change.
6. Click **Save**.
The new change is created and displayed in edit mode.
7. (Optional) Specify additional properties for the change, and click **Edit**  > **Save Edits**.
The edits are saved and the revision is displayed in read-only mode.

Copy a change item to a change notice working environment

If you are an authorized user, you can add a copied change object to the working environment of a change notice.

Procedure

1. If necessary, search or filter for the change object to copy.
2. Select the change object and click **Copy**  in the toolbar.
3. To include the copied change object in the working environment of a change notice, select a change notice from the list of **Changes** and click the **Reference Items** tab.
4. Select paste in the **Working Environments** section.



The change object is now included in the table. You can associate multiple change objects in the same manner.

Edit a change

During the change process, you can edit the properties of a change and relate other objects to it to aid in the analysis and resolution of the change. For example, you may need to add reference documents to a problem report or specify which objects are impacted by a change.

You must edit changes or add objects during different workflow process phases. For example, many workflow processes start with a **Prepare Package** phase that asks you to review the current change and add any objects needed for its analysis.

You can edit the properties of the change itself and its attachments. You can also edit in the table view.

Procedure

1. Search for the change object if it is not already displayed in your workspace.

2. In the **Overview** tab, edit the properties for the change. Click **Edit** in the top right.
3. The **Overview** tab contains an editable **Details** section.

Editable fields in the sections are surrounded by a bounding box.

The screenshot shows a 'Details' section with the following fields:

- Synopsis:** A text box containing 'Change baseplate material'.
- Description:** A text box containing 'Change base plate from metal to plastic'.
- Date Released:** A label.
- Effectivity:** A label.
- Fast Track:** A dropdown menu with 'No' selected.
- Implementation Priority:** A dropdown menu with 'Medium' selected.
- Customer Approval Required:** A dropdown menu with 'No' selected.
- Status of Customer Approval:** A dropdown menu with 'Approved' selected.
- Special Instruction:** A text box.

4. To add or remove associated items:
 - Click **References** tab for reference documents
 - Click **Impacted Items** tab for affected objects
 - Click **Attachments** tab for additional documents
5. Click **Save**.

The edits are saved, and the change is displayed in read-only mode.

Note

You can only modify change objects when the CMClosure status is open.

Create a change in the context of an object and attach it to the change

You can create a change directly from an object in a folder or a list to automatically add it during change creation.

Procedure

1. Select an object with which you want to associate a change.

You can use search to find and select multiple objects in any view.

2. Go to **More Commands** *** > **New** ✨ > **Create Change** ⚙️ from the **Page level** toolbar.

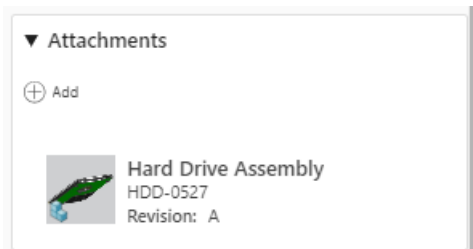
The common change types are displayed for the items you select. If only one type is available, the create box for that type is displayed. If you select multiple items for which there is no common change type, a message appears indicating that a change type cannot be created because no common change types are available.

3. Select the type of change you want to create.

Only changes that you can create are listed.

- **Change Notice**
- **Change Request**
- **Deviation Request**
- **Problem Report**
- **Simple Change**

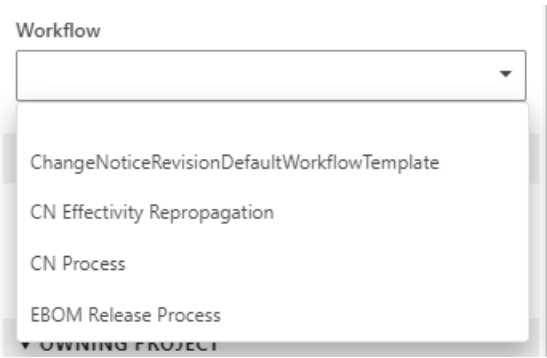
The object is automatically added as a reference document on the **Reference Items** or **Affected Items** tab, depending on preference settings.



4. Enter the properties for the object.

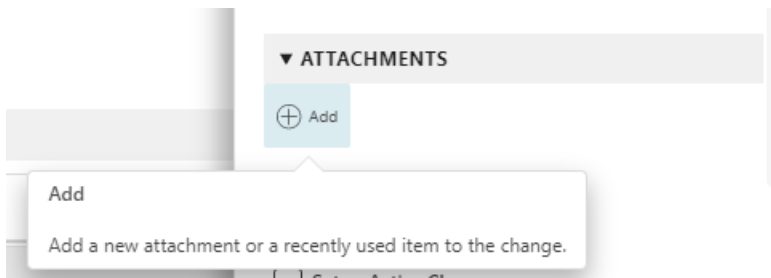
For example, specify the item ID, revision, synopsis, and description for a problem report.

5. (Optional) To submit the new change to a workflow, select the workflow template from the **Workflow** drop-down list. Templates display based on the type of the object selected.



If a default workflow exists for the content type, it is automatically specified as the workflow template.

6. (Optional) Click **Add**  under **Attachments** to add supporting objects and reference documents to the change.



7. If applicable, the project the object is owned by is listed under **Owning Project**.
To assign a new project to an object, click **Add Project** to search for and select a project.
8. Select **Open New Change** to open the change once it has been created.
Open New Change is checked if the preference **CM_open_change_on_create** is set to true by your administrator.
9. Select **Set as Active Change** to set a newly-created change notice as the active change. The current user is added as an **Analyst**.

Set as Active Change is checked if the preference **CM_set_active_change_on_create** is set to true by your administrator.

Once the change is created, the **Active Change** indicator is highlighted in the global header  and the newly created change is listed as the active change.



10. Click the **Participants** tab to include any members who participate in the review and execution of the change. The participant types are based on type of the change object selected in the first panel.

Note

This tab is not available for a simple change.

11. Expand the participant section, then click **Add Participants**  under each section to add the required participants.

12. Select one or more users to assign.

Optionally, you can type the name of the participant to whom you want to assign the change in the top search box and press **Enter** or click **Search**  to see results. Enter a partial name or wildcard characters.

Filters  are defaulted to **Type**, **Group**, or **Role**. A search box narrows your search to a specific property.

You can increase the size of the panel by dragging the two bars on the left side of the panel to the left.

13. Once selected, click **Add**.

14. Do one of the following on the **Create** tab:

- If you want to continue editing later, click **Create**.

The change is created and displayed in edit mode.

You can send it for resolution later.

- If you want to send it for resolution now, click **Create and Submit**.

The change is sent through the default workflow for resolution. Participants to review and approve the change may be automatically assigned depending on how your organization's change process is configured.

Depending on how your organization is configured, the **Create and Submit** or the **Submit** button may be the only button available. The business object constant **Awp0EnableCreateForCreatePanel** controls the display of the **Create** button. The business object constant **Awp0EnableSubmitForCreate** controls the display of the **Submit** button. The default value of either constant is **true**.

A thumbnail from the problem item is automatically attached to the change object, if it exists.

Note

If you prefer that the thumbnail is derived dynamically from the items related to change, your administrator can set the **CM_derive_thumbnail_from_relations** preference to true. **False** is the default option.

Cancel a change

You can cancel an engineering change notice (ECN) prior to its approval at any stage of the workflow. Before canceling an ECN, you can move any solution items that you wish to keep to another location to

clone and reuse. However, once an ECN is canceled, any remaining solution objects become canceled and cannot be reused.

Note

To be able to mark an ECN's solution Items as **Canceled**, the CM-cancel-change-notice-handler must be included on the workflow template that routes the ECN through the approval process. The **EBOM Release Process**, **Simple Change Workflow**, and **CN Process** workflow templates all contain this handler out-of-the-box to provide the ability to cancel solution items. Solution item types to be canceled can be specified as a handler argument.

Procedure

1. Open the ECN that you want to cancel.
2. To preserve any of the revised solution items it contains, do the following:
 - a. Click the **Affected Items** tab of the ECN.
 - b. Select the revised item(s) from the **Solutions Items** folder to keep, then click **Cut**  > **Paste**  from the primary toolbar to move the item to another location (such as **Explorer** or another change).

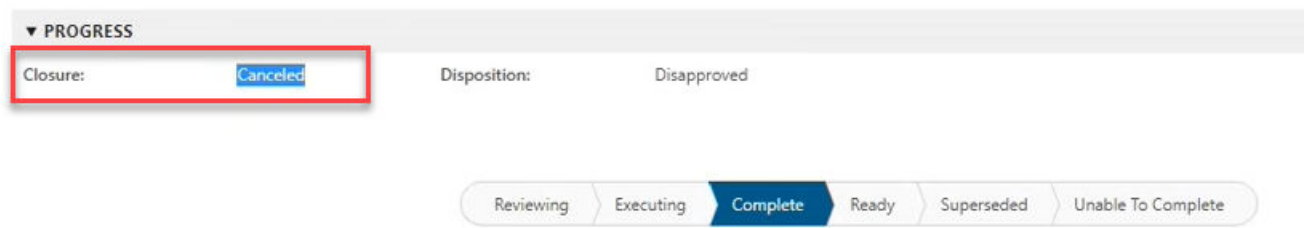
You can also **clone the canceled solution item** (click **New**  > **Save As**  and consume the newly created/revised non-canceled object.

3. From the ECN's **Overview** tab, click **Cancel Change**.



The screenshot shows the 'Overview' tab of an ECN. At the top, there are tabs for 'Overview', 'Affected Items', 'Reference Items', and 'Participants'. Below these are three buttons: 'Submit to...', 'Release Effectivity', and 'Generate Report'. A section titled 'TASK TO PERFORM' contains details about the workflow, including the name 'Validate change task' and task instructions. At the bottom, there are three buttons: 'Cancel Change', 'Repropagate Effectivity', and 'Validate Change'. The 'Cancel Change' button is highlighted with a red rectangular box.

Any objects that are part of the solutions items are permanently canceled and cannot be reused. The **Canceled** property is set as true on these solution items.



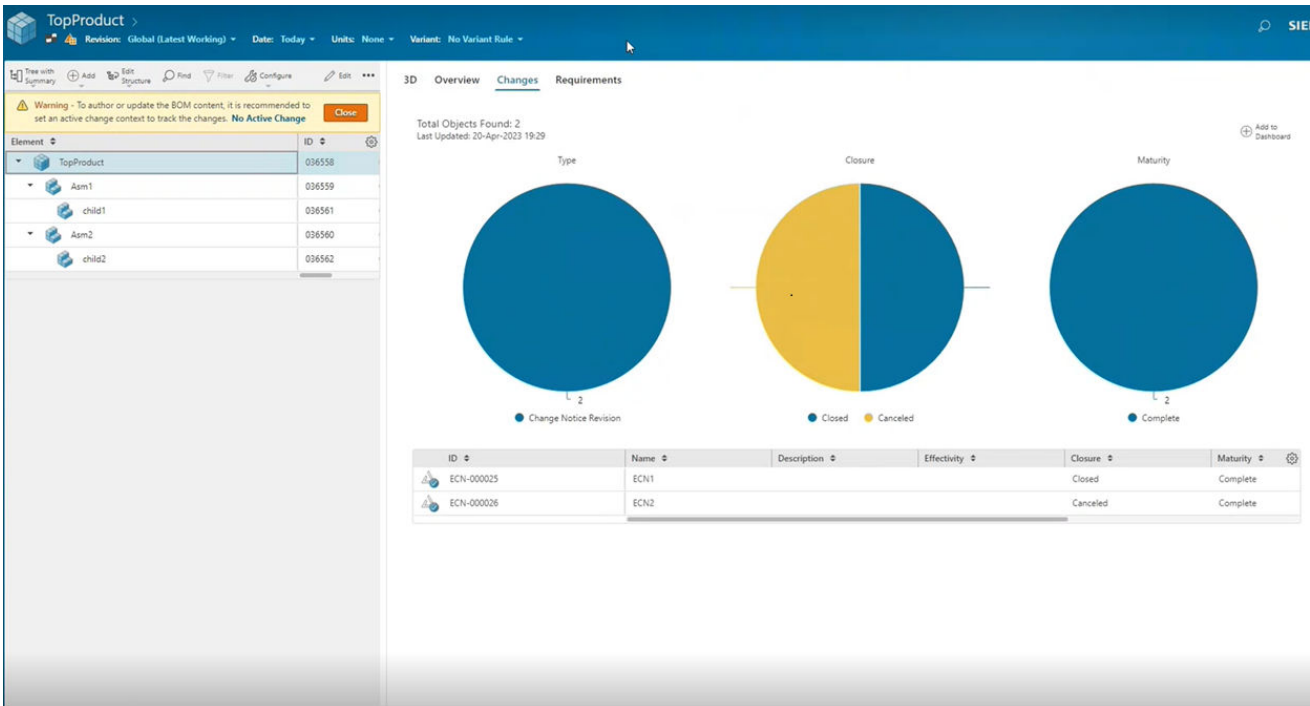
Solution items contained in the canceled change now contain a **Cancelled** state of **True** and cannot be modified.

The screenshot displays the 'SOLUTION ITEMS' section, which includes a table and a dropdown menu. The table has columns for 'Object', 'Lineage', 'Type', 'Release Status', 'Cancelled', and 'Date Released'. The first row is highlighted in light blue. A dropdown menu is open for the 'Cancelled' column, showing 'True' as the selected value.

Object	Lineage	Type	Release Status	Cancelled	Date Released
036568/A;1-New_child2		UB Part Revision		True	
PU_000291/B;Asm1	2	Part Usage		True	
PU_000294/B;child2	1	Part Usage		True	
PU_000295/A;Asm3		Part Usage		True	

The **Revision History** for the canceled item(s) also displays the **Cancelled** state of **True**. The canceled objects are always filtered out from all of the structure configuration.

In the [Changes Dashboard](#), the top-level product displays a pie chart that includes all of the ECNs associated with that collaborative product. Canceled changes are displayed here as well. Click a pie chart to filter your changes.



Effectivity dates are listed for any canceled ECN in the **Effectivity** column. You can compare dates to determine the progression of the ECN's effectivity.

Note

When an ECN contains solution items that include an effectivity range, and an ECN is canceled, any effectivity cutbacks on the canceled item are restored in full.

Send a change for resolution

Send a change for resolution

If you chose not to submit your change to a workflow process when you created it, you can send it later. For example, you can send the change later after you have added more supporting reference documentation. You must know the name of the workflow you want to submit it through.

Procedure

1. If necessary, find and display the change you want to send for resolution.
2. Go to **More Commands *** > Manage** **> Submit to Workflow** from the **Page level** toolbar. Follow the instructions to start a workflow for an object.
3. Click **Submit**.

Results

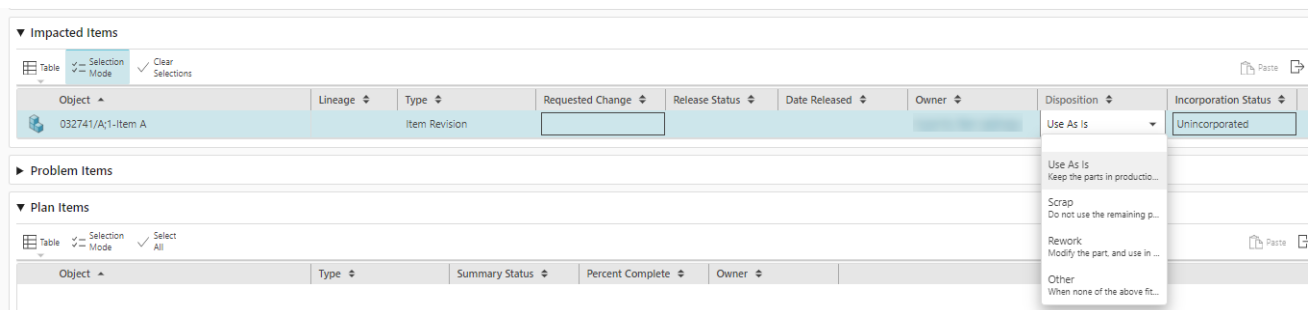
The change is sent through the workflow process. Check your Inbox for tasks to perform.

Specify impacted item disposition

If you are a user allowed to make changes to a change notice, you can specify the disposition of impacted items being revised or replaced in a change notice.

Procedure

1. Open the change notice you want to work with.
2. Click **Affected Items**.
3. Go to **More Commands *** > Edit**  **> Start Edit** from the **Page level** toolbar.
4. In the **Impacted Items** row, select the object and a disposition from the drop-down list:



- **Use As Is:** Keep the parts in production
- **Scrap:** Do not use the remaining parts.
- **Rework:** Modify the part, and use the modified parts going forward.
- **Other**

Other values may be available depending on your customization.

5. Go to **More Commands *** > Edit**  **> Save Edits** from the **Page level** toolbar.

Specify the requested change type for impacted items

The **Impacted Items** table contains an option to specify the **Requested Change** type.

Procedure

1. After adding an object to the **Impacted Items** table, go to **More Commands *** > Edit**  **> Start edit** from the **Page level** toolbar to add or modify the requested change type.
2. Click within the editable **Requested Change** column of the impacted item, and click the down arrow to view a list of the available change types.

The screenshot displays the 'Affected Items' tab in the software interface. It contains three tables: 'SOLUTION ITEMS', 'IMPACTED ITEMS', and 'PROBLEM ITEMS'. The 'IMPACTED ITEMS' table is the focus, showing a dropdown menu for the 'Requested Change' column. The dropdown menu lists three options: 'Revise', 'Create New', and 'Replace with Existing'. A tooltip for the 'Revise' option states: 'Requested change is to revise the impacted item.'

Object	Lineage	Type	Requested Change	Release Status	Date Released	Owner	Dispo
027187/B;1-Testchild8	2	Item Revision				ed (ed)	
027189/A;1-Testsubchild2		Item Revision				ed (ed)	
V8 ENGINE_DC2_SmartDiscovery_Ashika3/B;1-Test1b	1	Item Revision			06-May-2020	ed (ed)	
027187/A;1-Testchild1	2	Item Revision	Revise			ed (ed)	Use As Is
V8 ENGINE_DC2_SmartDiscovery_Ashika3/A;1-Test1	1	Item Revision	Revise			ed (ed)	Use As Is

Object	Type	Release Status	Owner
027189/A;1-Testsubchild2	Item Revision		ed (ed)
V8 ENGINE_DC2_SmartDiscovery_Ashika3/A;1-Test1	Item Revision		ed (ed)

3. Select the appropriate change type. Repeat for other impacted items.

The default list of values (LOV) includes:

- **Revise**
- **Create New**
- **Replace with Existing**

4. Go to **More Commands *** > Edit**  **> Save Edits** from the **Page level** toolbar. The updated **Requested Change** column displays the change types for the associated impacted items.

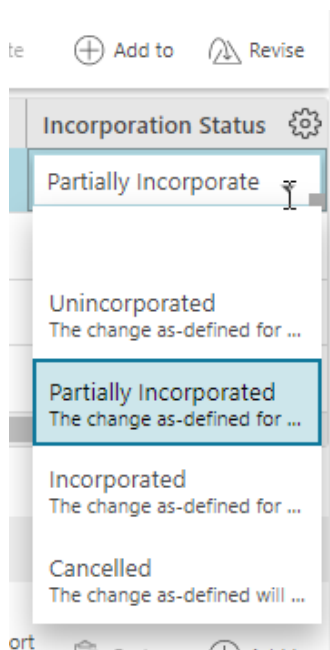
▼ IMPACTED ITEMS		
Table Selection Mode Select All		
Object ▲	Type ▼	Requested Change ▼
 027418/A;1-TestDoc	Document	Create New
 V8 ENGINE_DC2_SmartDiscovery_Ashika3/A;1-Test1	Item Revision	Revise

Specify the impacted item incorporation status

The **Impacted Items** table contains an option to specify the **Incorporation Status** of a change object in relation to a solution item.

Procedure

- After adding an object to the **Impacted Items** table, click the **Incorporation Status** column to show a list of incorporation statuses.



- Select the appropriate status type. Repeat for other impacted items.

The default list of values (LOV) includes:

- **Unincorporated**
- **Partially Incorporated**
- **Incorporated**

- Canceled

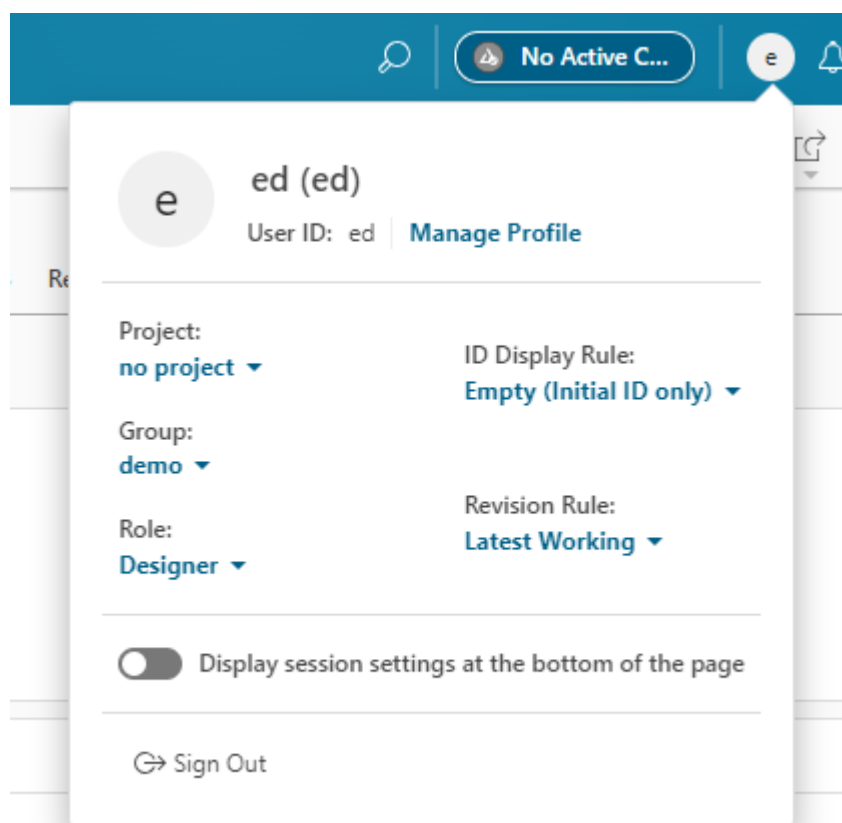
Note

For more information, refer to *About incorporating changes*.

Set change notice revision rule in user profile

Access the user profile by selecting the user icon or picture in the global navigation. The user profile shows the current user's role and capabilities, along with access to project and revision settings.

You can set the **Change Notice Revision** by selecting from the **Revision Rule** list in the user profile.

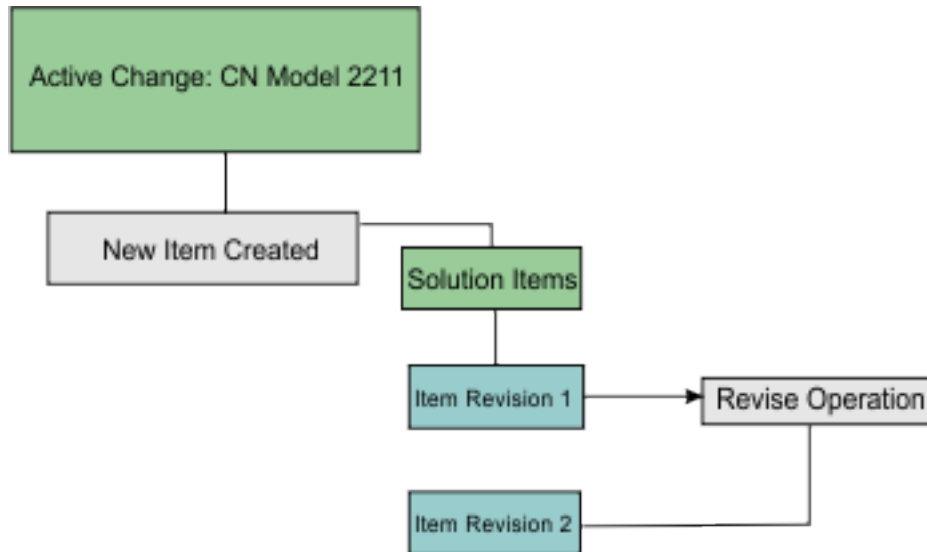


Procedure

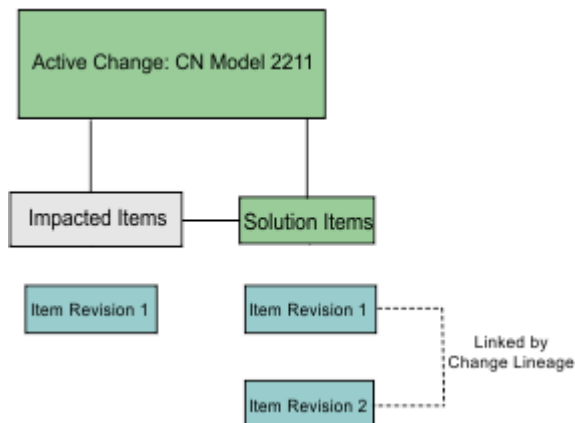
1. Open the user profile.
2. Under **Revision Rule**, select a rule from the list to apply the revision rule.

Examples

When a new item is created and CN Model 2211 is selected as the active change, its first **Item Revision** appears in the solution items of CN Model 2211. Likewise, if a revise operation is performed on an existing **Item Revision**, the newly-created item revision is also attached to CN Model 2211 as a solution item.



If the original item revision was already attached to CN Model 2211 as an impacted item, then in addition to adding the new item revision as a solution item, the original revision and the new revision are also automatically associated to each other with a change lineage.



Review impacted items

Impact analysis

Impact Analysis organizes the progression of a change object and all items associated with it. It automatically captures impacted candidates based on the problem items selected. This includes changes that are derived from another change containing impacted items. The derived change inherits all the impacted items.

Impacted items associated with a change are displayed in the **Overview**, **Affected Items**, and **Impact Analysis** tabs.

The **Impact Analysis** tab provides a view of all impacted items for a change object, in greater detail than the **Impacted Items** table in the **Overview** or **Affected Items** tabs. Impacted candidates are calculated for each problem item associated with a change, ensuring that all potential impacts are considered and addressed before proceeding with the change.

Each table contains details including type, relation, and status.

ECN-000409/A:1-Gimbal Spindle

OverviewAffected ItemsImpact AnalysisBOMParticipantsReference ItemsPlan ItemsImplementsWorkflowDependenciesRelationsReportsMass Update

Owner: charles (charles)Date Modified: 15-Oct-2024Release Status: Type Change Notice RevisionClosure: OpenDisposition: ApprovedECN-000409/A:1-Gimbal Spindle

Select a Problem Item and choose its candidates from the tables on the left to add impacted items. All impacted items will appear in the Impacted Items table.

Impact Analysis for Problem Item:
HDD-0540/A:1-Gimbal Spindle

▼ Impacted Candidates Items for HDD-0540/A:1-Gimbal Spindle

Revision Rule: Latest WorkingEffectivity: Parent ID:

Select ViewSelection ModeSelect AllConfigureImpactedNot ImpactedLegend: ImpactedNot Impacted

Object	Type	Source Object Type	Release
HDD-0540/A:1-Gimbal Spindle	Item Revision	Problem Item	
HDD-0511/A:1-RW Head Assembly	Item Revision	Assembly	
HDD-0538/A:1-Hard Drive Assembly 2	Item Revision	Assembly	
HDD-0527/B:1-Hard Drive Assembly	Item Revision	Assembly	
ACE_Master_Product/A:1	Item Revision	Assembly	

▼ Impacted Candidate Related Items for HDD-0538/A:1-Hard Drive Assembly 2

Select ViewSelection ModeSelect AllImpactedNot Impacted

Object	Type	Relation	Source Object	Rx
HDD_Model_Analysis_D.mp4	MISC	Specifications	HDD-0538/A:1-Hard Drive	
HDD_Model_Analysis_C.mp4	MISC	Specifications	HDD-0538/A:1-Hard Drive	
HDD-0538-A	UGMASTER	Specifications	HDD-0538/A:1-Hard Drive	
HDD Market Requirements.docx	MS WordX	Specifications	HDD-0538/A:1-Hard Drive	
Disk_Drive_assembly.mp4	MISC	Specifications	HDD-0538/A:1-Hard Drive	

▼ Impacted Items

Select ViewSelection ModeSelect AllRemove

Object	Type	Parent	Release
HDD-0538/A:1-Hard Drive Assembly 2	Item Revision		
HDD-0511/A:1-RW Head Assembly	Item Revision		

Note
Your administrator can enable and configure the preference **CM_WorkspaceObject_Impacted_Relations** to define items displayed in the **Related Items** table. See [Find related impacts for a problem item](#) for more information.

Impact analysis reduces the manual process of identifying potential impacts and the history related to a change object by assembling items from the **Affected Items** tab. The items added to an **Impacted Items** table are automatically added to all views.

Procedure

1. If there is more than one problem item for a change, select an item from the **Impact Analysis for Problem Item** list to view **Impacted Candidates Items**.
One or more items are displayed in the table below the list.

2. To add an **Impacted Candidates Item(s)** (*for the change notice*) to the **Impacted Items** table, select one or more objects in the **Impacted Candidates Items** table and click **Impacted** .

The **Impacted Items** table contains the impacted items for all problem items.

A yellow bar next to the name  indicates that it is added as an impacted item.

If you select an object in the **Impacted Candidates** table that has attached items, the attachment appears in the **Impacted Candidates Related Items** table below it. An example of an attached item is a reference document.

3. Click **Configure** . This lets you locate and view the impacted content by filtering the BOM structure by the problem item revision and its parent, and then configure a range of release effectivity (unit or date).

The **Configuration** panel opens. There are three filters available: **Revision Rule**, **Effectivity**, and **Parent ID**.

- a. Click the **Revision Rule** drop-down list box to select the revision rule that was set in the context of the user. It can be changed by selecting another revision rule from the drop-down list box.
- b. Click the **Effectivity** drop-down list box to select the range(s) of effectivity for the impacted content. The effectivity originates from the change notice.

Any assembly part that includes the specified range displays in the **Impacted Candidates Items** table. The effectivities shown are driven by the release effectivity set on the ECN revision/ECR revision. You can select multiple effectivity ranges from the drop-down list box.

- c. Click the **Parent ID** drop-down list box to select the ID of the level in the structure where you want to configure the impacted candidates. This is useful when your problem item is a part of multiple assemblies and you wish to configure the impacted candidates for a particular assembly. You can select multiple Parent IDs as well.

The **Parent ID** filter is only visible when Smart Discovery Indexing is enabled on the environment. This can be verified by the **QSEARCH_filter_queries_by_product_path** preference; your administrator has set it to true when Smart Discovery Indexing is enabled. The level on which Smart Discovery Indexing is done is shown in the drop-down list box. See [Enable the Parent ID selection](#) for more information.

The assembly parts that match all of the specified criteria then display in the **Impacted Candidates Items** table.

Configuration

Revision Rule: Latest Working

Effectivity: Click to add a value

Parent ID: HDD-0527/B-1-Hard Drive Asse... Clear (f)

Note: Configurations of multiple effectivities and multiple parent IDs are not yet supported.

Select a Problem Item and choose its candidates from the tables on the left to add impacted items. All impacted items will appear in the Impacted Items table.

Impact Analysis for Problem Item: HDD-0540/A-1-Gimbal Spindle

Impacted Candidates Items for HDD-0540/A-1-Gimbal Spindle

Object	Type	Source Object Type	Release
HDD-0540/A-1-Gimbal Spindle	Item Revision	Problem Item	
HDD-0511/A-1-RW Head Assembly	Item Revision	Assembly	
HDD-0538/A-1-Hard Drive Assembly 2	Item Revision	Assembly	
HDD-0527/B-1-Hard Drive Assembly	Item Revision	Assembly	
ACE_Master_Product/A/1	Item Revision	Assembly	

Impacted Candidate Related Items for HDD-0540/A-1-Gimbal Spindle

Object	Type	Relation	Source
HDD-0540-A	UGMASTER	Specifications	HDD-0540/A

Impacted Items

Object	Type	Parent
HDD-0538/A-1-Hard Drive Assembly 2	Item Revision	
HDD-0511/A-1-RW Head Assembly	Item Revision	

Note

The preference **QSEARCH_filter_queries_by_product_path** must be set to true (the default) by your administrator to view the **Configure** panel.

4. If a problem item is used in multiple impacted assemblies, not all of them may be impacted with a specific change. To indicate that an object is not an impacted candidate, select the object and click **Not Impacted** from the **Impacted Candidates Items** table.

A green-color bar next to the name  indicates that the candidate is a non-impacted item in the change notice.

If the item revision has any related items, you can also select non-impacted items from the **Impacted Candidates Related Items** table.

To revert the non-impacted status, click **Impacted**. A yellow indicator shows the object is impacted.

Relations view

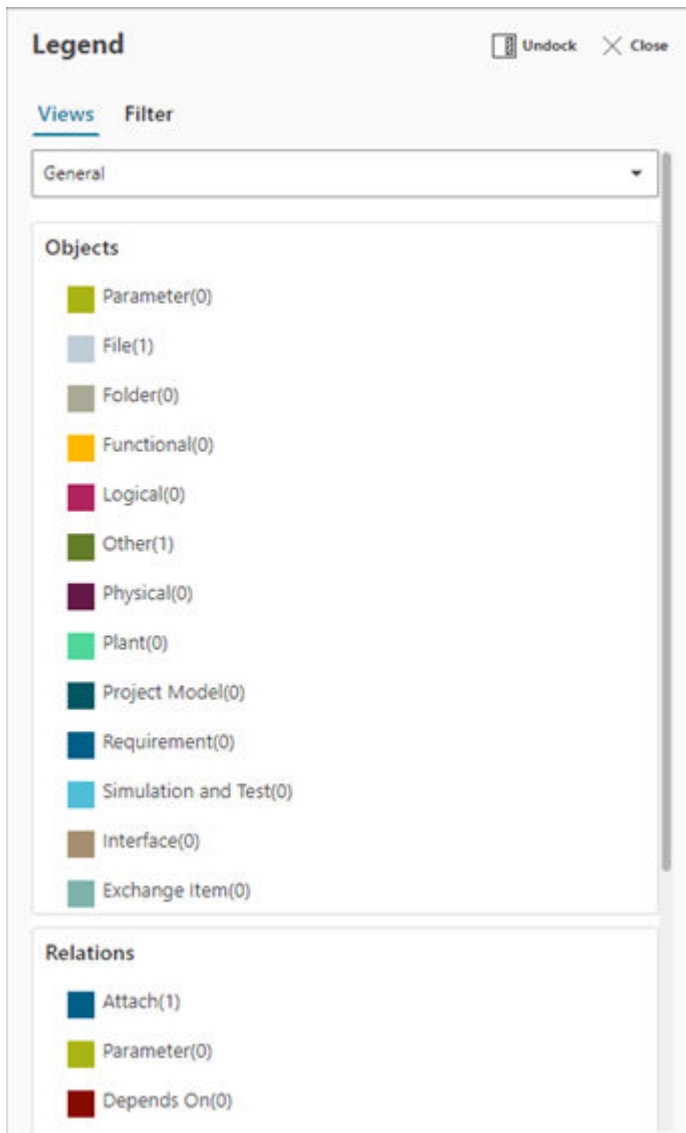
Procedure

1. Click **Select View**  to toggle between **Table View** or **Relations View**.

The **Relations View** is similar to the **Relations** tab, but displays the relations for individual **Problem Items**.

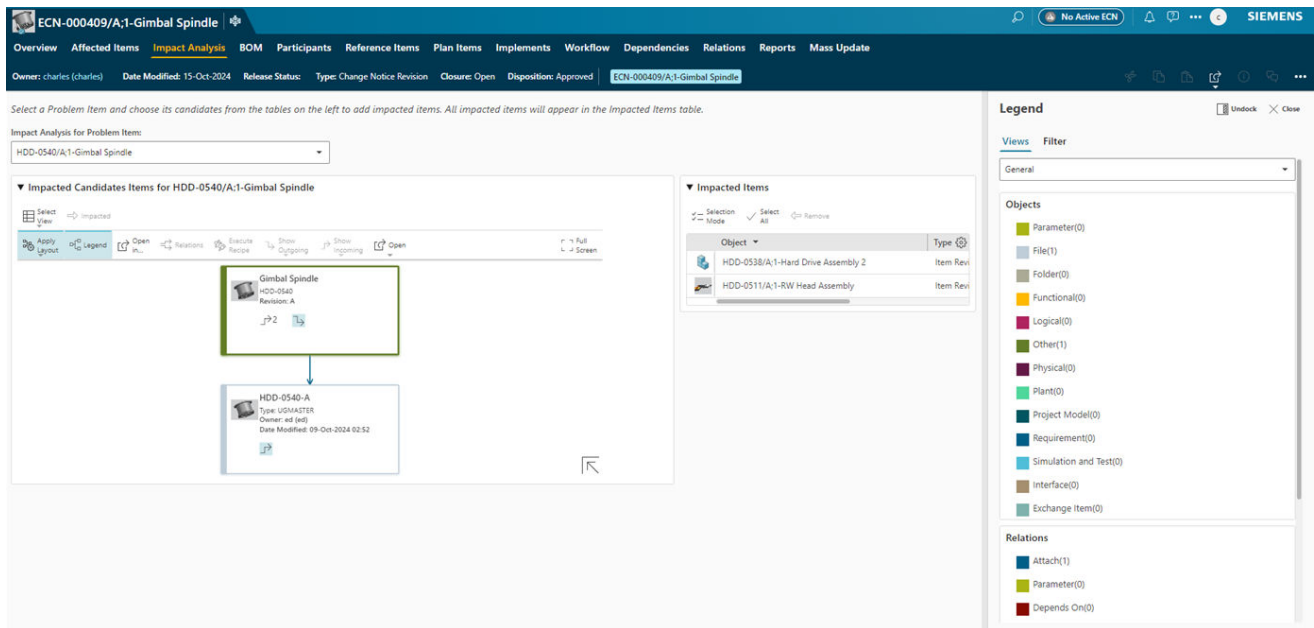
2. When in **Relations View**, click **Legend**  to filter and view the object relationships.

The **Legend** panel contains all object and relationship possibilities as determined by the type of impact analysis listed in the list.



The **Legend** displays the **Change Impact Analysis** view of the selected **Problem Item**. You can switch between the **Change** or **Change Impact Analysis** view of this item without having to navigate to the **Relations** tab.

3. Select a view from the list to alter the types of **Objects** and **Relations** listed and change the relationship view.

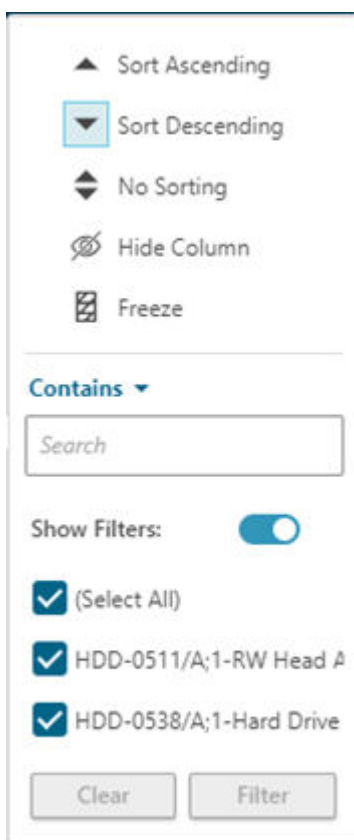


Filtering and Sorting

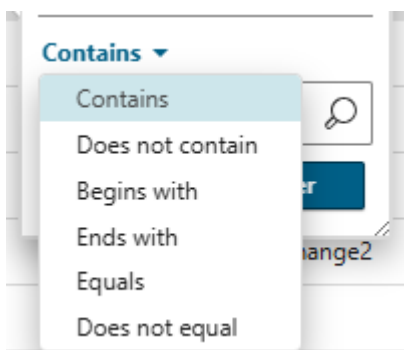
Each table in the **Impact Analysis** tab contains filtering and sorting capabilities.

Procedure

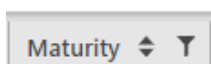
1. Click on a column heading to open the sorting and filtering panel for each column.



2. If available, click **Show Filters** to expand the available options.
Depending on the selected column, and your configuration, the options available for sorting and filtering will vary.
3. Select one or more items from the list and click **Filter**.
4. The **Search** list shows the parameters to search on specific criteria. The default value is **Contains**. Enter your text in the **Search** box and click **Filter**.



The filter icon  displays next to the column heading. Hovering over the icon shows the filter criteria.



Select the column and click **Clear** to remove the filter.

Identifying Part Usages

Part usage in a BOM refers to the instances or occurrences of a part within the BOM structure. In an engineering BOM (EBOM), there is only one actual part, and all copies assigned to the manufacturing BOM (MBOM) are considered part usages. These instances do not have separate lifecycles; any change to the actual part affects all of its occurrences in the MBOM.

On the **Impact Analysis** tab, the **Usage Part** column in the **Impacted Candidates Items** table reflects this information. Although hidden by default, you can add this column to the table.

Procedure

1. Click **Table Settings**  > **Arrange** .
The **Arrange** panel opens.
2. Select **Part Usage** from the **Available Columns** list, then click **Add** to move it to the **Displayed Columns** list.
3. Click **Save and Arrange**.

Results

When you move an impacted candidate to the **Impacted Items** table in an EBOM, the usage revision appears in the **Part Usage** column of the **Impacted Items** table.

In the **Impacted Items** table, the **Type** column identifies it as a part usage, and the **Part ID** listed in the column identifies the part usage between the parent and the child objects. The **Parent** column displays the Parent ID of the usages from which the impacted items originate.

When a usage part is included in multiple structures in a change notice, each part usage is listed under the **Parent** column. This lets you identify which usage is impacted by the part.

Enable the Parent ID selection

Smart Discovery Indexing must be enabled for each change notice to utilize the **Parent ID** drop-down list box. Once enabled, you must index each of your Parent IDs to view them in the **Configure** panel of the **Impact Analysis** tab.

Note

Learn more about Smart Discovery Indexing.

Procedure

1. Open the Teamcenter command prompt.

2. To enable smart_discovery_index service (Active Workspace Indexing Service), run the following command:
`smart_discovery_index -u=tcadmin -p=tcadmin -g=dba -enable`
3. To index a product ID of a BOM, run the following command:
`smart_discovery_index -u=tcadmin -p=tcadmin -g=dba -create -product=<prod ID>`
 For example, if product=209607 is the assembly ID in the BOM, run:
`smart_discovery_index -u=tcadmin -p=tcadmin -g=dba -create -product=029607`
4. After the Product IDs are identified, sync them by running the following command:
`cd C:\Apps\tc\tc2406\TR\TcFTSIndexer\bin`
`runTcFtsIndexer -task=discovery:sync`
5. To see a list of the indexed products, run the following command:
`smart_discovery_index -u=tcadmin -p=tcadmin -g=dba -list`

Find related impacts for a problem item

The **Impacted Candidate Related Items** table lets you associate various items or documents that are relevant to a particular change request or change order. It serves as a centralized location to track and manage all the items that are affected by or have a relationship with the change being proposed or implemented. By listing all of the related items, you can analyze the impact analysis more carefully since it shows how the change will affect other parts of the product.

Note

Your administrator can enable and configure the preference **CM_WorkspaceObject_Impacted_Relations** to define items displayed in the **Related Items** table. This preference only functions with revisable objects.

For example, if the problem item has a document revision attached via an IMAN_specification relation, upon selecting the problem item in the **Impacted Candidates Items** table, the document revision is displayed as a related object in the **Impacted Candidate Related Items** table.

Select the part or item in the **Impacted Candidate Related Items** table to view its related items. You can then add the related items to the **Impacted Items** as necessary.

▼ Impacted Candidates Items for HDD-0540/A;1-Gimbal Spindle

Revision Rule: Latest Working Effectivity: Parent ID:

Select View Selection Mode Select All Configure Impacted Not Impacted Legend: ■ Impacted ■ Not Impacted

Object	Type	Source Object Type	Release ...
HDD-0540/A;1-Gimbal Spindle	Item Revision	Problem Item	
HDD-0511/A;1-RW Head Assembly	Item Revision	Assembly	
HDD-0538/A;1-Hard Drive Assembly 2	Item Revision	Assembly	
HDD-0527/B;1-Hard Drive Assembly	Item Revision	Assembly	
ACE_Master_Product/A;1	Item Revision	Assembly	

▼ Impacted Candidate Related Items for HDD-0538/A;1-Hard Drive Assembly 2

Selection Mode Select All Impacted Not Impacted

Object	Type	Relation	Source Object	Re
HDD_Modal_Analysis_D.mp4	MISC	Specifications	HDD-0538/A;1-Hard Drive	
HDD_Modal_Analysis_C.mp4	MISC	Specifications	HDD-0538/A;1-Hard Drive	
HDD-0538-A	UGMASTER	Specifications	HDD-0538/A;1-Hard Drive	
HDD Market Requirements.docx	MS WordX	Specifications	HDD-0538/A;1-Hard Drive	
Disk_Drive_assembly.mp4	MISC	Specifications	HDD-0538/A;1-Hard Drive	

View multiple levels in the Related Items table using closure rules

You can view multiple levels of an assembly in the **Related Items** table by specifying closure rules. *Closure rules* are a set of predefined conditions that specify how to traverse the data structure and define relationships that are of interest and what to do when these relationships are encountered. They identify how to access and process certain relationships in the data structure, such as item revisions, datasets, variant expressions, and classifications.

When used in conjunction with an aligned design or part structure, closure rules let you display every related item of an impacted candidate, regardless of whether it is attached to a particular change notice. This additional data can help to identify possible impacted candidates for your problem item.

Select a Problem Item and choose its candidates from the tables on the left to add impacted items. All impacted items will appear in the Impacted Items table.

Impact Analysis for Problem Item:

052239/A;1-Bearing Part

▼ Impacted Candidates Items for 052239/A;1-Bearing Part

Revision Rule: Latest Working

Effectivity: Parent ID:

Legend: ■ Impacted ■ Not Impacted

Impacted Not Impacted Configure Selection Mode Select All Select View

Object	Type	Source Object Type	Release Status	
▼ 052239/A;1-Bearing Part	Part Revision	Problem Item		
052231/A;1-Bearing Design	Design Revision	Assembly		

▼ Impacted Candidate Related Items for 052231/A;1-Bearing Design

Legend: ■ Impacted ■ Not Impacted

Impacted Not Impacted Selection Mode Select All

Object	Type	Relation	Source Object	Rt
052249/A;1-Bearing Part Document	Document	Specifications	052231/A;1-Bearing Des...	
052244/A;1-Bearing Design Document	Document	Specifications	052231/A;1-Bearing Des...	
052231-Bearing Design	Design		052231/A;1-Bearing Des...	

In this example, problem item **Bearing Design** is a part revision in the **Impacted Candidates** table. Its design counterpart, **Bearing Design**, is then shown in the **Impacted Candidate Related Items** table. Since the **Bearing Design Document** and **Bearing Part Document** are related to the **Bearing Design** design revision, those documents are also automatically included in the table as a related item. Closure rules enabled these multiple related items to display.

Closure rules are created using the PLM XML application. Change Management provides the following closure rules OOTB to use with the **Related Items** table:

ClosureRule

Traversal Rule Name:CMImpactAnalysisClosureRule

Description:CMImpactAnalysisClosureRule

Scope of Traversal:☒ Export ☐ Import

Output Schema Format:PLMXML

Prim...	Primary Object	Seco...	Secondary Obj...	Relation Type	Related Property Or Object	Action Type	...	Predicate
CLASS	Design Revision	CLASS	Dataset	RELATIONP2S	IMAN_specification	PROCESS+ TRAVERSE		
CLASS	Dataset	CLASS	Dataset	RELATIONS2P	IMAN_UG_wave_part_link	PROCESS+ TRAVERSE		
CLASS	Dataset	CLASS	Dataset	RELATIONP2S	IMAN_UG_wave_part_link	PROCESS+ TRAVERSE		
CLASS	Dataset	CLASS	Design Revision	RELATIONS2P	IMAN_specification	PROCESS		
CLASS	Part Revision	CLASS	Design Revision	RELATIONP2S	TC_Primary_Design_Representation	PROCESS+ TRAVERSE		
CLASS	Design Revision	CLASS	ItemRevision	RELATIONP2S	IMAN_specification	PROCESS		
CLASS	Design Revision	CLASS	Dataset	RELATIONP2S	IMAN_specification	PROCESS		
CLASS	Design Revision	CLASS	Design	ATTRIBUTE	items_tag	PROCESS+ TRAVERSE		
CLASS	Design	CLASS	Design Revision	RELATIONS2P	IMAN_UG_wave_part_link	PROCESS		
CLASS	Design Revision	CLASS	Design	RELATIONP2S	IMAN_UG_wave_part_link	PROCESS+ TRAVERSE		
CLASS	Design	CLASS	Design Revision	REFBY	items_tag	PROCESS		
CLASS	Design Revision	CLASS	Dataset	RELATIONS2P	catiaV5_MML	PROCESS+ TRAVERSE		
CLASS	Dataset	CLASS	Design Revision	RELATIONP2S	catiaV5_MML	PROCESS		

You can modify/update the existing **CMImpactAnalysisClosureRule** closure rule to add or update closure rule clauses.

Note

Change Management supports both preference and closure rule mechanisms to retrieve related items for impacted candidates.

Revise problem or impacted items

You can revise impacted and problem items on a change notice without opening the objects separately.

Owner: ed (ed) Date Modified: 19-Sep-2025 Release Status: Type: Change Notice Revision Closure: Open Disposition: Approved

Overview

Affected Items

Impact Analysis

Participants

Reference Items

Workflow

Dependencies

Relations

Reports

Mass Update

Show in Single Table:☐

Solution Items (3)

Object	Lineage	Type	Proposed Desi...	Release Status	Cancelled	Date Released	Owner
047675/A/1-Bike		Item Revision					ed (ed)
047679/D/1-Seat - 1	1	Item Revision					ed (ed)
047680/C/1-Pedals - 4	2	Item Revision					ed (ed)

Impacted Items (3)

Object	Lineage	Type	Requested Change	Concurrent Changes	Release Status	Date Released	Owner	Disposition	Incorporation Status
047675/A/1-Bike		Item Revision					ed (ed)	Use As Is	Unincorporated
047679/C/1-Seat	1	Item Revision					ed (ed)	Use As Is	Unincorporated
047680/B/1-Pedals	2	Item Revision					ed (ed)	Use As Is	Unincorporated

Problem Items (1)

Object	Type	Release Status	Date Released	Owner
047675/A/1-Bike	Item Revision			ed (ed)

Note the following:

- When you revise an impacted item, the new revision becomes the solution item for the same change notice.
- If you revise a problem item, the new revision created becomes a solution item for the same change notice and the previous revision becomes an impacted item.
- When you perform a save as operation on a problem item or impacted item, the new item created becomes a solution for the change notice.
- Turn on **Show in Single Table** to consolidate all tables on the **Affected Items** tab into a single table, making it easier to track and update item statuses, changes, and related information without navigating through multiple sections. You can add additional impacted, problem, or solution items to the change notice via the **Add** ⊕ button.
- Icons indicate the relation type.

Overview	Affected Items	Impact Analysis	Participants	Ref
Show in Single Table: ☒				
This table contains Problems, Impacted and Solutions.				
Object ▲		Change Relation ⇅		
	035795/A;1-035795__PSI		Impacted	
	035795/A;1-035795__PSI		Problems	
	035795/B;1-035795__PSI		Solutions	

Revise Problem Items

Procedure

1. Open the change notice containing the affected items.
2. Click **Affected Items**.
3. Select the **Problem Item** to revise.
4. Click **Revise to Solution**  from the **Problem Items** toolbar. Or, right-click and select **Revise to Solution** .
Select the appropriate options and include a description, if necessary.
5. Click **Revise**.
A new item is created, which is a solution for the change notice.

Revise Impacted Items from the Affected Items tab

Procedure

1. Open the change notice containing the affected items.
2. Click **Affected Items**.
3. You can select one or more **Impacted Items** to revise.
4. Click **Revise to Solution**  from the **Impacted Items** toolbar. Or, right-click and select **Revise to Solution** .
If you select multiple items, a message appears, stating whether the revision is a success or a failure. Revisions that are successful appear in the **Solution Items** section.
5. Complete the options, and click **Revise**.
A new solution item is created for the change notice.

Revise Impacted Items from the Overview tab

Procedure

1. Open the change notice containing the affected items.
2. On the **Impacted Items** table on the **Overview** tab, select the **Impacted Item** to revise.
3. Click **Revise to Solution**  from the **Impacted Items** toolbar. Or, right-click and select **Revise to Solution** .
4. Complete the options, and click **Revise**.
A new solution item is created for the change notice.

Save a copy of an Impacted Item from the Affected Item tab

You can save a copy of an impacted item to create a new version of the item that reflects changes or updates needed for a change notice. This lets you maintain a record of the original item while creating a new version that incorporates necessary modifications. Advanced copy options let you specify which properties and related items to include in the new copy.

Procedure

1. Open the change notice containing the affected items.
2. On the **Impacted Items** table on the **Overview** tab, select the **Impacted Item** to revise.
3. Click **Save As to Solution**  from the **Impacted Items** toolbar. Or, right-click and select **Save As to Solution** .
4. In the **Save As** panel, specify the required properties.
5. Specify the required properties in **Advanced Copy Options** or retain the default values.

A new copy of the solution item is created for the change notice.

6. Click **Save**.

Perform a task in the Changes Overview tab

In addition to performing an assigned task from your Inbox, you can perform a task in the **Overview** tab of a change.

Procedure

1. Select **Changes**  to view all open changes.
2. Select a change from the list. If not already selected, click the **Overview** tab. The current assigned task is listed under **Task to Perform**.

▼ Task to Perform

Workflow:

CN Fast Track Process : ECN-000111/A;1-Change A

Name:

Elaborating

Task Instructions:

Open the Change Notice and on the Overview tab, do the following:

Describe the changes needed in the Synopsis and Description under Details

Set the Release Effectivity (optional)

Assign the Change Specialist, Contributor(s) and Approver(s) under Participants

Then, decide either to:

Cancel

Complete

Workflow Description:

CN Fast Track Process : ECN-000111/A;1-Change A

Comments:

Cancel

Complete

3. Review the instructions and complete the task. Similarly, complete any additional assigned tasks.

- (Optional) A change object could belong to more than one workflow. The workflow list displays the name of the currently selected workflow.

Select a workflow from the list to see the task to be performed for that workflow.

▼
TASK TO PERFORM

Workflow:

CN Process : ECN-000025/A;1-WF dropdown test3

CN Process : ECN-000025/A;1-WF dropdown test3

10.1.2 Simple Review No Profile : ECN-000025/A;1-WF dropdo...

Task Instructions:

Assign the Change Specialist 1 participant on the CN revision. Complete task when assignment done.

Workflow Description:

WF dropdown test3

Comments:

Complete

Make mass updates to a structure

You can perform mass updates to a structure by using a change notice to propose, manage, and apply changes to multiple impacted assemblies. You can perform actions like adding, substituting, or replacing items, and then submitting these markups through a workflow for approval and implementation.

Procedure

- From **Launcher**, click **Changes** .
- Search for the change notice you want to work with, select the change notice, and click **Open** .
- Click the **Affected Items** tab. In the **Problem Items** section, click **Add to** .
- Click the **Mass Update** tab.

The **Mass Update** tab is not visible when the change notice is in the **Cancelled** or **Closed** state.

Even if multiple problem items are associated with the same change notice or the same problem item is associated with multiple active change notices, you can propose mass update actions for the impacted items of all problem items through the **Mass Update** tab. These actions are applied together as part of a single workflow.

The problem item that you added from the **Affected Items** tab is shown in the **Problem Item** list under the **Impacted Assemblies** section.

All parent assemblies of the selected problem item are listed in a table under the **Impacted Assemblies** section.

5. In the list of impacted assemblies, select the row in which you want to make the update. Only assemblies with the status **Released** can be updated.
6. Choose **More Commands ***** > **Edit**  > **Start Edit**  from the **Page level** toolbar.
7. Click in the **Action** column and select the required action from the list. Only the actions available for the selected row are displayed. The selected action for the row is highlighted.
8. For **Add**, **Substitute**, or **Replace** actions, you can add the required item to the **Proposed Item** column using **Search** or **Palette**.
9. Click **Save Edits**  after updating the required rows.
10. To delete a saved markup, select the row, and select **None** from the **Action** column. The markup is deleted and the assembly is removed from the list of affected items.
11. Set the engineering change notice as Active Change.
12. (Optional) Click the **Affected Items** tab to view the list of items that are marked up for update.
13. Select the affected item that you want to verify from the list and click **Open**.
14. To verify the markup, choose **More Commands ***** > **Markup** .
15. Click the **Mass Update** tab. In the **Mass Update** tab, you can revert the markups before they are applied using the workflow.
16. To initiate applying the markup, click **Submit to Workflow** .

The **Submit to Workflow** panel with a list of workflow templates is displayed.
17. Select the **Review and Apply BOM Markups** workflow template and click **Submit**.
18. Assign the reviewers for the workflow.
19. Once the workflow is approved, the changes are applied. To verify the changes, go to **Affected Items**, select the row, and click **Show Markup** .

Derive a change and send for resolution

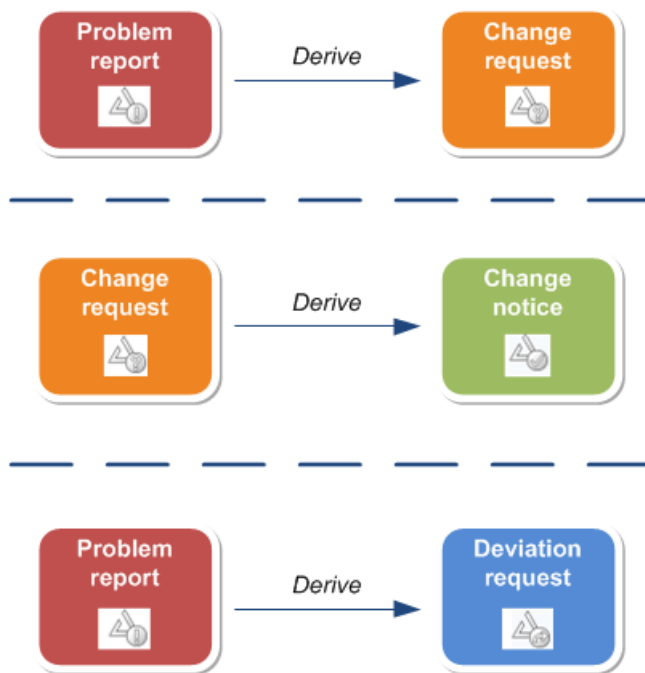
Derive a change from another change

Once you have identified a problem and investigated it, you can derive a change object into the next phase of the change process.

The following examples represent the **Derive** process:

- Derive a change request from a problem report to determine a solution for the problem.
- Derive a change notice from a change request to implement the solution to the problem.
- Derive a deviation request from a problem report to allow a deviation.

In addition, if your company has custom changes, you can derive one custom change from another custom change.



- The change must be in the correct change state for you to derive another change from it.
Permissible change states are controlled by condition rules and may be customized by a system administrator.
- Depending on your setup, the reference, problem, or impacted objects associated with the source change (its relationships) may be automatically added to the derived change. See your system administrator for more information.

Derive a Problem Report to a Change Request

You can derive a change request from a problem report to determine a solution for the problem.

Procedure

1. Select the relevant PR.
2. Go to **More Commands** *** > **New** ✱ > **Derive Change** ⚙ from the **Page level** toolbar.
3. In the **Derive Change** panel, select **Change Request** or **Deviation Request**, if available.
4. Enter the following information:
 - CR/DR number (if you want to change the number automatically displayed)
 - Revision number (if you want to change the number automatically displayed)
 - Synopsis
 - Description (optional)
 - For a deviation request, enter the deviation type.
 - (Optional) To submit the new change to a workflow, select the workflow template from the **Workflow** drop-down list. Templates display based on the type of the object selected.

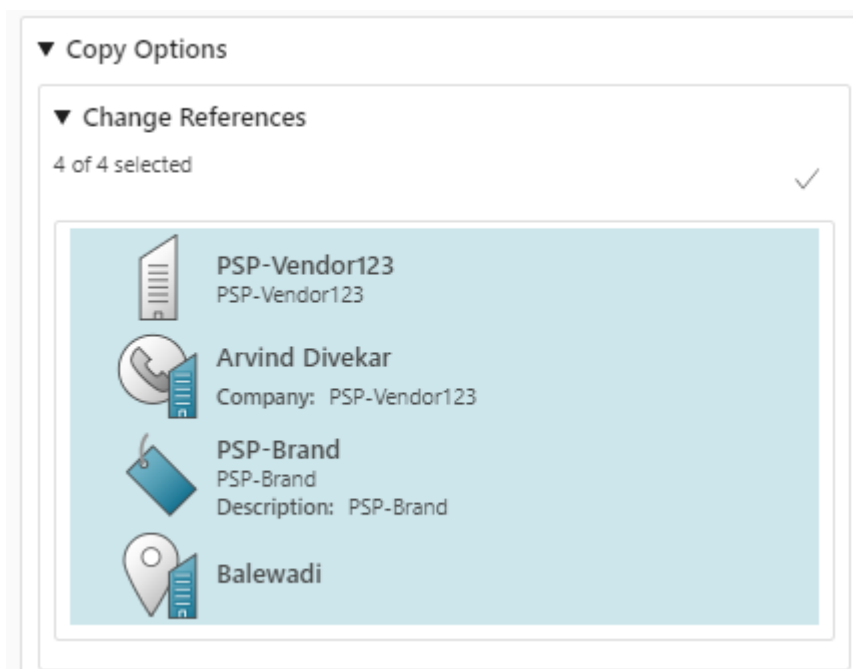
If a default workflow exists for the content type, it is automatically specified as the workflow template.

The Implements section shows the source change items.

5. (Optional) To copy **Affected Items** and **Reference Items** from a problem report to a change request, click the **Copy Options** label.

For more information on configuring **Copy Options**, refer to [Define deep copy rules for copying options from an ECR to an ECN](#).

When you click **Copy Options**, the section expands and shows each category of objects that are included in the problem report. Select or clear the items you wish to copy into the change request.



6. Click **Derive** to create the item, or click **Derive and Submit** to submit for resolution. If you select **Derive**, the new change opens in **Edit** mode. Make any necessary changes and click **Save Edits**.
If you select **Derive and Submit**, the new change is submitted to workflow.

Derive a single Change Request to a Change Notice

You can derive a change notice from a change request to implement the solution to the problem.

Procedure

1. Select the relevant change request.
2. Go to **More Commands** *** > **New** ✨ > **Derive Change** 📌 from the **Page level** toolbar.
3. Change the properties, if necessary.
The properties for the derived change are automatically filled based on the values of the initial change. In addition, any objects associated with the initial change (its relations) are automatically added to the derived change, depending on how Active Workspace is set up. Active Workspace also associates the objects appropriately.

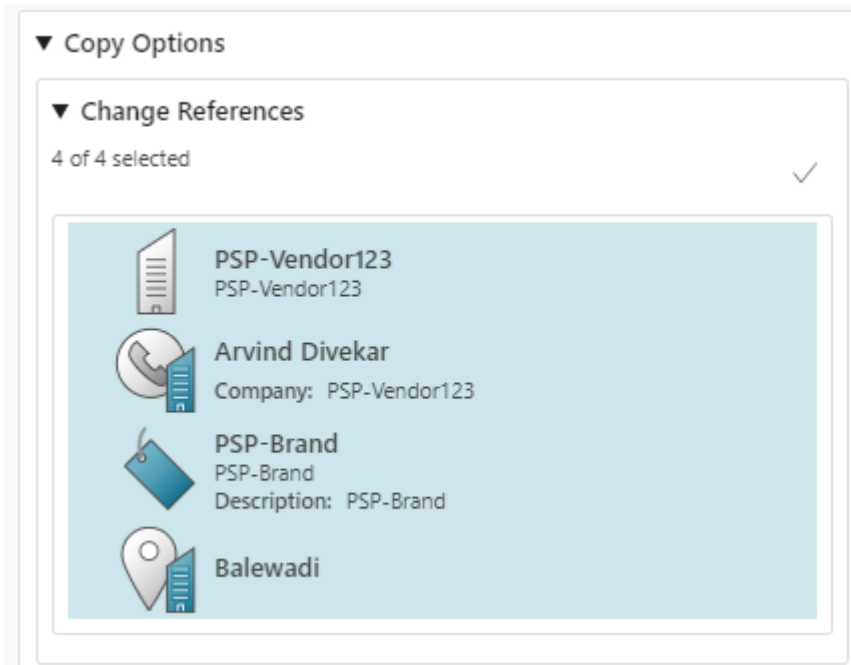
(Optional) To submit the new change to a workflow, select the workflow template from the **Workflow** drop-down list. If a default workflow exists for the content type, it is automatically specified as the workflow template.

The **Implements** section displays the source change request of the change notice.

4. (Optional) To copy **Affected Items** and **Reference Items** from a change request to a change notice, click the **Copy Options** label.

Refer to [Define deep copy rules for copying options from an ECR to an ECN](#) for more information on configuring **Copy Options**.

When you click **Copy Options**, the section expands and shows each category of objects that are included in the change request. Select or clear the items you wish to copy into the change notice.



5. Do one of the following:
- To continue editing, click **Derive**. You can send it for resolution later.
 - To send it for resolution immediately, click **Derive and Submit**.

If you select **Derive**, the new change will open in **Edit** mode. Make any necessary changes and click **Save Edits**.

If you select **Derive and Submit**, the new change is submitted to workflow.

Check your **Inbox** for tasks to perform.

Participants to review and approve the change may be automatically assigned depending on how your organization's change process is configured.

Results

The relationship between the initial change and the change it is derived from is shown in the **Implements** and **Implemented by** section of the **Reference Items** tab. The **Implements** object is the object this change was derived from, and the **Implemented By** object refers to a change object derived from this one.

You can click any of the changes to open and view them.

Derive a single change from multiple change objects

You can derive a single change from multiple, less-mature change objects.

Procedure

1. Select multiple change objects of the same type to use in creating the new change notice or change request.

Note

You cannot derive multiple change objects of different types.

The selected items are highlighted.

2. Go to **More Commands** *** > **New** ✨ > **Derive Change**.
3. In the **Derive Change** panel, select the type of change, if available.

Note

Change requests can only be derived into a change notice.

4. Enter the following information for a change request:
 - CR/CN number (if you want to change the number automatically displayed)
 - Revision number (if you want to change the number automatically displayed)
 - Synopsis
 - Description

Note

In Active Workspace 6.2/Teamcenter 14.2, the Description is optional. In all other versions, it is required.

- For a deviation request, enter the deviation type.
- (Optional) To submit the new change to a workflow, select the workflow template from the **Workflow** drop-down list. Templates display based on the type of the object selected.

Note

When you derive a change notice from multiple source change requests, you cannot edit the **Problem Items** or **Impacted Items** as you would when deriving a single change. All of the **Affected Items** are copied into the new change notice.

The **Implements** section shows all of the source change items.

5. Click **Derive** to create the item, or click **Derive and Submit** to submit for resolution.

If you select **Derive**, the new change opens in the **Edit** mode. Make any necessary changes and click **Save Edits**.

If you select **Derive and Submit**, the new change is submitted to the workflow.

Fast track a change notice

A fast track process eliminates the formal review process to implement a change. **Fast Track** is available from the editing menu or available during the change process workflow.

Refer to the *Change Manager* guide for more information and examples of both fast track and standard track.

Edit a change request manually to enable Fast Track

You can edit a change manually to send it through Fast Track.

Procedure

1. Click **Changes** , select a change notice, and expand the **Details** section.
2. Go to **More Commands** *** > **Edit**  > **Start Edit** or click **Start Edit**  from the **Page level** toolbar.
3. In the **Fast Track** list, select **Yes**.
If you select **No**, **Fast Track** is disabled when a change notice is derived from this change request. If left blank, Fast Track is still available during workflow.
4. Click **Save**.

Enable Fast Track during workflow processing

If a change notice was not derived from a change request, or if the change request did not have **Fast Track** enabled, **Fast Track** can be applied during the workflow process or as part of a custom workflow template.

Enabling **Fast Track** on a change request or change notice during workflow is built into the workflow templates for both change requests and change notices.

If a change request or change notice was sent to workflow, during the task **Specialist Planning Complete?** you can select **Plan OK - Fast Track**.

OverviewAttachmentsWorkflowAudit LogsAssignments

▼ ACTION

Name:Specialist: Planning Complete?

Description:Is enough information available to go to the CRB? Does the CR fit the Fast Track criteria? Please complete these fields:

- Fast Track
- Technical Review
- Priority
- Reason

Comments:

Plan Not OK - Reject

Plan Not OK - Rework

Plan OK - Fast Track

Plan OK - Standard Track

▼ TARGETS

✓

—

ECR Fast Track

ECR-000021

Revision: A

▼ PREVIEW

The result is shown in Workflow Viewer.

```
graph LR
    Start([Start]) --> Task1[Specialist: Planning Complete?]
    Task1 -- "Plan OK - Fast Track" --> Task2[Specialist: CRB Assignment]
    Task1 -- "Plan OK - Standard" --> Task2
    Task2 --> Task3[Change Review Board (CRB): Review Proposed Solution]
    Task3 --> Task4[Specialist: Set Change Review Board Decision]
    Task4 -- "Approve" --> Task5[Specialist: Derive CN, Submit CN]
    Task4 -- "Reject" --> Task6[Specialist: Close PRs]
    Task4 -- "Re-work" --> Task2
```

Change Summary

Change Summary overview

The **Change Summary** is included in the **Overview** tab of a change notice or simple change that contains **Solution Items** and revisions. Click  to view the **Change Summary** table full screen.

3-44
Software Version 2606

Beta Release

Change Management on Active Workspace — Usage
Unpublished work. © 2025 Siemens

Note

Your administrator can **configure the functionality** of the Change Summary, such as activating features like indicator columns, detailed Change panels, and filters for various change types.

The Change Summary table provides information about the ID, name, action, revision, and quantity, and also provides visual cues in red and green for modifications and revisions.

Change Summary Overview ECN-000011/A:1-ECN Truck											
Impacted Items: All				14							
Changes for		Merge Stat	Action	Element	ID	Revision	Revision Name	Substitute and substitute group	Description	Find Number	Quantity
			Modified	Truck Aux Engine Commercial Part	033887	B A	Truck Aux Engine Commercial Part				
			Modified	Truck Aux. and Power Supply (S)	3-004-006610	C A	Truck Aux. and Power Supply (S)				
			Added Existing	Air Intake Part	033886	A	Air Intake Part			90	
			Modified	Complete Battery Pack	033855	B A	Complete Battery Pack				
			Added Existing	Fuel Commercial Part	033871	C	Fuel Commercial Part			80	
			Removed	Fuel Commercial Part	033674	C	Fuel Commercial Part			60	
			Modified	Fuel and Ad-Blue Hoses (S)	3-004-006637	E A	Fuel and Ad-Blue Hoses (S)			25 20	10
			Property Modified	Fuel Line (4 Cyl) with End Stop	3-004-006149	A	Fuel Line (4 Cyl) with End Stop			180	10
			Property Modified	Fuel Line for 3Cyl Small Tank	3-004-005930	A	Fuel Line for 3Cyl Small Tank	033881/A 033056/A 033860/A		10	
			Substitute Added New	Fuel Line New Substitute	033881	A	Fuel Line New Substitute			10	
			Substitute Removed	Fuel Line Substitute For Small T	033056	A	Fuel Line Substitute For Small Tank			40	
			Set Primary	Fuel Line voor 4L Range Subs...	033864	A	Fuel Line voor 4L Range Substitute	3-004-005795/A		170	
			Unset Primary	Fuel Line voor 4L Range Exten...	3-004-005795	A	Fuel Line voor 4L Range Extender	033864-Fuel Line voor 4L Range Ex		170	
			New	Truck Power Supply Document	033884	A	Truck Power Supply Document				

For example, an assembly can be attached as an impacted item revision on a change notice, and that assembly is revised using **Revise Impacted** in the **Impacted Items** table. You can attach impacted and solution items and then select the solution item and relate it to the impacted item.

The **Change Summary** table is displayed if one of these operations occurs:

- Set lineage on a solution item.
- Perform **Revise Impacted** on an impacted item.
- Create, remove, change quantity or effectivity dates (BOM changes).

Note

Your administrator can modify the preference **CM_bomline_tracked_properties** to define BOM line properties for tracking. See [Configure Change Summary columns](#).

Depending on your organization, different column headings may appear. **ID** and **Name**, however, are always displayed.

If you select one or more items in the **Impacted Items** table, the Change Summary displays those changes.

Use the searchable text filter to type in keywords and narrow down the Change Summary, displaying only solution items related to the selected impacted items.

Use the drop-down menu to select the type of red line changes you want to view:



- **All:** View all changes (the default)
- **Merged:** View changes that have been merged from other change notices. By default, this option displays the revised assemblies. If merged changes exist, they will be shown; otherwise, the assemblies will appear without any children.
- **Current:** View changes made in the current (local) change notice

Your administrator can modify the preference **CM_change_summary_view_redlines** to specify whether the Change Summary table displays specific red lines.

Change Summary action types

The Change Summary table displays the following action types in the Action column:

- **Modified:** The item properties were modified
- **Added New:** A new item was added
- **Added Existing:** An existing item was added
- **Created:** A new item was created
- **Removed:** An item was removed
- **Replaced:** An item was replaced
- **Added New to Replace:** A new item was added to replace an existing item
- **Added Existing to Replace:** An existing item was added to replace another item
- **Save as:** Saves an assembly as a new item to create a solution item with automatic lineage and change tracking
- **Save as to Split:** Saves an assembly as a new item with replace and split effectivity
- **Save as to Replace:** Saves an assembly as a new item and replaces the original
- **Revert:** Undo redline changes to an item, returning it to the original impacted version.
- **Property Modified:** Item or occurrence properties were modified
- **Substitute Used:** A substitute was used
- **Substitute Use Removed:** A substitute use was removed
- **Substitute Added New:** A new substitute was added
- **Substitute Added Existing:** An existing substitute was added
- **Substitute Removed:** A substitute was removed
- **Set Primary:** A substitute was set as primary

- **Unset Primary:** A substitute was unset as primary

Change Summary indicators

Indicators help track modifications to your assembly, whether structural (like adding, removing, or replacing components, updating properties, or managing substitutes) or non-structural (such as changes to attachments and vendor parts).

These indicators take you directly to the corresponding details of the changes made for:

- Properties (item revision properties and occurrences) 
- Substitutes 
- Attachments 
- Vendor parts 



Click any of the indicators to open its corresponding **Details of Change** pane.

Change Summary | Overview | ECN-000034/A;1-Truck_Change_Pahal

Exit
Full...

Impacted Items: ALL
Add to
Details of...
Update Alignment
Remove Repla...

Changes for	Merge Status	Action	Element	ID	Revision
		Modified	Beta25_Truck_3 Axle	Rev1_3-004-006660_Charn	C A
		Modified	Touchscreen Display	033669_Deloo_TC	C A
		Modified	Truck Front Axle (r5)	Rev1_3-004-006604_Charn	C A
		Modified	Front (Bare) Axles	Rev1_3-004-006639_Charn	C A
		Modified	Front Axle (20L steered, freewheel, lon...	Rev1_3-004-006455_Charn	C A
		Property Changed	Shock absorber	Rev1_3-004-005979_Charn	A
		Substitute Removed	Shock absorber sub4	033713_sub4-Deloo_TC	A
		Substitute Added Exi	Shock absorber sub6	033870	A
		Modified	Torsian Sway Bar (extra bended)	Rev1_3-004-006378_Charn	C A
		Modified	laminated spring leafspring	Rev1_3-004-005190_Charn	C A
		Manufact	laminated spring leafspring		

Details of Changes
Undo
Close

Rev1_3-004-006378_ChangeECI_Deloo_TC/C;1-Torsian Sw...

Properties Substitutes Attachments Vendor Parts

Files (3)

Documents (6)

Object	References
033704_Doc1/A;1-Torsian Sway Bar Doc1	
033705_Doc2/A;1-Torsian Sway Bar Doc2	
033706_Doc3/A;1-Torsian Sway Bar Doc3	
033707_Doc4/A;1-Torsian Sway Bar Doc4	
033868/A;1-Torsian Sway Bar Doc5	
033872/A;1-Torsian Sway Bar Doc6	

The **Details of Changes** pane highlights what has been altered in the Change Summary.

Compare item revisions in the Change Summary

Solution items that are revisions and contain a lineage to impacted items can be compared in the **Change Summary**.

Procedure

- 1. Select a solution item from the **Change Summary** table to view a comparison of the properties between the **Impacted Item** and the **Solution Item**.
- 2. Expand assemblies to compare the properties of individual structure components.

Refer to **Customize your change summary or comparison view** to select or customize the display.

The **Action** column displays whether the solution item was added, modified, or replaced.

Components that were replaced show both the original and the replacement in the summary. Selecting either the original or the replacement component shows the comparison.

 044878	 Adaptor-V3 Adaptor-V2	Modified	C B		
 044920	<i>Tour wrap</i> Tour-soft	Modified	B A		
 030-12212-1-n	030-12212-1-n	Replaced	A	+	each
 047132	<i>Shaft-Stiff</i>	Added New	A	1	each
 044878	Adaptor-V3 Adaptor-V2	Modified	C B		
 044920	<i>Tour wrap</i> Tour-soft	Modified	B A		

COMPARISON

		
044878/B;1-Adaptor-V2	044878/C;1-Adaptor-V3	
Name	Adaptor-V2	Adaptor-V3
Description	Adaptor-V2	Adaptor-V3, Draw
Release Status	TCM Released	
Checked-Out		Y
ID	044878	044878
Revision	B	C
In Process	False	False
Classified in		
Checked-Out By		ed (ed)

How release effectivity displays in the Change Summary

While defining a product managed with occurrence effectivity, the effectivity decisions often evolve as the product content itself is developed. The design engineer makes decisions for effectivity ranges when additional or replacement parts are included in an assembly BOM in the context of an ECN. Sometimes these effectivity dates ranges can overlap partially (a split), or have the same effectivity date range that overlaps entirely (a complete overlap).

Effectivity ranges differ in appearance in the Change Summary when updates are made to a BOM item in the context of an ECN.

Complete overlaps of effectivity are represented in the Change Summary on a single line. Here, a sequence change was made to the **Magnet Frame Lower** part to include overlapped (the same) effectivity dates. Since there is no effectivity change for the item in the context of the ECN, the sequence change is displayed on one line.

Change Summary Overview ECN-000017/A;1-Baseplate Assembly Change for future release						
Element	ID	Revision	Revision Name	Action	Occurrence Effectivity	Sequence
Baseplate Assembly	031460	B	Baseplate Assembly	Modified		
High power Magnet	031475	A	High power Magnet	Added New to Re...	31-Aug-2023 14:30 to U...	20
Magnet	031462	A	Magnet	Replaced	31-Aug-2023 14:30 to U...	20
Magnet Frame Lower	031461	A	Magnet Frame Lower	Property Changed	31-Aug-2023 14:30 to ...	12

When there is no complete overlap and modifications are made to an assembly that results in a split or cutback effectivity in the context of an ECN, two lines display the changes in the Change Summary. Here, a sequence change was made to a Lock Gimbal, which is indicated by the two lines displaying the different effectivity dates.

Change Summary Overview ECN-000018/A;1-Baseplate Assembly Change for fasttrack release							
Element	ID	Revision	Revision Name	Action	Occurrence Effectivity	Sequence	Quantity
▼ Baseplate Assembly	031460	C	Baseplate Assembly	Modified			
Base Plate	031471	A	Base Plate	Replaced	31-Aug-2023 14:30 to 10-Sep-2023 14:29 (NONE) 31-Aug-2023 14:30 to U...	70	
Enhanced Base Plate	031477	A	Enhanced Base Plate Base Plate	Added New to Re...	10-Sep-2023 14:30 to UP (NONE)	70	
Lock Gimbal	031463	A	Lock Gimbal	Property Changed	31-Aug-2023 14:30 to 10-Sep-2023 14:29 (NONE) 31-Aug-2023 14:30 to U...	30	
Lock Gimbal	031463	A	Lock Gimbal	Property Changed	10-Sep-2023 14:30 to UP (NONE)	33	2
Magnet	031472	A	Magnet	Removed	31-Aug-2023 14:30 to 10-Sep-2023 14:29 (NONE) 31-Aug-2023 14:30 to U...	80	
New Magnet	031476	A	New Magnet	Added New	10-Sep-2023 14:30 to UP (NONE)	100	

Customize your Change Summary or comparison view

You can configure the **Change Summary** table or the **Comparison** table to hide or display columns or to rearrange columns.

The default column name order is **Changes for Properties, Substitutes, Attachments, and Vendor Parts; Merge Status; Action; Element; ID, Revision Name; and Quantity**.

Note

You cannot change the position of the **ID** column, but the other columns can be moved to suit your working preferences.

Additionally, you cannot hide the **Name** column, but you can change its position.

Procedure

1. From the **Overview** of a Change Notice Revision, click **Settings**  to open the **Arrange** panel.
The **Arrange** panel for the **Comparison** table has a similar list of options as the **Change Summary**.
2. Select a column name to rearrange. You can change the column order by clicking  to move up or  to move down.
3. Click  to reset the list to the default configuration.

Note

This resets your user-specific column configuration preferences and restores the original columns as defined by your administrator in the site-level preference. For more information about how column configuration works, see [Configure Change Summary columns](#).

4. To add or remove column headings, select a name in the appropriate column and click **Add**  or **Remove** .
5. (Optional) Select **Save as new arrangement**. Enter a unique name and click **Save and Arrange**.
(Optional) You can use a previously created column arrangement by clicking **Column Arrangements**  and selecting a saved arrangement from the list.
6. Click **Arrange** to apply your changes.

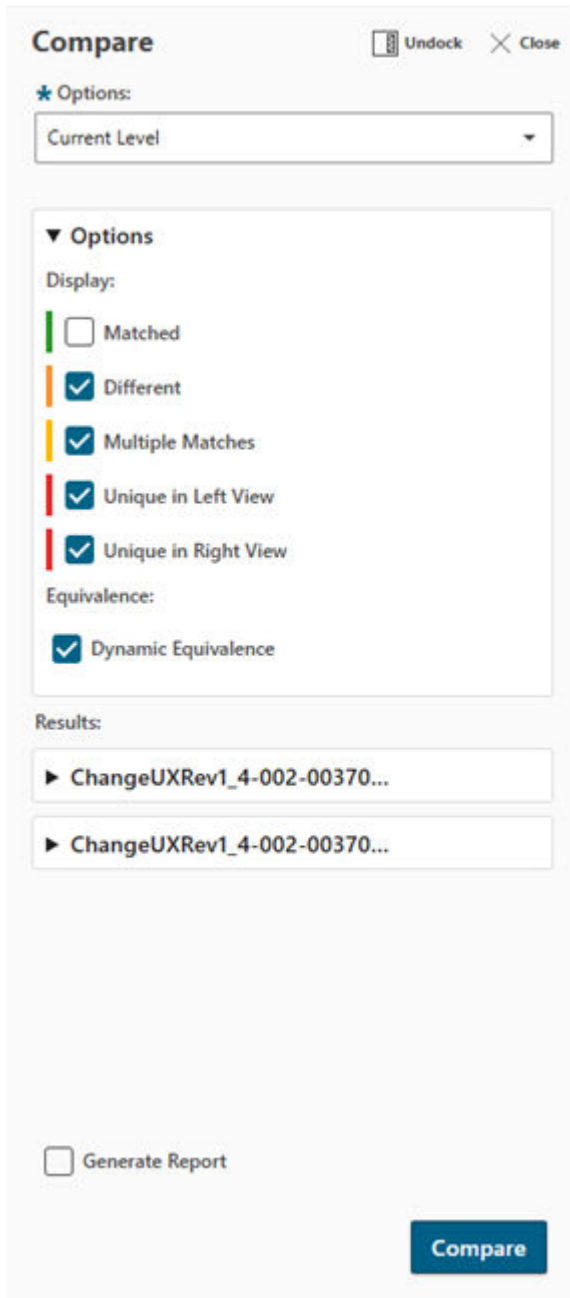
Compare modifications in the Change Summary

You can compare assemblies with revisions from the **Change Summary** table.

Procedure

1. Open a change notice.

Go to the **Overview** tab and in the Change Summary, click **Compare**  on any modified assembly. The comparison window opens, showing the different revisions and the **Compare** panel.



The screenshot shows the 'Compare' panel with the following elements:

- Header:** 'Compare' title, 'Undock' button, and 'Close' button.
- Options:** A dropdown menu currently set to 'Current Level'.
- Options Section:** A collapsed section titled 'Options' containing:
 - Display:**
 - ☐ Matched
 - ☒ Different
 - ☒ Multiple Matches
 - ☒ Unique in Left View
 - ☒ Unique in Right View
 - Equivalence:**
 - ☒ Dynamic Equivalence
- Results:** Two expandable list items, both labeled 'ChangeUXRev1_4-002-00370...'.
- Footer:** A 'Generate Report' checkbox and a large blue 'Compare' button.

2. Select the different levels for comparison in the list, and expand the **Options** section to view and change the options to compare.
3. Click **Compare** to run the comparison.

The time taken for the comparison depends on the size of the assembly and the levels and options selected in the **Compare** panel.

An alert  appears when the comparison is complete.

View 4G data for a change

4G data can be viewed for a change. In the following example, the change summary shows the structure that is part of the change. The reuse design element is the solution for the change. Note, however, that subordinate design elements do not appear. The underlying assembly and components are the solution items for the change notice.

ID	NAME	ACTION	REV	QTY
 OPV079282	shifter assembly	Modify	002:1 001:1	1
 OPJ90319_002	shifter assembly	Modify	002:1 001:1	1
 017750	shifter mechanism	Modify	002:1 001:1	1
 007385	fastener	Qty Change	001:1	3 4
 017753	shifter reverse lever	Modify	002:1 001:1	1
 017754	shifter reverse rod attachment	Remove	001:1	1
 017765	shifter shaft short throw	Replace	001:1	1
 017755	shifter shaft long throw		001:1	1
 007386	shifter knob	Create	001:1	1
 OPV079475	another assembly	Modify	002:1 001:1	1
 OPV079486	single part reuse bracket	Modify	002:1 001:1	1
 017768	bracket	Modify	002:1 001:1	1

Track Engineering BOM effectivity cutbacks in the Change Summary

When you introduce a new part to replace an existing part in an Engineering BOM structure and change its effectivity range, that modification is called an *effectivity cutback*. You reduce (or cut back) the effectivity range of the original part so that it does not overlap with the effectivity of the new, or replacing, part.

When a user opens a change notice for the BOM and **updates its effectivity** after the usages or parts are released, the Change Summary in the **Overview** tab shows the modifications made (via redlines) in the context of the change notice.

For example, an engineer has a released BOM and wants to set the configuration of the BOM window to the change notice effectivity date. The engineer inserts a new level within the BOM structure, and this new part (solution item) inherits its element effectivity from the active change notice. To avoid an effectivity conflict between the impacted item and solution item, the effectivity of the impacted item is cut back to end before the solution item becomes effective.

Here, modifications made to both the existing part (the impacted item: **Glass** in this example) and the new part (the solution item: **Insert Level for Glass**) are shown in the Change Summary structure. The **Occurrence Effectivity** column shows the effectivity dates for both the old and new occurrence.

▼ CHANGE SUMMARY					
ID	Name	Action	Occurrence Effectivity	Usage Name	Usage Revision
029892	Insert Level for Glass	New			
029891	Glass	Added Existing	01-May-2022 00:00 to UP (NONE)	Glass	A
029889	RH Mirror				
029891	Glass	Removed	01-Feb-2022 00:00 to 30-Apr-2022 23:59 (NONE) 01-Feb-2022-00:00 to UP (NONE)	Glass	B A
029892	Insert Level for Glass	Added New	01-May-2022 00:00 to UP (NONE)	Insert Level for GL...	A

Along with showing the effectivity cutbacks for a structure when a level is added, the Change Summary also identifies these split occurrences when a BOM engineer:

- Removes a level in a structure
- Adds a level in a structure
- Adds a child or sibling structure component
- Removes a structure component
- Replaces a structure component
- Move a usage/occurrence using drag and drop within same BOM structure
- Edits the properties of a part or an assembly: sequence, quantity, occurrence name, all notes, or custom property
- Adds a copy of the existing objects to a structure
- Copies and pastes a component within a structure

All of these changes are represented in the Change Summary for the change notice.

Create a Simple Change

Introduction to Simple Change

Simple change simplifies the process for creating and performing change revisions. It is designed for situations that do not require the detailed features of a change notice or detailed workflows.

Simple change  is a subtype of a change notice.

The screenshot shows the SAP Change Management interface for a 'Simple Change' (ECN-000005/A;1-Simple Change). The interface is divided into several sections:

- Header:** Displays the change ID, owner, date modified (16-Apr-2025), release status, type (Change Notice Revision), closure (Open), and disposition (Approved).
- Overview Tab:** Contains a 'Task to Perform' section with workflow details and a 'Task Instructions' section. It also includes a 'Comments' field and buttons for 'Cancel Change' and 'Release Change'.
- Impacted Items (1):** A table showing the impacted item: 034753/A;1-Bike, with lineage 1 and type Item Revision. It also shows the requested change (SC-000002/A;1-Simple ...).
- Change Summary:** A table showing the change summary for the impacted item. It includes columns for Changes for, Merge Status, Action, Element, ID, and Revision. The table shows a single row for the item 034753, which is modified and has a revision of 1.
- Details:** A section providing a synopsis (Simple Change), description (For bike), date released, effectivity, fast track, and implementation priority.

Example

Simple change is intended for minor changes handled by a single change analyst or a small team.

A change analyst is tasked with a minor change to a part. Instead of managing the change using a standard, elaborate change notice, the team uses *simple change* to streamline the process.

Simple change uses a simplified workflow, reducing the number of steps to release a change. Unlike a standard change notice where the change is created separately from the workflow process, a *simple change* is submitted automatically to the workflow.

When you revise an impacted item, it is automatically added to the Change Summary as a solution.

States of a simple change

A change has two key states that capture where in the change process it is and what decisions have been made about the change.

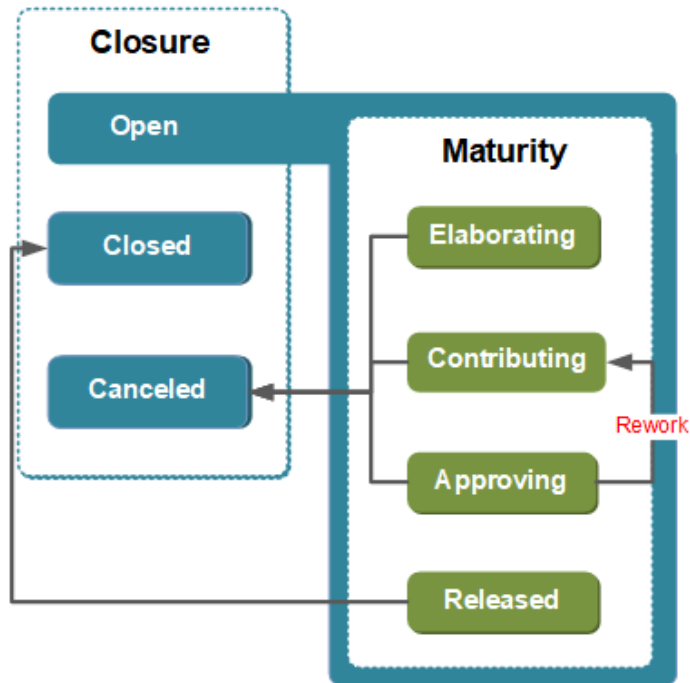
- Its status or *Closure* (for example, Open or Canceled).
- The degree of completion of the overall change process, that is, its *Maturity*.

The change states interact with each other and are dependent on the other change states. For example, *Maturity* is a substate of the *Closure* state. The states of a change are set during the workflow process.

Tip

You can view the states of a change in the **Overview** tab.

The following graphic illustrates the simple change states. For detailed information on change states, see *Change Management — Deployment and Rich Client Usage* in the Teamcenter help.



Create and resolve a simple change

Create a simple change

Use simple change as an alternative for making minor changes in smaller teams or organizations.

Procedure

1. Select an object and go to **More Commands** *** > **New** ✨ > **Create Change** ⚙ from the **Page level** toolbar. Select **Simple Change** from the list.
Alternatively, from the **Launcher**, click **Changes**.
If this option is unavailable, search for it. You can pin it from the search results for future access.
2. Click **Create Change** ⚙ from the top toolbar, then select **Simple Change**.

3. **Fill in the required information** and click **Create and Submit**.
The change is created and automatically submitted to the simple change workflow.

Add and revise impacted items

When you start a simple change process from an object, the object is automatically added as an **Impacted Item**.

Procedure

1. Open the simple change and in the **Impacted Items** section, click **Add** ⊕.
2. Select an object and click **Add** ⊕.
Repeat as necessary.
3. (Optional) Select an object from your folders.
 - a. Go to **More Commands *** > Manage** ✂ > **Add to My Changes** ↵ from the **Page level** toolbar.
 - b. Select a simple change from the list of changes, choose **Impacted Items** from the **Relation** list, and click **Add**.
4. Select any object in the **Impacted Items** table and click **Revise to Solution** ↵.

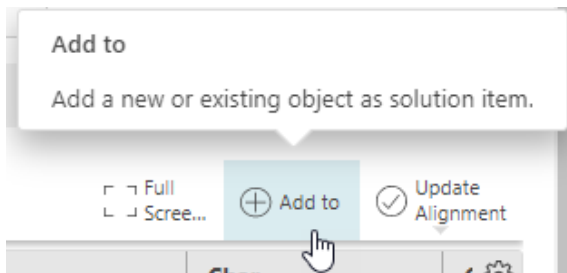
ID	Name	R...	Type	Release Status	Requested Change	Lineage
0015...	Item A	A	Item Revision			
0...	Item B	A	Item Revision			

Add solution items

When you revise an impacted item, the new revision becomes the solution item for the same simple change.

Procedure

1. Revise any impacted items to create a solution item in the **Change Summary**.
The relationship between an impacted item and a solution is identified by the **Lineage** column. For more information regarding lineage, refer to the [Change lineage section](#).
2. Click **Add to** ⊕ to add a solution item that is not a revision of an impacted item.

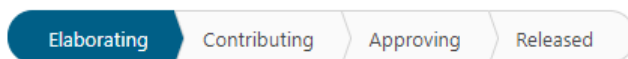


Completing simple change tasks

If a task is assigned to you as part of the change workflow, you can access the task and mark its completion from your inbox.

If you are an authorized user, you can complete the tasks from the *simple change Overview* in **Change Manager**.

Completed tasks are displayed with the status **Released** in the progress bar.



Manage the participants in a Simple Change

The **Overview** tab contains a **Participants** section to manage the participants of a *Simple change*.

▼ PARTICIPANTS		
⊕ Add ⊖ Remove ↔ Replace		
Type ▾	User ▾	⚙
Requestor	ed (ed)	
Contributor	bob (bob)	
Contributor	ed (ed)	
Approver	ed (ed)	

Simple change consists of three types of change participants.

- Requestor
- Contributor
- Approver

Tasks are automatically sent to the assigned participant's inbox. You can add participants (as contributors and approvers). You can also replace and remove participants.

For more information on change participants refer to [Who are the participants in a change?](#)

Coordinating multi-site changes

Authoring changes at multiple sites

Change Management provides a centralized system that streamlines the process of implementing changes across multiple sites. This process uses a master change at a central (or engineering/EBOM) site, which creates and submits subordinate changes to a distributed (or remote, or DBOM) design site through automated workflow handlers. Design engineers then act on the design data subset owned by their respective remote sites and sync back their modifications to the central site when complete.

This orchestration of the engineering tasks from the central EBOM site with the design work at the remote sites can help greatly with collaboration and project management. Steps are organized using a workflow template created by the central site. This template includes tasks that utilize the **CM-create-multisite-change-notice-objects** and **CM-transfer-multisite-parent-structure** workflow handlers.

This introduction will guide you through the key aspects of coordinating change notices to achieve efficient execution across multiple locations.

Orchestrate change notices across multiple locations

Orchestrating the engineering tasks and design work across multiple locations is performed through a workflow template with tasks that include specific Change Management handlers.

A master change at a central (or engineering) site triggers these handlers to create and submit subordinate changes to remote design sites. Design engineers work on their assigned data subsets and synchronize their modifications back to the central site upon completion.

The key aspects of coordinating change notices across multiple locations are as follows.

Part 1: Central site tasks

An engineer at the central site performs the following tasks.

Procedure

1. Create and assign the change notice

[Create a change notice](#) and [add participants](#) for an engineering bill of materials (EBOM). This change notice is created using a workflow template with the CM-create-multisite-change-notice-objects and CM-transfer-multisite-parent-structure workflow handlers and set as an **active change**.

2. Create the EBOM

Add new usages to the new or existing EBOM within the context of the change notice. The **Change Summary** tracks all of the modifications and their respective usages as solution items.

See the Engineering BOM Management-Usage guide for more information about creating BOM assemblies.

3. Assign the solution items to remote sites

Now that the EBOM is assembled, you can assign the solution items to the remote (or design) site.

- In the change notice, click the **Affected Items** tab.
- In the **Proposed Design Owning Site** column, use the drop-down list to select the remote site for the solution item components.

The screenshot displays the SAP Change Management interface for change notice **ECN-000279/A;1-EBOM_PETER**. The **Affected Items** tab is active. The **Solution Items** table is shown with columns: Object, Lineage, Proposed Design Owning Site, and Type. The item **028804/A;1-EBOM-Part1** is selected, and its **Proposed Design Owning Site** dropdown menu is open, listing various sites. The **Type** for this item is **UB Part Revision**.

Object	Lineage	Proposed Design Owning Site	Type
028803/A;CP-EBOM-Peter 2			Collaborative Product EB...
028804/A;1-EBOM-Part1		[Dropdown Menu]	UB Part Revision
028805/A;1-EBOM-Part2			UB Part Revision
028806/A;1-EBOM-Part3			UB Part Revision
PU_002481/A;EBOM-Part1			Part Usage
PU_002482/A;EBOM-Part2			Part Usage
PU_002483/A;EBOM-Part3			Part Usage

The dropdown menu for **Proposed Design Owning Site** includes the following options:

- adw_ref1
- adw_ref2
- allegro_ref1
- allegro_ref2
- AltiumDesigner_App
- bcz_BB_site1_5255451
- bcz_BB_site2_5255452
- bcz_Offline_site1_5455451
- bcz_Offline_site2_5455452

Site assignments are then used to push dependent change notices to the remote sites via a task using the CM-transfer-multisite-parent-structure handler.

4. Automated dependency

Once assignments have been made, the central site creates a new DBOM change notice and establishes a dependency between the master EBOM and DBOM change notices. The system replicates the EBOM change notice items to the respective DBOM change notice and transfers ownership of these changes to the users at the remote sites via the CM-create-multisite-change-notice-objects and CM-transfer-multisite-parent-structure workflow handlers.

The dependency between the EBOM and DBOM changes is visible on the **Dependencies** tab.

Designers can then start claiming and reviewing those DBOM changes at their remote sites.

The workflow using the two handlers will wait to continue at the engineering site until all subordinate DBOM change notices are released.

Part 2: Remote Site tasks

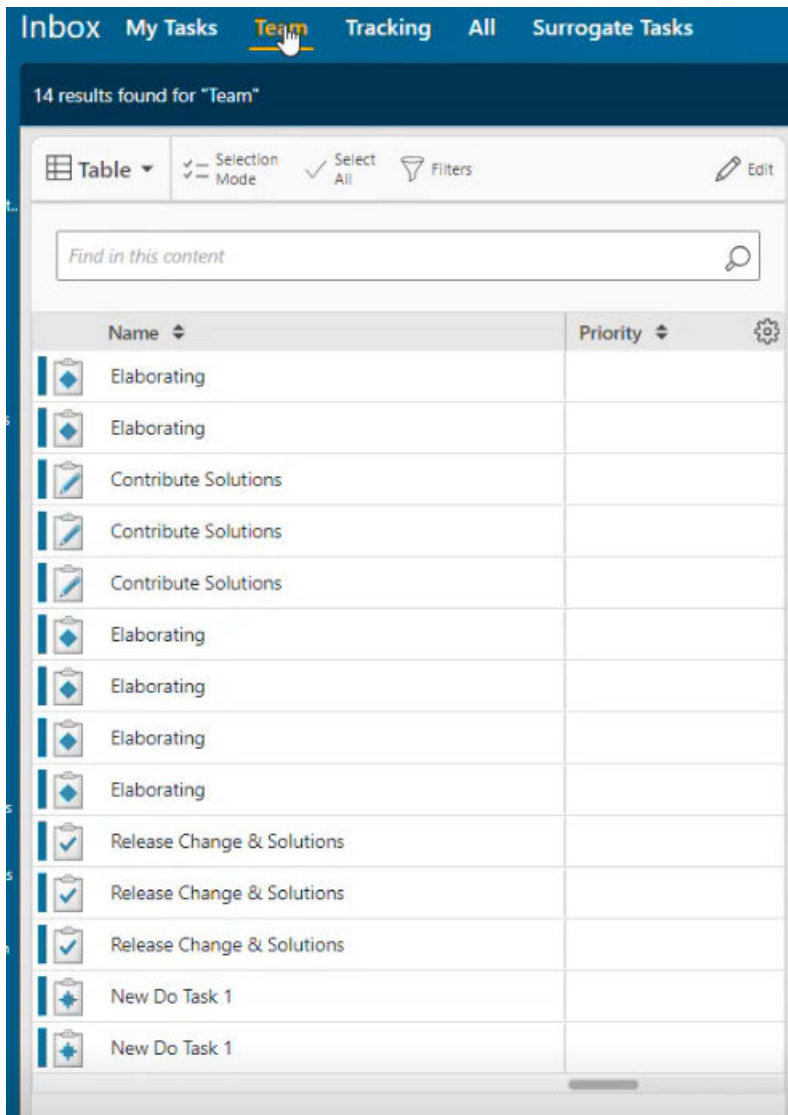
At the remote site, the system replicates the EBOM Change Notice items to the respective DBOM change notice.

When the designer opens the change notice and clicks the **Dependencies** tab, the change notice created specifically for the remote site is listed.

A designer at the remote site performs the following tasks.

Procedure

1. The Inbox contains one team task that lists all the subtasks required of the change notice. Claim a task once it is in the Inbox.



Once the change noticed is opened, all of the solution items that came from the engineering site are included in the designer's **Impacted Items** automatically.

2. Update the DBOM within the context of the change notice.
3. Align the DBOM with the EBOM.
4. Perform a part CAD alignment.
5. Once the components are aligned, release the DBOM change notice.
All data created from the DBOM at the remote site is then synchronized automatically back to the central site.

Part 3: Back at the central site

Once the remote engineer releases the DBOM change notice data and it is automatically synchronized back at the central site, the change progress shows as **Complete**.

All of the solutions that the remote site added in the DBOM change notice can be seen on the **Dependencies** tab's **Change Summary**.

The screenshot shows the Siemens Change Management interface for ECN-000280/A:1. The 'Dependencies' tab is active, displaying the 'Change Summary' section. The 'Progress' bar indicates the change is 'Complete'. The 'Change Summary' table lists the following items:

ID	Action	Revision	Element	Revision Name	Variant Formula
027504-PH1	New	A	DBOM-Pahal	DBOM-Pahal	
027506-D2	Added New	A	DBOM-D1	DBOM-D1	
027508-D2	Added New	A	DBOM-D2	DBOM-D2	
027509-D3	Added New	A	DBOM-D3	DBOM-D3	

If you open the EBOM structure, you can view the alignments with the corresponding design in the **Aligned Designs** column.

The screenshot shows the Siemens Change Management interface for the EBOM structure 'CP-EBOM-Peter 2'. The 'Aligned Designs' column displays the alignment between the EBOM structure and the DBOM change notice. The table below shows the alignment:

Element	Usage Own...	Usage Last ...	Usage Last ...	Usage Gro...	Part Maturity	Aligned Designs	Usage Maturity	ECN In	ECN Out
CP-EBOM-Peter 2					Concept	027506-D2/A:1-DBOM-D1	Concept	ECN-000279/A:1-EBOM_PETER	
EBOM-Part1	Peter Project (...)	06-Jun-2024	Peter Project (...)	demo	Concept	027508-D2/A:1-DBOM-D2	Concept	ECN-000279/A:1-EBOM_PETER	
EBOM-Part2	Peter Project (...)	06-Jun-2024	Peter Project (...)	demo	Concept	027509-D3/A:1-DBOM-D3	Concept	ECN-000279/A:1-EBOM_PETER	
EBOM-Part3	Peter Project (...)	06-Jun-2024	Peter Project (...)	demo	Concept				

Now the EBOM Change Notice can be released.

Non-functional changes

What is a non-functional change?

You can modify parts in an assembly without impacting their form, fit, or function through non-functional changes (NFC).

What is form, fit, and function?

- **Form:** This refers to the physical characteristics of an object, like its shape, size, weight, and how it looks. Imagine a smartphone; its form includes its rectangular shape, the size of the screen, and its color.
- **Fit:** This is about how well a part connects or fits with other parts. It is like making sure a puzzle piece fits perfectly with the others. In a smartphone, fit would be how the battery fits inside the phone or how the phone fits in a case.
- **Function:** This describes what the object or its parts can do. It is about the performance and capabilities. For a smartphone, function includes things like how well the camera takes pictures or how fast the phone runs apps.

NFCs avoid a major revision of the part, bypassing a lengthy and complex review.

NFCs are also useful for updating specific attributes like the vendor parts or custom descriptions. Consider a scenario where a company has a product assembly in their PLM system, and its release status is marked as **Released** within the system. Initially planned for a holiday season release, the business decides to release it in the spring instead. This change can be tracked in a custom description attribute. Another example is when a vendor part needs to be updated from one supplier to another. These changes do not affect the part's form, fit, or function and can be processed under an NFC.

Create a non-functional change

Use non-functional changes to track updates without altering the original base revision ID.

Procedure

1. Select an object and go to **More Commands *** > New ✨ > Create Change**  from the **Page level** toolbar. Select **Non-Functional Change** from the list.

Alternatively, from the **Launcher**, click **Changes**.

If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Click **Create Change**  from the top toolbar, then select **Non-Functional Change**.
3. **Enter the required information** and click **Create and Submit**.
The change is created and submitted to the Non-Functional Change Workflow.

Note

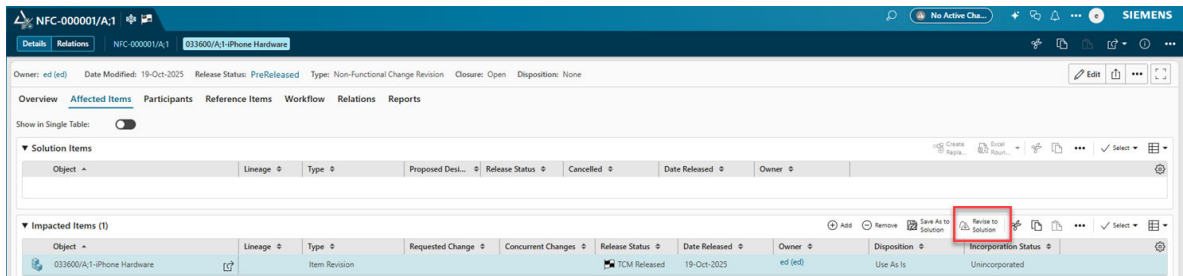
You cannot set an NFC as an active change.

Add and revise impacted items in a non-functional change

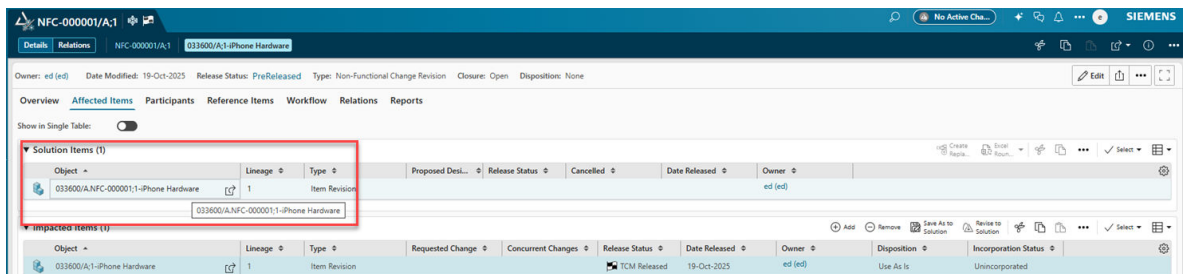
When you start a non-functional change process from an object, the object is automatically added as an Impacted Item.

Procedure

1. Open the non-functional change and click the **Affected Items** tab.
2. In the **Impacted Items** section, click **Add** ⊕.
3. Select an object and click **Add** ⊕.
Repeat as necessary.
4. (Optional) Select an object from your folders.
 - a. Go to **More Commands** *** > **Manage** ✂ > **Add to My Changes** ↩ from the **Page level** toolbar.
 - b. Select a non-functional change from the list of changes, choose **Impacted Items** from the **Relation** list, and click **Add**.
5. Select any object in the **Impacted Items** table and click **Revise to Solution** ↻.



The **Impacted Items** move to the **Solution Items**, but the base revision ID stays at **A** (not **B** as in a regular change notice) and the NFC ID is appended to the revision ID:



Note

Turn on **Show in Single Table** to consolidate all tables on the **Affected Items** tab into a single table, making it easier to track and update item statuses, changes, and related information without navigating through multiple sections. You can add additional impacted, problem, or solution items to the NFC via the **Add** ⊕ button. Icons indicate the relation type.

Edit solution items in a non-functional change

You can perform minor modifications to a solution item for the non-functional change without triggering a major revision.

Prerequisites

The required Attribute Level Security (ALS) configuration is performed by your administrator.

Preferences are available for tracking the vendor part modifications.

CM_item_revision_tracked_properties displays specific item revision properties and **CM_bomline_tracked_properties** tracks and displays specific occurrence properties.

The restrictions on the modifications that can be made to the solution items of an NFC are governed by Access Manager (AM) rules. Non-functional changes offer two AM Rules (**In Active NFC** and **Has NFC Class**), as well as a new accessor (**Active NFC Participant**).

Procedure

1. Open the line item.
The **Base Revision ID** remains unchanged at **A**, while the **Revision ID** changes to **.NFC** to indicate that it is a non-functional change.

The screenshot displays the Siemens PLM software interface for a non-functional change (NFC) on the 'iPhone Hardware' item. The top navigation bar shows the item path '033552/A.NFC-000006;1-iPhone Hardware' and the status 'No Active ECN'. The main table lists the components of the iPhone Hardware, all with revision .NFC-000006. The details pane on the right shows the properties of the selected item, including the ID 033552, Base Revision ID A, and Revision .NFC-000006.

Element	ID	Revision	Base Revision ID	Revision Name	Description	NFC Demo Description	Find Number	Quantity	Release Status
iPhone Hardware	033552	A.NFC-000006	A	iPhone Hardware		Christmas Edition			
Screen and Input x1	033553	A.NFC-000006	A	Screen and Input		Christmas Edition	10	1	
Lightning to C cable x2	033554	A.NFC-000006	A	Lightning to C cable		Christmas Edition	20	2	
Li Battery 2716 mAh x1	033555	A.NFC-000006	A	Li Battery 2716 mAh		Christmas Edition	30	1	
Audio and Output x1	033556	A.NFC-000006	A	Audio and Output		Christmas Edition	40	1	
Camera x1	033557	A.NFC-000006	A	Camera		Christmas Edition	50	1	
Sim Card Slot x1	033558	A.NFC-000006	A	Sim Card Slot		Christmas Edition	60	1	

The details pane shows the following properties for the selected item:

- ID: 033552
- Revision: A.NFC-000006
- Base Revision ID: A
- Revision Name: iPhone Hardware
- Description:
- NFC Demo Description: Christmas Edition
- Occurrence Name:
- Reference Designator:
- Find Number:
- Quantity:

2. Click **Edit**  to make any modifications.
Only minor, non-functional edits can be made to the change notice, such as:
 - Changes to the configured custom **Description**.
 - Adding and removing a vendor part

- Modifications to the relation property between the part and the vendor part (for example, preferred status)

Tip

You can use the **Excel Round-trip**  > **Export to Excel**  feature in the Solution Items table to track the modifications done in Excel for the custom **Description** property. For example, in Excel, change the components' custom description from **Christmas Edition** to **New Year Edition**. Then, from the Solution items table, select all items and use the **Excel Round-trip** feature to import the changes. As a result, the selected Solution Items are updated with the new custom description.

You cannot modify the assembly by adding or removing items.

The **Change Summary** reflects the changes via redlines made to the non-functional change.

4. Viewing changes

View your changes

You can view all changes and see their various states on the **Changes** page or the **Changes** tab on applicable objects. For example, you can view the completion of prior tasks, the current task, and pending decisions for the current task's associated workflow.

If an object has multiple changes, the changes are listed chronologically, with sortable columns.

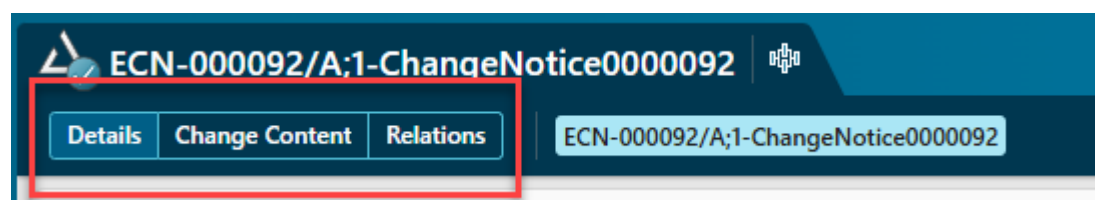
3D Overview Finishes Classification Made From Where Used Changes Attachments History Relations Collaboration Participants Simulation Reports

▼ CHANGES

Object	Description	Relation Type Name	Closure	Disposition	Maturity	
ECN-000008/A;1-Wheel...	WheelRev2	Problems	Open	None	Elaborating	
ECR-000004/A;1-WheelR...	WheelRev1	Problems	Open	None	Elaborating	

Change notice work modes

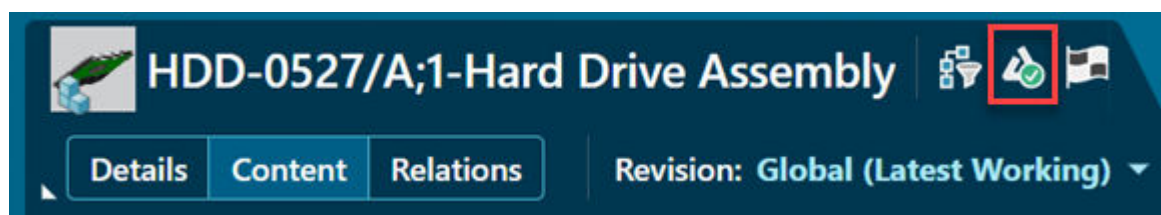
When you open a change notice, you can access various work modes that offer different perspectives.



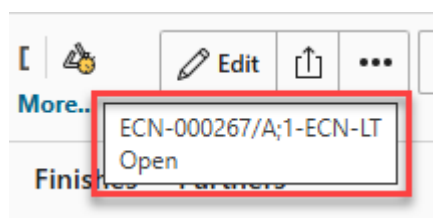
- **Details** displays the **Overview**, **Change Summary**, **Affected Items**, **Impact Analysis**, **Attachments**, and other information.
- **Change Content** lets you modify and perform structural operations on your solution Items but also displays multiple top-level impacted and solution Items all at once. This differs from the **Content** view, which displays just one top-level assembly at a time.
- **Relations** lets you view relationships between this change notice and other items.

Change notice indicators

When an object in an assembly is included as a solution item in an authoring change notice, an icon appears next to its name. This icon, for example, signifies that the associated change notice is closed.



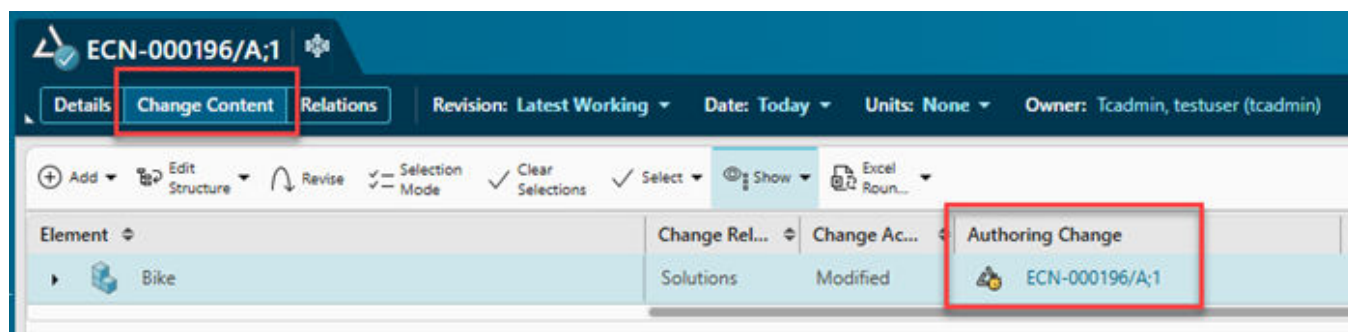
The indicator's tooltip reveals the specific change notice that includes the object.



The following icons represent the status of a change notice:

- In Progress
- Pause
- Done/Complete/Closed
- Canceled

Indicators also appear in the **Authoring Change** column of an assembly within the **Content** and **Change Content** views. This provides a quick and easy way to identify which authoring change the object is part of and its status.



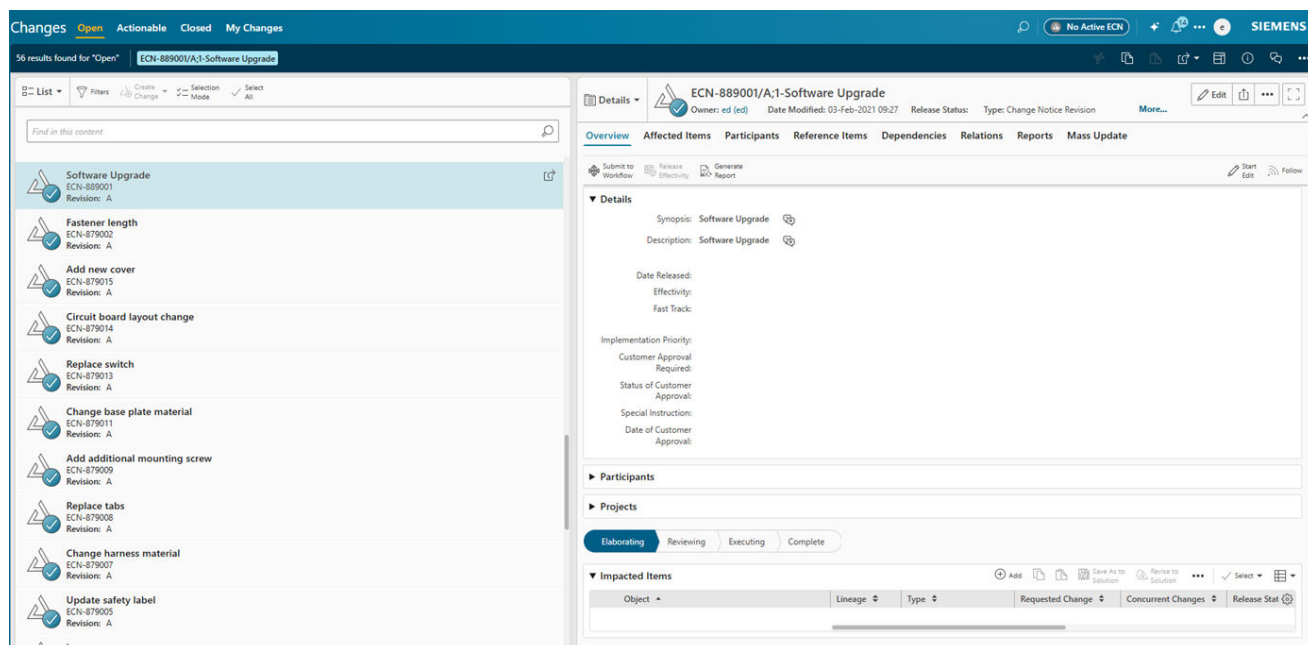
Note

An item is specific to a particular revision and can only be part of one ECN.

Procedure

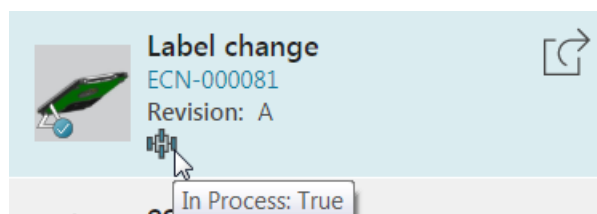
1. To view all changes, click **Changes**

Select a change and use the navigation tabs along the top to see the information associated with the change.



Navigation tabs along the top show the state of the different changes where you are the owning user. If you do not select a specific change item, a bar chart appears.

Changes submitted to workflow are identified by an in-process indicator.



2. Select **Filter**  to open the **Filters** panel. Select one or more filter categories. Selected items are identified with a check mark. Results are automatically displayed.
3. Review the tabs.

Overview

The **Overview** tab provides comprehensive information about the change.

A visual status bar indicates where the change is in its process.

This example shows four standard maturity values, and two additional custom values.



The details under the change notice name display the value of the most relevant properties.

The **Tasks to Perform** section contains active tasks for the assigned user. The assigned user can perform these tasks without navigating to the inbox. This section automatically reflects the current active tasks.

The **Details** section provides comprehensive information about the change being proposed or implemented.

If applicable, the project details are listed under the **Projects** section.

The **Participants** section lists the users directly involved in processing the change.

The **Impacted Items** section lists specific items that will be affected by the proposed change.

The **Change Summary** section displays BOM edits. Additional references are provided for less mature change objects from which this change was derived (implements) or more mature change objects derived from this one (implemented by).

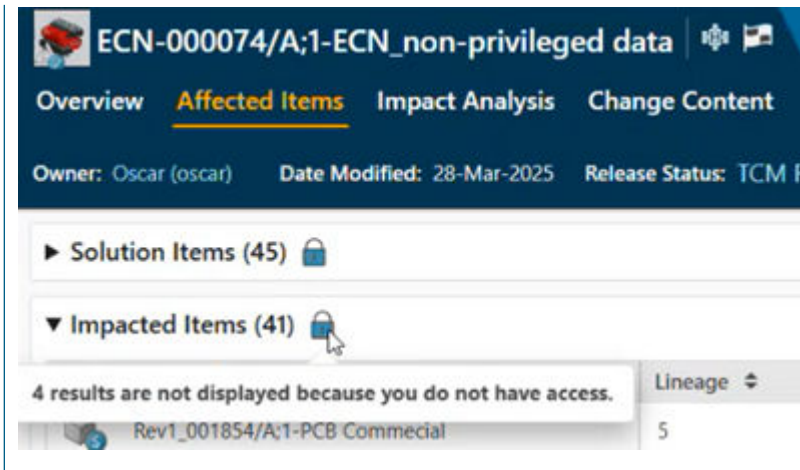
Affected Items

The **Affected Items** tab shows objects impacted by the change. They include:

- The object identified as having a problem.
- Those objects that are being changed (impacted) as a result of the change process.
- (Change notice or change request only) Schedules or plan items to manage the change process.
- (Change request only) Changes that implemented the identified problem as well as all the solutions of the current change notices in the **Solution** column of the **Implemented By** table.

Note

Administrators can configure the system by setting up the preferences **AW_Display_Not_Readable_Count** for search results and **AWC_display_unreadable_count_in_objectset** for tables. This ensures that only users with the appropriate authority can access privileged data. When users view changes, they can see the total number of affected items, but items they lack read access to will be marked with a lock icon in the table . This approach informs users of the existence of non-privileged data without granting full visibility. The choice to display the total count or the lock icon must be configured according to company policies, and these settings are disabled by default.



Impact Analysis

The **Impacted Items** table appears in the **Overview**, **Affected Items**, and **Impact Analysis** tabs.

Participants

The **Participants** tab includes members who participate in the review and execution of the change.

Reference Items

The **Reference Items** tab includes objects supporting the investigation of the change. By default, when you create a change in the context of an object, the object is placed in the **Reference Items** tab.

Workflow

The **Workflow** tab shows the workflow associated with the change and its current status, if any.

Dependencies

The **Dependencies** tab allows you to order and sequence change notices with respect to each other across change requests, by designating a change notice as before, after, or concurrent with respect to another change notice. There are three dependency types: **Preceding**, **Concurrent**, and **Succeeding**.

Relations

View the relations link to the change in the **Relations** tab, such as attachments or reference documents and network of related data.

By default, the change and its problem item appear.

Reports

You can generate and view reports for change objects using the **Reports** tab, depending on the configuration of your organization.

View and perform tasks using the Actionable tab

Actionable tab

The **Actionable** tab consolidates all the changes with active tasks assigned to the current user.

Similar to the **Open** tab in **Changes**, the **Actionable** tab provides the full change summary of a selected change with any assigned actions in the **Overview** section. This eliminates the need to navigate to the inbox to perform tasks.

When you complete a task, the **Overview** updates with the next active task.

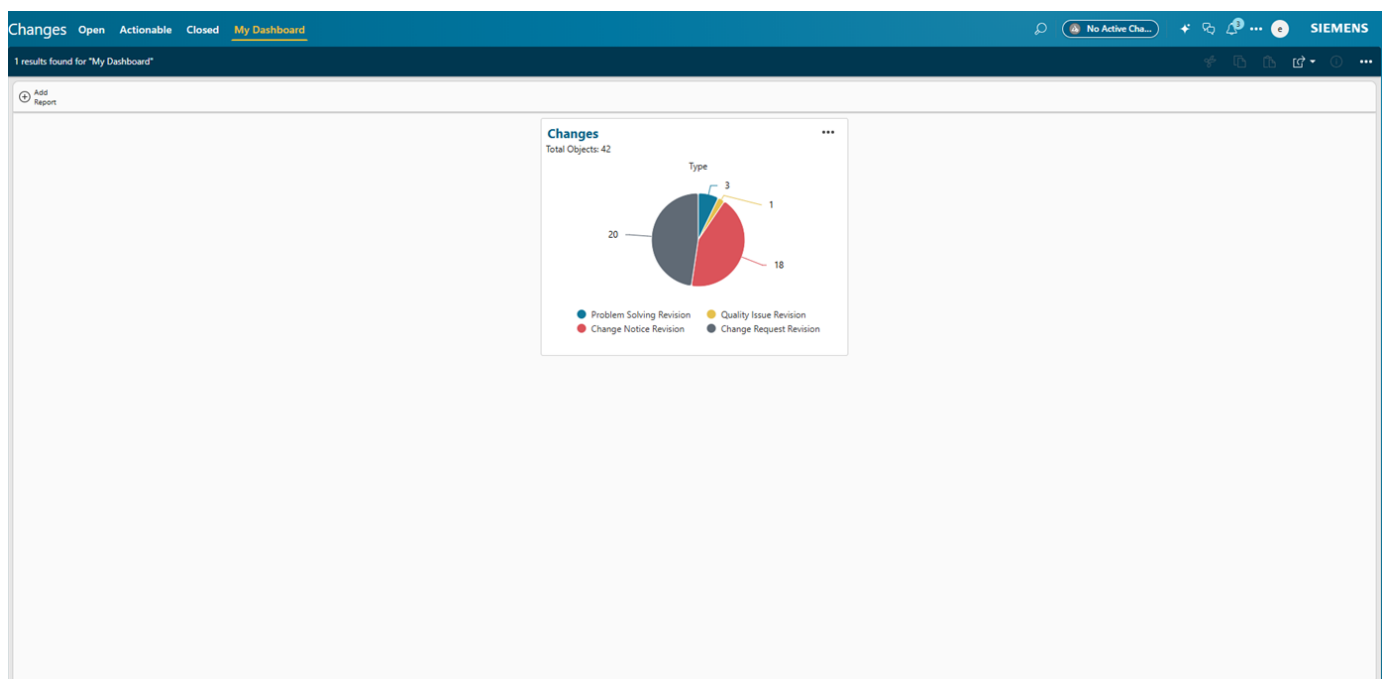
Perform assigned tasks using the Actionable tab

Procedure

1. Click **Changes** , and select the **Actionable** tab.
All changes with tasks assigned to the user are listed.
2. Select a change from the list. The **Action** section lists the current task assigned to the user.
3. Perform the task according to the details and instructions listed.
The **Action** section is updated with the next task in the workflow.

View changes using the Changes Dashboard

The Changes dashboard organizes and displays all change information in a pie chart and tabular interface that is filterable and sortable.



Procedure

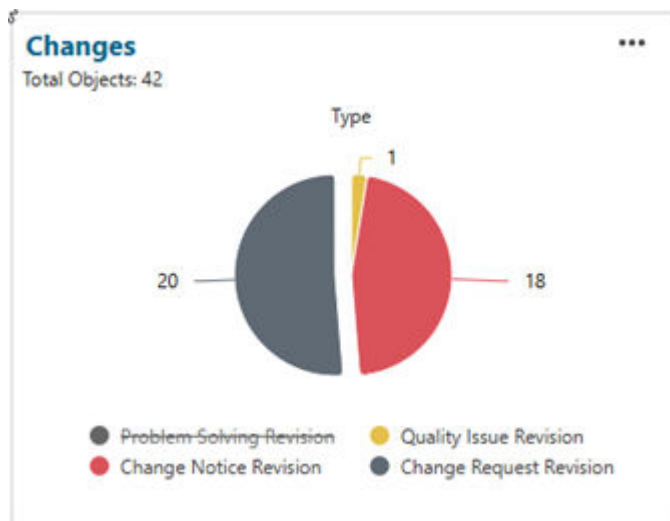
1. From the **Launcher**, click **Changes**.

If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Click **My Dashboard**.

The out-of-the-box report displays all of the change types. The pie charts are color coded with the type and quantities of the specific changes.

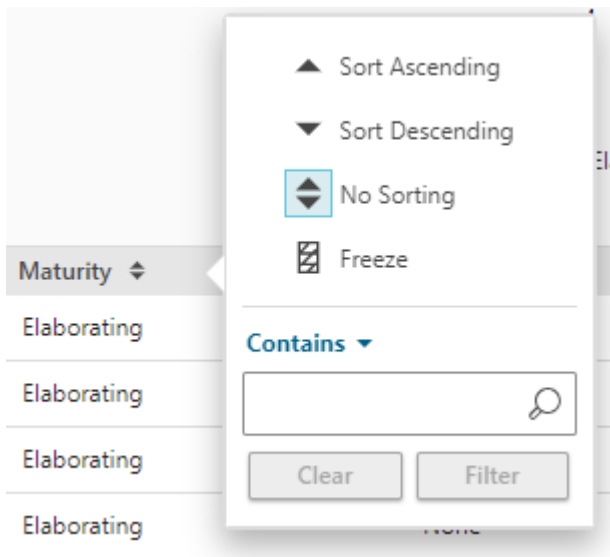
You can click on a change type to remove it from the pie chart, allowing you to focus on the remaining types.



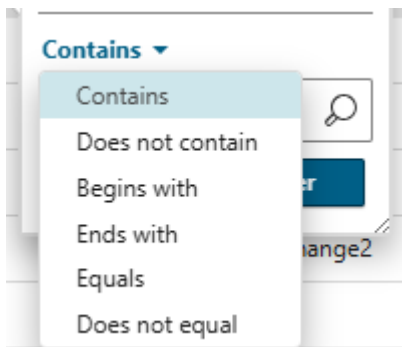
3. Click **Changes**.
You continue to the **Changes** dashboard. This dashboard displays a list of all the out-of-the-box changes, their properties, and statuses.
4. Depending on the quantity, types, and roles in your organization, the pie charts contain different information and color codes.
5. Click the legend below the **Type** graph to clear and remove items from the chart. Cleared items are grayed out. Click a grayed out item to include it in the pie chart again.
6. The table reflects the selections and filtering applied on the pie charts.

The options available for sorting and filtering vary depending on the column selected, and your configuration.

Click on a column heading to open the sorting and filtering panel for each column.

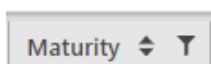


- The **Search** list shows the parameters to search on specific criteria. The default value is **Contains**.



Enter your text in the **Search** box and click **Filter**.

- Hover over the filter  to see the filter criteria.



Select the column and click **Clear** to remove the filter.

Add a custom report to the Changes Dashboard

Customize your Changes dashboard by including a custom report to monitor and track specific change notices.

Note

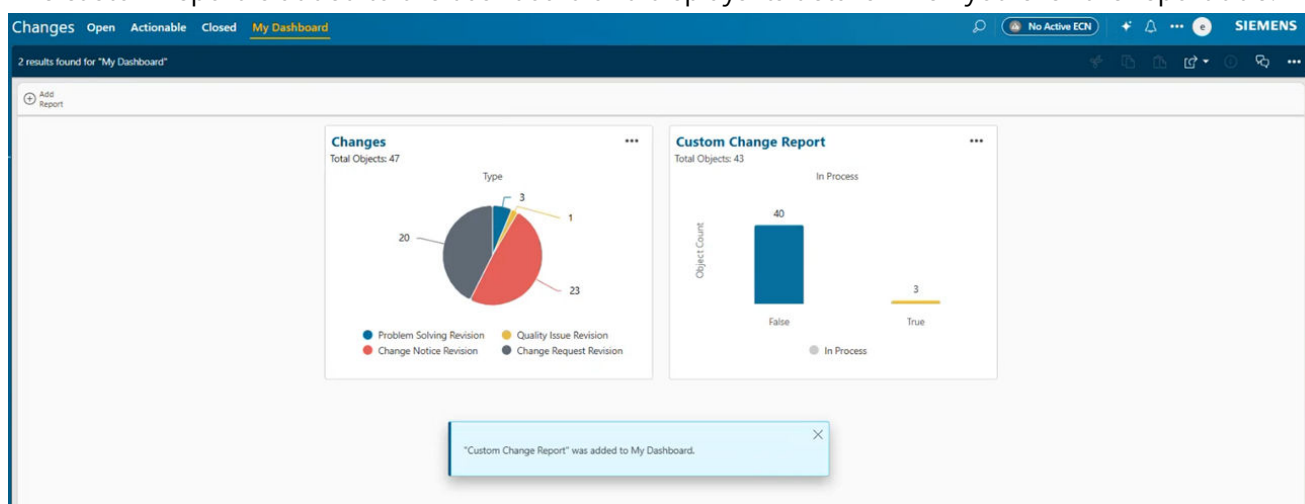
Custom reports are unique to each user and remain private unless an administrator configures them to be shared across different user accounts.

Prerequisites

Ensure you have a custom report available. Administrators configure the data displayed in custom reports.

Procedure

1. From the **Launcher**, click **Changes**.
If this option is unavailable, search for it. You can pin it from the search results for future access.
2. Click **My Dashboard**.
3. Click **Add Report** .
The **Add Report** panel opens.
4. Select your custom report from the **Report Templates** list.
5. Click **Add Report**.
The custom report is added to the dashboard and displays its details when you click the report title.



View change history

Both revision and change history are available for change notice impacted or solution items and appear in the same window.

Procedure

1. Open a change notice.
2. From the **Affected Items** tab, select an impacted item or solution item and open it.
3. Click **History**.

For 4GD business objects (partitions and design elements), you can use the **History** tab to view information about all versions of the object and the corresponding change:

- For each major revision of an object attached to a change notice, a separate row is displayed for each change notice in which it is a solution item or impacted item.
- When a major revision of a 4G object is pasted to the impacted item folder of any change (without an associated solution item), a row is displayed in the **Change History** section with the associated authorizing change, but the **Revision** and **Solution** columns are blank.
- Each line of the change history for an item contains values for **Closure**, **Maturity**, and **Disposition**, except for item revisions that are not impacted or solution items.
- When a solution item revision is removed from a change notice, the revision still appears in the **Change History** section, but all columns after the **Solution** column are cleared for that revision.

View change history for an item revision

When you select to view the Incorporation status of an item in the **Change History** tab of the **Summary** tab, the **Change History** dashboard appears with the following columns.

The column	Displays
Solution Item	The ID and name of the solution item.
Impacted Item	The ID and name of the impacted item.
Markup	The name and ID of any markup.
Authorizing Change Notice	The name of the change notice revision.
Closure	The Closure of the change notice, including open, closed, cancelled, or hold.
Maturity	The Maturity of the change notice revision, including elaborating, reviewing, executing, and completion.
Disposition	The Disposition of the change notice revision, including none, investigate, disapproved, deferred, and approved.
Incorporates Changes of	IDs of the item revision or the name of the markup being incorporated by the change notice revision.
Incorporated by	The name of the incorporating change notice revision.
Incorporated into	The ID of the revision the change was incorporated into.
Incorporation Status of Change	Shows the Incorporation status set on the impacted item, except if the Incorporation status is set to Partially Incorporated . Then, this indicates that another change notice has incorporated the change fully.

Display program relations for a change

You can view the programs and events related to a change request or change notice on the **Reference Items** tab. When the program planning template is installed, you can search and add programs, projects, and sub-projects to change requests; while change notices can be associated only to events.

To add a program:

Procedure

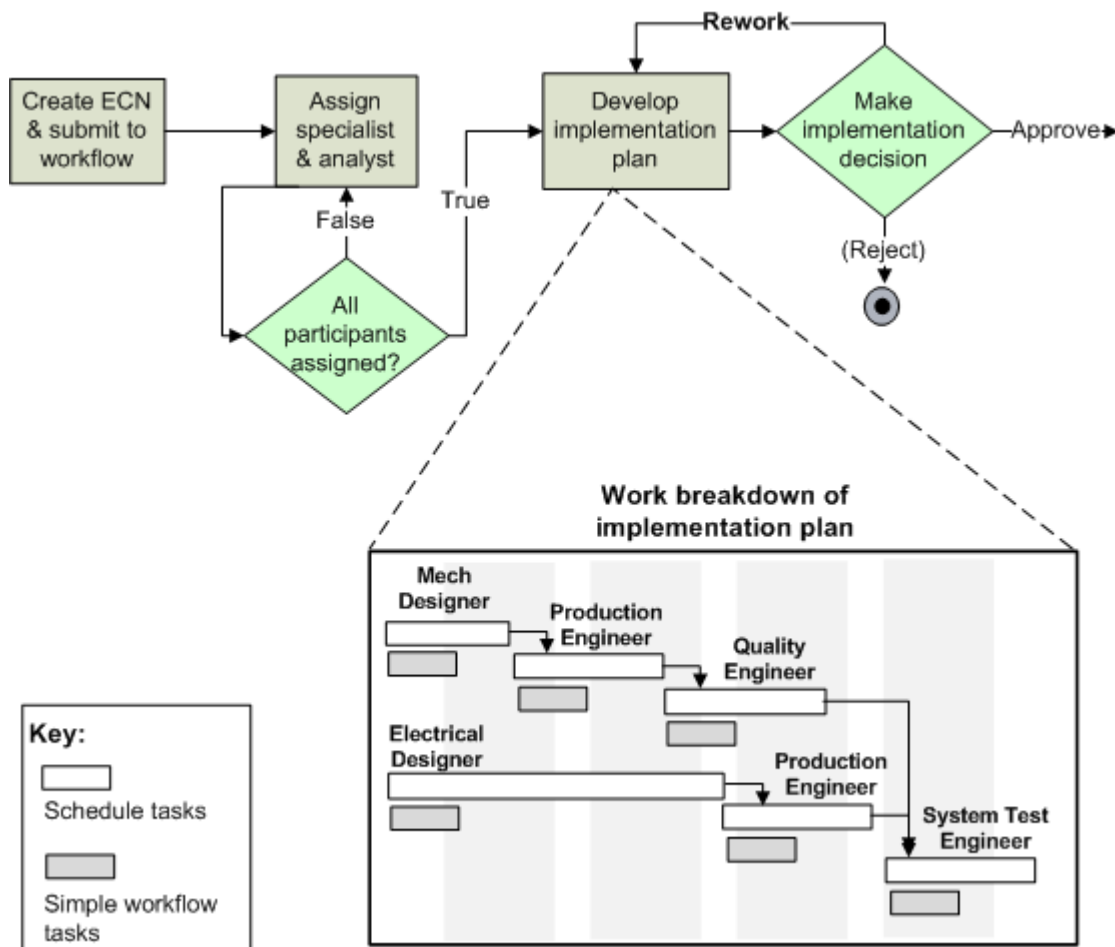
1. Click **Add to**  under **Programs**.
2. For a change request, search or select the program, project or sub-project. For a change notice, search or select the event.
3. Click **Add** .

5. Managing plan items and schedule tasks

About scheduling implementation activities

Schedule Manager is integrated into Change Manager so you can use it to schedule implementation activities (work breakdowns) associated with a change. The **Plan Items** folder stores the work breakdowns. Creating work breakdowns are useful to:

- Analyze the impact of the change to determine the amount and scope of the work required for the change. It can also identify the parts and documents impacted by the change.
- Plan the implementation of the change to specify the actions, or tasks, that address the change.



Example of a schedule work breakdown

Whether you create a work breakdown and define tasks depends on the complexity of the change and the conventions at your site. A simple change request, for example, may require relationships only to affected parts, affected documents, and supporting information, while a more complex change requires information about subassemblies or other components that only becomes clear after detailed analysis. In

addition, during the change creation, you may not know what other items must be added until the tasks are built and executed.

Schedules can be created in an ad-hoc manner to meet the requirements of each individual change object, or they can be created (copied) from a schedule template. A schedule template is a pre-defined schedule of tasks that is established for changes of a specific type. When you plan the implementation for such a change, you start with the corresponding schedule template. Any number of schedules can be defined and related to a change object.

The schedule can have tasks for each discipline or user to create new revisions or items and then make the changes to implement the proposed solution. Depending on the change work to be done, you could have one task per assignee that covers the changes for any number of items assigned to that user, or you could have one task per problem item or impacted item, or any other style of work breakdown you choose.

Because schedule tasks are not routed to users, it is good practice to have a simple one-step workflow task associated with each schedule task, as shown in the figure. The workflow routes the schedule task to the assignee's inbox and manages the implementation of the task. The resource associated with the schedule task is automatically assigned as the user to perform the workflow task. If more than one user is assigned to the schedule task, the single privileged user is assigned to the workflow. The workflow could include tasks to approve the changes made to each item revision, or this type of validation can be left to a later step (a later task in the schedule or a later workflow task in the ECN workflow). This would apply a status (for example, **Pre-Released**).

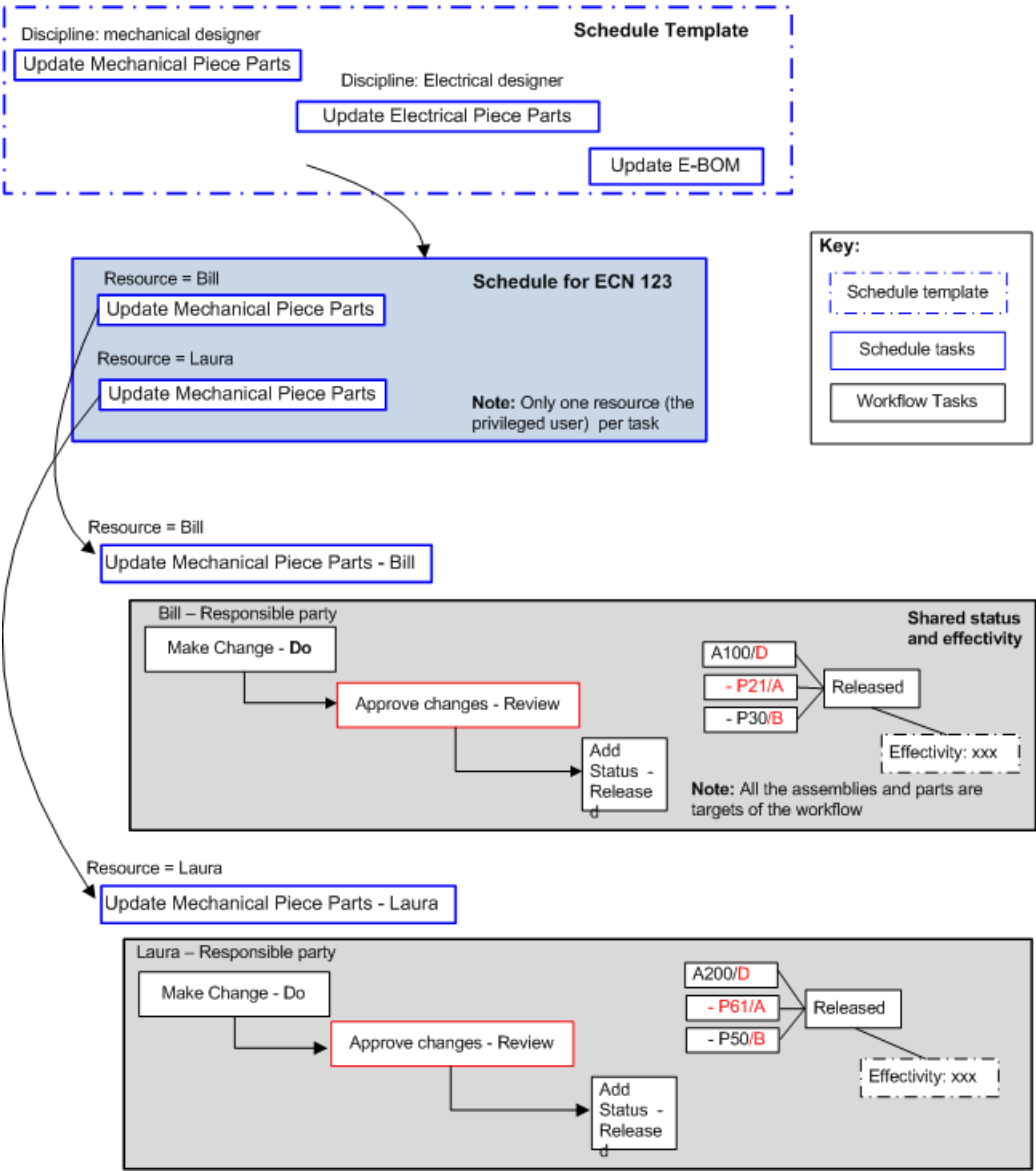
Flexibility of schedules

The advantage of a schedule is its flexibility. A schedule can easily be made larger or smaller, depending on the need. By contrast, workflows are much more rigid as they control the process and decision-making authority. Workflows should be kept simple, and schedules used when more complexity is required. Typically schedule templates are used to make defining the schedule for a particular ECR or ECN much quicker.

An analyst usually creates a work breakdown schedule, but another type of user can create it, and the analyst can then relate the schedule to the change object through the **Plan Items** or **Work Breakdown** relationship.

Example of a schedule work breakdown

The following is a simple example of a work breakdown for an engineering change notice (**ECN 123**). The work breakdown was created based on a standard schedule template (shown in the figure with dashed lines). The schedule was then modified to meet the ECN's requirements, creating copies of the Update Mechanical Piece Parts task for the two designers involved (Bill and Laura). Each of those tasks has the same small workflow template attached, which routes the task to the assignee (for example, Bill), then routes it for review of the changed and new items, and finally adds a status to them.



Use schedule tasks to manage changes

A complex change can have dozens or hundreds of problem, impacted, and solution items to manage. You can efficiently manage these as change items under a single change object by using a schedule task in one or more schedules (also known as plan items).

Schedule Manager helps with change-related work by creating a schedule task as a change object. This way, you can add the change items—problem, impacted, and solution items—to the schedule task, which in turn can be assigned to individual users for quick access. Distributing work across different users with different timelines can help to track completion at an individual change artifact level.

The change items associated with a schedule task appear in the **Affected Items** tab where you can do the following:

- Add and remove items from the change object
- Work on multiple items at once
- Propagate change items to a schedule task

Add plan items to a change object

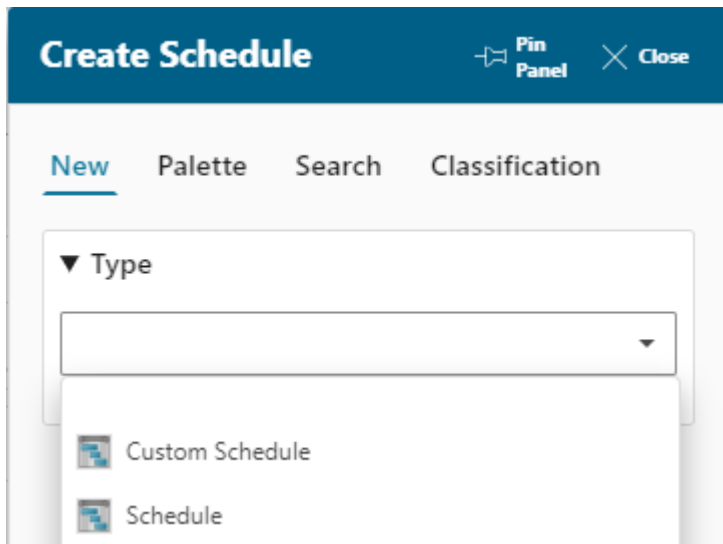
The **Plan Items** section of the **Reference Items** tab contains schedules for a change object. You have the option to add a schedule for a change request or change notice or create a schedule from a template.

Schedules created in the **Plan Items** tab are editable in Schedule Manager.

Create a schedule

Procedure

1. Open the change notice you want to work with.
2. Click the **Reference Items** tab.
3. In the **Plan Items** section, click **Add to** ⊕.
4. Select **Create Schedule** to manually create the schedule structure.
Select the type of schedule and enter the schedule information. Items marked with an asterisk are required.



Select **Open On Create** to open the schedule in Schedule Manager.

5. Click **Create**.

Create a schedule from template

Procedure

1. Open the change notice you want to work with.
2. Click the **Reference Items** tab.
3. In the **Plan Items** section, click **Add to** ⊕.
4. Select **Create Schedule From Template**.
Enter the information for the new schedule and select a template from the list. Items marked with an asterisk are required.

Create Schedule From Template × Close

▼ **Template**

 **Analog_IC_Development_Sch...**
Schedule
Start Date: 09-Apr-2019 08:00
Owner: Peter Project (peter)

▼ **Schedule Properties**

✳ **Name:**

Required

Start Date:

DD-MMM-YYYY  HH:MM:SS 

☐ Template

☐ Open On Create

Selecting **Open On Create** opens the schedule in Schedule Manager.

5. Click **Create**.

Roll up relations from a schedule task to a change object

You can select a plan item and commit the schedule task relations to the change notice. Change Manager creates the same relation between the object and the change notice.

Procedure

1. Open the change notice you want to work with.
2. Click **Reference Items**.
3. Select a schedule from **Plan Items**.
The columns display status and completed status for each schedule.
4. Go to **More Commands *** > Manage**  **> Rollup**  from the **Page level** toolbar.

The page refreshes after the action is completed, and a message appears at the bottom of the window to indicate the action is complete.

Propagate relations from a change object to a schedule task

You can selectively propagate (or associate) the items related to a change object (problem, impacted, solution, and reference items) to related plan items (schedule tasks) associated with the change object. This allows you to quickly correlate the change items to be worked on during a task with the task so a user can easily access them. For example, you can propagate the assembly with the problem bumper to the scheduled task so the user can replace the bumper with a new one.

Procedure

- 1. Open the change notice you want to work with.
- 2. Click **Affected Items**.
- 3. Select a schedule from **Plan Items**.
- 4. Go to **More Commands *** > Manage  > Propagate **.
The change objects and the schedule tasks display next to each other for comparison.

Propagate Change Relations | From: ECN-000142 - To: Schedule A

General - Associate items related to a change object (problem, impacted, solution, and reference items) to the schedule tasks associated with the change.

From: ECN-000142

Select All

Clear Selections

Object	Type	Relation	Owner
038458/A/1-Bike	Item Revision	Impacted	Kusnirik, Pam (je3noq)
038458/A/1-Bike	Item Revision	Problems	Kusnirik, Pam (je3noq)
038458/B/1-Bike	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038458/A/1-Frame	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038461/A/1-Standard Wheel	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038468/A/1-Subframe	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038462/A/1-Gears	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038463/A/1-Seat	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038469/A/1-Padding	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038460/A/1-Front Fork	Item Revision	Solutions	Kusnirik, Pam (je3noq)
038467/A/1-Extra small rims	Item Revision	Solutions	Kusnirik, Pam (je3noq)

To: Schedule A

Select All

Clear Selections

Name	Description	Start Date	Finish Date
Task		11-Oct-2023 09:00	10-Nov-2023 18:00
Milestone C		07-Nov-2023 08:00	10-Nov-2023 08:00
Milestone B		31-Oct-2023 08:00	10-Nov-2023 08:00
Milestone A		24-Oct-2023 08:00	10-Nov-2023 08:00

Close

Propagate

Note

You can propagate the change objects into multiple schedules (one schedule at a time).

- 5. Select the problem, impacted, solution, and reference item(s) from the change object on the left, then select the schedule task to associate it with on the right, and click **Propagate**.
The impacted contents that were propagated from the change notice to the schedule display in the **Change Relations Propagated to Schedule Tasks** table.

Propagate relations from a change object to a schedule task

Propagate Change Relations | From: ECN-000142 - To: Schedule A

General - Associate items related to a change object (problem, impacted, solution, and reference items) to the schedule tasks associated with the change.

From: ECN-000142

Select All

Clear Selections

Object	Type	Relation	Owner
038458/A/1-Bike	Item Revision	Impacted	Kusnirik, Pam (xe3noq)
038458/A/1-Bike	Item Revision	Problems	Kusnirik, Pam (xe3noq)
038458/B/1-Bike	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038459/A/1-Frame	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038461/A/1-Standard Wheel	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038466/A/1-Subframe	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038462/A/1-Gears	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038463/A/1-Seat	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038469/A/1-Padding	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038460/A/1-Front Fork	Item Revision	Solutions	Kusnirik, Pam (xe3noq)
038467/A/1-Extra small rims	Item Revision	Solutions	Kusnirik, Pam (xe3noq)

To: Schedule A

Select All

Clear Selections

Name	Description	Start Date	Finish Date
Task		11-Oct-2023 09:00	10-Nov-2023 18:00
Milestone C		07-Nov-2023 08:00	10-Nov-2023 08:00
Milestone B		31-Oct-2023 08:00	10-Nov-2023 08:00
Milestone A		24-Oct-2023 08:00	10-Nov-2023 08:00

Change Relations Propagated To Schedule Tasks

Object	Type	Relation	Owner
038461/A/1-Standard W...	Item Revision	Solutions	Kusnirik, Pam (xe3noq)

Close

Propagate

Continue propagating items as needed.

- 6. Click **Close** to save and exit.

Now, if you open a schedule task with the propagated change items, the **Affected Items** tab displays those items.

6. Using Active Change

What is Active Change?

An *active change* applies a context to your current working session that associates revisions with a specific change notice.

When you make revisions to an impacted item within a change notice, it automatically becomes a solution item with lineage. These solution items appear in the change summary of the **Overview** tab. If you create or revise other objects outside of the change, they are not associated or automatically applied in the context of the change notice and are not tracked in the change summary.

Setting a change notice as an active change lets you create revisions or property changes that are automatically applied to the active change and are tracked in the change summary. This is most important in a BOM context where you add, remove, replace, or modify properties. With an active change context set, these property changes are tracked in the change summary. If active change is not set, these property changes are not tracked.

The default setting is **No Active Change**. Only a current, open change notice or simple change is available to set as an active change.

A business object constant is added to facilitate this function, **Cm0ChangeManaged**, and its default setting is **true** for **ItemRevision** and **Configurator** objects. If this constant is set to **true**, the new object is tracked against the global change context (set in the **User profile**).

Note

The active change list contains change notices which are open and where the current user is an Analyst or Change Contributor.

Understand active change and authoring change

In Change Management, you can record change notices using two primary methods: authoring changes and active changes.

Both methods follow a similar process. In an authoring change, any edits you make to the opened item revision become part of the change notice as a solution item. In contrast, you must manually set an active change. Once an active change is established, it automatically records all newly created objects within that change.

To designate a change notice as an active change, the change must be in the Open state, and the current user must be either an analyst or a change contributor.

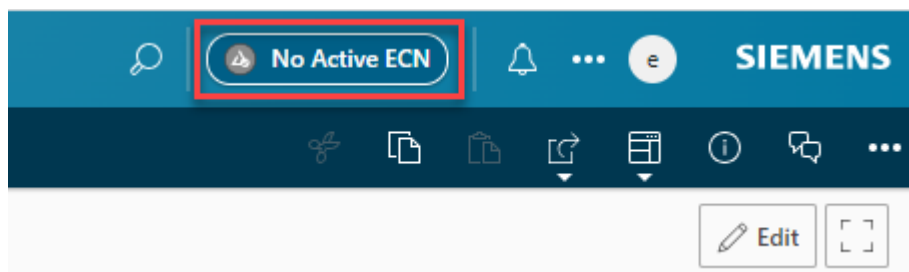
Are my edits going into an active or authoring change?

Tracking changes can be complex at times, as team members frequently work on different items simultaneously and update multiple change notices at once. To manage these changes effectively, consider the following guidelines:

- When you edit from an open change notice, any modifications to the affected items are recorded in the open authoring change.
- When you edit a revisable workspace object, changes apply to the active change. For example, adding or revising an attachment will apply to the active change.
- When you edit an assembly BOM structure, edits are directed to the authoring change.

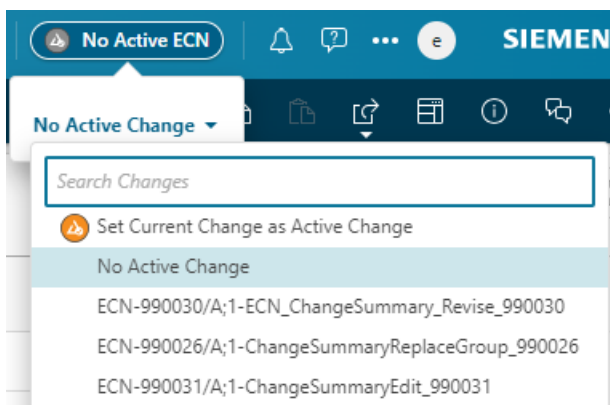
Set a change notice or a simple change as an Active Change

The **Active Change** setting is located in the global navigation area.



Procedure

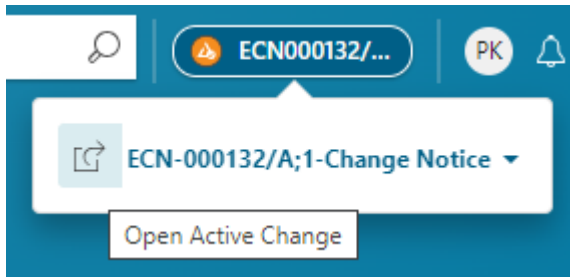
1. Select a change notice or simple change from the list to apply it as the active change, or choose to set the current change as the active change. Use the search box to filter the list to locate a specific change.



The **Active Change**  icon is highlighted indicating a change notice or simple change is selected.

The active change setting remains applied even after you log off.

2. To open the active change, click the change, then the open icon .



3. If you want to remove the applied active change, select the active change > ▼ > **No Active Change**.
4. Complete the required tasks in the change notice workflow. Once all tasks are complete and the change notice is closed, the Active Change is unset and the status is **No Active Change**.

Set a change notice or a simple change as an Active Change

7. Change tracking

Review active or closed changes for a structure

You can track changes (added, modified, replaced, revised, or deleted) to the assemblies using a change request or a change notice. When the ECN or ECR is an Active change, redlines are shown in the structure. To view the closed changes, use the **Show Redlines** command.

Prerequisites

Your administrator can configure the various methods to effectively track and display relevant changes to an assembly.

Procedure

1. Search for and open the structure. If the structure has changes, they are indicated by redlining. Expand the assembly to view the details of the change.

Element	ID	Revision
Engine	035611	B
Piston	035612	A
<i>Enhanced Valve</i> Valve	<i>035617</i> 035613	A
Connecting Rod	035614	A
<i>Connecting Rod Assembly</i>	<i>035621</i>	<i>A</i>

- Deleted parts are indicated by a red strikethrough.
- The added parts display in green and are italicized.
- For replaced or revised parts, the old and the new values are shown side by side.

After the structure is revised, the redlines are shown for all changes including the edits that you make for the part attributes.

2. (Optional) To disable highlighting the changes in the structure, go to **More Commands *** > View** and turn off **Show Redlines** from the **Page level** toolbar.
3. Only active changes are shown by default. To view the closed changes, click **More Commands *** > View** > **Show Redlines** from the **Page level** toolbar.

You can return a redline change back to its original impacted version by [reverting the BOM modifications](#).

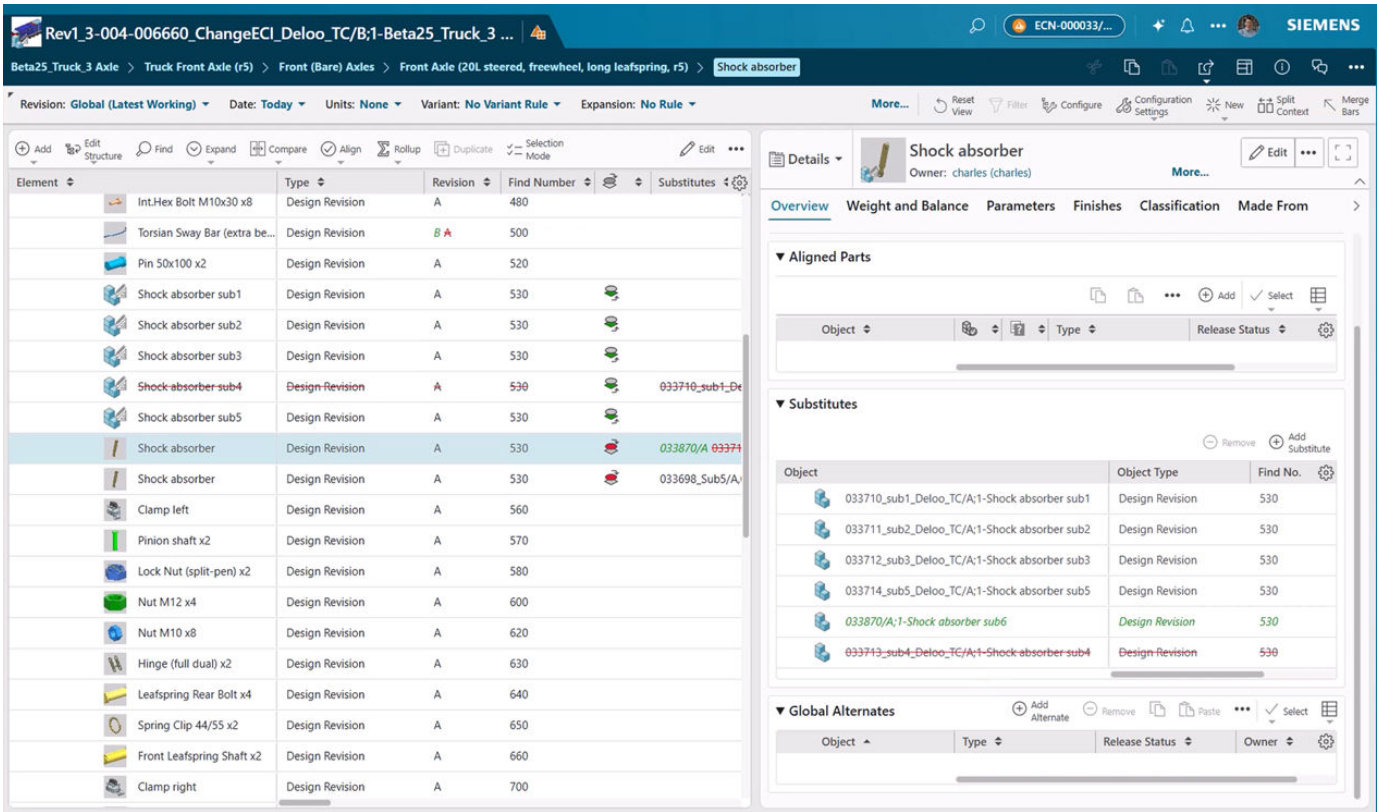
Track BOM modifications

Track BOM modifications in the Change Summary

Applying an active change context for BOM revisions automatically tracks the modifications in the Change Summary.

See [Set a change notice or a simple change as an Active Change](#) for more information.

With the change notice set as an active change, BOM modifications and properties are identified by redlines. For more information on the **Show Redlines** feature, refer to [Review active or closed changes for a structure](#).



These modifications, not the full assemblies, are tracked in the [Change Summary](#) of the change notice.

ECN-00033/A:1-Truck_Change_Charles

Overview Affected Items Change Content Participants Reference Items Workflow Dependencies Relations Reports Mass Update

Owner: charles (charles) Date Modified: 21-Mar-2025 Release Status: Type: Change Notice Revision Closure: Open Disposition: Approved ECN-00033/A:1-Truck_Change_Charles

Task to Perform: Workflow: NX_ChangeNoticeRevisionDefaultWorkflowTemplate : ECN-00033/A:1-Truck_Change_Charles

Comments:

Impacted Items (7)

ID	Object	Vendor Part Name
Rev1_3-004-006660_ChangeECL_Deloo_TC	Rev1_3-004-006660_ChangeECL_Deloo_TC/A:1-Beta25_Truck_3 Axle	Beta25_Truck_3 Axle
Rev1_3-004-006604_ChangeECL_Deloo_TC	Rev1_3-004-006604_ChangeECL_Deloo_TC/A:1-Truck Front Axle (r5)	Truck Front Axle (r5)
033669_Deloo_TC	033669_Deloo_TC/A:1-Touchscreen Display	Touchscreen Display
Rev1_3-004-006639_ChangeECL_Deloo_TC	Rev1_3-004-006639_ChangeECL_Deloo_TC/A:1-Front (Bare) Axles	Front (Bare) Axles
Rev1_3-004-006455_ChangeECL_Deloo_TC	Rev1_3-004-006455_ChangeECL_Deloo_TC/A:1-Front Axle (20L steered, freewh...	Front Axle (20L steered, ...)
Rev1_3-004-005190_ChangeECL_Deloo_TC	Rev1_3-004-005190_ChangeECL_Deloo_TC/A:1-Leafspring	Leafspring
Rev1_3-004-006378_ChangeECL_Deloo_TC	Rev1_3-004-006378_ChangeECL_Deloo_TC/A:1-Torsian Sway Bar (extra bended)	Torsian Sway Bar (extra ...)

Change Summary

Impacted Items: ALL

Changes for	Merge Status	Action	Element	ID	Revision
		Modified	Beta25_Truck_3 Axle	Rev1_3-004-006660_Char	B A
		Modified	Touchscreen Display	033669_Deloo_TC	B A
		Modified	Truck Front Axle (r5)	Rev1_3-004-006604_Char	B A
		New	Torsian Sway Bar Doc5	033868	A
		New	TouchScreen_Xenarc	033869	A

Manage and track vendor parts

Vendor parts manage changes to commercial parts within a BOM context. When a change notice is issued, specific actions related to vendor parts can be tracked and displayed in the system.

Note

When your administrator sets the **CM_keep_solution_item_after_revert** preference to false, vendor part changes are reverted along with structural and property changes. When set to true (default), vendor part changes are preserved during revert operations.

More information about vendor parts can be found here. You can also find details on how vendor parts are displayed in the [Change Summary](#).

Procedure

1. Open the change notice and click the **Affected Items** tab.
2. Open the assembly BOM from the **Solution Items**.
3. Select the item in the BOM for which you want to make vendor part modifications to open **Details** pane.

4. The Vendor Parts tab displays the associated vendor parts. You can add, remove, or modify the vendor part in the **Vendor Parts** table.

Modifications include editing the vendor part description and its preferred status. To edit the preferred status, select either **Preferred** or **Backup** from the drop-down list in the **Preferred Status** column.

Results

Adding a vendor part

When you add a vendor part to a revised commercial part within a BOM, the system displays the new vendor part as green lines in the change notice. In the **Details** pane, the new vendor part is listed under the **Vendor Parts** tab.

In the Change Summary, this action is labeled as **Property Modified**.

Removing a vendor part

When you remove a vendor part from a revised commercial part within a BOM, the system displays the removed vendor part as a red line in the **Vendor Parts** tab of the BOM's **Details** pane.

In the Change Summary, this action is labeled as a property modification.

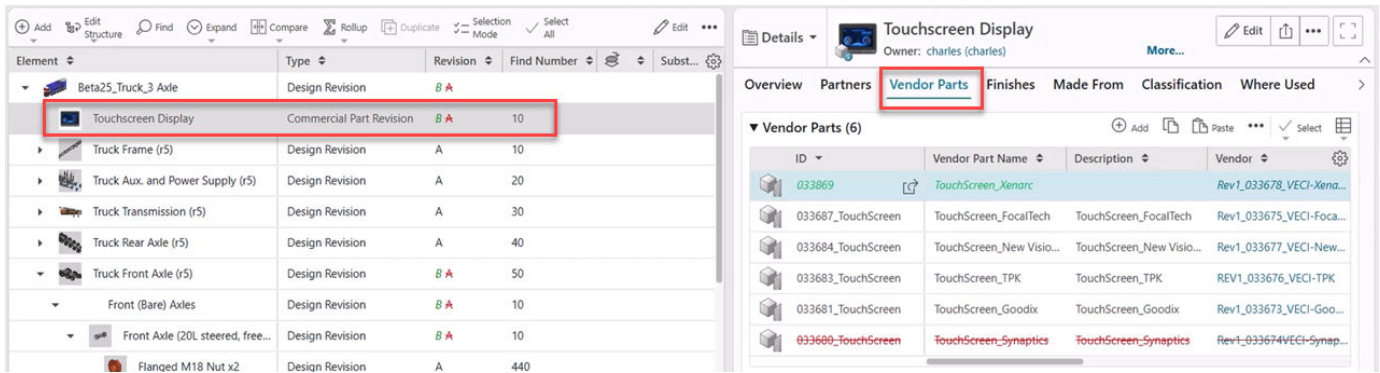
Modifying the vendor part description

When you modify the vendor part description, the system displays the old and new descriptions as red lines in the change notice. In the **Details** pane, the modified description is listed under the **Vendor Parts** tab.

In the Change Summary, this action is labeled as a property modification.

Modifying the Preferred Status

When you modify the preferred status of a vendor part by changing it to either **Preferred** or **Backup**, the previous preferred status displays as a red line, while the new preferred status is shown as a green line in the **Vendor Parts** tab of the BOM's **Details** pane.

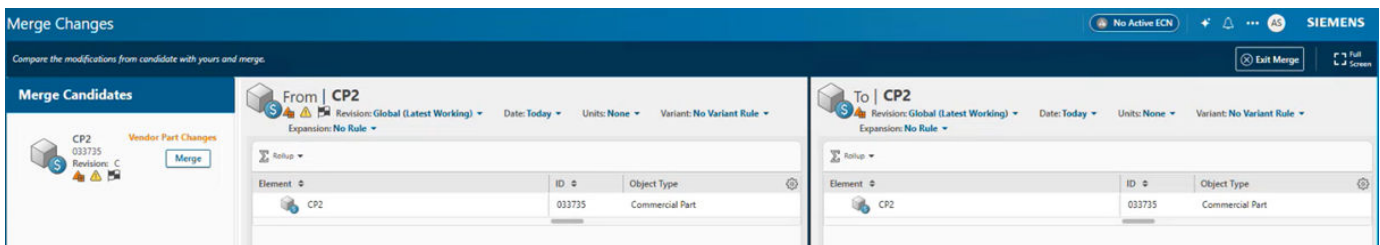


In the Change Summary, this action is labeled as a property modification.

Merging vendor parts

When you release a change notice, any work-in-progress concurrent change notices that share the impacted items will receive a merge indicator in their Change Summary ➡.

In a manual merge, you can click the merge indicator in the Change Summary to view a side-by-side presentation of the two assembly revisions. You have complete control over what to **merge**, allowing you to quickly see the differences as merge candidates and merge the vendor parts into the final change notice.



For guidance on how to resolve vendor part conflicts during an auto-merge, refer to [Managing change notice conflicts during an auto-merge](#).

Manage and track item revision attachments

When a change notice is issued, specific actions related to attachments can be tracked and displayed.

The **Attachments** panel lets you view attachments (forms or datasets) to the item revision that corresponds to the selected line. It also allows you to add attachments, create new attachments, remove attachments, and change attachment relation properties.

Note

When your administrator sets the **CM_keep_solution_item_after_revert** preference to false, attachment changes are reverted along with structural and property changes. When set to true (default), attachment changes are preserved during revert operations.

Additional information about attachments can be found [here](#).

Procedure

1. Open the change notice and click the **Affected Items** tab.
2. Open the assembly BOM from the **Solution Items**.
3. Select the item in the BOM for which you want to make attachment updates to open **Details** pane.
4. The **Attachments** tab displays the associated vendor parts. You can add, remove, or modify the attachments in the associated attachment tables, such as **Files** or **Documents**.

Results

Adding an attachment

When you add an attachment (either a file or a document) to a revised item revision, the system displays the newly added attachments as green lines in the change notice. In the **Details** pane, the new attachment is listed under the **Attachments** tab (under **Files** and **Documents**).

In the Change Summary, this action is labeled as **Property Modified**.

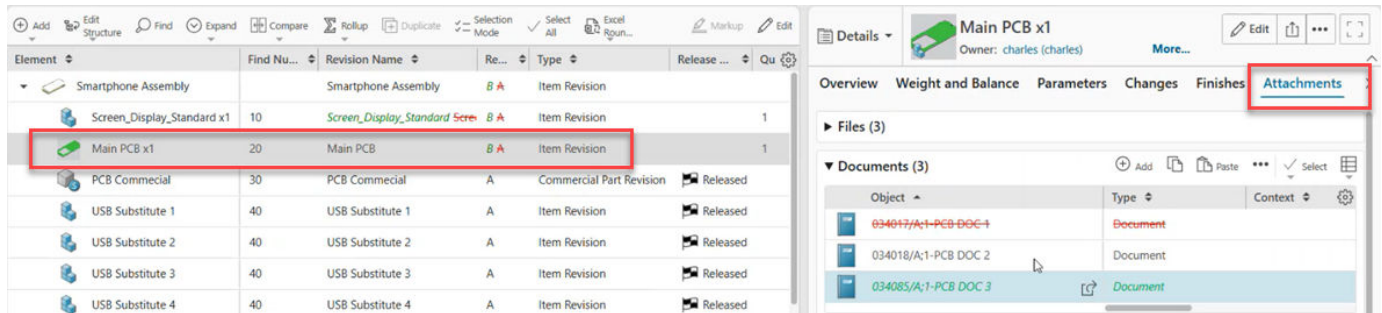
Removing an attachment

When you remove or cut an attachment (either a file or a document) from a revised item revision on the **Attachments** tab, the system displays the removed attachments as red lines in the change notice. In the **Details** pane, the removed attachment is listed under the **Attachments** tab (under **Files** and **Documents**).

In the Change Summary, this action is labeled as **Property Modified**.

Modifying relational properties of an attachment

When you modify a custom relational property of an attachment (either a specification relation or reference relation), the system displays the previous relation as red lines in the change notice and the new property on relation as a green line. In the **Details** pane, the previous relation displays with red lines in the change notice and the new property as a green line under the **Attachments** tab (under **Files** and **Documents**).

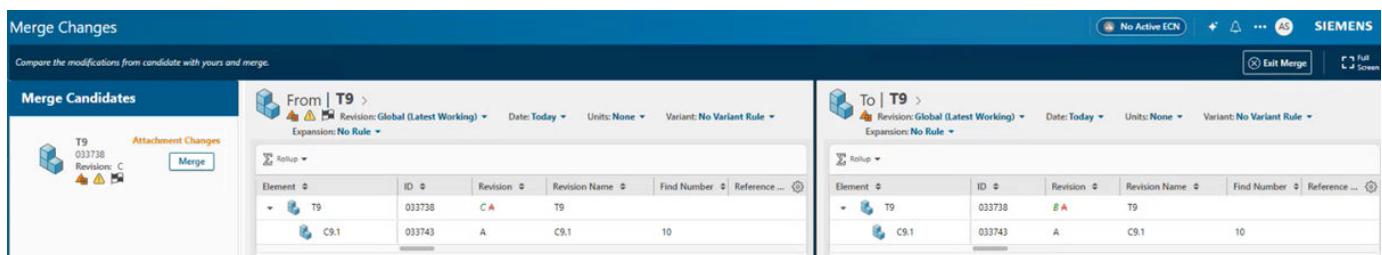


In the Change Summary, this action is labeled as **Property Modified**.

Merging attachments

When you release a change notice, any work-in-progress concurrent change notices that share the impacted items will receive a merge indicator in their Change Summary .

In a manual merge, you can click the merge indicator in the Change Summary to view a side-by-side presentation of the two assembly revisions. You have complete control over what to **merge**, allowing you to quickly see the differences as merge candidates and merge one or more required attachments into the final change notice.



For guidance on how to resolve attachment conflicts during an auto-merge, refer to [Managing change notice conflicts during an auto-merge](#) for more information.

After merging, the target object panel reflects the combined changes, while the Change Summary reflects the merged updates.

Manage and track substitutes

Substitute components are alternative items that can replace the original components in an assembly when preferred parts are unavailable. Engineers/Designers often define substitutes to use when preferred parts are unavailable, such as during a stock shortage. You can define substitutes for specific occurrences in an assembly and add or remove multiple substitutes for a part.

A *substitute group* consists of components that can be interchanged with another group of components within a structure.

When a change notice is issued, specific actions related to substitutes can be tracked and displayed.

You can add and remove substitutes in a change notice for both the Substitute Occurrence model and the Substitute Item model. For more information about substitutes, see [Defining substitute components](#).

Note

The **PSEnableOccToOccSubstituteGroups** preference, set by an administrator, controls whether substitutes added to an item revision are shared across all occurrences of that item in the assembly. See [Configure occurrence substitutes and substitute groups](#).

Substitute Occurrence model

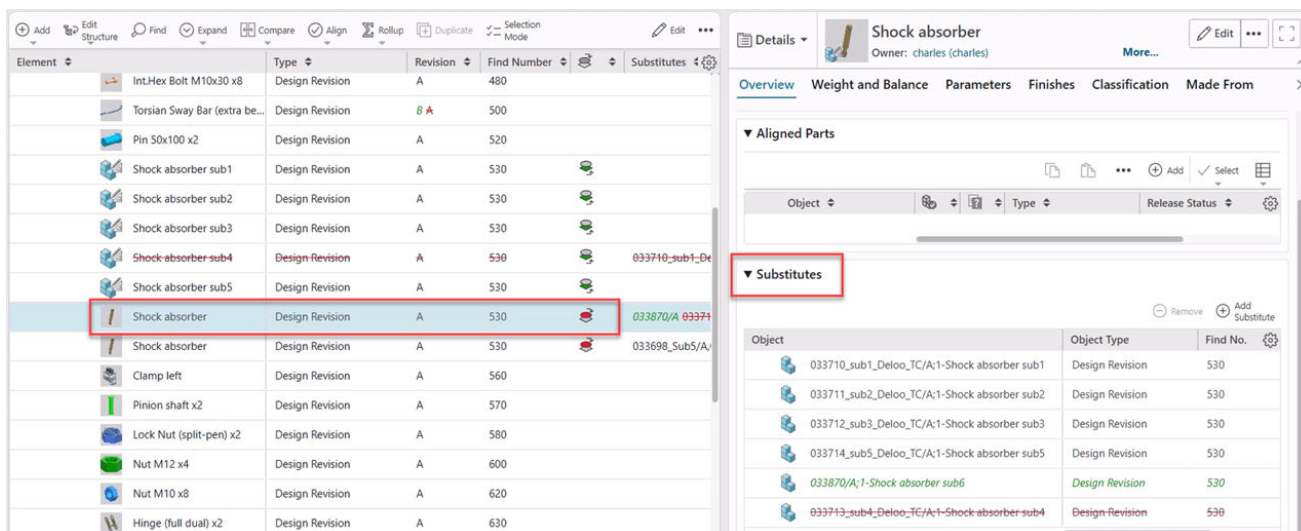
Procedure

1. Open the change notice and click the **Affected Items** tab.
2. Open the assembly BOM from the **Solution Items**.
3. Select the part for which you want to create or remove a substitute.
4. **To add a substitute:**
 - a. Click **Add**  **> Substitute** .
 - b. Enter a **Name**.
 - c. Click **Add**. The substitute is added to the **Substitutes** list in the **Details** pane, added to the BOM assembly, and tracked in the Change Summary with its corresponding indicator and green lines.

In the Change Summary, the substitute line displays the action labeled as **Substitute Added New** or **Substitute Added Existing**, along with the action **Property Modified** for the line where the substitute was added. These two lines are grouped together.
5. **To remove a substitute:**
 - a. In the BOM, identify a line item that contains a substitute to remove by its corresponding icon.
 - b. Right-click the line item and select **Remove**. You can also select the substitute in the **Details** pane and click **Remove** . The **Details** pane reflects the removed substitute. The substitute is added to the Substitutes list in the **Details** pane, removed from the BOM assembly, and tracked in the Change Summary with redlines.

In the Change Summary, the substitute line displays the action labeled as **Substitute Removed** and the action **Property Modified** for the line from which the substitute was removed. These two lines are grouped together.
6. **To set and unset a primary substitute:**
 - a. Right-click the selected substitute item and click **Set Primary** . This replaces the object in the structure with the selected substitute.

The **Occurrence Type** column indicates that the line modified line item is now the primary substitute , and the Change Summary shows the actions labeled as **Set Primary** and **Unset Primary** that are grouped together..



Object	Object Type	Find No.
033710_sub1_Deloo_TC/A;1-Shock absorber sub1	Design Revision	530
033711_sub2_Deloo_TC/A;1-Shock absorber sub2	Design Revision	530
033712_sub3_Deloo_TC/A;1-Shock absorber sub3	Design Revision	530
033714_sub5_Deloo_TC/A;1-Shock absorber sub5	Design Revision	530
033870/A;1-Shock absorber sub6	Design Revision	530
033713_sub4_Deloo_TC/A;1-Shock absorber sub4	Design Revision	530

Substitute Item model

Procedure

1. Open the change notice and click the **Affected Items** tab.
2. Open the assembly BOM from the **Solution Items**.
3. Select the part for which you want to create or remove a substitute.
4. **To add a substitute**
 - a. In the **Substitutes** section of the **Details** pane, click **Add** .
 - b. In the **Add Substitute** panel, select the **Type**.
 - c. Enter a **Name**.
 - d. Click **Add**. The substitute is added to the **Substitutes** list in the **Details** pane and added to the BOM assembly. The substitute part is added to the change notice as a solution. The line where the substitute is added is labeled as **Property Modified**, and an indicator is visible for this line .
 - e. To make use of the substitute elsewhere in an assembly, select the substitute and click **Use**. This option replaces the object in the structure with the selected substitute. The replaced substitute is indicated in the **Change Summary** by **Substitute Used** and **Substitute Use Removed**, respectively.
5. **To remove a substitute:**
 - a. In the BOM, identify a line item that contains a substitute to remove by its corresponding icon.

- b. Right-click the line item and select **Remove**. You can also select the substitute in the **Details** pane and click **Remove** . The Details pane reflects the removed substitute. The substitute is removed from the **Substitutes** list in the **Details** pane, removed from the BOM assembly, and tracked in the Change Summary with redlines.

Results

Adding a substitute

The substitute is added to the **Substitutes** list in the **Details** pane, added to the BOM assembly, and tracked in the Change Summary with its corresponding indicator and green lines.

Removing a substitute

The **Details** pane reflects the removed substitute. The substitute is removed from the **Substitutes** list in the **Details** pane, removed from the BOM assembly, and tracked in the Change Summary with redlines.

Merging substitutes

When you release a change notice, any work-in-progress concurrent change notices that share the impacted items will receive a merge indicator in their Change Summary .

In a [manual merge](#), you can click the merge indicator in the Change Summary to view a side-by-side presentation of the two assembly revisions. You have complete control over what to merge, allowing you to quickly see the differences as merge candidates and merge one or more required substitutes into the final change notice.

For guidance on how to resolve substitute conflicts during an auto-merge, refer to [Managing change notice conflicts during an auto-merge](#) for more information.

After merging, the target object's substitute changes are overridden by the source, and the target object panel displays these updates. Additionally, the Change Summary reflects the merged updates.

Manage and track item revision properties

You can change object revision properties for existing objects in a BOM that is included in a change notice. For example, you can change the object name, description, revision name, and other revision-level attributes, as well as occurrence property changes such as quantity and sequence number.

When you issue the change notice, the system tracks and displays specific actions related to these revisions in different views.

You can revert object revision property changes by using the Revert command. The [CM_keep_solution_item_after_revert](#) preference, which your administrator sets, controls the revert behavior only for attachments and vendor parts, and whether the solution item remains in the change

after the revert. Reverting structural changes and object revision property changes is independent of this preference.

Procedure

1. Open the change notice and click the **Affected Items** tab.
2. Open the assembly BOM from the **Solution Items**.
3. Select and revise the item in the BOM for which you want to make item revision updates to open **Details** pane.
4. The **Overview** tab displays the item properties available to edit.

Results

Changes made to item revision properties are tracked and displayed as red lines on the item revision's **Overview** page, in the BOM context, and in the Change Summary.

The screenshot shows the SAP Change Management interface. On the left, a BOM table lists various components. The 'laminated spring x2' item is highlighted with a red box. On the right, the 'Details' pane is open for 'laminated spring x2'. The 'Overview' tab is selected, showing properties for revision 'B'. The 'Revision' field is highlighted with a red line, indicating a change from 'A' to 'B'. The 'Revision Name' is 'laminated spring Leefspring', also with a red line. The 'Find Number' is 470 and 'Quantity' is 2.

Element	Type	Revision	Find Number	Subst...	Release Status
Truck Aux. and Power Supply (r5)	Design Revision	A	20		Released
Truck Transmission (r5)	Design Revision	A	30		Released
Truck Rear Axle (r5)	Design Revision	A	40		Released
Truck Front Axle (r5)	Design Revision	B A	50		
Front (Bare) Axles	Design Revision	B A	10		
Front Axle (20L steered, free...)	Design Revision	B A	10		
Flanged M18 Nut x2	Design Revision	A	440		Released
Bolt for Torsion Bar x2	Design Revision	A	450		Released
Spring Clip 33mm x8	Design Revision	A	460		Released
laminated spring x2	Design Revision	B A	470		

Merging properties

When you release a change notice, any work-in-progress concurrent change notices that share the impacted items will receive a merge indicator in their Change Summary .

In a manual merge, you can click the merge indicator in the Change Summary to view a side-by-side presentation of the two assembly revisions. You have complete control over what to merge, allowing you to quickly see the differences as merge candidates and merge one or more properties into the final change notice. However, you cannot pick and choose individual properties for the merge; it must be all or none.

For guidance on how to resolve item revision property conflicts during an auto-merge, refer to [Managing change notice conflicts during an auto-merge](#).

After merging, the target object panel reflects the combined changes, while the Change Summary reflects the merged updates.

Administrators can choose which **properties to exclude** during a merge by creating a site-level preference or updating an existing preference value, ensuring these exclusions are applied generically across all relevant items.

Change tracking for reference designators

Reference designators clearly identify the location of a component within an electrical schematic or on a printed circuit board. Commonly used in electronics, reference designators typically consist of a letter (or letters) followed by a number (for example, R1 for a resistor, C2 for a capacitor).

In Change Management, *reference designators* are a common component used by engineers to record position details of components on a printed circuit board (PCB) in a PCB assembly BOM. When a change is proposed, the reference designators use a unique text field to identify the specific components affected. There can often be a very high number of reference designators in a change notice, especially in complex designs.

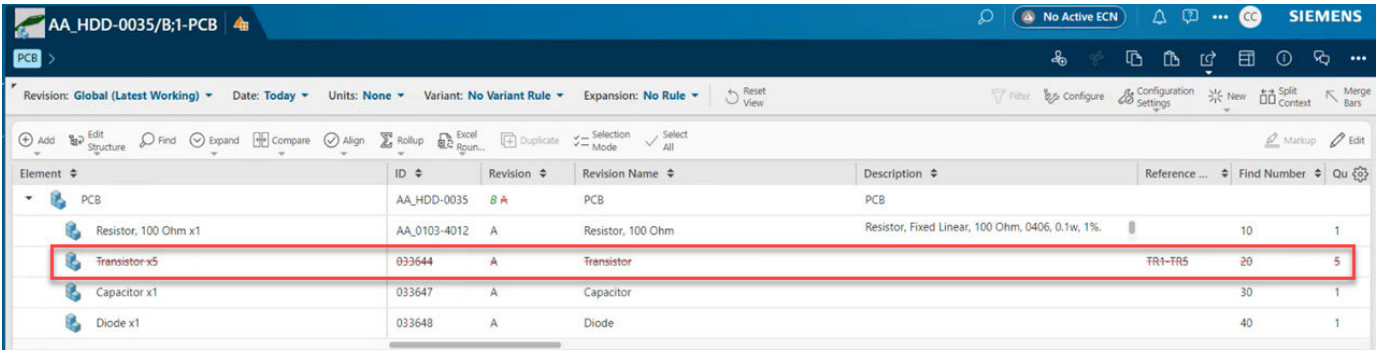
The change notice highlights any modified reference designators by showing redlines in both packed and unpacked mode. For instance, if you pack five occurrences of the same part with different reference designators in a change notice, the **Reference Designator** column displays them as a concatenated set, such as **TR1-TR5**. To edit reference designators, you must first unpack them. After making any edits, you can repack the reference designators and identify the changes in a single row using redlines. The **Quantity** and **Reference Designator** columns are automatically updated on the packed line to reflect the modifications.

Note

The BOM shows both packed and unpacked modifications, but the Change Summary only displays modifications in unpacked mode.

Remove all reference designators

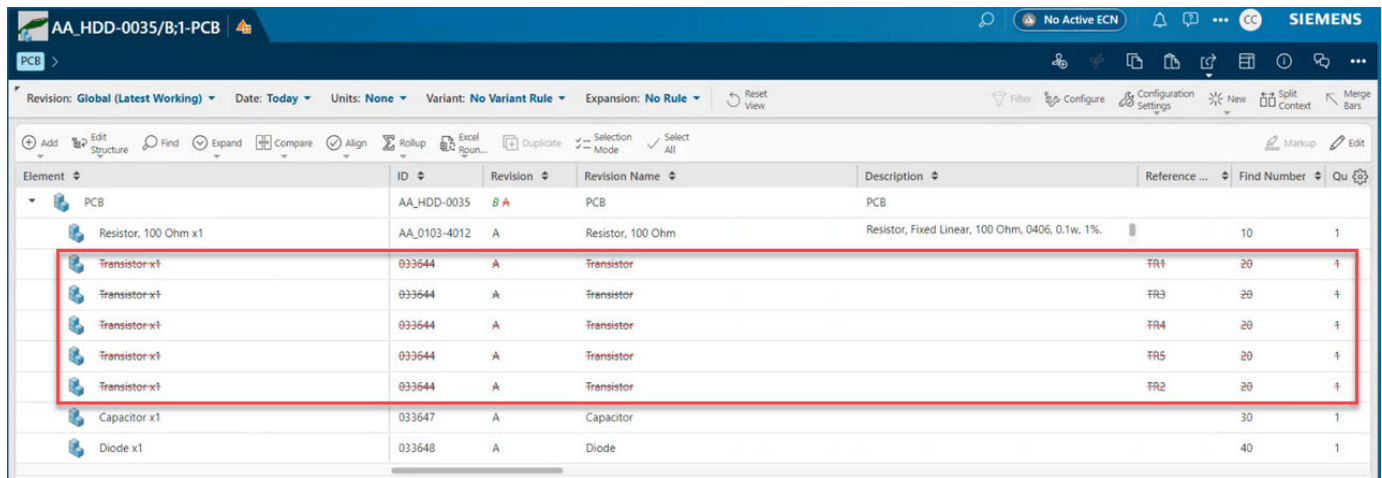
Here, five transistors were removed, which the **Reference Designator** column reflects in a range as **TR1-TR5**.



The screenshot shows the Siemens Change Management interface. At the top, the project is identified as 'AA_HDD-0035/B;1-PCB'. Below the header, there are various filters and toolbars. The main table displays a Bill of Materials (BOM) with columns: Element, ID, Revision, Revision Name, Description, Reference, Find Number, and Quantity. The 'Transistor x5' row is highlighted with a red border, showing a concatenated reference designator 'TR1-TR5' and a quantity of 5.

Element	ID	Revision	Revision Name	Description	Reference	Find Number	Quantity
PCB	AA_HDD-0035	B	PCB	PCB			
Resistor, 100 Ohm x1	AA_0103-4012	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100 Ohm, 0406, 0.1w, 1%		10	1
Transistor x5	033644	A	Transistor		TR1-TR5	20	5
Capacitor x1	033647	A	Capacitor			30	1
Diode x1	033648	A	Diode			40	1

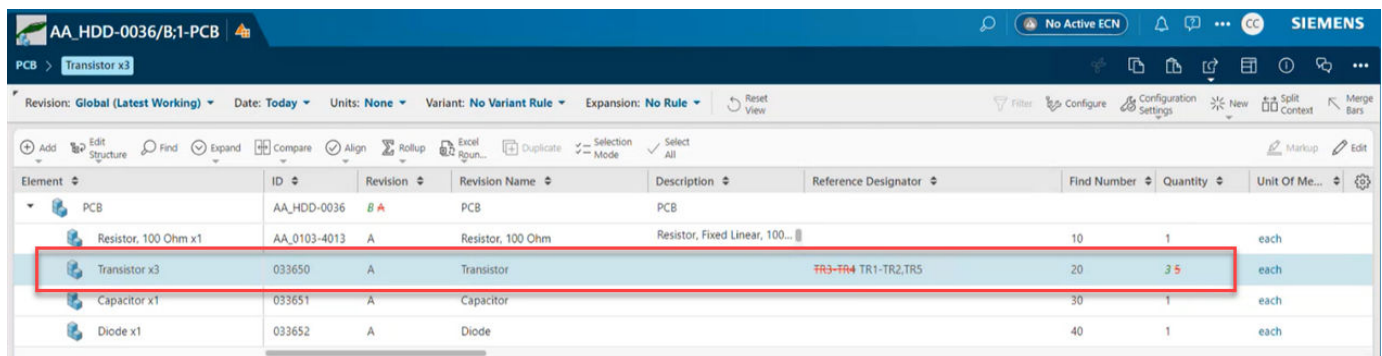
Click **Expand, Unpack All** to view the unpacked transistors:



Element	ID	Revision	Revision Name	Description	Reference ...	Find Number	Qu
PCB	AA_HDD-0035	B A	PCB	PCB			
Resistor, 100 Ohm x1	AA_0103-4012	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100 Ohm, 0406, 0.1w, 1%		10	1
Transistor x1	033644	A	Transistor		TR1	20	1
Transistor x1	033644	A	Transistor		TR3	20	1
Transistor x1	033644	A	Transistor		TR4	20	1
Transistor x1	033644	A	Transistor		TR5	20	1
Transistor x1	033644	A	Transistor		TR2	20	1
Capacitor x1	033647	A	Capacitor			30	1
Diode x1	033648	A	Diode			40	1

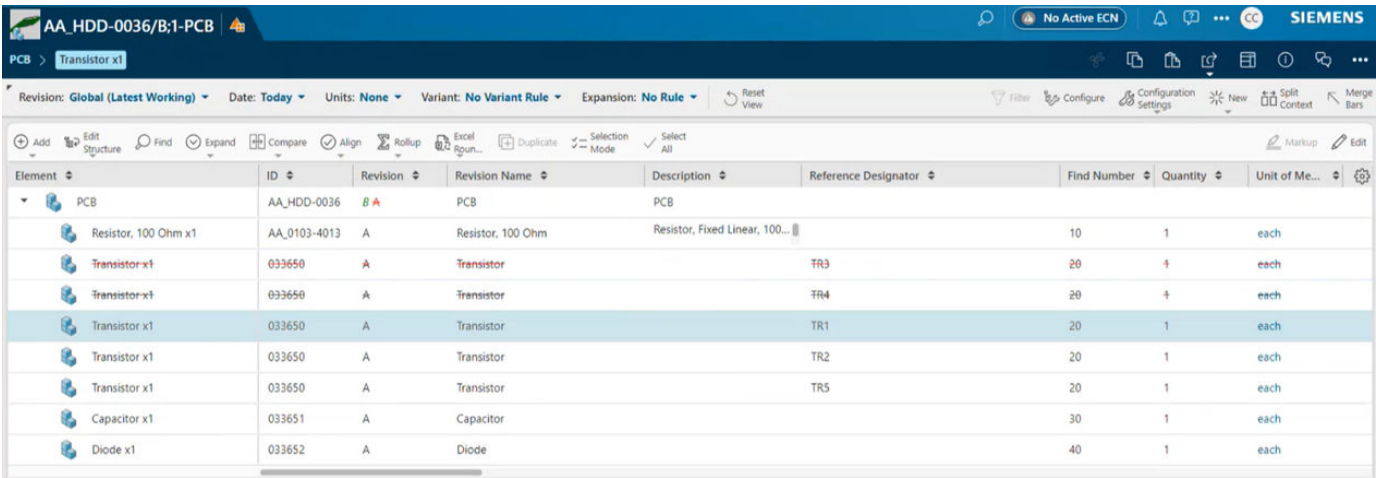
Perform a partial removal

If you perform a partial removal in a change context, the removed items will combine with the remaining items in packed mode. The **Reference Designator** and **Quantity** property will update and show as redlined. For example, if you remove two transistors (TR3 and TR4) out of five, the remaining transistors will appear in a single row in packed mode. The **Reference Designator** will show TR3 and TR4 as removed with a red strike-through, while TR1, TR2, and TR5 will remain unchanged. The **Quantity** will display 3 in green and 5 with a red strike-through.



Element	ID	Revision	Revision Name	Description	Reference Designator	Find Number	Quantity	Unit Of Me...
PCB	AA_HDD-0036	B A	PCB	PCB				
Resistor, 100 Ohm x1	AA_0103-4013	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100...		10	1	each
Transistor x3	033650	A	Transistor		TR3-TR4 TR1-TR2,TR5	20	3 5	each
Capacitor x1	033651	A	Capacitor			30	1	each
Diode x1	033652	A	Diode			40	1	each

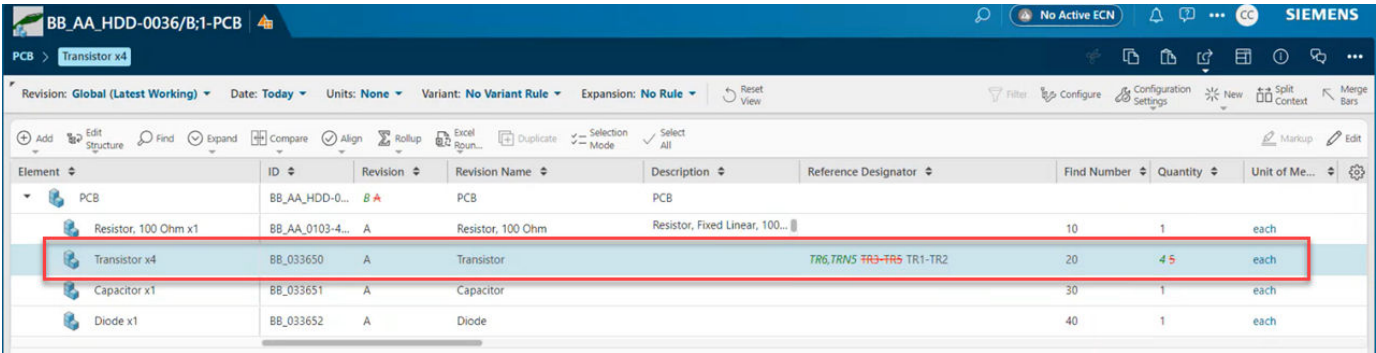
Click **Expand**, **Unpack All** to view the unpacked modifications:



Element	ID	Revision	Revision Name	Description	Reference Designator	Find Number	Quantity	Unit of Me...
PCB	AA_HDD-0036	B	PCB	PCB				
Resistor, 100 Ohm x1	AA_0103-4013	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100...		10	1	each
Transistor x1	033650	A	Transistor		TR3	20	1	each
Transistor x1	033650	A	Transistor		TR4	20	1	each
Transistor x1	033650	A	Transistor		TR1	20	1	each
Transistor x1	033650	A	Transistor		TR2	20	1	each
Transistor x1	033650	A	Transistor		TR5	20	1	each
Capacitor x1	033651	A	Capacitor			30	1	each
Diode x1	033652	A	Diode			40	1	each

Perform multiple modifications

When you make multiple modifications to an assembly, such as property edits, adding and removing lines, and other changes, all those adjustments pack into a single line. For example, if an assembly contains multiple removed lines, added lines, and property edits, the packed line for the **Reference Designator** property will display the new values and added lines in green. The old values and removed lines will appear as red strike-through, followed by unchanged values.



Element	ID	Revision	Revision Name	Description	Reference Designator	Find Number	Quantity	Unit of Me...
PCB	BB_AA_HDD-0...	B	PCB	PCB				
Resistor, 100 Ohm x1	BB_AA_0103-4...	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100...		10	1	each
Transistor x4	BB_033650	A	Transistor		TR6, TR5, TR3, TR5, TR1, TR2	20	4	each
Capacitor x1	BB_033651	A	Capacitor			30	1	each
Diode x1	BB_033652	A	Diode			40	1	each

Click **Expand, Unpack All** to view the unpacked modifications:

Element	ID	Revision	Revision Name	Description	Reference Designator	Find Number	Quantity	Unit of Me...
PCB	BB_AA_HDD-0036/B:1-PCB		PCB	PCB				
Resistor, 100 Ohm x1	BB_AA_0103-4...	A	Resistor, 100 Ohm	Resistor, Fixed Linear, 100...		10	1	each
Transistor x1	BB_033650	A	Transistor		TR3	20	1	each
Transistor x1	BB_033650	A	Transistor		TR4	20	1	each
Transistor x1	BB_033650	A	Transistor		TR1	20	1	each
Transistor x1	BB_033650	A	Transistor		TR6	20	1	each
Transistor x1	BB_033650	A	Transistor		TR2	20	1	each
Transistor x1	BB_033650	A	Transistor		TR5 TR5	20	1	each
Capacitor x1	BB_033651	A	Capacitor			30	1	each
Diode x1	BB_033652	A	Diode			40	1	each

New or updated values will always display before removed items in the **Reference Designator** column.



Packed modifications display in a BOM as follows:

Modification	Reference Designator column displays items as:
Additional items added	Green
Items removed	Red, strike-through
Property changes	New values in green, old values in red strike-through

Consolidate multiple reference designator modifications in a single line

You can choose to keep multiple values of reference designators in a single line. Instead of having to unpack multiple lines, redlines will display on the same line, making it easier to identify different quantities. This is especially useful when dealing with a large number of modified reference designators.

For example, you can add a single line with multiple values, such as **CR1-CR5**. If you later edit **CR2** to **CRN2**, the change notice will display this on one (unpacked) line:

Element	ID	Revision	Revision Name	Description	Reference Designator	Quantity	Unit of Me...	Find Number	Release Effectivity
011019/B:1-T7	011019	A	T7						
011020/A:1-CT.1 x 3	011020	A	CT.1		TRN2 TR2 TR1 TR3	3	each	10	
011023/A:1-CT.2 x 3	011023	A	CT.2		CRN2 CR2 CR1 CR3	3	each	40	

- CRN2 is green, since that is the new value.

- CR2 is red, since that is the removed value.
- CR1, CR3 are unchanged since they had no modifications.

Modifications are reflected the same way in the [Change Summary](#).

Save As operations in change tracking

Using Save As, you can create a new item or assembly with a new Item ID and its own lifecycle. This lets you clearly compare versions, maintain traceability, and track redline changes between product variants during a change process. When you perform a Save As operation a change notice, the new item inherits any effectivity from the active change notice, and all modifications are tracked with redlines in the Change Summary.

Save As supports dash-roll behavior. You can use this to introduce new product types or variants while still keeping a clear distinction between the original product and the new one. Lineage is retained between the original and new part for change tracking. The Change Summary displays specific action labels based on the operation performed: Save As, Saved As to Replace, or Saved As to Split. When effectivity is applied, the Change Summary shows the effectivity date ranges for both the original and new items.

After performing a Save As operation and making BOM modifications to the design assembly, you can align the changes with the collaborative product BOM. The guided update process transfers the modifications from the design BOM to the product BOM while preserving the effectivity dates from the ECN.

What's the difference between Save as and Revise?

Revise: Creates a new revision of an existing item with the same Item ID (for example, Rev A becomes Rev B). The new revision preserves its lineage to the previous revision, enabling change tracking in the Change Summary within the context of a change notice.

Save As: Creates a new item or assembly with a new Item ID and its own lifecycle, so you can compare, track, and redline changes between product variants in the context of a change.

What's the difference between Save As vs. Save As and Replace?

Save As

- Creates a new item with a new Item ID
- Does *not* automatically replace the original item in the BOM structure
- The new item exists independently
- Change Summary action: **Saved As**
- Example: Creating a product variant that exists alongside the original (not replacing it)

Save As and Replace

- Creates a new item with a new Item ID *and* replaces the original item in the BOM structure
- The original item is marked as **Replaced** (shown with red strikethrough in redlines)
- The new item takes the place of the original in the assembly
- Change Summary action: **Saved As to Replace**
- Example: Replacing an existing part with a completely new part (different Item ID; the dash roll behavior)

Occurrence effectivity with Save As

When performing a Save As and Replace, you can manage occurrence effectivity in two ways:

Without Split:

When you perform Save As and Replace without split, the new item completely replaces the original item with the same effectivity range. The effectivity dates overlap entirely (complete overlap), and the Change Summary displays the action as **Saved As to Replace** on a single line.

With Split:

When you perform Save As and Replace with split, the effectivity range is divided between the original and new items to avoid overlap. This creates different effectivity ranges where the original item's effectivity is cut back to end before the new item becomes effective. The Change Summary displays the action as **Saved As to Split** and shows two lines with a border to indicate the split effectivity ranges.

For example, if the original part is effective from 01-Feb-2022 to UP (unlimited), and you save as and replace with split effective 01-May-2022:

- The original part's effectivity is cut back to 01-Feb-2022 to 30-Apr-2022
- The new part's effectivity begins at 01-May-2022 to UP

This split effectivity prevents overlap and ensures clear traceability of when each variant is effective in the product structure.

Using **Save As and Replace** (with or without split) lets you create product variants while maintaining their independent lifecycles. For example, when shifting from a Rolls Royce engine to a GE Jet engine, the engine mount assembly must be completely replaced with a new assembly rather than a revised version of the existing assembly. **Save As and Replace** enables this replacement while maintaining traceability between the original and new product variants.

Change Summary redlines

All BOM modifications made to the saved-as assembly are tracked with redlines in the Change Summary, including:

- Adding new parts
- Replacing parts
- Removing parts
- Property changes
- Insert level and remove level operations
- Substitutes and effectivity changes

The Change Summary displays the appropriate action label based on the type of Save As operation:

Save As: When you save an assembly as a new item

Saved As to Replace: When you save an assembly as a new item and replace the original

Saved As to Split: When you save an assembly as a new item with replace and split effectivity

The Change Summary displays the results of your actions based on the operation you perform:

- **Save As and Replace:** Change Summary displays one line with redlining and the action Saved As to Replace. When effectivity exists on the selection, Change Summary displays two lines with a border: one with the action Saved As to Replace and one with the action Saved As.
- **Save As and Replace with Split:** Change Summary displays one line with redlining and the action Saved As to Split. When effectivity exists on the selection, Change Summary displays two lines with a border: one with the action Saved As to Split and one with the action Saved As.
- **Save As:** Change Summary displays one line with redlining and the action Saved As. When effectivity exists on the selection, Change Summary displays two lines with a border, both with the action Saved As.
- **Create Replace Group (new solution item):** When you select a new solution item and the replaced item (shown in red strikethrough) and choose **Create Replace Group**, the Change Summary displays one line with redlining and the action **Added New to Replace**.
- **Create Replace Group (existing solution item):** When you select an existing solution item and the replaced item (shown in red strikethrough) and choose **Create Replace Group**, the Change Summary displays one line with redlining and the action **Added Existing to Replace**.

Perform a Save As operation in a change context

Using the Save As functionality, you can create a new item or assembly with a new Item ID and an independent lifecycle, allowing controlled comparison, traceability, and redline tracking of changes between product variants in the context of a change.

Prerequisites

An active change context is set.

Procedure

1. Open the active change notice with the BOM to edit, and click the **Affected Items** tab.
2. Open the BOM from solution item to modify.
3. From the Page Level toolbar, click **New > Save As and Replace** .
4. In the **Save As And Replace** panel, configure the new item properties:
 - (Optional) Check **Clone Children** to create new, independent item revisions for all child components (parts and subassemblies) referenced by the top-level assembly, instead of reusing the original components. This breaks the relationship to the source structure so that you can modify the cloned assembly and its children without impacting the original assembly.
 - (Optional) Check **Split Occurrence** to create a parallel subassembly structure for the replacement assembly with non-overlapping effectivity ranges. The occurrence splits automatically using the effectivity dates or units from the active change notice. The original item's effectivity is cut back and the new item's effectivity begins where the original ends.

If automatic splitting is disabled, the occurrence still splits, but you need to set the effectivity dates or units yourself.

Note

If the new occurrence does not appear in the structure, use **Show Excluded by Effectivity** to display it (in the work area toolbar, click **Configure**  > **Show Excluded by Effectivity**).

If you do not select **Split Occurrence**, the new item completely replaces the original item with the same effectivity range.

- Enter the item **Name** and **Description**.
 - If you are creating a design, select the **Copy Part Alignment** check box. If you select this check box, another revision of the existing aligned part is automatically created and aligned with the design. If you do not select this check box, a new revision of the existing aligned part is created at the time of alignment.
 - Specify the **Advanced Copy Options** as needed.
5. Click **Save As And Replace**.
The system creates the new assembly and automatically:
 - Adds it as a solution item to the active change
 - Creates lineage between the original (impacted) and new (solution) assembly
 - Enables change tracking for all BOM modifications
 6. Make any required BOM modifications to the saved-as assembly.
All modifications are tracked with redlines and displayed in the Change Summary.

7. (Optional) Perform a **guided update** to align the design BOM changes with the collaborative product BOM.
 - a. From the Change Summary, select the modified design assembly and click **Align > Guided Update**.
 - b. In the **Guided Update** panel, review the changes listed.

The design structure and its aligned engineering BOM are displayed side by side.
 - c. Click **Update** to transfer the modifications to the engineering BOM.

The effectivity dates from the change notice are preserved in the product BOM assembly.

Results

The saved-as assembly appears in the **Solution Items** with lineage to the original assembly. The Change Summary displays the action as **Saved As**, **Saved As to Replace**, or **Saved As to Split** based on the options selected:

- **Save As:** When you save an assembly as a new item
- **Saved As to Replace:** When you save an assembly as a new item and replace the original
- **Saved As to Split:** When you save an assembly as a new item with replace and split effectivity

The Change Summary tracks all BOM modifications with redlines and displays the effectivity date ranges for both the original and new items (under the **Occurrence Effectivity** column). Property changes, replacements, and additions are indicated with specific action labels and effectivity dates.

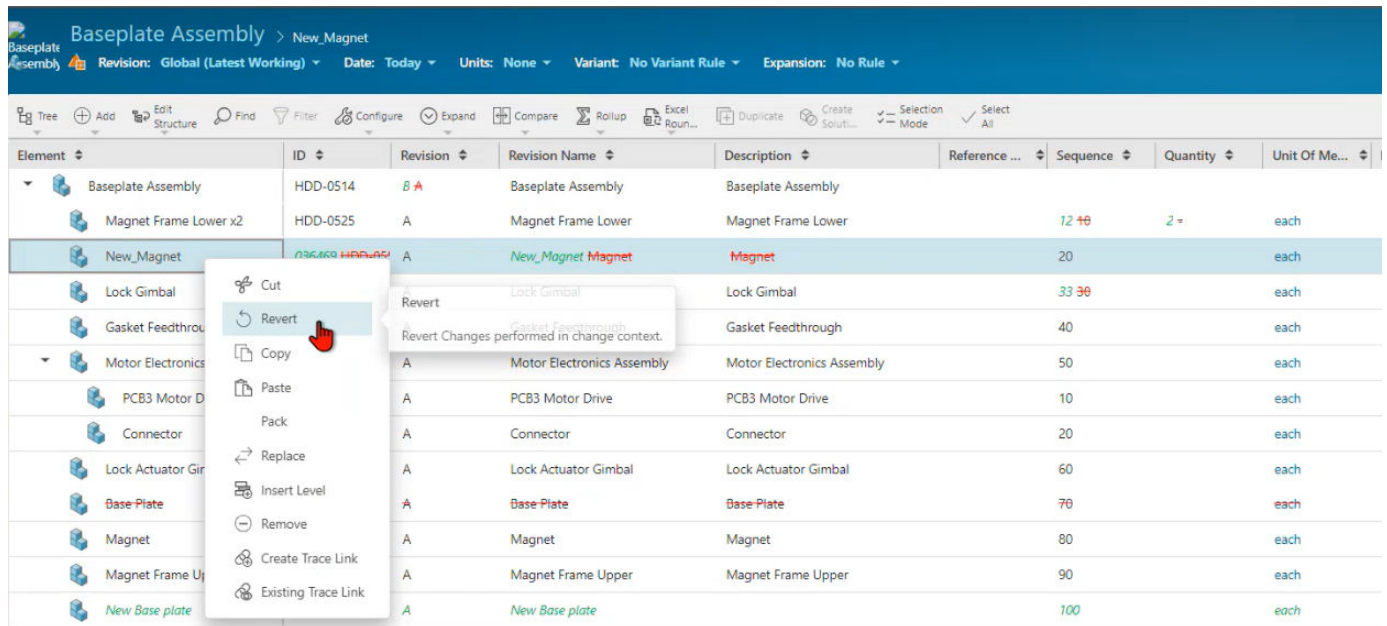
Revert BOM changes

Revert BOM modifications

The revert functionality applies to those ECNs that include redline changes. When a user is authoring a BOM in the context of an ECN and has either replaced, removed, or added a new line item(s), sometimes unintentional changes can occur. An incorrect entry might unintentionally be made.

The revert functionality lets you return one or multiple redline changes to their original impacted version (the baseline state of the item before the change notice was created, typically a released revision such as revision A), with or without release effectivity dates. You can revert structural changes (add, remove, replace), including occurrence property changes (for example, quantity and sequence number), as well as object revision property changes, attachment changes, and vendor part changes. The **CM_keep_solution_item_after_revert** site preference controls whether attachments and vendor parts are reverted and whether the solution item is removed from the change object. When reverting changes to

commercial parts, the revert functionality handles vendor part modifications based on this preference setting.



Revert a BOM without release effectivity dates

The revert functionality lets you return a redline change back to its original impacted version. This procedure explains how to revert structural changes in items in BOMs without any release effectivity dates applied.

Object revision property changes can always be reverted. Attachment changes and vendor part changes can be reverted based on the **CM_keep_solution_item_after_revert** preference, which your administrator sets.

Procedure

1. Open the ECN with the BOM to edit, and click the **Affected Items** tab.

Note

You cannot revert a line item in a closed ECN.

2. Open the BOM from solution item to modify and make any changes to your BOM items.
3. To revert a change, select the line item(s) and click **Edit Structure** > **Revert** or right-click and select **Revert**. A confirmation message displays indicating the revert status and the number of items successfully reverted; click **Revert** at the confirmation message. The line item(s) returns to its original impacted version in the BOM.

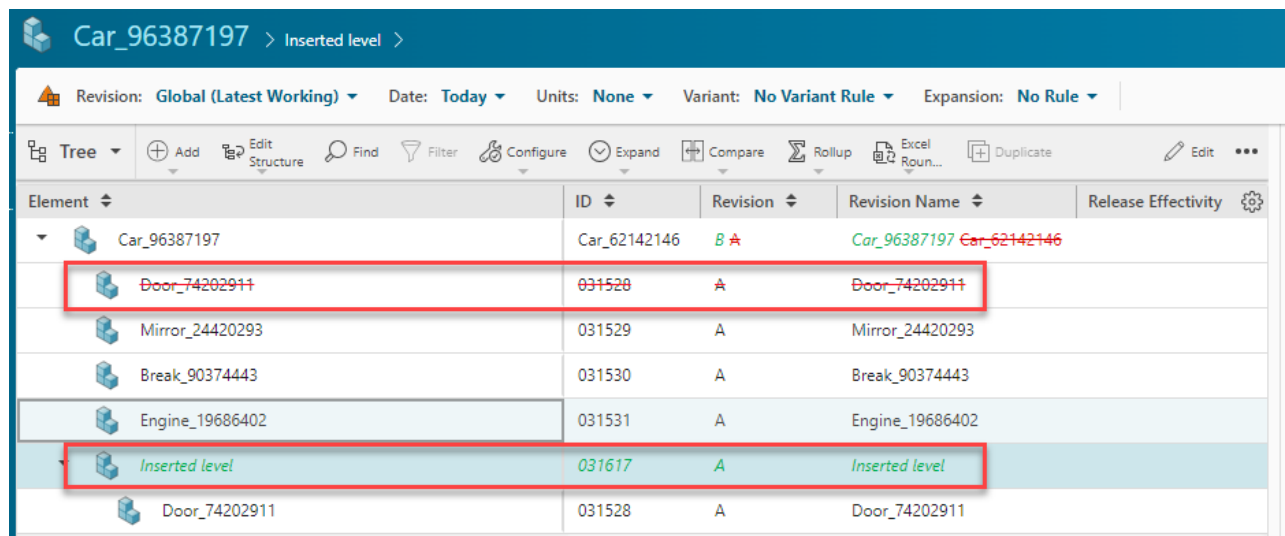
Note the following:

- You can revert more than one line item at a time. When you select multiple items for revert and some items have no redlines, the system reverts only the items with modifications. A message displays showing how many items were successfully reverted and lists the specific items that were not reverted because they had no changes.
- If you add a new line item and revert, the line item is removed from the BOM. However, the new (unwanted) line item still displays in the Change Summary for tracking purposes, even though it is not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.
- If you remove a line item and revert, the line item returns to the BOM.
- If you replace a line item with a new item and revert, the original line item returns to the BOM.

However, the new (unwanted) line item still displays in the Change Summary for tracking purposes, even though it is not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.

If you replace a line item with an existing line item and revert, that existing line item will not show in the Change Summary.

- If you update any line item(s) properties and revert (**Quantity**, **Sequence**, and so on), those edits are returned to the original properties in the BOM.
- When you insert a level and revert, you must revert both the newly inserted level as well as the removed (red-lined) level to return the structure to the original.



Element	ID	Revision	Revision Name	Release Effectivity
Car_96387197	Car_62142146	B	Car_96387197 Car_62142146	
Door_74202911	031528	A	Door_74202911	
Mirror_24420293	031529	A	Mirror_24420293	
Break_90374443	031530	A	Break_90374443	
Engine_19686402	031531	A	Engine_19686402	
Inserted level	031617	A	Inserted level	
Door_74202911	031528	A	Door_74202911	

However, the new (unwanted) inserted level still displays in the Change Summary for tracking purposes, even though it is not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.

- If you remove a level and revert, each line item that was associated with the removed level must be reverted, along with the removed (red-lined) line item, to return the structure to the original.

Element	ID	Revision	Revision Name	Release Effectivity
Car_62142146	Car_62142146	C A	Car_62142146	
Mirror_24420293	031529	A	Mirror_24420293	
Break_90374443	031530	A	Break_90374443	
Engine_19686402	031531	A	Engine_19686402	
Insert-level	031610	A	Insert-level	
Door_74202911	031528	A	Door_74202911	

- If you modify a packed line item and revert, each line in the pack is reverted to the original. If you unpack a line item, each line can be modified and then reverted individually.
- If you add or remove an attachment to a line item revert, those changes are reflected in the BOM.
- In the case of an assembly BOM, a line item that is a subassembly with redlines cannot be reverted unless its children are reverted first. Once you revert the children, you can revert the subassembly line. If you select the subassembly line and revert, the revision is no longer associated with the change context and is no longer redlined. The line item is also not shown in the Change Summary as that revision is no longer in the **Solution Items** or **Impacted Items** and will not be released in the current ECN. This behavior also applies to the top line in an assembly BOM.

If you attempt to revert a parent item that has redlined children without first reverting the children, the system displays an error message indicating that the item cannot be reverted because it has redlined children. You must revert the children first, then revert the parent. When this occurs, no changes are reverted for the parent item, including property changes and attachment changes.

- When you select multiple line items and **Revert**, the confirmation message shows the number of line items that were reverted.

If not all selected items are redlined (meaning there are no modifications to revert) and you choose to revert, the confirmation message still shows the total number of line items selected. Upon continuing and selecting **Revert**, the error message displays how many items were successfully reverted and provides reasons for any items that were not reverted.



Note

You cannot revert a line item that is defined as an absolute occurrence.

Revert a BOM with release effectivity dates

The revert functionality lets you return a redline change back to its original impacted version. This procedure explains how to revert items in a BOM with release effectivity dates applied.

If you add release effectivity, the date range displays in the **Occurrence Effectivity** column of the Change Summary. For example, in this Change Summary, 031983 was added as a new line item to *replace* 031976 and the date range for 031983 now reflects the new release effectivity of **15-Jan-2025 to UP**:

Change Summary | Overview | ECN-000095/A;1-car-ecn2

ID	Action	Revision	Element	Revision Name	Occurrence Effectivity	Quantity	Sequence
031975	Modified	B A	Car	Car			
031979	Property Changed	A	Break	Break	01-Jan-2025 01:00 to 15-Jan-2025 00:59 (NONE) 01-Jan-2025 01:00 to UP (NONE)	30	
031979	Property Changed	A	Break	Break	15-Jan-2025 01:00 to UP (NONE)	66 - 300 30	
031976	Replaced	A	Door	Door	01-Jan-2025 01:00 to 15-Jan-2025 00:59 (NONE) 01-Jan-2025 01:00 to UP (NONE)	10	
031983	Added Existing to Repla...	A	Replace-Part Door	Replace-Part Door	15-Jan-2025 01:00 to UP (NONE)	10	
031982	Added Existing	A	New-Part	New-Part	15-Jan-2025 01:00 to UP (NONE)	50	
031977	Removed	A	Window	Window	01-Jan-2025 01:00 to 15-Jan-2025 00:59 (NONE) 01-Jan-2025 01:00 to UP (NONE)	20	

However, once a revert is performed on the replaced item, the line item reflects the original release date range, or removes the date range if one was not present previously.

Procedure

- 1. Open the ECN with the BOM to edit that includes effectivity dates, and click the **Affected Items** tab.

Note

You cannot revert a line item in a closed ECN.

- 2. Open the solution item to modify and make any changes to your BOM items.
To show any occurrences that are excluded by the currently applied effectivity, in the work area toolbar, click **Configure** > **Show Excluded by Effectivity**. This will let you compare all of the effectivity date ranges between items in the BOM.
- 3. To revert a change, select the line item(s) and right-click and select **Revert**. A confirmation message displays, indicating the revert status and the number of items successfully reverted. Click **Revert** at the confirmation message.
The line item(s) returns to its original impacted version in the BOM. The Change Summary displays the line item with reverts incorporated.

Note the following for items with release effectivity applied:

- You can revert more than one line item at a time.
- If you add a new line item and revert, the line item is removed from the BOM. However, the new (unwanted) line item still displays in the Change Summary for tracking purposes, even though it is

not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.

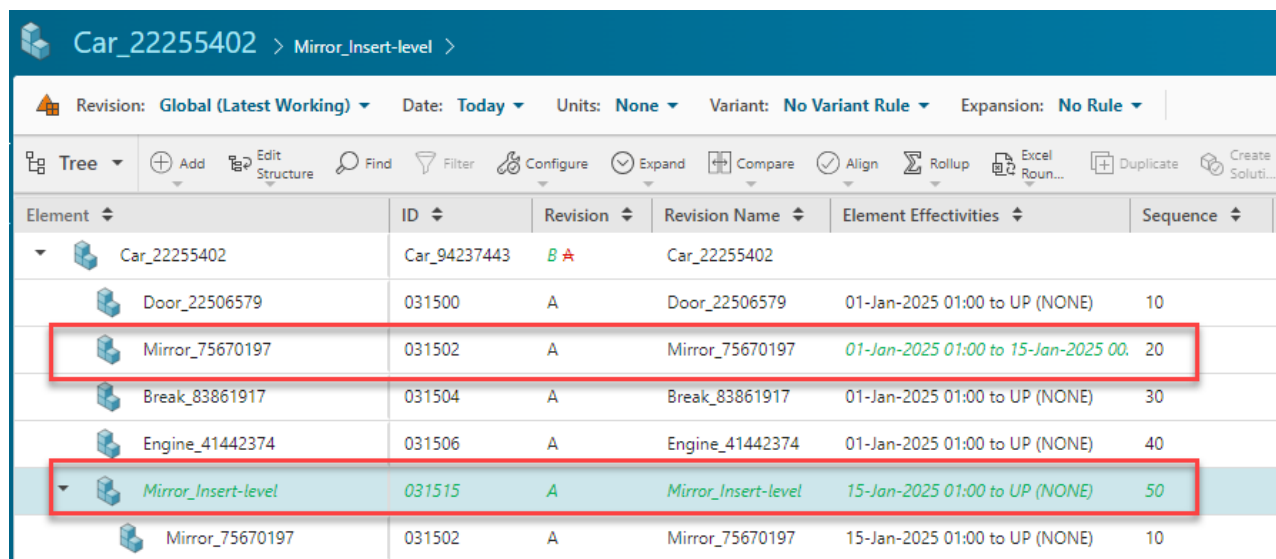
Once reverted, the occurrence effectivity date range returns to the date range specified in the original ECN.

- If you revert after removing a line item, the line item returns to the original date range.
- If you replace a line item with a new item and revert, the original line item returns to the BOM.

However, the new (unwanted) line item still displays in the Change Summary for tracking purposes, even though it is not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.

If you replace a line item with an existing line item and revert, that existing line item will not show in the Change Summary.

- If you revert after updating any line item properties (**Quantity**, **Sequence**, and so on), the new line item with the updated properties is removed, and the original line item returns to the original effectivity date range in the BOM.
- When you insert a level and revert, you must revert the newly inserted level and the original line item (on which the inserted level was performed) to return the structure to the original.



Element	ID	Revision	Revision Name	Element Effectivities	Sequence
Car_22255402	Car_94237443	B	Car_22255402		
Door_22506579	031500	A	Door_22506579	01-Jan-2025 01:00 to UP (NONE)	10
Mirror_75670197	031502	A	Mirror_75670197	01-Jan-2025 01:00 to 15-Jan-2025 00:00	20
Break_83861917	031504	A	Break_83861917	01-Jan-2025 01:00 to UP (NONE)	30
Engine_41442374	031506	A	Engine_41442374	01-Jan-2025 01:00 to UP (NONE)	40
Mirror_Insert-level	031515	A	Mirror_Insert-level	15-Jan-2025 01:00 to UP (NONE)	50
Mirror_75670197	031502	A	Mirror_75670197	15-Jan-2025 01:00 to UP (NONE)	10

However, the new (unwanted) inserted level still displays in the Change Summary for tracking purposes, even though it is not associated with the BOM structure. You can cut the line item from the **Affected Items** tab of the solution item to completely remove it from the Change Summary.

Once reverted, the occurrence effectivity date range returns to the date range of the line item specified in the original ECN.

- If you remove a level and revert, each line item that was associated with the removed level must be reverted, along with the original line item, to return the structure to the original.

Once reverted, the occurrence effectivity date range returns to the date range of the line item specified in the original ECN.

- If you modify a packed line item and revert, each line in the pack is reverted to the original. If you unpack a line item, each line can be modified and then reverted individually. When the **CM_keep_solution_item_after_revert** preference is set to false, if you add or remove an attachment to a line item and revert or make a property change to the relation of an existing attachment, those changes are reverted. When the preference is set to true, attachment changes are preserved. The same behavior applies to vendor part changes..

If you unpack a line item, each line can be modified and then reverted individually.

- If you add or remove an attachment to a line item revert, those changes are reflected in the BOM.
- In the case of an assembly BOM, a line item that is a subassembly with redlines cannot be reverted unless its children are reverted first. Once you revert the children, you can revert the subassembly line. If you select the subassembly line and revert, the revision is no longer associated with the change context and is no longer redlined. The line item is also not shown in the Change Summary as that revision is no longer in the **Solution Items** or **Impacted Items** and will not be released in the current ECN. This behavior also applies to the top line in an assembly BOM.
- When you select multiple line items and **Revert**, the confirmation message shows the number of line items that were reverted.

If not all selected items are redlined (meaning there are no modifications to revert) and you choose to revert, the confirmation message still shows the total number of line items selected. Upon continuing and selecting **Revert**, the error message displays how many items were successfully reverted and provides reasons for any items that were not reverted.



Note

You cannot revert a line item that is defined as an absolute occurrence.

Revert object revision property changes

The revert feature allows you to undo changes to object revision properties made during an active change. When you are updating item revisions in a change notice, you can adjust properties like the item name, description, revision name, quantity, sequence, and other revision-level attributes. If you make any accidental changes, you can revert them to restore the original property values.

The **CM_keep_solution_item_after_revert** site preference controls the revert behavior.

Prerequisites

- An active change notice must be set as the active change context.

- Object revision property changes must have been made to item revisions in the BOM.
- The change must not be released.

Procedure

1. Open the change notice with the active change context.
2. Open the BOM from solution item to modify and make any changes to your BOM items.
3. Locate the item revision with property changes (indicated by redlines).
4. To revert a change, select the line item(s) and right-click and select **Revert** , then click **Revert** at the confirmation message.
5. Review the confirmation message that displays which changes will be reverted (based on the preference setting).
6. Click **Revert** to confirm.
A confirmation message indicates which attachment changes were reverted.

Results

The object revision property changes are reverted to their original values. Depending on the **CM_keep_solution_item_after_revert** preference setting, attachments and vendor parts may be preserved or reverted, and the solution item may remain in or be removed from the change object.

Revert attachment changes

The revert feature allows you to undo attachment changes made to item revisions in the context of an active change. When working with a change notice, you can add, remove, or modify attachments on item revisions. If you make any accidental attachment changes, you can revert these changes.

The **CM_keep_solution_item_after_revert** preference controls whether attachment changes are reverted.

Prerequisites

- An active change notice (ECN) must be set as the active change context.
- Attachment changes (add, remove, or modify) must have been made to item revisions.
- The change must not be released.
- Your administrator must set the **CM_keep_solution_item_after_revert** preference to false to enable attachment revert

Procedure

1. Open the change notice with the active change context.

2. Open the BOM from solution item to modify and make any changes to your BOM items.
3. Locate the item revision with attachment changes.
4. Right-click the item revision and select **Revert** .
5. Review the confirmation message that displays which changes will be reverted (based on the preference setting).
6. Click **Revert** to confirm.
A confirmation message displays indicating which attachment changes were reverted.

Results

When the preference is set to false, attachment changes are reverted to their original state. Added attachments are removed, removed attachments are restored, and modified attachments return to their previous state. When the preference is set to true, attachment changes are preserved.

Revert vendor part changes

The revert feature lets you to undo vendor part changes made to item revisions in the context of an active change. When working with a change notice, you can add, remove, or modify vendor parts and their properties (such as Preferred Status). If unintentional vendor part changes occur, you can revert these changes.

Vendor parts show the connection between commercial parts and vendors, known as the Approved Manufacturer List (AML). Supply Change Program Managers (SCPM) handle tasks to keep supply data organized in PLM. They create vendors, link vendor parts to vendors, update details for both vendors and vendor parts, and adjust vendor part settings.

The [CM_keep_solution_item_after_revert](#) site preference controls whether vendor part changes are reverted.

Prerequisites

- An active change notice must be set as the active change context.
- Vendor part changes (add, remove, or property modifications) must have been made to item revisions.
- The change must not be released.
- Your administrator must set the [CM_keep_solution_item_after_revert](#) preference to false to enable vendor part revert.

Procedure

1. Open the change notice with the active change context.
2. Open the BOM from solution item to modify and make any changes to your BOM items.
3. Locate the item revision with vendor part changes.

4. Right-click the item revision and select **Revert** .
5. Review the confirmation message that displays which changes will be reverted (based on the preference setting).
6. Click **Revert** to confirm.
A confirmation message displays indicating which vendor part changes were reverted and the number of items successfully reverted.

Results

When the preference is set to false, vendor part changes are reverted to their original state. Added vendor parts are removed, removed vendor parts are restored, and modified vendor part properties (such as Preferred Status) return to their previous values. When the preference is set to true, vendor part changes are preserved.

Note

You can revert multiple vendor parts at once using multi-select. The confirmation message indicates the number of selections being reverted. If not all selected items can be reverted, the confirmation message displays how many items were successfully reverted and provides reasons for any items that were not reverted. Property of relation changes are also included in the revert process.

Understanding revert confirmation messages

Confirmation messages display before and after revert operations to inform you which changes will be reverted and the status of the revert operation.

How the preference setting affects messages

The **CM_keep_solution_item_after_revert** preference controls which changes are reverted. Your administrator configures this preference. The confirmation messages you see vary based on this preference setting.

For information about how administrators configure the **CM_keep_solution_item_after_revert** preference, see [Configure revert behavior](#).

Pre-revert confirmation messages

Before the revert operation executes, a confirmation message displays which changes will be reverted. This message provides an opportunity to review the scope of the revert operation before proceeding.

The pre-revert confirmation message indicates:

- The number of items selected for revert
- Which types of changes will be reverted (structural updates, property edits, attachments, vendor parts)

- Whether the solution item will remain in or be removed from the change object

During multi-select operations, there is no dynamic check of the redlines for the selected data. The confirmation message displays only the changes that can be reverted according to the preference setting.

Confirmation message examples

The pre-revert confirmation message displays specific information based on your selection and the preference setting:

When **CM_keep_solution_item_after_revert** is set to **false**:

- The message indicates that all changes will be reverted including structural updates, property edits, attachments, and vendor part modifications.
- The message indicates that the solution item will be removed from the change object.
- **Example:** *Reverting 2 selections. All changes including structural updates, property edits, attachments, and vendor parts will be reverted. The solution item will be removed from the change object.*

When **CM_keep_solution_item_after_revert** is set to **true** (default):

- The message indicates that structural updates and property edits will be reverted.
- The message indicates that changes to attachments and vendor parts will be preserved.
- The message indicates that the solution item will remain in the change object.
- **Example:** *Reverting 2 selections. Structural updates and property edits will be reverted. Attachments and vendor parts will be preserved. The solution item will remain in the change object.*

Review the confirmation message carefully to ensure the revert scope matches your intent before clicking **Revert** to confirm.

Post-revert confirmation messages

After the revert operation completes, a confirmation message displays the revert status. This message confirms which items were successfully reverted and identifies any items that were not reverted.

Full success scenario

When all selected items have modifications and are successfully reverted, the post-revert confirmation message displays the total number of items successfully reverted. For example, if you select two items with redlines and both are reverted, the message confirms that two items were reverted.

Partial success scenario

When some selected items have no modifications, the system reverts only the items with changes. The post-revert confirmation message displays:

- The number of items successfully reverted out of the total number selected.
- A list of specific items that were not reverted because they had no changes.

For example, if you select three items but only one has redlines, a message indicates that one of the three selections is reverted and lists the two items not reverted due to no changes.

Error messages for parent items with redlined children

When you attempt to revert a parent item that has redlined children, the system displays an error message. The message indicates that the item cannot be reverted because it has redlined children. The error message instructs you to revert the children first before reverting the parent. No changes are reverted for the parent item, including property changes and attachment changes, until the children are reverted.

For example, if you select an assembly with a modified description and an added attachment, but the assembly has redlined children, the error message indicates that zero of one selections were reverted and explains that the item cannot be reverted because there are children with redlines.

Multi-selection with parent items

When performing a multi-selection revert that includes a parent item with redlined children and other valid items, the system processes each item independently. The parent item will not be reverted and will appear in the error message. Other selected items without redlined children dependencies will be successfully reverted. The post-revert confirmation message displays the count of successfully reverted items and lists items that could not be reverted with specific reasons.

Multi-selection revert scenarios

When selecting multiple items with different types of changes (structural, property, attachment, vendor part), the system evaluates each item independently based on the **CM_keep_solution_item_after_revert** preference setting.

The confirmation message indicates which types of changes will be reverted for the selected items based on the preference setting. The system demonstrates both full success and partial success scenarios:

- **Full success:** When all selected items have modifications (such as selecting only items with redlines), the system reverts all items and displays a confirmation message showing the total number of items successfully reverted
- **Partial success:** When you select items with and without modifications, the system reverts only those with changes. For instance, if you select Assembly Part A without redlines, Assembly Part B with redlines, and Assembly Part C without redlines, only Assembly Part B will be reverted. Assembly Part A and Assembly Part C will appear in an error message, indicating they were not reverted due to the absence of changes.

Example: Reverting changes for commercial parts

This example demonstrates how to revert object revision property changes, attachment changes, and vendor part changes for commercial parts in an **ECU Cooling Fan Module Assembly**.

Scenario

The ECU cooling fan module assembly at **Revision A** serves as the baseline state before any changes are made. This released assembly contains multiple components, including three commercial parts that demonstrate vendor part revert capabilities:

- Controller PCB (commercial part with attachments and vendor parts)
- High speed fan driver IC (commercial part with attachments and vendor parts)
- Additional components for structural change demonstrations

Attachments and vendor parts have been added to the controller PCB and high speed fan driver IC. The assembly requires updating from **Revision A** to **Revision B**.

The screenshot displays the Siemens Tc software interface. The main window shows the 'ECU Cooling Fan Module Assembly' structure. The 'Adv Controller PCB x2' part is highlighted in the tree view. The right-hand pane shows the 'Part Properties' for this component, including its ID (ICMP-001), revision (B A), and description: 'Info Compact printed circuit board managing fan control, speed regulation, and diagnostic functions using integrated electronic components.' The 'Design Required' checkbox is checked, and the 'Type' is set to 'Commercial Part'.

Element	Revision	Quantity	Revision Name	Find Number
ECU Cooling Fan Module Assembly	B A		ECU Cooling Fan Module Assembly	
Cooling Fan Housing	A		Cooling Fan Housing	10
Brushless DC Fan Motor	B A	2 -	Brushless DC Fan Motor	20
ECU Interface Connector	B A		ECU Interface Connector	30
Rubber Shock Mounts x4	A	4	Rubber Shock Mounts	40
Thermal Shield Plate x3	A	3 -	Thermal Shield Plate	50
EMI/EMC Protection Cover	A		EMI/EMC Protection Cover	60
Wiring Harness (Fan to ECU)	A		Wiring Harness (Fan to ECU)	70
Adv Controller PCB x2	B A	2 -	Adv Controller PCB Controller-PCB	88 89
Adv High-Speed Fan Driver IC x3	B A	3 -	Adv High-Speed Fan Driver IC High-Sp...	95 99
Thermal Sensor Module	A		Thermal Sensor Module	100
Mounting Brackets	A		Mounting Brackets	110

Types of changes made to commercial parts

The controller PCB and high speed fan driver IC components include all types of changes that can be reverted:

- **Property changes:** Find number, quantity, description, and revision name modifications
- **Attachment changes:** Attachments added and removed from the item revisions

- **Vendor part changes:** Vendor parts added, removed, and property modifications (such as Preferred Status changes)

These changes are made in the context of the change notice's Revision B upgrade without disturbing the ongoing change process.

Reverting property and attachment changes

When reverting changes to commercial parts like the controller PCB:

1. The system identifies all modifications including property changes (quantity, description, revision name) and attachment changes (added or removed files).
2. Based on the [CM_keep_solution_item_after_revert](#) preference setting that an administrator set, the system determines which changes to revert.
3. A confirmation message displays which changes will be reverted before the operation proceeds.
4. After the revert operation completes, a confirmation message displays the revert status.

Reverting vendor part changes for commercial parts

How vendor parts can be modified, added, or removed in commercial products like the controller PCB and high-speed fan driver IC are based on your preference settings. If set to false, all vendor part modifications, along with structural and property changes, are reverted. If set to true (the default setting), vendor part changes remain intact during revert operations.

Parent-child revert restrictions

In an assembly BOM, a parent item that is a subassembly with redlined children cannot be reverted unless the children are reverted first. For example, if the ECU cooling fan assembly has property changes (such as a modified description) and an added attachment, but also has redlined children, attempting to revert the parent displays an error message indicating that the item cannot be reverted because it has redlined children. The error message instructs you to revert the children first before reverting the parent. No changes are reverted for the parent item, including property changes and attachment changes, until the children are reverted.

Multi-selection revert scenarios

The assembly demonstrates both full success and partial success scenarios for multi-selection revert:

- **Full success:** When all selected items have modifications (such as selecting only items with redlines), the system reverts all items and displays a confirmation message showing the total number of items successfully reverted.
- **Partial success:** When you select items with and without modifications, the system reverts only those with changes. For instance, if you select a thermal shield, a brushless DC fan motor with redlines, and a cooling fan without redlines, only the brushless DC fan motor will be reverted. The thermal shield and cooling fan will appear in an error message, indicating they were not reverted due to the absence

of redlines. A confirmation message will show the number of items successfully reverted and list the specific items that were not reverted.

Create a product change after the fact for change tracking

You can start product modifications outside of a change context in a rapid and agile manner before the change notice is available. Once the change notice is available, you can add the modified assemblies *after the fact* to the change notice.

Adding a modified assembly as a **Solution Item** to a change notice automatically generates tracking in the change summary, displayed with redlines.

For more information on the **Show Redlines** feature, refer to [Review active or closed changes for a structure](#).

Consider the following points when performing this agile development.

- The product revision must not be a solution item to an existing change notice.
- Only assigned participants can add an assembly to a change notice.
- A change notice must be in a state of executing maturity prior to adding the revised assembly as a solution item.

To add a modified assembly as a solution item to a change notice:

Procedure

1. Open a previously released assembly and make revisions as required.
2. To add the assembly to a change, go to **More Commands ***** > **Manage**  > **Add to My Changes** from the **Page level** toolbar.

3. Select a change notice from the list.
4. Select **Solutions** from the **Relations** list and click **Add**.
The original version of the structure automatically populates the **Impacted Items** section of the **Affected Items** tab.

The contents of the modified structure are shown in the **Change Summary**. BOMEdits are automatically created for all BOMline changes.

Note

Your administrator can enable and configure the **Cm0ImpactedItemReleasedStatuses** preference to customize how **Impacted Items** are created when adding revisions as **Solution Items** using **Add to My Changes**. When you add a **Solution Item** with a revision that matches the custom release status

value in the **Cm0ImpactedItemReleasedStatuses** preference, it is automatically added as an **Impacted Item**. This includes revisions that contain multiple release statuses. If a **Solution Item** contains multiple released revisions, only the most recent revision with that custom release status is added as an **Impacted Item**.

If the preference is not set, the **Impacted Item** is derived from the last released revision of the **Solution Item**.

Note

Your administrator can customize how **Impacted Items** are created when adding revisions as **Solution Items** using **Add to My Changes**. When you add a **Solution Item** with a revision that matches the defined custom release status value, it is automatically added as an **Impacted Item**. This includes revisions that contain multiple release statuses. If a **Solution Item** contains multiple released revisions, only the most recent revision with that custom release status is added as an **Impacted Item**.

If not set, the **Impacted Item** is derived from the last released revision of the **Solution Item**.

Get guidance on updating an aligned engineering BOM for a change context

When a design structure is updated, you must analyze the impact of the design changes on the corresponding engineering BOM (Usage BOM or Assembly BOM) . Subsequently, you must update the engineering BOM so that the part occurrences are correctly aligned with the design occurrences. If you perform modifications followed by guided update under an active change then only modified assemblies are taken for alignment updates ignoring rest of the assemblies.

Procedure

1. From **Inbox** or **Changes**, open the active change.
2. Take a bottom-up approach while updating the corresponding engineering BOM. Perform level-by-level updates until you reach the top product.

From **Change Summary**, select the innermost modified design assembly and click **Update Alignment**  > **Guided Update** .

The design structure and its aligned engineering BOM are displayed side by side. The **Guided Update** panel opens in **Design Part Alignment** mode. The panel shows the context of alignment (design and part level selected for alignment) along with the direction of alignments from the design structure to the engineering BOM using an arrow in the alignment context.

3. (Optional) Select the **Show Alignment Status** option to see alignments as they are done.
4. (Optional) Select the **Show Redlines** option to see the changes in the structure.
5. Ensure the **BOM Updates** option is selected.

The **BOM Updates** option is selected by default and enables structure modifications along with alignments. If you disable this mode for design-part alignment, you cannot enable it for design occurrence-part occurrence alignment after design-part alignments are complete.

6. Review the required changes in the **Guided Update** panel.

In the **Guided Update** panel, all the required changes are listed. At the selected level, only its first-level modified designs are shown in the panel.

The design assembly on which you initiate a guided update is shown in focus mode, while the rest of the assemblies are grayed out.

7. (Optional) Select the **All Levels** option if you want to do design-part alignments on all the structure levels for the selected context. Only design-part alignments can be done at all levels at the same time. Irrespective of the status of the **All Levels** option, design occurrence-part occurrence alignments are always performed level by level.

The panel displays various filters, for example, **Matches**, **Add New**, **Remove**, **Replace**. Each filter shows the count of designs it filters, if it is selected. The filter selections do not impact alignment operation. By default, the panel selects the filters on which your action is requested.

8. (Optional) Select one or more filters to see the respective designs and their possible alignments.
9. Review the possible alignments and available actions in the panel.

The panel shows possible alignments along with different actions you can take and information about that action. Select each design from the panel to highlight it within the structure.

You can take the following actions:

- **Align**—Align the part and design. Select a design from the existing list. Preferred parts appear at the top, separated from the rest of the structure.
- **Defer**—Take action at a later stage.
- **Modification Actions (Add, Remove, Replace, Modify)** — For the Add action, you can add a new part or search for an existing part for alignment. If you do not add manually, **Teamcenter** adds it for you.

10. Once you have taken the required actions, click **Update Design Part**.

Your parts and designs alignments are updated now. The panel goes into **BOM Alignment** mode for design occurrence-part occurrence alignment with the initial context node in focus.

11. The panel again shows the context, direction, and filter tiles for design occurrence-part occurrence alignment with possible matches based on your design-part alignment. Verify the alignments and click **Update**.
12. (Optional) If you have deferred any alignments during the design-part alignment stage, you can align them now or defer them again.

Design occurrence-part occurrence alignment is updated for the selected level.

13. Click the **Next** button to initiate design occurrence-part occurrence alignment for the next modified level under selected context.
14. Repeat steps 11 through 13 until all the levels are design occurrence-part occurrence aligned.

Your design structure and engineering BOM are alignments are updated now.

What to do next

After updating the aligned engineering BOM, first perform an alignment check to verify if the engineering BOM is updated and aligned correctly. Next, release the active change notice.

Specify release effectivity

A product structure goes through many changes during the evolution of its product definition. Using *structure effectivity*, you can capture how the product structure has evolved over a period. One type of effectivity, *release effectivity*, helps in configuring different revisions of the same element. It denotes from which date or for which unit (and end item) an element revision is effective.

Note

Learn more about structure effectivity here.

You can create a new release effectivity for a particular date or a range of dates, or a unit number or a range of units. For example, in an engine cooling block, revision A has 5 cooling fins (a unit) and in revision B it is changed to 6. Release effectivity is used when deciding which revision to use. It is necessary to have a release status associated with an element to author release effectivity on it.

Release effectivity can be applied to change notices created in the context of an object or by clicking the **Launcher**, then clicking **Changes**.

Procedure

1. Search the element (having some release status associated with it) and go to **More Commands** ******* > **Manage**  > **Release Effectivity**  from the **Page-level** toolbar.
Or, click **More Commands** from the **Details** toolbar and select **Release Effectivity**.
2. In the **Release Effectivity** panel, select the appropriate release status.
3. In the **Release Effectivity** panel, click **Add Effectivities** .
4. Add a new effectivity or locate existing effectivities from the **Search** tab. You can add multiple effectivities to the selected element.
5. To specify the dates for release effectivity:
 - a. In the **New** tab, select **Date**.
 - b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.
The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.

- c. Select the **Start** date from the calendar.
 - d. Select the **End** date from the calendar. If applicable, you can select **UP** (all future dates) or **SO** (stock out).
 - e. Select the **Protect** check box if you want to control access to the effectivity using Access Manager.
 - f. Click **Add** to create the effectivity.
After release effectivity is added, the occurrence is configured based on the current revision rules.
6. To specify units for release effectivity:
- a. In the **New** tab, select **Unit**.
 - b. Select the **Share** check box to create a shared effectivity. Specify the effectivity **Name**.
The **Name** field is displayed only for shared effectivities. You can search for the effectivity by name.
 - c. Specify the desired unit or a range of units in the **Unit** field.
 - d. Click the **Replace**  icon to add a new **End Item** or search for an existing **End Item**.
 - e. Select the **Protect** check box if you want to control access to the effectivity using Access Manager.
 - f. Click **Add** to create the effectivity.
After the release effectivity is added, the element is configured based on the current revision rules.

Merging unincorporated changes

About unincorporated changes

An *unincorporated change* is a change that is approved but not yet incorporated in the design of a product. Unincorporated changes enable a change to be processed and completed, even though not all of the required updates have been added into the main release. Usually, changes are not incorporated fully because there is not enough time, importance, or budget to incorporate the changes completely through the development process.

You can merge unincorporated changes to existing designs using Active Workspace. By viewing a side-by-side presentation of these two assembly versions, you can quickly see the differences and merge any required portions into the final ECN.

The merge functionality occurs when a released change acts on the same impacted revision as a work-in-progress change; in that instance, you can view the merge candidates. In other words, a merge happens from a released change to a work-in-progress change where the impacted revision is common to both.

Note

Merge does not occur in cases where the different changes act on different impacted revisions, or when both changes are work-in-progress changes.

Example

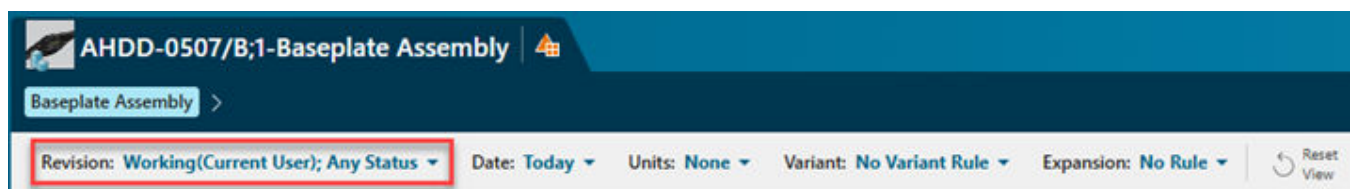
For example, during the manufacturing phase of production, an emergency revision of a product part is released by a designer. To fully incorporate this change into the assembly, the following process takes place:

1. The change specialist responsible for managing the change process creates an ECN to implement the change and adds the items of the part that needs to be updated to the **Impacted Items** folder. The change specialist then starts a workflow for the ECN and assigns the necessary participants.
2. The analyst, assigned to implement the ECN, modifies the item with Revision A and creates a new revision, which appears in the **Solutions Items** folder as Revision B.
3. The analyst then creates a second ECN to maintain the emergency change, and the change specialist starts a workflow and assigns participants.
4. The analyst creates a new revision and makes the change, called Revision C.
5. Revision C, which includes the emergency update, is then merged into the source ECN via [the merge process](#). Once complete, all of the merged updates are reflected by redlines in the Change Summary.
6. To finish, the analyst releases the merged ECN.

BOM Considerations

Merge functionality performs the same for assembly and usage BOMs, but note the following:

- Since the topline of a usage BOM is not revised, the usage assembly has its own independent lifecycle and is released as part of a change object. This merge scenario assumes that there are at least two users who are making a change to the same part in different browsers.
- The topline of an assembly BOM can be altered since its top node is classified as an item revision.
- The business object constant **Fnd0MaxAllowedWorkingRevisions** (defined on **Fnd0AbstractOccRevision**) must be modified in BMIDE for **Ebm0AbstractPartOccRevision** to include a value as **-1**. This allows for multiple working revisions of EBOMs (the default value is 1). For example, if the value of **Fnd0MaxAllowedWorkingRevisions** is set to **-1**, more than one user can make changes to the same structure with the revision rule set as **Working (Current user); Any status**.



For a full example of incorporating changes, see [Example of incorporating changes](#).

The following procedures describe how to merge unincorporated changes with and without release effectivity.

Related Concepts

[What happens when there are concurrent changes to the same item across multiple revisions?](#)
[What happens when concurrent change notices introduce conflicting changes to the same item?](#)

Related Tasks

[Merge an unincorporated change for a BOM into an existing change notice](#)
[Merge an unincorporated change for a BOM with effectivity into an existing change notice](#)
[Automatically merge BOM updates from a released change into an ongoing change](#)

What happens when there are concurrent changes to the same item across multiple revisions?

When multiple users work concurrently on different revisions of the same base item, automatic merge capabilities propagate changes across the revision chain.

Overview

When multiple concurrent modifications affect different revisions of the same base item (also known as stacked changes), Teamcenter prevents the loss of BOM redlines when changes span across revisions. It assesses concurrent status by focusing on the base item, rather than requiring identical revisions of the impacted item. All ongoing changes are considered for the same impacted item as concurrent and allows them to be merged, regardless of the specific revision each change targets.

This functionality helps manage concurrent changes across multiple revisions by ensuring that BOM redlines are preserved. It enables auto-merge to propagate changes to all related work-in-progress items, regardless of the revision of the impacted item. You do not need to manually update impacted items to the latest released revision, as warnings and notifications about concurrent changes across revision boundaries are provided. Additionally, it maintains the change lineage and incorporation status throughout the entire chain of concurrent modifications.

How concurrent changes are detected

When a change releases an item revision, that revision becomes the latest released revision. Teamcenter recognizes new changes created using this latest released revision as concurrent with other pending changes that are based on earlier revisions of the same base item.

This enables the following behaviors:

- Auto-merge propagates changes to all work-in-progress changes for the same base item, regardless of which specific revision each change is modifying.

- Concurrent change warnings and properties reflect the concurrent relationship across revision boundaries.
- The change history displays which changes have been merged into other concurrent changes.

Merge behavior

When you release a change notice that involves concurrent changes across multiple revisions, Teamcenter identifies all work-in-progress changes for the same base item, evaluates merge candidates across all concurrent revisions, applies auto-merge rules based on administrator-configured preferences, and updates the Change Summary with merge indicators for all affected changes. Auto-merge integrates the released changes into all concurrent work-in-progress changes for the same base item.

The **Concurrent Changes** column in the **Impacted Items** table displays all change notices that are modifying the same base item, enabling you to track concurrent activity across revision boundaries.

Merge candidates appear in the Change Summary for all affected changes. You can click the merge indicator to view a side-by-side comparison of the source and target changes. The released change (source) overrides any conflicting changes in the concurrent changes (target) based on configured preferences.

Here, a released concurrent change has happened elsewhere in Teamcenter by another user, but it affects this current change, as indicated by the **Concurrent Changes** column and the merge indicators. Two items have been changed (two merge indicators).

The screenshot displays the Teamcenter interface for change management. The top section, 'Impacted Items (1)', shows a table with columns: Object, Lineage, Type, Requested Chan..., and Concurrent Changes. The 'Concurrent Changes' column contains the text 'ECN-000210/A;1-ECN_Ed_V', which is highlighted by a red rectangular box. Below this is the 'Change Summary' section. It includes a dropdown for 'Impacted Items' set to 'ALL' and a 'Current' filter. The main table has columns: Changes for, Merge Status, Action, Element, ID, and Revision. The 'Merge Status' column contains two green icons representing merge candidates, also highlighted by a red rectangular box. The table lists three items: 'Beta25_Smartphone Assembly' (Modified), 'Main PCB' (Property Modified), and 'SIM Nano' (Property Modified).

Click Merge to view the side-by-side change notices to identify the updated items.

Note

Your administrator can enable and API to programmatically detect whether a change is auto-merge related. This API can be used to determine if auto-merge is enabled before creating related change objects.

Smartphone assembly with concurrent changes

This example demonstrates how concurrent changes work when multiple engineers modify the same assembly simultaneously.

Three engineers work on different parts of the Beta25_Smartphone Assembly at the same time, creating concurrent changes that the system must track and merge.

1. Beta25_Smartphone Assembly/Rev A is released.
2. Charles creates ECN-000207 to modify the battery, creating Rev B from Rev A.
3. Oscar creates ECN-000208 to modify the speaker and camera, creating Rev C from Rev A. When Oscar adds the assembly to ECN-000208, the system issues a concurrent change warning because both engineers are modifying the same base revision. The **Impacted Items** table displays ECN-000207 in the **Concurrent Changes** column.
4. Ed creates ECN-000210 to modify the PCB, creating Rev D from Rev A.
5. Charles releases ECN-000207, which creates Rev B. Auto-merge automatically propagates the battery changes to Oscar's Rev C and Ed's Rev D. The **Change Summary** in Oscar's ECN-000208 displays merge indicators showing which changes have been auto-merged from the released change.
6. Pahal creates ECN-000209 based on Rev B and makes additional modifications.
7. Oscar reviews ECN-000208 before release. A warning displays on the **Overview** tab indicating that releasing may overwrite changes from other concurrent modifications. Oscar can click the merge indicator to view a side-by-side comparison of the source change and target change, with conflicting items highlighted.
8. Oscar releases ECN-000208. Auto-merge propagates the speaker and camera changes to Ed's Rev D and Pahal's ECN-000209.

Note

Your administrator can enable an API to programmatically detect whether a change is auto-merge related. This enables custom workflows to check if auto-merge is enabled before creating related change objects, preventing the creation of unnecessary change items when auto-merge will handle the propagation automatically.

Related Concepts

About unincorporated changes

[What happens when concurrent change notices introduce conflicting changes to the same item?](#)

Related Tasks

[Merge an unincorporated change for a BOM into an existing change notice](#)

[Merge an unincorporated change for a BOM with effectivity into an existing change notice](#)

[Automatically merge BOM updates from a released change into an ongoing change](#)

What happens when concurrent change notices introduce conflicting changes to the same item?

When working with collaborative systems, managing concurrent changes is crucial to ensure data integrity and consistency. Merging helps streamline this process by integrating updates from released changes into ongoing work.

The system considers all work-in-progress changes for an impacted item as concurrent, irrespective of the item's specific revision. This implies that any change involving an impacted item is recognized as concurrent with other pending changes, even if those changes stem from earlier revisions of the same item.

A manual merge requires users to view both the released change and the in-progress change side-by-side and select the updated items to merge, with the option to ignore any changed items. This method can allow for more control over which changes are integrated but requires more manual effort and time. In contrast, an auto-merge automatically integrates updates from a released change into all ongoing work-in-progress changes as soon as the change is released, reducing manual intervention and speeding up the development cycle.

While auto-merge can be highly beneficial, it also presents challenges when there is a large volume of data being merged. Automatic merging may combine conflicting changes based on the configuration, and with too much data, these conflicts can go unnoticed by users during the process.

For example, if Team Member A and Team Member B both modify the same part or assembly in different ways, the auto-merge results might be confusing, especially with a large amount of data. Additionally, when changes are stacked across multiple revisions (for example, Change 1 releases Part A/B, Change 2 and Change 3 are working on Part A/C and Part A/D, and Change 4 is created using Part A/B as the impacted item), the system recognizes all these changes as concurrent and applies auto-merge across the entire chain of revisions. However, warnings and notifications alert users to potential concurrent changes, helping prevent conflicts before they occur. In such scenarios, changes need to be carefully managed and reviewed.

Since automatic merging runs behind the scenes and is not controlled by a user, warnings/notifications can be configured by an administrator, based on business needs. For example, routine updates might enable auto-merge, but warnings can be configured for critical changes that require careful review. Preferences are available to determine when auto-merge should be applied and when warnings should be issued.

Prerequisites

Your administrator determines the course of action for handling conflicting changes during an automatic merge. **Specific settings** must be configured to describe the behavior needed during an auto-merge of conflicting changes. Settings can also be configured to enable notifications and warnings for concurrent changes.

Warnings and notifications

Users can be made aware of concurrent changes through the following methods:

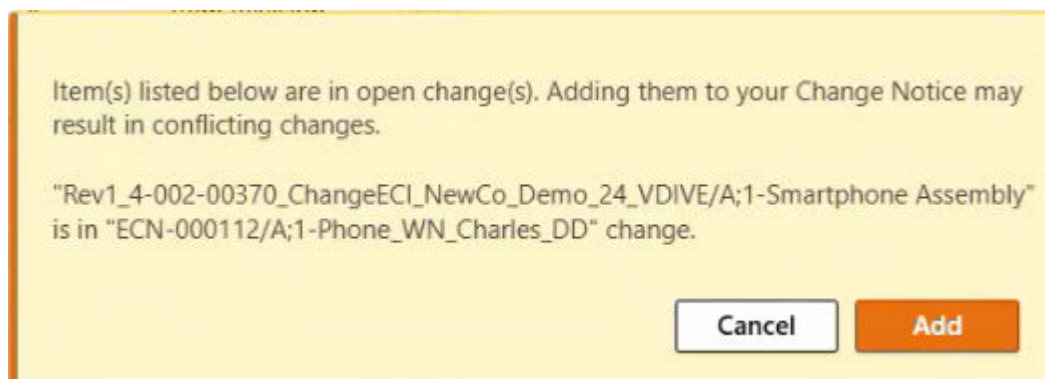
- An actionable warning is issued when an impacted item is added to a change notice that is already an impacted item in another change notice.
- A user notification is issued when another user adds the same impacted item to their change.
- A notification is sent to the user, along with a warning to the Change Implementation Board, when another user's concurrent change is validated, reviewed, and approved.
- Users receive an actionable warning when they release their concurrent change.
- Users receive a notification when another user releases their concurrent change.

When releasing a change that involves different item revisions in concurrent changes (stacked changes), users receive enhanced warnings. For example, if you release a change where the impacted item is Part A/C, and there are concurrent changes using Part A/B (the previously released revision) as their impacted item, the system alerts you that releasing your change may overwrite redlines from other concurrent changes, even though the impacted item revisions differ.

Example

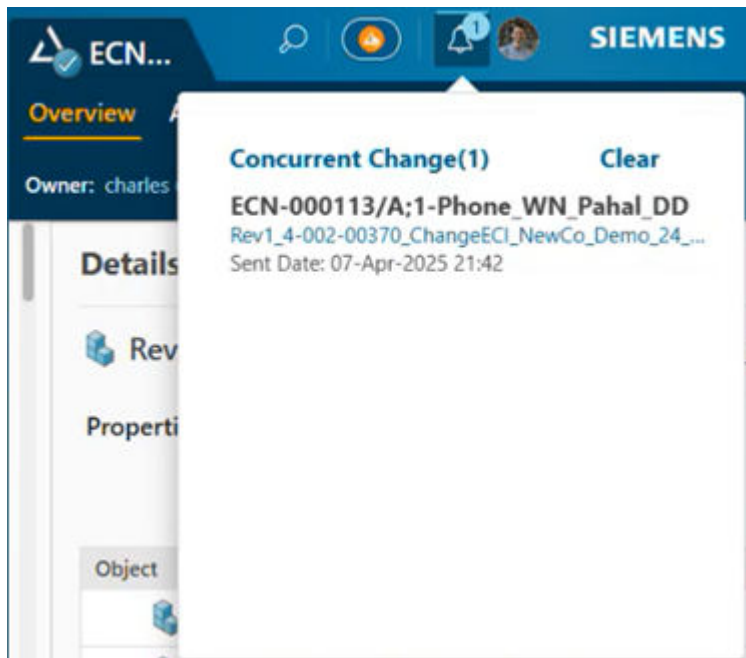
In this example, there are three users: Charles, Pahal, and Oscar. Charles and Pahal are change analysts, while Oscar is part of the Change Implementation Board (CIB).

Charles initiates a change notice for a smartphone assembly and makes modifications. At the same time, Pahal creates a concurrent change notice, which involves a shared impacted item revision. Pahal receives a warning that this item revision is already being modified by Charles.

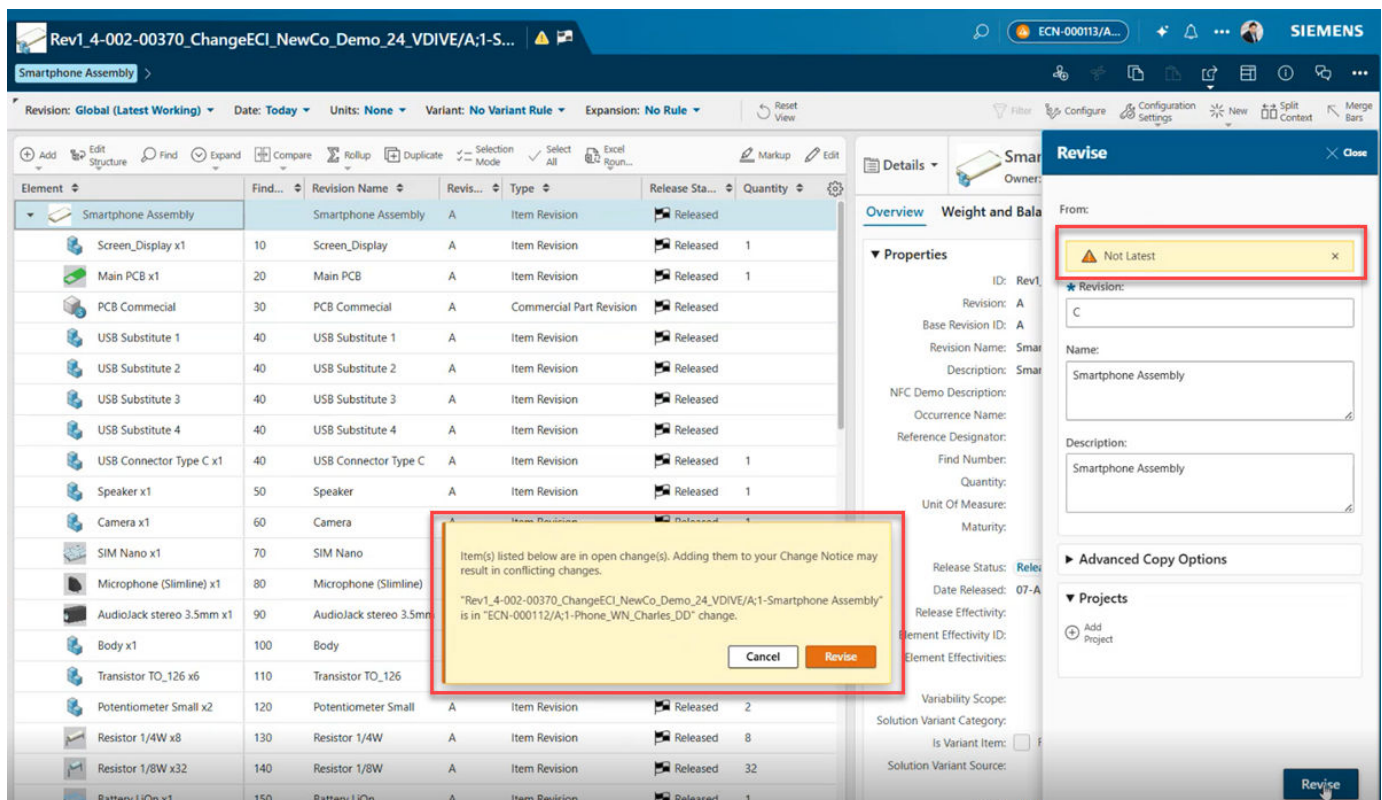


Upon confirming, Pahal proceeds with his change notice, triggering an alert back to Charles that his assembly is now included in another change.

What happens when concurrent change notices introduce conflicting changes to the same item?



Pahal then continues to revise his assembly, but is alerted that the item he is revising is not the latest and that he is about to create a concurrent change. He is always aware of any conflicts before proceeding.



Now there are two concurrent changes ready to be released.

Meanwhile, Charles acknowledges the parallel change and submits his change notice to Oscar at the CIB. Oscar then receives a warning in his Inbox indicating that Charles's change conflicts with Pahal's change.


Reviewers Perform Signoff

[Overview](#)
[Attachments](#)
[Workflow](#)
[Assignments](#)

Type: Signoff
Oscar (oscar) - Engineering/Designer

▼ Action

Warning - Approving this may overwrite redlines from other concurrent changes. To see all pending changes, open the Change Notice, "ECN-000112/A;1-Phone_WN_Charles_DD" in Targets and click the link in the Concurrent Changes column in the Impacted Items table (Overview).

ECN-000113/A;1-Phone_WN_Pahal_DD:
Rev1_4-002-00126 ChangeECI NewCo Demo
24_VDIVE/A;1-Main PCB
Rev1_001852.NewCo_Demo_24_VDIVE/A;1-PCB
Commercial
Rev1_001852.NewCo_Demo_24_VDIVE/A;1-PCB
Commercial
Rev1_4-002-00370_ChangeECI
_NewCo_Demo_24_VDIVE/A;1
-Smartphone Assembly

Name:

Reviewers Perform Signoff

Task Instructions:

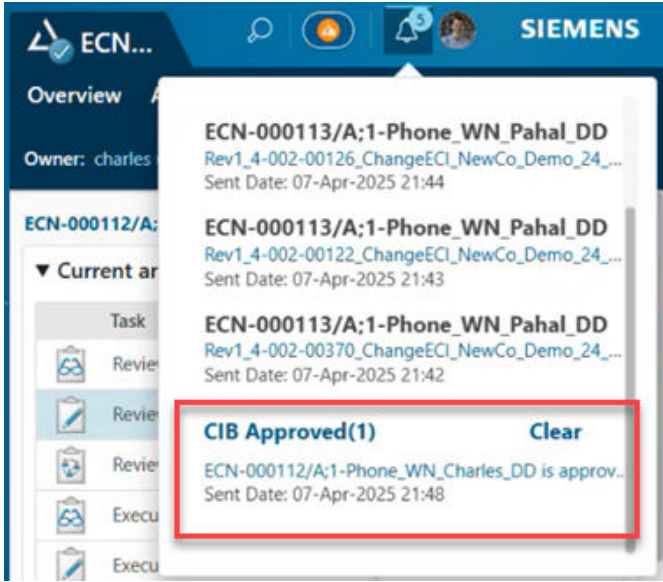
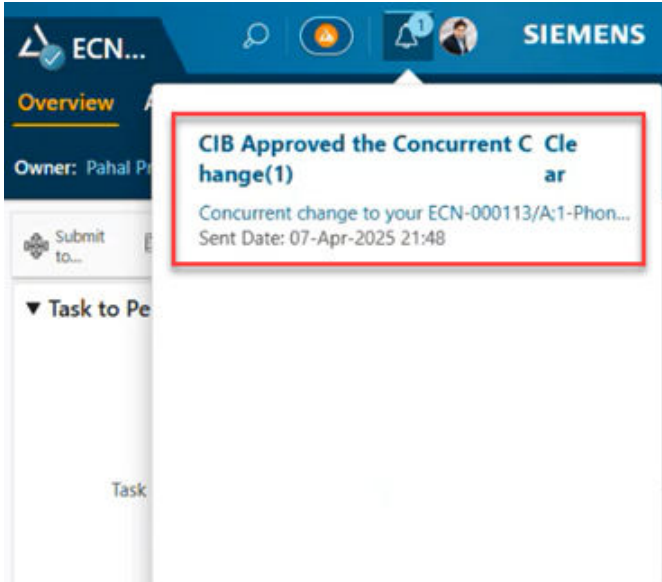
Open the Change Notice. Review the Details and the solutions in the Change Summary on the Overview tab.

Then, comment and decide either to:

- Approve
- Reject

After discussing the situation with Pahal, Oscar approves the change and sends alerts to both Charles and Pahal.

What happens when concurrent change notices introduce conflicting changes to the same item?

Charles	Pahal
 The screenshot shows Charles's workspace with a list of tasks. A red box highlights a warning message: "CIB Approved(1) Clear". The message text reads: "ECN-000112/A;1-Phone_WN_Charles_DD is approv... Sent Date: 07-Apr-2025 21:48". Above this warning, several ECN entries are listed, including "ECN-000113/A;1-Phone_WN_Pahal_DD" with a "Sent Date: 07-Apr-2025 21:44".	 The screenshot shows Pahal's workspace with a red box highlighting a warning message: "CIB Approved the Concurrent C Change(1) Clear". The message text reads: "Concurrent change to your ECN-000113/A;1-Phon... Sent Date: 07-Apr-2025 21:48".

Before releasing the change notice, Charles receives a similar warning so he can be absolutely sure he wants to proceed.

 **ECN-000112/A;1-Phone_WN_Charles_DD** 

Overview Affected Items Change Content Participants

Owner: charles (charles) Date Modified: 07-Apr-2025 Release Status:

 Submit to...  Release Effectivity  Generate Report

▼ Task to Perform
Workflow: Concurrent Change Notice Warnings
Notifications Auto Merge Process : ECN-000112/A;1-Phone_WN_Charles_DD

Warning - Releasing this may overwrite redlines from other concurrent changes. To see all pending changes, open the Change Notice, "ECN-000112/A;1-Phone_WN_Charles_DD" in Targets and click the link in the Concurrent Changes column in the Impacted Items table (Overview).

ECN-000113/A;1-Phone_WN_Pahal_DD:
Rev1_4-002-00126 ChangeECI NewCo Demo 24_VDIVE/A;1-Main PCB
Rev1_001852.NewCo_Demo_24_VDIVE/A;1-PCB Commercial
Rev1_001852.NewCo_Demo_24_VDIVE/A;1-PCB Commercial
Rev1_4-002-00370_ChangeECI _NewCo_Demo_24_VDIVE/A;1-Smartphone Assembly

Name: Review Concurrent Change
Workflow Description: Concurrent Change Notice Warnings
Notifications Auto Merge Process : ECN-000112/A;1-Phone_WN_Charles_DD

Comments:

Cancel

Release

Upon release, the system automatically merges Charles's changes into Pahal's change notice, resulting in the green merge icon  in Pahal's Change Summary, indicating a successful merge.

This process shows how important warnings and notifications are for helping users make the right decisions and manage changes accurately.

Related Concepts

About unincorporated changes

What happens when there are concurrent changes to the same item across multiple revisions?

Related Tasks

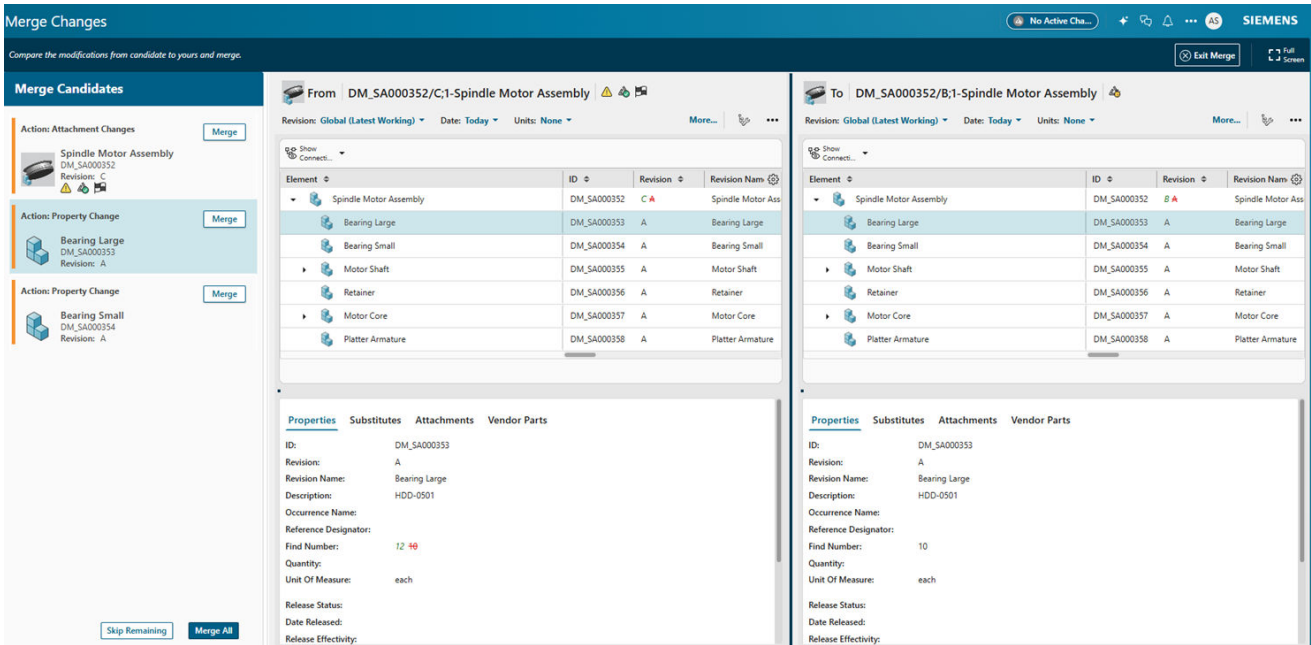
- Merge an unincorporated change for a BOM into an existing change notice
- Merge an unincorporated change for a BOM with effectivity into an existing change notice
- Automatically merge BOM updates from a released change into an ongoing change

Merge an unincorporated change for a BOM into an existing change notice

You can compare and merge unincorporated changes to existing designs. Through this process, you can integrate an emergency released ECN into an in-progress ECN by viewing both structures side-by-side and selecting updated items to merge.

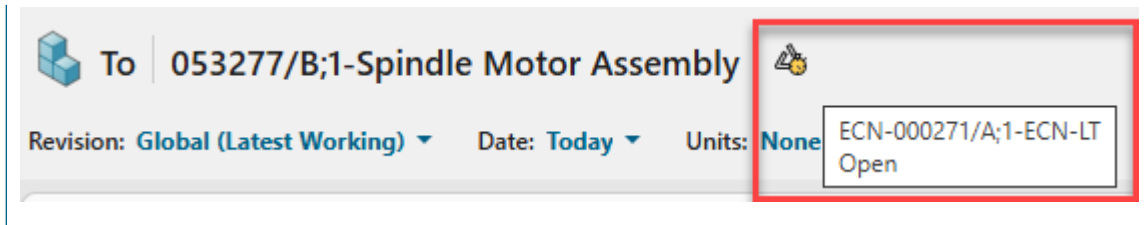
Procedure

- Open the work-in-progress ECN that is to incorporate the emergency released ECN, and view its Change Summary.
- In the Change Summary, under the **Merge Status** column for the updated object, click the merge required indicator . This displays both the completed emergency ECN (**From**, on the left) with the work-in-progress ECN (**To**, on the right). You can resize panels by dragging their borders left or right. Selecting the merge candidate highlights it within the assembly, making it easy to locate.



Note

Hover over the change icons to view their respective ECN numbers.



3. Review the **Merge Candidates** and select:

- Click **Merge** next to each revision to select that specific item.
- Click **Skip Remaining** to mark the merge status as complete. You can continue the merge later as needed.

If you prefer not to merge some or all of the changes displayed in the **Merge Candidates** side-by-side panels, you can select this option to set the merge status to complete (green) without actually merging the candidates.

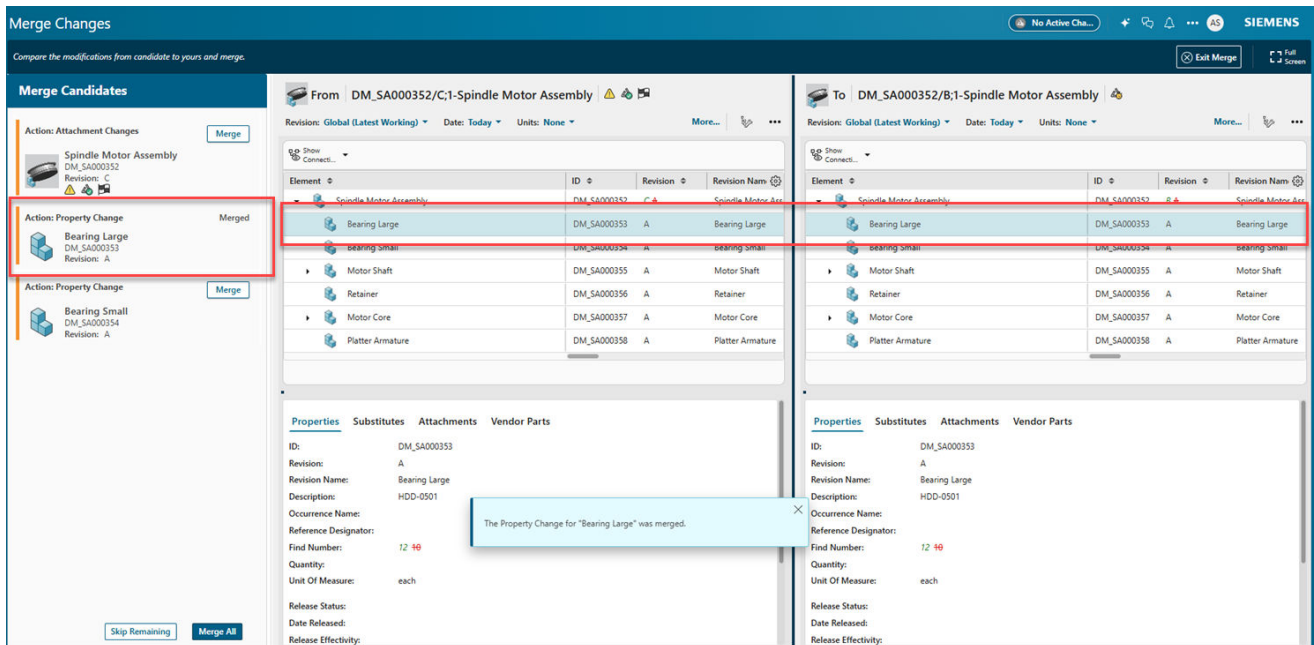
You can confirm the merge status as complete once the released task in the workflow is finished, ensuring that all potential merges have been thoroughly reviewed.

- Click **Merge All** to select all of the changes to incorporate into the **Target** panel.

Note

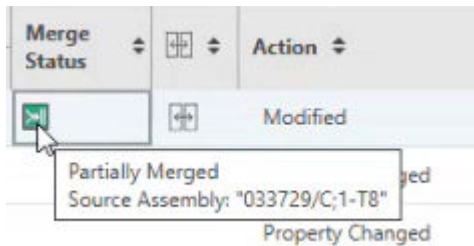
The ECN can still be released without accepting all the merge candidates.

After merging, the **Target** object panel reflects the combined changes. Here, a property change was merged into the target assembly.

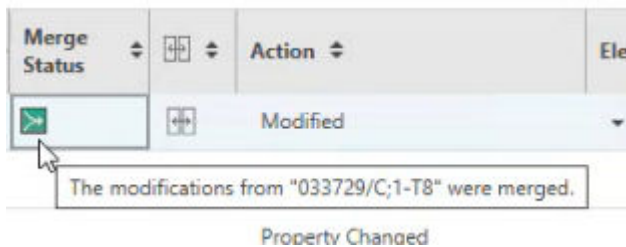


- ### 4. Click **Exit Merge** (top right).
- The Change Summary reflects the merge updates.

Here, the **Merge Status** icon  indicates that only one merge candidate was selected, and not all of the modifications that were merged, resulting in a partial merge.



If you chose **Merge All** or **Skip Remaining**, the Change Summary will display this status through the **Merge Status** icon. 



5. Release the merged change notice.

Related Concepts

About unincorporated changes

What happens when there are concurrent changes to the same item across multiple revisions?

What happens when concurrent change notices introduce conflicting changes to the same item?

Related Tasks

[Merge an unincorporated change for a BOM with effectivity into an existing change notice](#)

[Automatically merge BOM updates from a released change into an ongoing change](#)

Merge an unincorporated change for a BOM with effectivity into an existing change notice

Active Workspace lets you compare and merge unincorporated changes within existing designs. Through this process, you can integrate an emergency released ECN into a work-in-progress ECN by viewing both structures side-by-side and selecting updated items to merge. In this process, an ECN that contains a BOM with effectivity dates is merged into an existing ECN.

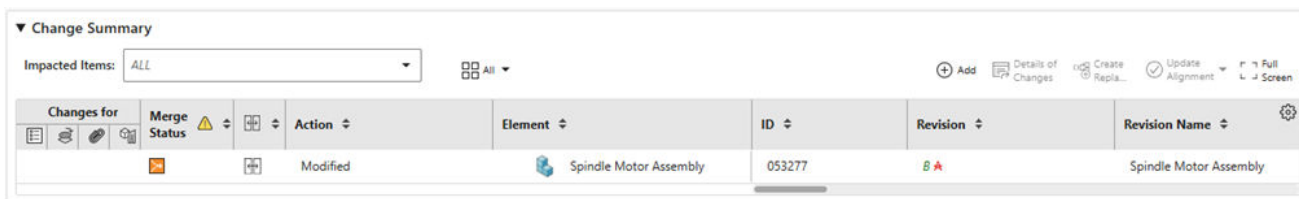
Element effectivity lets you configure distinct revisions of the same design element by defining the planned timing and conditions under which a change is implemented and becomes active. You can apply an effectivity cutback to an assembly item to modify its scope or timing by setting specific dates or a date range, or by specifying a unit number or range of units for the change's implementation. Effectivity sets

the original plan for a change notice, while an effectivity cutback lets you adjust that plan if unexpected problems arise or priorities shift.

For more information, see [About structure effectivity](#).

Procedure

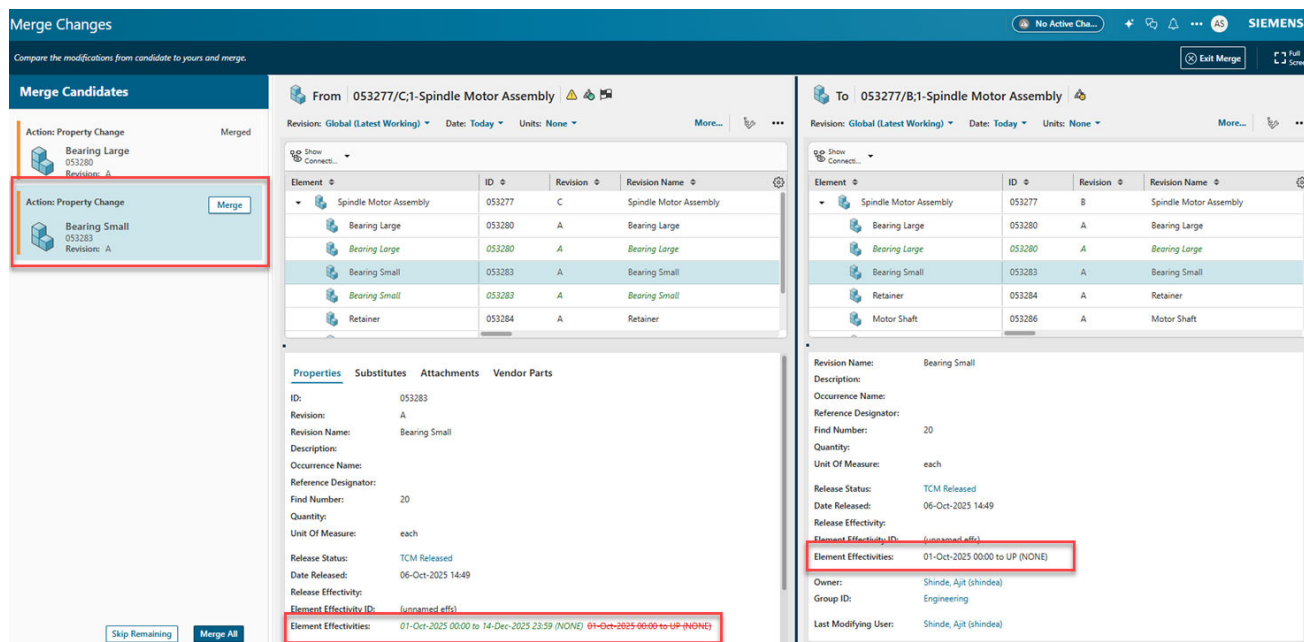
1. Open the work-in-progress ECN that is to incorporate the emergency released ECN, and view its Change Summary.
2. In the **Change Summary**, under the **Merge Status** column for the updated object, click the merge required indicator. 



This displays the **Merge Changes** screen, with both the completed short term, emergency ECN (**From**, on the left) with the long-term, work-in-progress ECN (**To**, on the right). You can resize panels by dragging their borders left or right. Selecting the merge candidate highlights it within the assembly, making it easy to locate.

Here, the short-term ECN contains new updates (two property changes) to merge into the long-term ECN. One property change has already been merged and one remains to be merged that includes effectivity dates.

When an assembly part has two distinct effectivity dates (as shown here in the **Element Effectivities** dates for **Bearing Small**), the merge process includes an effectivity cutback. This ensures that the short-term ECN does not overlap with the effectivity of the new, or replacing, part.



Merge Candidates

Action	Item	Revision	More...
Property Change	Bearing Large	053280	Revision: A
Property Change	Bearing Small	053283	Revision: A

From 053277/C:1-Spindle Motor Assembly

Element	ID	Revision	Revision Name
Spindle Motor Assembly	053277	C	Spindle Motor Assembly
Bearing Large	053280	A	Bearing Large
Bearing Large	053280	A	Bearing Large
Bearing Small	053283	A	Bearing Small
Bearing Small	053283	A	Bearing Small
Retainer	053284	A	Retainer

Properties Substitutes Attachments Vendor Parts

ID: 053283
Revision: A
Revision Name: Bearing Small
Description:
Occurrence Name:
Reference Designator:
Find Number: 20
Quantity:
Unit Of Measure: each
Release Status: TCM Released
Date Released: 06-Oct-2025 14:49
Release Effectivity:
Element Effectivity ID: (unnamed effs)
Element Effectivities: 01-Oct-2025 00:00 to 14-Dec-2025 23:59 (NONE) 01-Oct-2025 00:00 to 14-Dec-2025 23:59 (NONE)

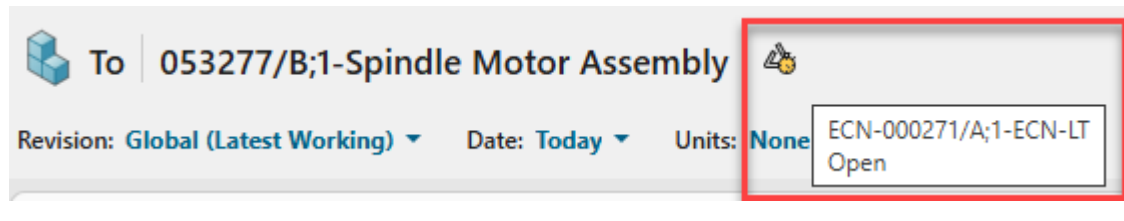
To 053277/B:1-Spindle Motor Assembly

Element	ID	Revision	Revision Name
Spindle Motor Assembly	053277	B	Spindle Motor Assembly
Bearing Large	053280	A	Bearing Large
Bearing Large	053280	A	Bearing Large
Bearing Small	053283	A	Bearing Small
Retainer	053284	A	Retainer
Motor Shaft	053286	A	Motor Shaft

Revision Name: Bearing Small
Description:
Occurrence Name:
Reference Designator:
Find Number: 20
Quantity:
Unit Of Measure: each
Release Status: TCM Released
Date Released: 06-Oct-2025 14:49
Release Effectivity:
Element Effectivity ID: (unnamed effs)
Element Effectivities: 01-Oct-2025 00:00 to 14-Dec-2025 23:59 (NONE)
Owner: Shinde, Ajit (shinde)
Group ID: Engineering
Last Modifying User: Shinde, Ajit (shinde)

Note

Hover over the change icons to view their respective ECN numbers.



To select a pre-determined effectivity range for each ECN, under **Date**, select **Date Range** from the drop-down to display the **Date Range Effectivity** dialog. This lets you specify the time period during which an item in the assembly is effective.

3. Review the **Merge Candidates**.

- Click **Merge** next to each revision to select that specific item.
- Click **Skip Remaining** to incorporate some of the changes.

If you prefer not to merge some or all of the changes displayed in the **Merge Candidates** side-by-side panels, you can select this option to set the merge status to complete (green) without actually merging the candidates. You can continue the merge later as needed.

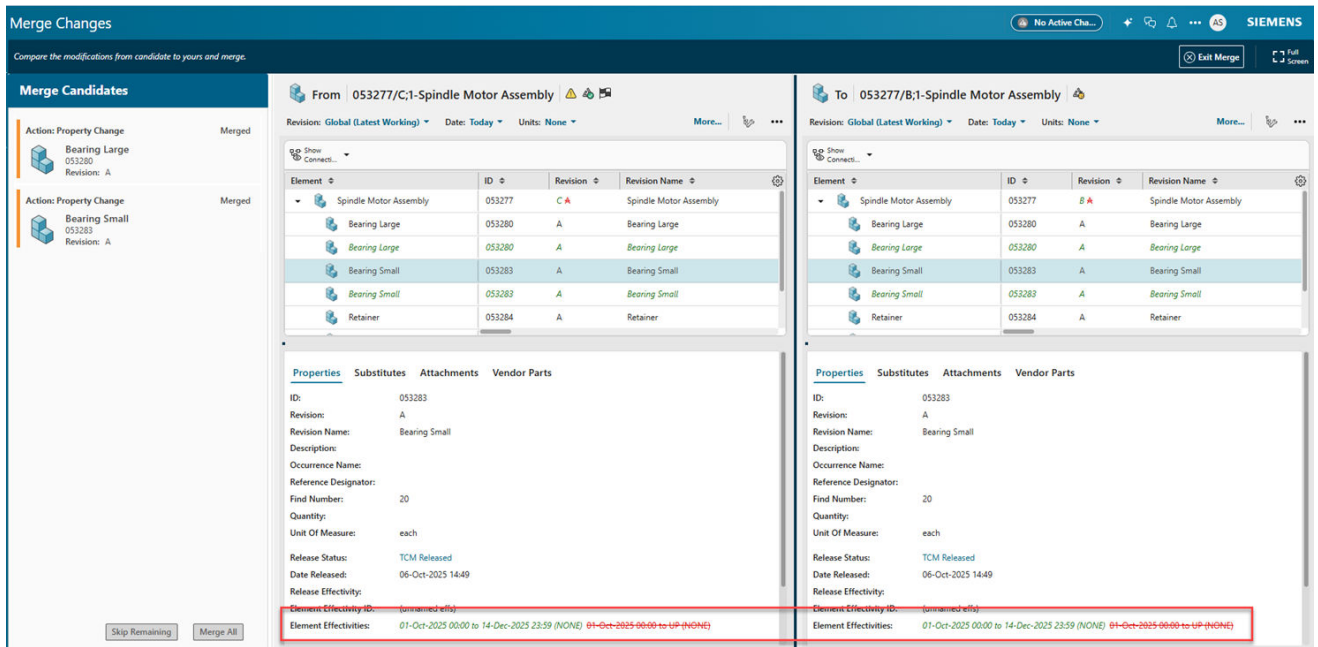
You can confirm the merge status as complete once the released task in the workflow is finished, ensuring that all potential merges have been thoroughly reviewed.

- Click **Merge All** to select all of the changes to incorporate into the **To** (target) panel.

Note

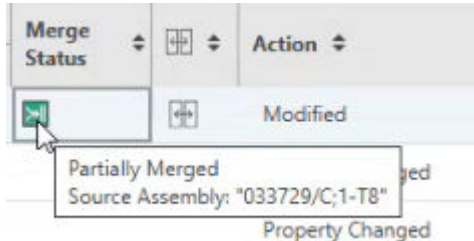
The ECN can still be released without accepting all of the merge candidates.

After merging, the **To** panel reflects the combined changes. Here, the property change was merged and now includes the new effectivity date (as indicated by redlines).

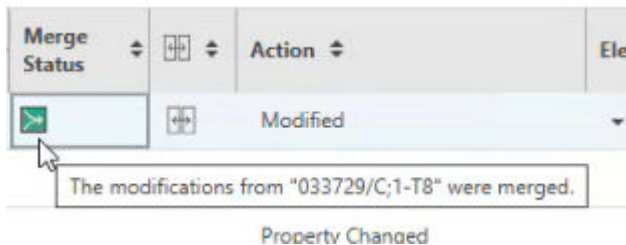


4. Click **Exit Merge**.
The **Change Summary** reflects the merged updates.

Here, the **Merge Status** icon  indicates that only one merge candidate was selected, and not all of the modifications that were merged, resulting in a partial merge.



If you chose **Merge All** or **Skip Remaining**, the Change Summary will display this status through the **Merge Status** icon. 



Effectivity dates are displayed under the **Occurrence Effectivity** column and you can identify the effectivity cutbacks that are now in place.

▼ Change Summary

Impacted Items:

Changes for	Merge Status	Action	Element	Find Number	Occurrence Effectivity	Quantity
	Modified		Spindle Motor Assembly			
	Property Modified		Bearing Large	10	01-Oct-2025 00:00 to 14-Dec-2025	
	Property Modified		Bearing Large	15 40	15-Dec-2025 00:00 to UP (NONE)	
	Property Modified		Bearing Small	20	01-Oct-2025 00:00 to 14-Dec-2025	
	Property Modified		Bearing Small	25 20	15-Dec-2025 00:00 to UP (NONE)	

Merging effectivity ranges in change notices works by combining the time periods during which the changes are valid. If two change notices have different effectivity dates, merging them covers the entire range of both original periods. This ensures that all changes are effective for the combined duration.

- Release the merged change notice.

Related Concepts

About unincorporated changes

What happens when there are concurrent changes to the same item across multiple revisions?

What happens when concurrent change notices introduce conflicting changes to the same item?

Related Tasks

Merge an unincorporated change for a BOM into an existing change notice

Automatically merge BOM updates from a released change into an ongoing change

Automatically merge BOM updates from a released change into an ongoing change

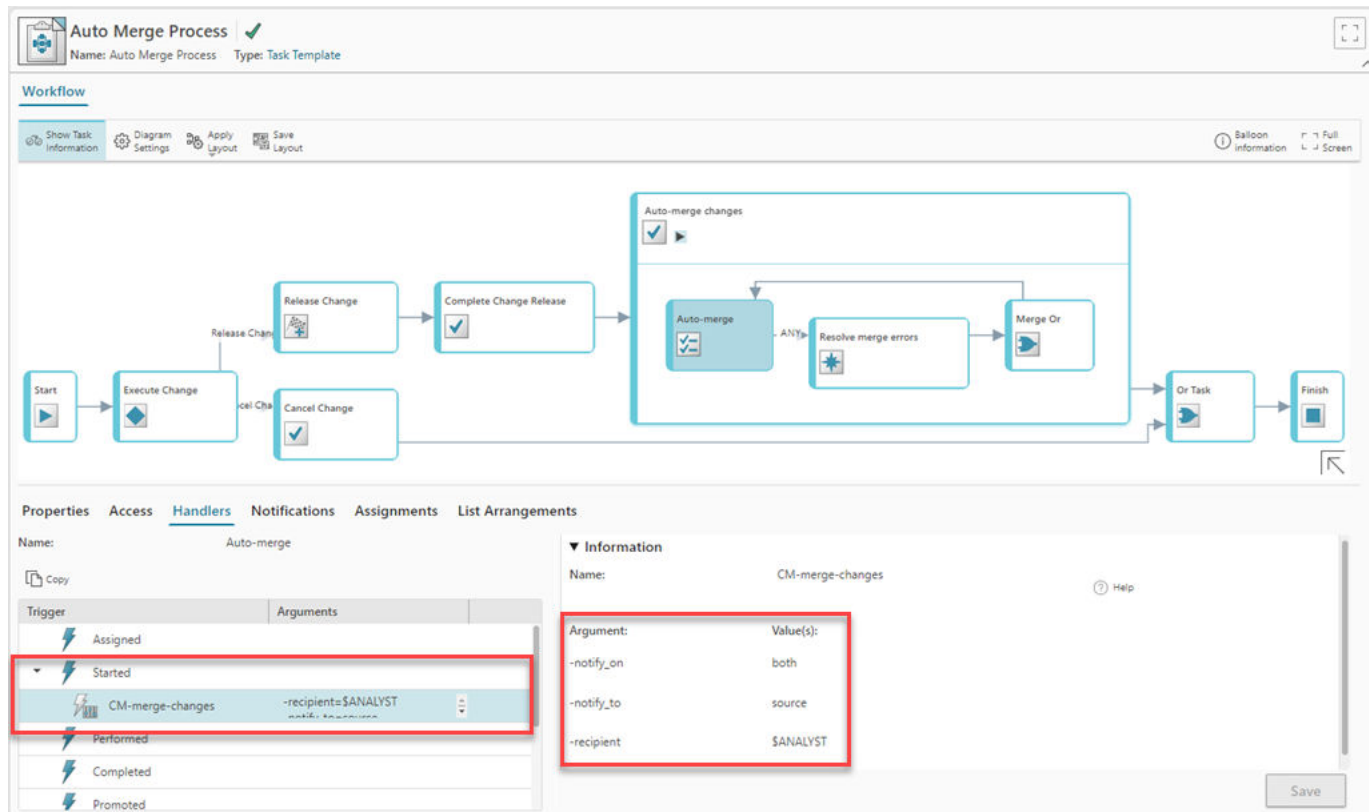
Managing large merges can often be stressful, with the constant risk of missing some changes. Users can modify numerous BOM properties, as well as add, remove, or replace many items. Implementing automatic merging for updated assemblies from a released change into ongoing concurrent work can speed up the development cycle and reduce manual intervention. This is especially useful in environments with frequent updates or multiple contributors.

In a manual merge, you can integrate a released change into an in-progress change by viewing both structures side-by-side and selecting the updated items to merge. However, you also have the ability to ignore any changed items, making merging optional. During an auto-merge, when a change is released, all ongoing work-in-progress changes are immediately updated to incorporate the released change.

The auto-merge functionality recognizes concurrent changes across different revisions of the same impacted item, ensuring that changes are merged even when subsequent changes are created using the latest released revision as their baseline.

Prerequisites

To auto-merge change notices, the change release workflow must include a task with the **CM-merge-changes** workflow handler attached. This handler identifies the specific task(s) that use the auto-merge capabilities and determines if and when other users are notified.



Note

Teamcenter provides a sample out-of-the-box workflow that demonstrates how to use this handler in the change release process. See the **Auto Merge Process** template.

Since the **CM-merge-changes** handler is configured and running in the background on the task, a notification summarizing the auto-merge details is sent to the users specified in the **-recipient** argument of the handler. Your administrator can configure the notification mode (email, alert, or both) through the user-based **WRKFLW_notification_mode** preference. See the section **Results** for more information.

Procedure

1. Open the fast track change notice that includes the assembly to update. This change notice will eventually be released and auto-merged into the other work-in-progress change(s).

When another user makes a concurrent change to the assembly being updated, the **Concurrent Changes** column in the **Impacted Items** table on the **Affected Items** tab shows the change notice(s) happening simultaneously for the affected assembly.

This indicates to each user that another change notice has been opened for the assembly. For example, User A can see that there are two concurrent changes opened for the Baseplate Assembly and one for the Spindle Motor Assembly, in addition to his own change notice.

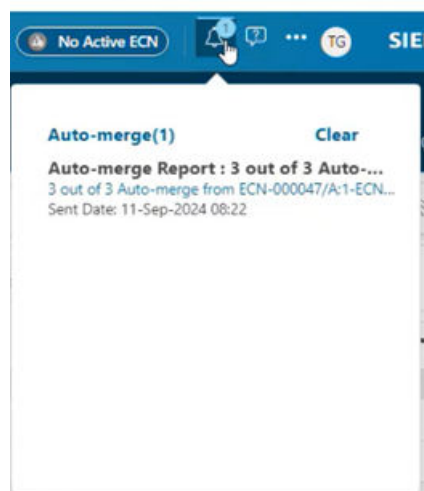
▼ Impacted Items				
Object	Lineage	Type	Requested Change	Concurrent Changes
 SL_TG_HDD-0507/A:1-Baseplate Assembly	2	Item Revision		ECN-000047/A:1-ECN-FT1, ECN-000050/A:1-ECN-LT
 SL_TG_SA000316/A:1-Spindle Motor Assembly	1	Item Revision		ECN-000047/A:1-ECN-FT1

- Once the assembly is ready, click **Release Change** on the **Overview** tab. The released change notice's modifications are automatically merged into all other applicable open concurrent changes. This includes concurrent changes made using different revisions of the same impacted item. All modifications are treated as concurrent adjustments to the same foundational item, ensuring seamless integration across different revisions.

The Change Summary for the released change notice and the concurrent change notice(s) displays the updated assembly. A green icon  indicates merged assembly items. Click this merge icon to view a side-by-side comparison of the source and target merges. The released change (source) will override any changes made to the same assembly item(s) in the concurrent changes (target).

Results

Since the **CM-merge-changes** handler is configured and running in the background on the task, a notification summarizing the auto-merge details is sent to the users specified in the **-recipient** argument of the handler.



Clicking the link in the notification takes you to the summary page, which lists details about the solution items that were auto-merged. The summary page also includes a **Related Objects** link to the **Auto-merge Report**.

Successful auto-merge:

Related Objects: Auto-merge Report : 1 out of 1 solution item(s) from ECN-000080/A;1-Change2_AM were auto-merged successfully.

Failed auto-merge:

Related Objects: Auto-merge Report : 0 out of 1 solution item(s) from ECN-000080/A;1-Change2_AM were auto-merged successfully.

This table shows that the updates were successfully integrated into the same assemblies within the concurrent change notices of both users.

▼ Preview

Auto-merge Report : 3 out of 3 solution item(s) from ECN-000092/A;1-Change4_AM were auto-merged successfully.
HTML
17-Oct-2024
4371 Bytes

Show Markups
Edit Markups
Highlight Text
Symbols
Stamps
Symbols
Print Markups
Checkout
Cancel Checkout
Checkin
Full Screen

Auto-merge of solution item(s) from ECN-000092/A;1-Change4_AM with concurrent change(s) has been completed. Open each concurrent change to view the details in the Change Summary.

From Solution Item	Concurrent Change	To Solution Item	Status	Failure Reason	Analyst
034781/D;1-Baseplate Assembly	ECN-000090/A;1-Change2_AM	034781/B;1-Baseplate Assembly	Successful		
034781/D;1-Baseplate Assembly	ECN-000091/A;1-Change3_AM	034781/C;1-Baseplate Assembly	Successful		
034785/C;1-Spindle Motor Assembly	ECN-000091/A;1-Change3_AM	034785/B;1-Spindle Motor Assembly	Successful		

This report lists:

- From Solution Item**
The solution item of the source/released change notice.
- Concurrent Change**
 - The target/open change notice in the source change's auto-merge report.
 - The source/released change notice in the target change's auto-merge report.
- To Solution Item**
The solution item of the target/open change.
- Status**
The state of the merge(s): **Successful** or **Failed**.
- Failure Reason**
A specific explanation or cause for why the merge was unsuccessful.
- Analyst**
The Change Analyst of the concurrent change.

Change Management generates a unique auto-merge report for each source and target change notice. Each report lists the concurrent change(s) and corresponding analyst for that change. For example, this report was received for the target/open changes for the same instance as in the previous report:

▼ Preview

Auto-merge Report : 2 out of 2 solution item(s) from ECN-000092/A;1-Change4_AM were auto-merged successfully. ▼

HTML17-Oct-20243500 Bytes

Show Markups

Edit Markups

Highlight Text

Symbols

Stamps

Symbols

Print Markups

Checkout

Cancel Checkout

Checkin

FullScreen

Auto-merge of solution item(s) from ECN-000092/A;1-Change4_AM with concurrent change(s) has been completed. Open your change ECN-000091/A;1-Change3_AM to view the details in the Change Summary.

From Solution Item	Concurrent Change	To Solution Item	Status	Failure Reason	Analyst
034781/D;1-Baseplate Assembly	ECN-000092/A;1-Change4_AM	034781/C;1-Baseplate Assembly	Successful		Tanay Gupta
034785/C;1-Spindle Motor Assembly	ECN-000092/A;1-Change4_AM	034785/B;1-Spindle Motor Assembly	Successful		Tanay Gupta

This report is also available to open or download on the **Reference Items** tab of the released (source) and open (target) change notice (appended by the date, which provides a history for the auto-merges).

ECN-000030/A;1-ECN-FT1

No Active ECN

SIEMENS

Overview

Affected Items

Impact Analysis

Participants

Reference Items

Workflow

Dependencies

Relations

Reports

Owner:

Date Modified: 03-Oct-2024

Release Status: Released

Type: Change Notice Revision

Closure: Closed

Disposition: Approved

ECN-000030/A;1-ECN-FT1

▼ Reference Items

Object

Type

Release Status

Date Released

Owner

AutoMergeReport03_10_2024_11_37_38

HTML

Shinde, Ajit (shindea)

▼ Working Environments

Object

Status

Date Released

Owner

Download File

Download the file to your device.

If an error occurs, a task is sent to the owner of the released change in the Inbox on the **Overview** tab. Instructions are provided to address the errors and restart the auto-merge by clicking **Complete**.

▼ Task to Perform

Workflow:

Auto Merge Process : ECN-000032/A;1-ECN-ERROR-FT1

Name:

Resolve merge errors

Task Instructions:

Open this change's References tab and open the latest Auto-Merge report. After addressing the errors listed, restart the Auto-Merge by selecting this task's complete button.

Workflow Description:

Auto Merge Process : ECN-000032/A;1-ECN-ERROR-FT1

Comments:

Complete

Examples

When performing an auto-merge of change notices, Change Management handles modifications based on whether they are conflicting or non-conflicting. For example, when the source and target change notice have non-conflicting modifications made (edits made to different items in the BOM), all edits are auto-merged.

However, when two change notices have conflicting changes (different values made to the same BOM line) the source's change notice modifications take precedence. For example, if a source change notice increases a BOM line part's **Quantity** value from **1** to **2**, and the target change notice changes the **Quantity** value from **1** to **3**, the auto-merge will take the source's value and keep it as **2**.

Related Concepts

About unincorporated changes

What happens when there are concurrent changes to the same item across multiple revisions?

What happens when concurrent change notices introduce conflicting changes to the same item?

Related Tasks

Merge an unincorporated change for a BOM into an existing change notice

Merge an unincorporated change for a BOM with effectivity into an existing change notice

Automatically merge BOM updates from a released change into an ongoing change

8. Managing change lineage

Change lineage overview

Change lineage tracks the evolution of impacted items through a relationship between **Solution Items** and the corresponding **Impacted Items**. It helps ensure that the change impact can be understood by downstream systems such as manufacturing and purchasing, and provides audit information.

Impacted Items are the old revisions or versions of content (parts, design elements, part usage, document and item revision) that are revised, versioned, or replaced as part of the change.

Solution Items are the new revisions or versions of content that are released by this change.

Capture change lineage information

Manually set change lineage

Procedure

1. From the **Change Summary**, select one item from the **Solution Items** table.
2. Click **Create Replace Group** .
3. In the **Set Lineage** dialog, selected the object to link to the **Solution Item**.
The lineage number is assigned, and it appears in the Lineage column for both the **Solution** item and the **Impacted** item, indicating that lineage is established.

Automatically set change lineage

Change lineage is automatically set if you create a revision to an object when a change notice is selected as an **Active Change**.

A revision created from a change object is automatically added to the **Solution Items** table and the lineage is set and displayed in the **Lineage** column.

Delete change lineage information

You can delete the change lineage by deleting relations between a group of solution items and their respective impacted items associated with the lineage group.

Procedure

1. From the **Change Summary** tab, select one or more objects in the **Solution Items** table for which to delete lineage.
2. Click **Remove Replace Group** .

A message indicates that the lineage was unset. The **Lineage** column is now empty for the **Solution** items and their respected **Impacted** items.

Replacement groups

Parts and assemblies are often added to and deleted from a product structure during its construction or modification. Some of these additions and deletions can be grouped together as a *replacement* action. For example, two added parts may replace the form and functionality of one canceled component, making a replacement. When you create and replace groups in a product structure, a graphical representation (often referred to as a supercedure) is created that represents the replacement history of a given occurrence in the product structure.

Oftentimes, various intricate relationships are formed during a change to an object. For example, in Change Management, a solution to an impacted item is not always a revision of the impacted item. It can be a separate item (for example, a markup). In addition, there can be more than one revision of the same item in either or both the **Impacted Items** and **Solution Items** folders.

Creating replacement groups can assist with simplifying these complex relationships by tracking the evolution of impacted items through a relationship between **Solution Items** and the corresponding **Impacted Items**. This helps ensure that the change impact can be understood by downstream systems such as manufacturing and purchasing, and provides audit information.

Create replacement groups

Create replacement groups (or a supercedure) to represent the replacement history of a given occurrence in the product structure.

Procedure

1. In the **Change Summary**, select multiple rows with the **Action** columns listed as **Added New** and **Removed** to create a new replacement group. All selected rows must be under the same parent object.
2. Click **Create Replace Group** .
A message indicates that the replacement group was set successfully. The **Added New** items display as **Added New to Replace** and the removed item displays as

Replaced in the **Action** column. A black outline borders the replacement group.

▼ CHANGE SUMMARY

Full Screen Create Repla... Add to

ID	Action	Revision	Element	Revision Name	Char
037335	Modified	B ★	Item A	Item A	
037348	Added New to Replace	A	Child C	Child C	
037340	Replaced	★	Item C2	Item C2	

Delete replacement groups

You can remove the create-replace group relationship by deleting the interaction between a group of solution items and their respective impacted items associated with the lineage group. This removes future traceability between the **Impacted Items** and the **Solution Items**.

Procedure

1.

In the **Change Summary**, select the **Added New to Replace** object.
2.

Click **Remove Replace Group**.
A message indicates that the replacement group was removed. The replacement group and **Action** column display the previous action types.

Genealogy

In a parent assembly, *genealogy* provides a visual representation of how a particular occurrence in an assembly evolves over multiple ECNs.

Genealogy for any part correlates to a parent assembly that was successively revised with a property change or a **replacement group** of changes. Genealogy can *only* be viewed in the context of the parent assembly.

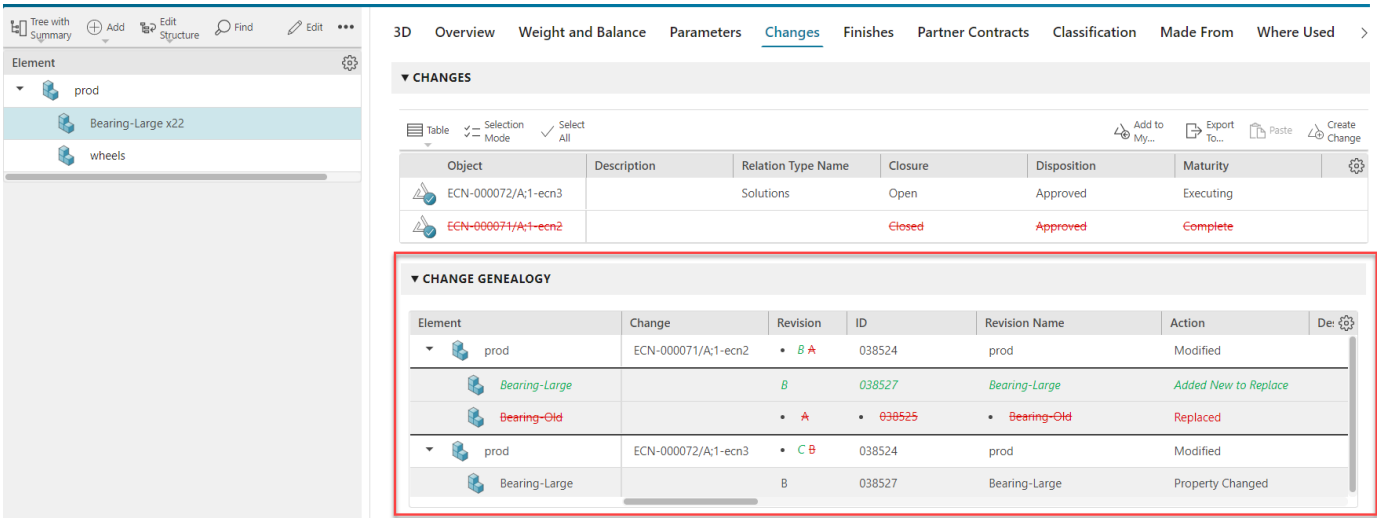
For example, our manufacturer creates the Spindle Motor Assembly. The Spindle Motor Assembly includes a part called Bearing-Old. This assembly has gone through a number of replacement group changes via the following ECNs:

- ECN1

Spindle Motor Assembly is revised from A to B. Bearing-Old is replaced with Bearing-Large, and the ECN is released.
- ECN2

Spindle Motor Assembly is revised from B to C. The Quantity property of Bearing-Large is increased and the part is replaced.

Now, the **Change Genealogy** displays this evolution of the Bearing in the context of the Spindle Motor Assembly.

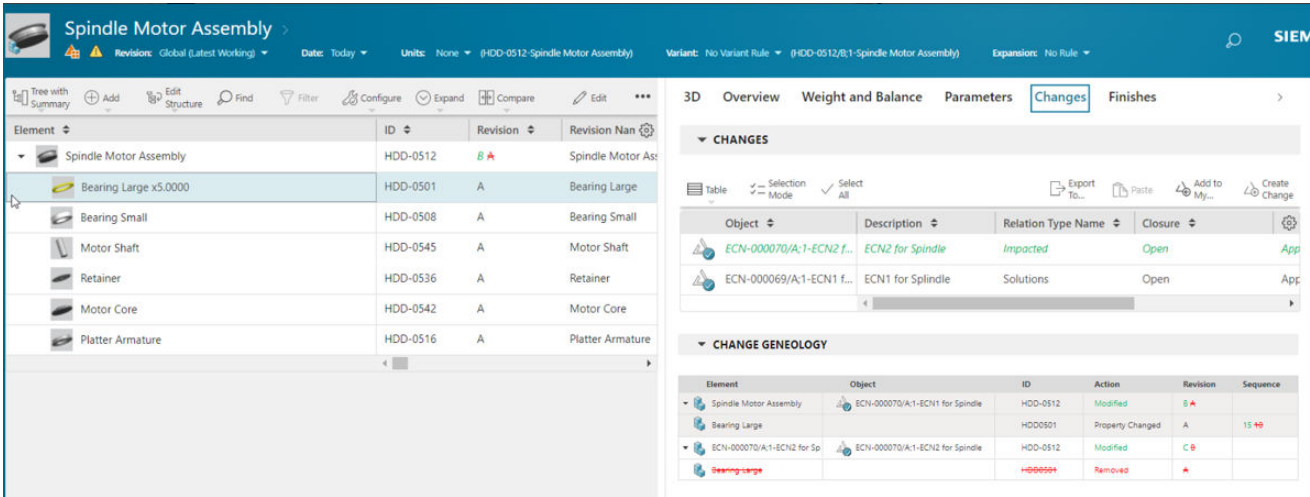


View the genealogy of a part or assembly

Genealogy is the evolution of the part and changes. It provides a visual representation of how a particular part in an assembly structure evolves over multiple ECNs in the context of its parent.

Procedure

1. Open any revision of an assembly and select a part that has multiple ECNs opened against it.
2. Click the **Changes** tab.



- The ECN assembly on the left panel lists changes from top to bottom, identifying how the part has evolved from when it was first created to its current state.
- On the right panel, the top **Changes** table shows the list of ECNs associated with the selected part.

- The **Change Genealogy** table shows how the selected part evolved and highlights the changes that were made for each ECN in the assembly, in the context of the parent revision. Use the scroll bar to view all the revisions in the table.

9. Generating change reports

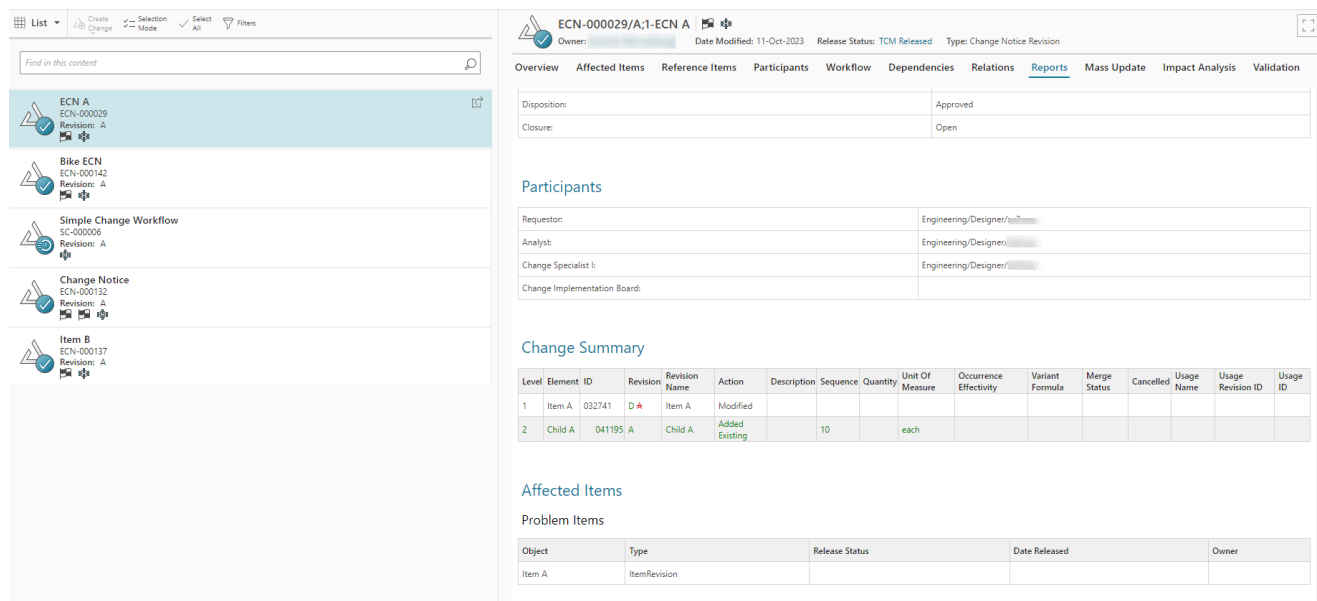
Generate the Change Item Report

The Change Item Report summarizes the information in the **Overview**, **Affected Items**, **Reference Items**, and **Participants** tabs of a change notice. This also includes revision information from the **Change Summary**, if available.

Procedure

1. Select a change notice.
2. Go to **More Commands *** > New ✨ > Generate Report**  from the **Page level** toolbar on any tab. Or, click **More Commands** from the **Details** toolbar and select **Generate Report**.
3. Select **Change Item Report**.
4. (Optional) On the **Generate Report** panel, enter a unique name for **Save to FileName** to save the report.
5. (Optional) Generating the report could take several minutes. You can select **Run in Background** to continue working while the report is generated.
6. Click **Generate**.

The **Change Notice Report** opens in the **Reports** tab of the change notice and saves a copy of the report in your My Teamcenter **NewStuff** folder.



The screenshot displays the 'Reports' tab for a specific change notice (ECN-000029/A;1-ECN A). The interface includes a left-hand navigation pane with a list of change notices. The main content area shows the 'Reports' tab, which contains several sections:

- Disposition:** Approved
- Closure:** Open
- Participants:** A form with fields for Requestor, Analyst, Change Specialist, and Change Implementation Board, all populated with 'Engineering/Designer/...'.
- Change Summary:** A table with 2 rows and 14 columns (Level, Element ID, Revision, Revision Name, Action, Description, Sequence, Quantity, Unit Of Measure, Occurrence Effectivity, Variant Formula, Merge Status, Cancelled, Usage Name, Usage Revision ID, Usage ID). The first row shows 'Item A' with revision 'D' and action 'Modified'. The second row shows 'Child A' with revision 'A' and action 'Added Existing'.
- Affected Items:** A section titled 'Problem Items' with a table containing columns for Object, Type, Release Status, Date Released, and Owner. The table shows 'Item A' as an 'ItemRevision'.

Generate the As Planned/As Released baseline report

The **As Planned/As Released baseline report** displays the structural hierarchy of an object and provides details about attached documents, the authorizing change notice, effectivity, and release date. It also displays the planned change notice for in-process documents as well as change effectivity.

Example

Run the report on an assembly with multiple levels, attachments, and revisions. The report identifies affected items with document attachments and their respective change notices.

To see the structure of a change notice, generate the [Change Item Report](#).

Procedure

1. Select an object.
2. Go to **More Commands ***** > **New**  > **Generate Report**  from the **Page level** toolbar.
3. Select **As Planned As Released Baseline Report**.
4. (Optional) enter a unique name for **Save to FileName** in the **Generate Report** panel.
5. Click **Generate**.

Generating the report could take several minutes. You can optionally click **Run in Background** to continue working while the report is generated.

The report automatically opens in the **Reports** tab.

A copy of the report is available in **Reports > Saved Files**.

The report's table is broken down into three main sections. The contents of the columns are dependent on the structure configuration, whether or not an attachment is revised or released, and if it is part of a change notice.

Note

Revisions to document attachments are not automatically updated in the **Attachments** tab. When a document attachment is revised, manually add it to the **Attachments** in order to accurately display the revision history in the **As Planned As Released** report.

Items

Contains all of the items in an assembly.

Documents As Planned/As Released

Contains documents and other attachments.

If an attachment is released as part of a change with a **Proposed Effectivity** this is listed in the Effectivity Date column with the related change notice number in the Change Notice column. If the attachment was not released as part of a change, those columns are empty.

Planned Changes

Contains only items that currently part of an unreleased change. Solution items that are revisions of an impacted item are added and include the change notice. The related impacted items show that it will be deleted.

The **CM_PRIMARY_DOCUMENT_RELS** preference must be configured by your administrator in order for the baseline report to run. It defines the relations through which a document is attached to an Item Revision. If a document is primary and the Item Revision is secondary in the relationship, the prefix **S2P** must be provided.

The default settings are as follows:

IMAN_specification

Fnd01isDescribedByDrawing

FND_TraceLink

FND0DesignToBomRelation

IMAN_METarget

Allowable values are any relation that can be associated with an ItemRevision.

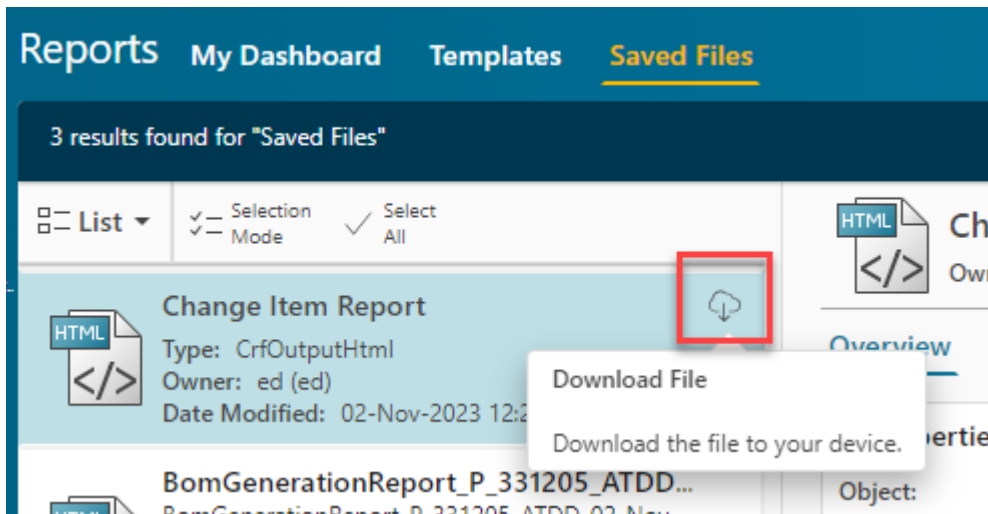
View and share reports

Procedure

1. From the **Launcher**, click **Reports**. 

If this option is unavailable, search for it. You can pin it from the search results for future access.

2. Select a report from the list to preview it.
3. Select a report and click download .



The file is saved as an HTML file in the default downloads location.

4. Go to **More Commands** *** > **Share**  > **Copy Link**  from the **Page level** toolbar to copy the report link to your clipboard and paste the link into an email or any other message for distribution.
5. (Optional) Open the downloaded file in a browser to view or print the report.

10. Setting change relations

Set change dependency

You can order and sequence change notices with respect to each other across change requests, by designating a change notice as before, after, or concurrent with respect to another change notice. There are three dependency types: **Preceding**, **Concurrent**, and **Succeeding**.

Only privileged users can manage change sequencing. More information on the following conditions can be found in *Business Modeler IDE* in the Teamcenter collection.

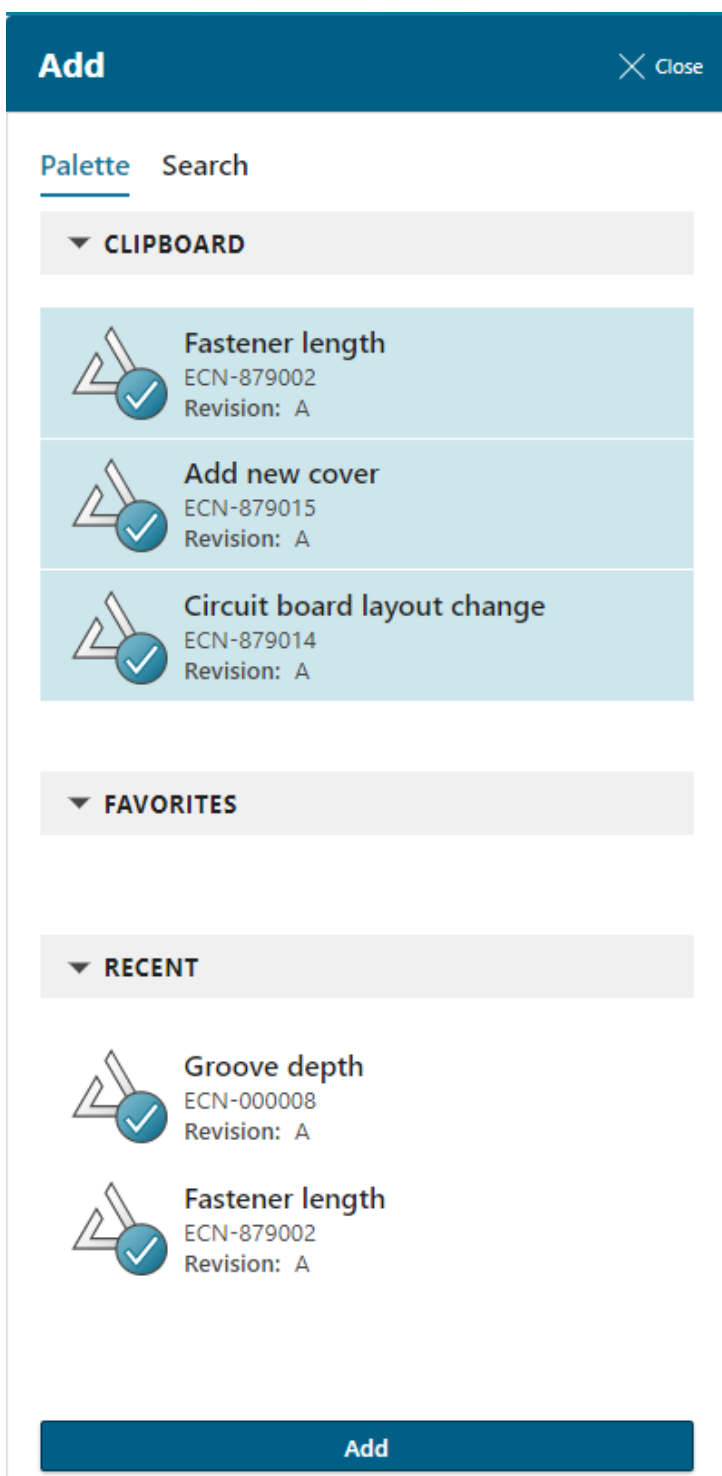
- **Cm0IsCmDependencyEditableForPrimary**
- **Cm0IsCmDependencyEditableForSecondary**
- **Cm1Awp0CutCmdCond**
- **Cm1SequenceChangeCmdCond**

Note

Your administrator can set the **CM_validate_changeNotice_dependency** preference. This preference controls the change notice dependency when adding a release status to a change notice revision. The default value is **false**.

Procedure

1. Copy the change notices to sequence by selecting them and clicking **Copy** .
2. Select the change notice you want to sequence the copied change notices with, and click **Dependencies**.
3. In the row containing the dependency type (**Preceding**, **Succeeding**, or **Concurrent**), click **Add** . The **Add** palette appears.
Alternatively, you can select **Search** from the **Add** panel, and search for the specified change notice.



4. Select the copied change notices from the palette clipboard, or those returned from the search, and click **Add**.

The relations are created and appear as sequenced under the appropriate heading.

Overview Affected Items Reference Items Participants <u>Dependencies</u> Relations Reports Audit Logs Mass Update						
▼ PRECEDING						
Table Selection Mode Clear Selections Export To... Paste Sequence Change						
ECN Number	Revision	Synopsis	Closure	Disposition	Maturity	
 ECN-879015	A	Add new cover	Open	None	Elaborating	
 ECN-879002	A	Fastener length	Open	None	Elaborating	
 ECN-879014	A	Circuit board layout chan...	Open	None	Elaborating	
▼ CONCURRENT						
Table Export To... Paste Sequence Change						
ECN Number	Revision	Synopsis	Closure	Disposition	Maturity	
▼ SUCCEEDING						
Table Export To... Paste Sequence Change						
ECN Number	Revision	Synopsis	Closure	Disposition	Maturity	

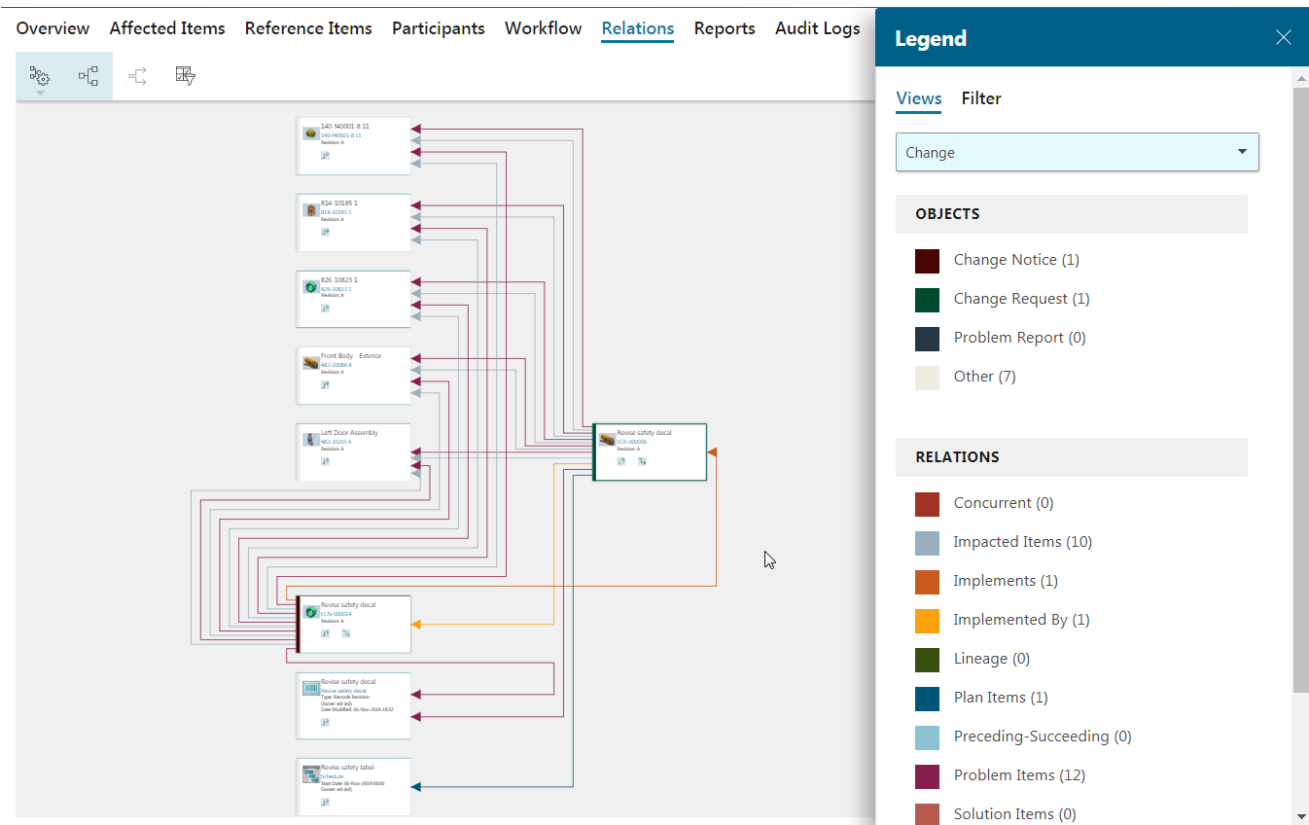
View change relations

Improve efficiency by graphically viewing and exploring change dependencies using the relations diagram. This provides a graphical representation of the dependencies in a change sequence.

Procedure

1. Open the change notice to view.
2. Click the **Relations** tab and then **Legend**  to open the **Legend** panel.
3. Select **Change** from the list of available views.
4. Select the **Object** or **Relation** you want to view from the **Legend** panel.

The **Legend** displays the full **Change** view. You can switch between the **Change** or **Change Impact Analysis** view without navigating to the **Impact Analysis** tab.



The diagram is arranged to show preceding through succeeding items, based on diagram layout. The legend appears on the right side of the window. In addition to the legend, the number of objects or relations of each type is indicated.

Color	Represents
Objects	
Burgundy	Change notice
Forest green	Change request
Navy blue	Problem report
Pale gray	Other
Relations	
Rust	Concurrent
Gray	Impacted items
Orange	Implements
Yellow	Implemented by
Olive Green	Lineage
Turquoise	Plan items

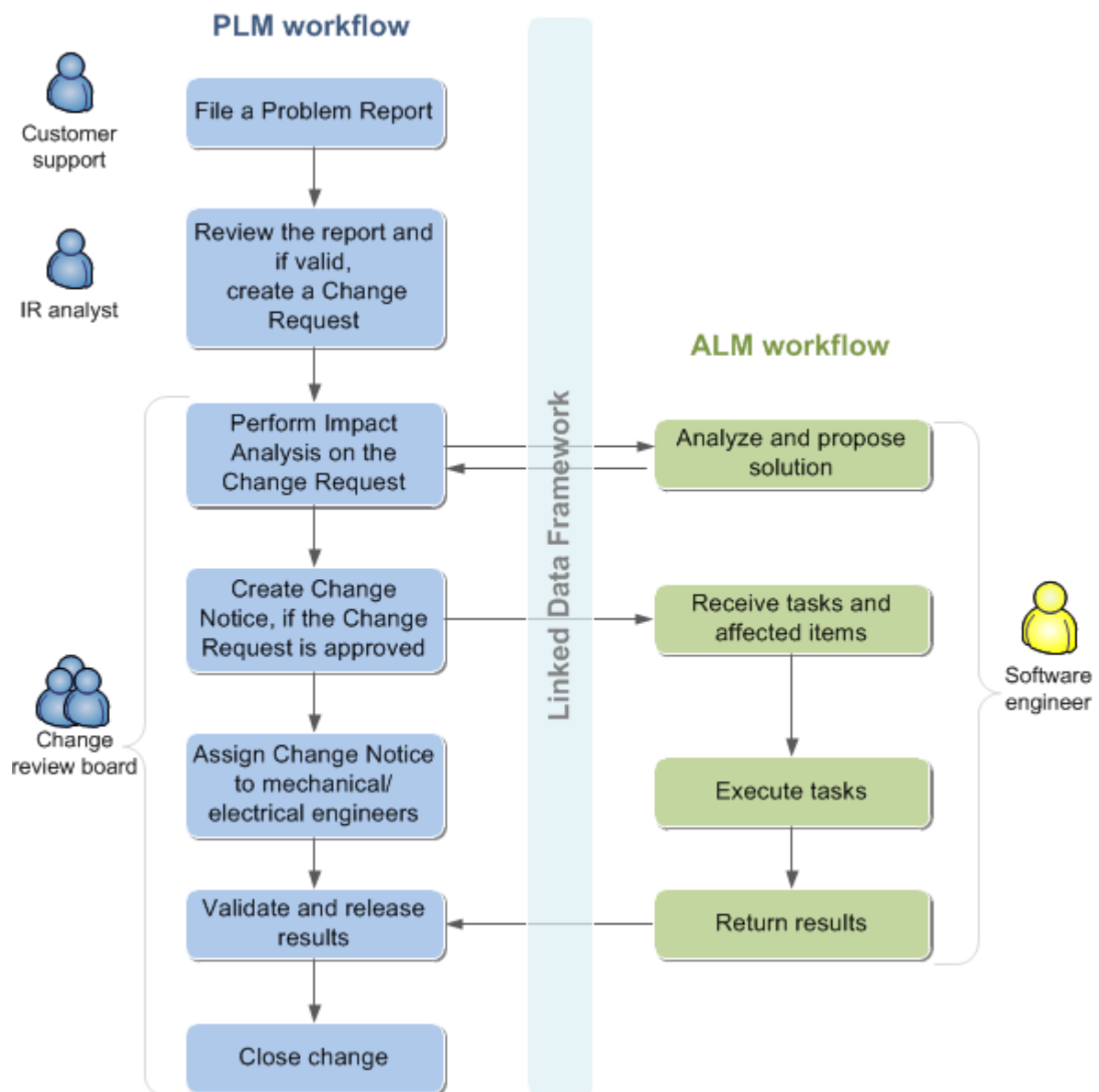
Color	Represents
Light blue	Preceding-succeeding
Magenta	Problem items
Salmon pink	Solution items

11. Linking Active Workspace change elements to external application elements

Overview of linking changes using Linked Data Framework

Linked Data Framework is an integration framework based on linked data standards. Using Linked Data Framework, customizers can integrate Teamcenter with external applications such as Polarion.

The following process depicts an integration between Teamcenter, which is a PLM application, and an ALM application by using Linked Data Framework.



Elements in Teamcenter are associated with the external elements using HTTP links or remote links. Once a remote link is created, you can view the external element within Teamcenter. For example, you can create a link from a change request in Teamcenter to a software defect in the external application. Once this is done, you can view the software defect from Teamcenter.

Customizers must set up the integration before you can use it.

Creating and deleting links

Link to a change by creating a resource in the external application

By default, *items*, *change*, and *requirements* are the only elements that Linked Data Framework supports. Contact your administrator if you wish to create links for other Teamcenter elements.

Procedure

1. Select a change and then click the **Reference Items** tab.
2. In the **Remote Links** section, click **Add** ⊕.
3. In the **Add** panel, update the properties as follows:
 - a. Select the project from the **Project** list.
 - b. Click the **New** option.
 - c. To create a resource in the external application, click **Add** ⊕ next to the **Remote Reference** label.
Log on to the external application if prompted and create the resource in that application.
 - d. Click **Create** to go back to the **Add** panel.
 - e. From the **Relation Type** list, select the relation you want to create between Teamcenter and the external element.

Note

If only one relation is applicable, this relation is used automatically. Manual selection is not required.

- f. In the **Add** panel, click **Add** to create the link.
The new links appear in the **Remote Links** section.

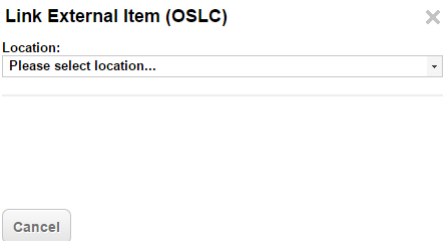
Create a link from the external application by selecting a change in Teamcenter

The exact method for creating a link to a Teamcenter resource from an external application differs across applications. Refer to your application documentation for more information.

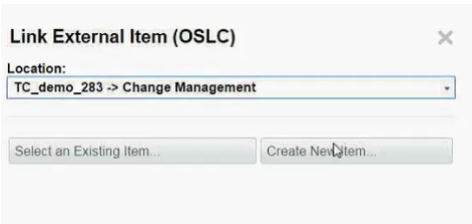
In Polarion, you can create a link to a Teamcenter change request as follows:

Procedure

- 1. In Polarion, navigate to a project and select a work item such as Issue.
- 2. In the **Linked Work Items** section, click **Edit** ..
- 3. Select a relationship for the link from the **Role** list.
- 4. Click **Select Work Item from Linked Data Friend Server** ..
- 5. In the **Link External Item (Linked Data)** dialog box, select a service provider from the **Location** list.



- 6. Click **Select an Existing Item**.



If this is a new session, you must log on to Teamcenter.

- 7. From the **Select** dialog box, select the change, and click **OK**.

The link to the Teamcenter Change Request is created in Polarion.

Linked Work Items				
Suspect	Role	Title	Project	Revision
☐	relates to	Power window issue	pol_proj	

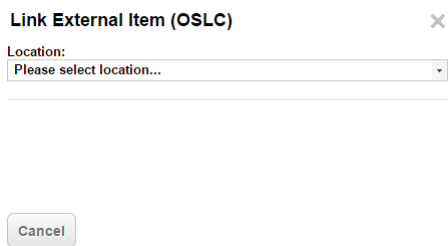
Create a link from the external application by creating a change in Teamcenter

The exact method for creating a link to a Teamcenter resource from an external application differs across applications. Refer to your application documentation for more information.

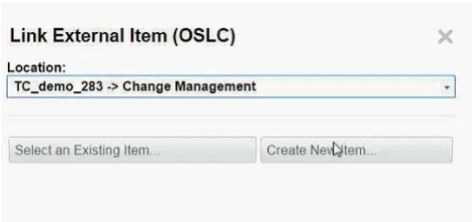
In Polarion, you can create a link to a Teamcenter change request as follows:

Procedure

- 1. In Polarion, navigate to a project and select a work item such as Issue.
- 2. In the **Linked Work Items** section, click **Edit** .
- 3. Select a relationship for the link from the **Role** list.
- 4. Click **Select Work Item from Linked Data Friend Server** .
- 5. In the **Link External Item (Linked Data)** dialog box, select a service provider from the **Location** list.



- 6. Click **Create New Item**.



If this is a new session, you must log on to Teamcenter.

- 7. From the **Create** dialog box, select the *Change type*, for example, **Change Request**.
- 8. Update the properties of the change and click **Add**.

The link to the Teamcenter Change Request is created in Polarion.

Linked Work Items				
Suspect	Role	Title	Project	Revision
☐	relates to	Power window issue	pol_proj	

Link to a change by selecting an existing resource in the external application

By default, *items*, *change*, and *requirements* are the only elements that Linked Data Framework supports. Contact your administrator if you wish to create links for other Teamcenter elements.

Procedure

1. Select a change and then click the **Reference Items** tab.
2. In the **Remote Links** section, click **Create Remote Link** ⊕.
3. In the **Add** panel, update the properties as follows:
 - a. Select the project from the **Project** list.
 - b. Click the **Existing** option.
 - c. To select a resource in the external application, click **Add** ⊕ next to the **Remote Reference** label.

Log on to the external application if prompted and select an existing resource in that application. This action takes you back to the **Add** panel.

- d. From the **Relation** list, select the relation you want to create between Teamcenter and the external element.

Note

If only one relation is applicable, this relation is used automatically. Manual selection is not required.

- e. Click **Add** to create the link.
The new links appear in the **Remote Links** section.

Delete remote links

Procedure

1. Select a remote link.
2. Go to **More Commands** *** > **Edit** ✎ > **Delete Remote Link**.
When you delete a remote link, backlinks are also deleted.

12. Viewing external links

View and edit resource information for external applications in Active Workspace

Procedure

1. Select a remote link and click **Open**  to see information about the link in the **Preview** tab.
2. To edit the external resource, go to **More Commands** ******* > **Edit**  > **Start Edit** from the **Page level** toolbar and update the resource.
3. To save the edits, use the save functionality of the external application. This is available as an embedded section in Active Workspace.

View links to objects in external applications using the Relations tab

The Relations tab allows you to view and navigate the relations between different objects in a graph view. With Linked Data Framework, you can view the relations between Teamcenter objects and objects in external applications.

A relations graph shows two elements: objects and relations. When you click the **Relations** tab or expand relations, you may need to log on to external applications.

You can perform all the operations that Relations supports such as expand, collapse, filter content, and change layout with Linked Data Framework objects. You can also expand external objects.

You can also expand to all the levels below the Requirement Collections artifact in Polarion, for example, a Polarion Live Document.

View links to objects in external applications using the **Relations** tab

13. Change management configuration

Change management configuration tasks

What is change management?

Change management is an organized way to implement changes to products and ensure the quality of every change. Users can access the changes they have implemented through the **Launcher**, and clicking **CHANGES** to display the **Changes** page.

Why configure change management?

How your organization processes changes is unique, and you can configure aspects of how changes are handled to match your organization's process. For example, you can set a different default workflow to be initiated when a user submits a change request.

What can I configure?

You can configure the following aspects of change management:

- **Set the default workflow.**
- **Configure how changes are derived.**
- **Define deep copy rules.**
- **Setting up filtering in the Changes page.**
- **Configure the contents of tabs in the Changes page.**

What do I need to do before configuring?

Before you can configure change management, you must add the application to your existing Teamcenter environment.

1. Log on to Deployment Center and select the environment to which you want to add Change Management.
2. Go to the **Applications** tab. Click **Add or Remove Selected Applications** .
3. In the **Available Applications** panel, use the web browser search to find the **Change Management** application. Select the application, and then click **Update Selected Applications**.

Deployment Center automatically selects any additional dependent applications.

4. Go to the **Deploy** tab. Click **Generate Install Scripts** to generate deployment scripts you will use to update affected machines.

When script generation is complete, note any special instructions in the **Deploy Instructions** panel.

5. Locate deployment scripts, copy each script to its target machine, and then run each script on its target machine.

For more information about running deployment scripts, see *Deployment Center — Usage*.

Or, if you see Teamcenter Environment Manager (TEM), add the following features in the **Features** panel:

- **Change Management** (server)

Installs the server-side definitions for changes.

Select **Active Workspace**→**Server Extensions**→**Change Management**.

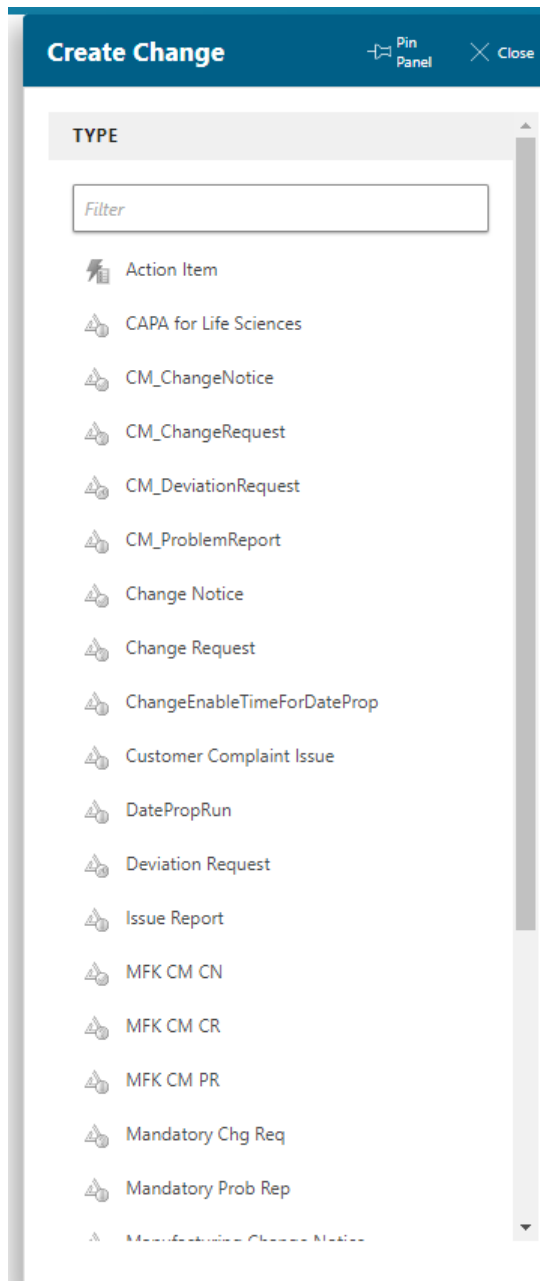
- **Change Management** (client)

Installs the user interface elements for viewing changes in Active Workspace.

Select **Active Workspace**→**Client**→**Change Management**.

Configuring the availability of Changeltem custom objects

User-defined child objects of the **Changeltem** business object are displayed in the **Create Change** panel (**More Commands** *** > **New** > **Create Change** from the **Page level** toolbar). These are based on the creatable conditions defined in BMIDE.



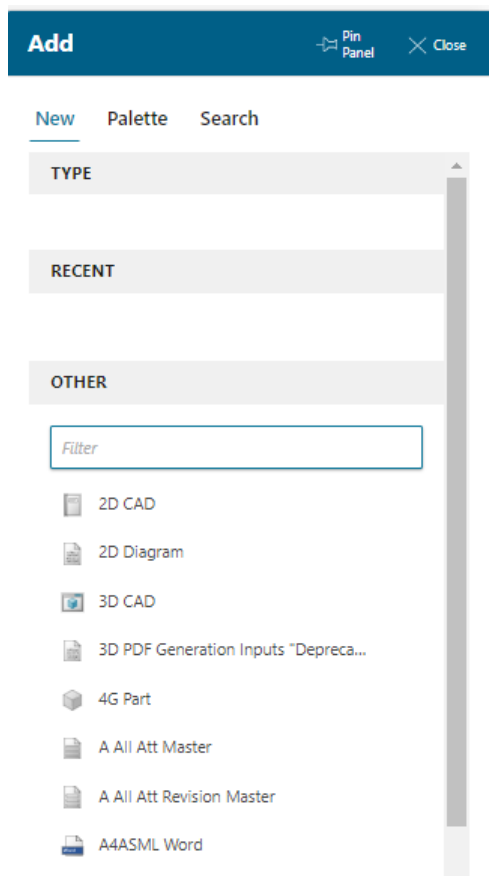
Custom child objects can also be made available on the **Home** page (click **Folders**). Click **Add to** (above the **Contents** table):

▼ CONTENTS

Table Selection Mode Select All Export To... Paste Add to

Object	Type	Checked-Out	Owner	Group ID	Date Modified	Release Status	
031100/A1-SimpleChan...	Item Revision		ed (ed)	demo	19-Apr-2022		
Rush_Test	Folder		ed (ed)	demo	19-Apr-2022		
Newstuff	Newstuff Folder		ed (ed)	demo	06-Mar-2022		
Mailbox	Mail Folder		ed (ed)	demo	06-Mar-2022		

This displays the **Add** panel containing the custom child objects.



To implement this behavior, an administrator can add the internal names of the custom child objects of the **ChangeItem** business object to the **AWC_TypeSelectorExclusionTypeList** site-level preference.

The internal names of the following child objects are included in the **AWC_TypeSelectorExclusionTypeList** site-level preference:

- **ChangeNotice**
- **ChangeRequest**
- **ProblemReport**
- **Cm0GnWorkOrder**

Automating the submission of changes to workflow

Use the following preferences to set the default workflow that should start when a user submits a problem report, change notice, or change request. The default workflow is **ChangeItemRevisionDefaultWorkflowTemplate**, which is a simple process to select a signoff team and then have each participant of the team perform his/her signoff task to approve the change.

For a	Use the preference	Its default is
Change notice revision	ChangeNoticeRevision_default_workflow_template	CN Fast Track Process
Change request revision	ChangeRequestRevision_default_workflow_template	ChangeItemRevisionDefault WorkflowTemplate
Problem report revision	ProblemReportRevision_default_workflow_template	ChangeItemRevisionDefault WorkflowTemplate

Configuring how changes are derived

When deriving a change from another use the **CM_automate_derive_propagation** preference to enable the automatic propagation of the relations (such as reference items and problem items) from the source change to the derived change. You configure which relations to propagate using the following preferences. For example, for a problem report enable the propagation of its problem items (**CMHasProblemItem**) and its reference items (**CMReferences**).

When deriving a change object from a	Set the relations propagated using
Problem report	CM_ProblemReportRevision_Relations_To_Propagate
Deviation request	CM_Cm0DevRqstRevision_Relations_To_Propagate
Change request	CM_ChangeRequestRevision_Relations_To_Propagate

You configure which change object users can derive from another using the **CM_change_derivations** preference.

For more information, refer to *Teamcenter Environment Variables* in the Teamcenter help.

Defining deep copy rules for creating changes

Use the Teamcenter Business Modeler IDE deep copy rules to set what objects and attributes are copied when a user creates a copy of a change. Deep copy rules define whether objects belonging to a business object instance can be copied when a user performs a save as or revise operation on that instance. Deep copy rules can be applied to any business object type and are inherited by children business object types.

Using deep copy rules, you can configure whether the following are copied for a change:

- Name, subject, description

- Problem and Impacted Items
- Referenced or related documents

The figure shows the deep copy rules defined by default for a **ChangeltemRevision** in the **Deep Copy Rules** editor. The rules define that when copying a change, copy the problem, impacted, and reference objects, but do not copy the incorporates and solution items.

Business Object : ChangeltemRevision

Main | Properties | Operations | Display Rules | **Deep Copy Rules** | GRM Rules | Operation Descriptor

☒ Show Inherited Rules
☒ Organize By Inheritance

Target Business Object	Target Primary	Operation	Type	Relation Type/Reference P...	Attached Business O...	Condition	Action	Required	Secured	Copy Properties on Rela...	COTS	Template
ItemRevision	✓	SaveAs	Relation	Fnd0DigitalSignatureRel	Fnd0DigitalSigna...	isTrue	NoCopy	✓	✓	✓	✓	foundation
ItemRevision	✓	SaveAs	Relation	Fnd0DigitalSignObsole...	Fnd0DigitalSigna...	isTrue	NoCopy	✓	✓	✓	✓	foundation
ChangeltemRevision	✓	SaveAs	Relation	CMSolutionToImpacted	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CM0Incorporates	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMHasImpactedItem	Match All	isTrue	CopyAsReference	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMHasProblemItem	Match All	isTrue	CopyAsReference	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMHasSolutionItem	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMHasWorkBreakdown	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMImplements	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	CMReferences	Match All	isTrue	CopyAsReference	✓	✓	✓	✓	cm
ChangeltemRevision	✓	SaveAs	Relation	HasParticipant	Match All	isTrue	NoCopy	✓	✓	✓	✓	cm

Note

Validate that the setting **copyFromOriginal** is set to true on persistent properties to copy them during a save as or revise operation. If **copyFromOriginal** is set to false on persistent properties, the properties are not copied during save as or revise.

Refer to the *BMIDE for Data Model Design* in the Teamcenter help for more information.

Note

Copying changes is not available in the Teamcenter rich client.

Define deep copy rules for copying options from an ECR to an ECN

Use Teamcenter Business Modeler IDE deep copy rules to configure the copying options when a user derives an ECN from an ECR. Deep copy rules define whether objects belonging to a business object instance can be copied when a user performs a **Derive** operation on that instance. Deep copy rules configure the copy option, providing the ability to select individual objects from the **Affected Items** and **Reference Items** in an ECR.

You can create rules for the default relation types **CMHasImpactedItem**, **CMHasProblemItem**, **CMReferences**. Refer to *BMIDE for Data Model Design* in the Teamcenter help for more information.

Note

Validate that the setting **copyFromOriginal** is set to true on persistent properties to copy them during a save as or revise operation. If **copyFromOriginal** is set to false on persistent properties, the properties are not copied during save as or revise.

Using deep copy rules, you can configure whether the following are copied into the ECN:

- Impacted Items
- Problem Items
- Reference Items

Create Deep Copy Rules in BMIDE

Procedure

1. Create a custom template based on the **Change Manager** template.
2. Open the **Change Item Revision Business Object**.

You can create a **Derive** deep copy rule for **GnChangeRequestRevision** and **GnProblemReportRevision**.

- Select the **Show Inherited Rules** check box to display all rules inherited from parent business objects.
 - Select the **Organize by Inheritance** check box to sort the rules by parent business object names.
 - Use the **Add**, **Edit**, or **Remove** buttons to work with the deep copy rules.
3. Select the **Deep Copy Rules** tab and click the **Add** button to add a row for each rule.
 4. Choose the business object that the deep copy rule is applied to.

Specify the parameters for each rule.

Parameter	Description
Target Primary?	Mark or clear the check box as appropriate.
	☒ When the checkbox is marked, Target Business Object is the primary object of the relationship specified in the Relation Type box. When the business object instance is revised or saved, the secondary objects are carried forward and related using the relation in the Relation Type box.
	☐ When the checkbox is clear, Target Business Object is the secondary object of the relationship specified in the Relation Type box. When the business object instance is revised or saved, the

Parameter	Description
	primary objects are carried forward and related using the relation in the Relation Type box.
Operation Type	Select Derive .
Type	Relation creates the deep copy relationship
Relation / Reference Property	Select CMHasImpactedItem , CMHasProblemItem , or CMReferences . You will supply a rule for each relation.
Attached Business Object	For CMHasImpactedItem and CMHasProblemItem types select ItemRevision . For CMReferences select WorkspaceObject .
Condition	Select the condition isTrue .
Action	Choose the kind of copying to be allowed for the business object. The available options differ depending on the type of target business object. Select CopyAsReference . This creates a new relation between the new revision and the related object. Therefore, modifications performed on the copied object are propagated to the source object.
Required	Leave blank.
Secured	Select if you want to prevent the deep copy rule from being modified or overridden by another template.
Copy Properties on Relation	Select if you want persistent properties on relation objects carried forward when the primary objects participating in relations are revised or saved as new objects. If not selected, only mandatory properties are carried forward.

5. Click **Finish**.

The rule is created and appears in the table in the **Deep Copy Rules** editor.

Configuring requested change type for impacted items

As an administrator, you can configure the list of values (LOV) for the requested change types available for the **Impacted Items** table.

The **Cm0RequestedChangeLOV** is attached to the **Cm0RequestedChange** property and contains the list of requested change types.

The default LOV includes:

- **Revise**
- **Create New**
- **Replace with Existing**

Modify the list and the description, as necessary.

Managing change notice conflicts during an auto-merge

To perform a merge successfully, administrators must carefully configure specific settings. These settings determine whether the source or target change takes precedence in an automatic merge when there are conflicting changes. Additionally, administrators can set up notifications and warnings for users to make them aware about any concurrent changes.

Auto-merge configuration

Since automatic merging runs behind the scenes and is not controlled by a user, administrators must determine the course of action for change notices when conflicts arise, based on business needs.

- The **CM_automerge_conflicting_changes** preference specifies whether a conflicting change will merge automatically or manually. True is the default (conflicting changes are automatically merged based on the preference **CM_automerge_overwrite_conflicting_changes**).
- The **CM_automerge_overwrite_conflicting_changes** preference specifies whether the merge process will overwrite the modifications from the released change or retain the work in progress modifications to overwrite modifications (**source** is the default).

If the value of **CM_automerge_overwrite_conflicting_changes** is set to **target**, the conflicting change(s) in the target assembly will remain unchanged. However, the merge status will be marked as complete (green), an auto-merge notification will be sent, but the merge candidate(s) will be enabled.

How do I determine the "winning" change notice when conflicts arise?

During an auto merge, you can configure the released change (source) to override or "win" any changes made to the same assembly items in concurrent changes (target) using a combination of the **CM_automerge_conflicting_changes** and **CM_automerge_overwrite_conflicting_changes** preferences.

By default, these preferences enable auto merge to overwrite modifications from a released change notice to work-in-progress changes, known as a source win.

Handlers and workflows

The following handlers must be configured to receive notifications/warnings about concurrent changes:

- **CM-show-concurrent-warning-on-task**
Action handler which calculates concurrent changes and issues warning for CIB reviewers.
- **CM-validate-concurrent-change-exists**
Rule handler that validates the presence of a concurrent change.
- **CM-concurrent-notification**
Action handler that calculates concurrent changes and identifies associated participants. Depending on its placement and the corresponding JSON file arguments, it sends alerts when a reviewer approves a concurrent change or when the change is released.

Two out-of-the-box sample workflows are provided for the warning process as guidance:

- Auto Merge Process
- Concurrent Change Notice Warnings Notifications Process

Notifications and warnings

Warnings/notifications can be configured by an administrator, based on business needs.

The **CM_participants_notify_conflicting_changes** preference defines the types of the participants to receive the concurrent change alert notification. This preference includes the default participant values as requestor and analyst.

The **CM_show_concurrent_change_warning** preference lets users enable or disable display warning message when concurrent changes are created (default is true).

The **CM_show_concurrent_active_change_warning** preference lets users enable or disable the display of warning message in an Active Change context when concurrent changes are created (default is true).

The **CM_show_concurrent_change_notification** preference lets users enable or disable the display of notifications when concurrent changes are created (default is true).

Attachment and vendor part conflicts

Additional configuration is required when using vendor parts and attachments in a change notice.

It is important to note that while an automatic merge of all types of conflicts (occurrence properties, substitutes, item revision properties, attachments, and vendor parts) depends on only two preferences (**CM_automerge_conflicting_changes** and **CM_automerge_overwrite_conflicting_changes**), a manual/ auto merge of different kind of changes (item revision changes, attachments, and so on) depends on other preferences.

To manage the merging of attachments and vendor parts, use the preferences **ChangeSummaryTableColumns_Indicator_HasAttachment** and **ChangeSummaryTableColumns_Indicator_HasVendorPart**. These preferences specify the types of relations and relation properties to be merged.

Some attachment types are associated with GRM relations that have an out-of-the-box (OOTB) cardinality of 1-1. For example, an item revision (primary) associated with UGMASTER (secondary) has a 1-1 cardinality. In these cases, auto merge overwrites modifications from a released change to work-in-progress changes, regardless of the preference value, because this scenario is not considered conflicting. Consequently, attachments from work-in-progress changes are lost, so caution is advised to prevent unintended overwriting.

In scenarios where multiple datasets are attached to item revisions tracked in both released changes (IR/B) and work-in-progress changes (IR/C), the merge process results in all datasets from IR/B being attached to IR/C, in addition to the existing datasets.

These examples demonstrate how the **ChangeSummaryTableColumns_Indicator_HasAttachment** and **ChangeSummaryTableColumns_Indicator_HasVendorPart** preferences handle conflicts, specifying whether the changes from the source or target win in each scenario. Each example includes two changes, Change A and Change B.

Example 1 conflict

In this scenario, the released item Item1/A - Doc1 undergoes an attachment relation property change in Change A and the removal of an attachment in Change B. This results in a conflict.

The merge candidates are the attachment changes from Item1/B. If the source wins, the attachment in Item1/C is reinstated with the property change. If the target wins, the attachment in Item1/C is removed.

Example 2 conflict

In this scenario, the released Item1/A - Doc1 involves removing an attachment in Change A and an attachment relation property change in Change B. This results in a conflict.

The merge candidates are the attachment changes from Item1/B. If the source wins, the attachment in Item1/C is removed. If the target wins, the attachment in Item1/C is reinstated with the property change.

Example 3 conflict

In this scenario, the released Item1/A - doc1 experiences an attachment relation property change in Change A and the removal of an attachment in Change B, leading to a conflict. The merge candidates are the attachment changes from Item1/B.

If the source wins, the attachment in Item1/C is reinstated with the property change. If the target wins, the attachment in Item1/C is removed.

Exclude specified data types from being merged in attachments and vendor parts

Use the **CM_merge_exclude_datatypes** site preference (default: UGMASTER) to exclude specified data types from being merged in attachments and vendor parts, whether manually or automatically. After making changes to the preference value, it is recommended to restart the Active Workspace session for the modified values to become applicable.

- If only excluded types are available for merging, the Merge tile  and indicators do not appear, but redlining will still be visible for these types in the **Attachments** tab (in table mode).
- If attachments include both excluded and non-excluded types, the Merge icon  and indicators will show as usual.

After merging, only non-excluded attachments will be merged, the tile will be disabled, and the Merge Complete icon  will be displayed in the [Change Summary](#).

Item revision exclusions

In an automatic or manual merge, the **ItemRevision_merge_change_prop_exclusion_list** preference defines the list of item revision properties to exclude while merging item revision properties changes. You can also create a site-level preference **<ObjectType>_merge_change_prop_exclusion_list** to exclude properties as well. For example, if you wanted to make a specific property exclusion list for a document revision merge, you would create a **DocumentRevision_merge_change_prop_exclusion_list**.

Disable the workflow selection property on the Create Change dialog

To activate the **Create and Submit** button when creating a change notice, the workflow property must be visible on the Create stylesheets and contain a value.

Procedure

1. Locate the section in the stylesheet that resembles the following:

```
<content
visibleWhen="{pref:CM_process_template_visible_on_create}==true">
  <htmlPanel declarativeKey="Cm1CreateChangeProcessTemplates"/>
</content>
```

2. Add **context="disabled"** to the **declarativeKey** attribute to make the workflow property visible but read-only. The updated section should appear as follows:

```
<content  
  visibleWhen="{pref:CM_process_template_visible_on_create}==true">  
    <htmlPanel declarativeKey="Cm1CreateChangeProcessTemplates"  
      context="disabled"/>  
  </content>
```

Using **context="disabled"** lets workflows display as read-only, which is helpful for users who prefer to maintain a fixed workflow template without any modifications.

Note

This modification is applicable from Teamcenter 2412 and onward.

Configure change tracking for revisable workspace objects

Revisable workspace objects (WSO) are workspace objects that can be revised. These are identified by the Business Object (BO) type constant **Fnd0AllowReviseOperation**, set to true. These objects can be tracked to manage changes and updates efficiently.

Use BMIDE to enable change tracking for any revisable workspace object.

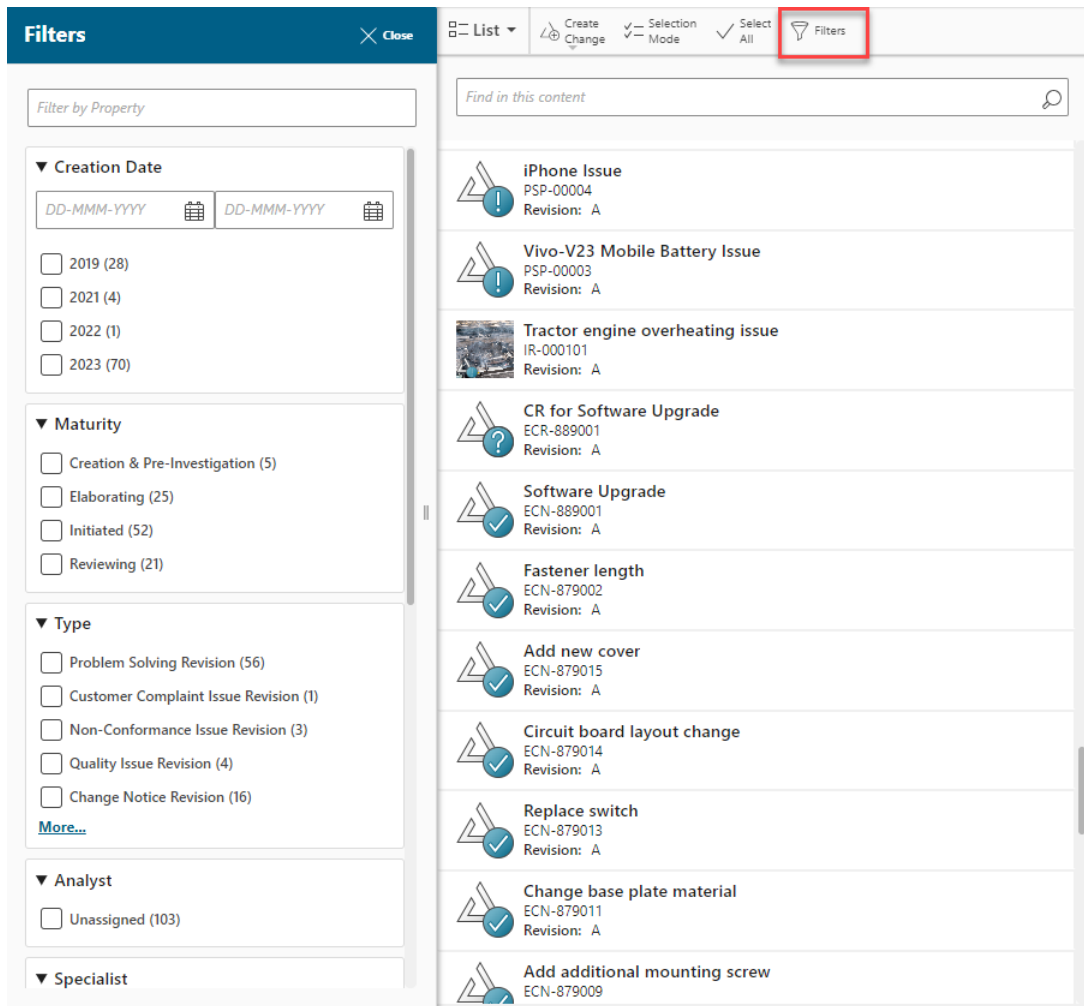
Procedure

1. In BMIDE, locate the BO type constant **cm0ChangeManaged** for the object you want to track.
2. Change it to **true** to enable change tracking.

Configuring the Changes page

Setting up filtering in the Changes page

You can set the properties that filter the changes that appear in the **Changes** page for changes found when selecting the **Submitted** tab. The changes are change business objects of the **ChangeltemRevision** type and its subtypes.



To set the filter, in the Business Modeler IDE, set the following property constants on the property of the change object on which you want to filter.

- **Cm1ChangeCanFilter**

Indicates that change business objects can be filtered on the property.

- **Cm1ChangeFilterPriority**

Indicates the priority of the property that determines its order in the list of filters displayed in the **Changes** page. The lower the value, the higher its priority and, therefore, the higher its position in the list of filters.

Siemens Digital Industries Software recommends that you assign values from a range to accommodate additional properties in the future. For example, assign priorities such as 100, 200, and 300, instead of 1, 2, and 3.

By default, the following properties are shown as filters for Change business objects:

- **Creation Date**

- **Maturity**
- **Type**
- **Analyst**
- **Specialist**
- **Requestor**
- **In Process**
- **Closure**
- **Fast Track**
- **Disposition**

Change filters can only be set on persistent and compound properties.

Properties supported for filtering	Properties not supported for filtering
• Date • String • References • Logical	• String properties with long string as storage • Numeric properties • Array properties

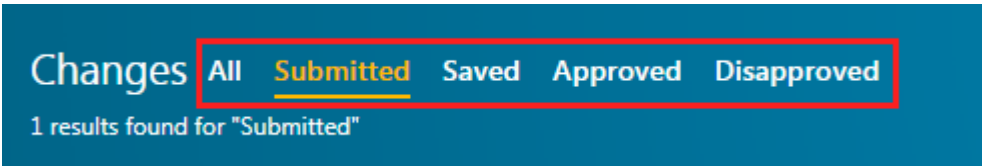
Refer to *BMIDE for Data Model Design* in the Teamcenter help for more information.

Configuring the contents of tabs in the Changes page

Note

For more information on creating a sublocation tab, refer to Using configuration builder to manage routes.

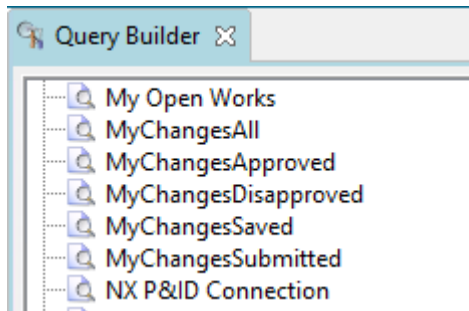
You can define which queries are used for each sublocation within the **Change** location.



The queries used for each sublocation *must* be the server-side Teamcenter platform saved queries. Saved queries in Active Workspace will not work in this case.

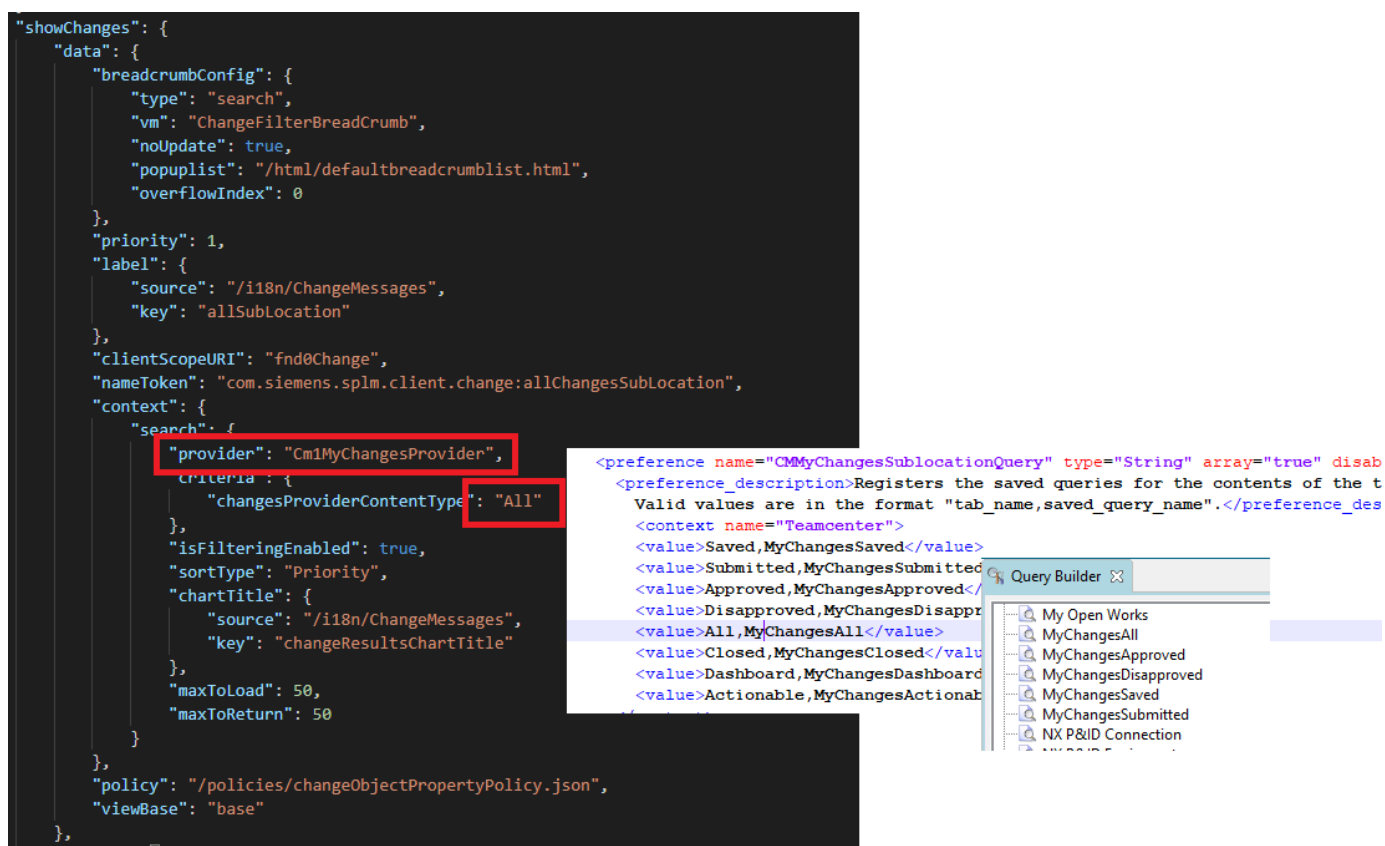
You can see a list of server-side saved queries (and execute them manually) by using **Advanced Search** `qid1928856`.

You can view the definition of and create new server-side Teamcenter platform saved queries by using the Query Builder application in the rich client.



Redefining the queries of the tabs is particularly helpful when you have defined custom participants or changes. You can replace your custom participants and changes with those in the default queries so the tabs show your company's content.

You must modify the **CMMMyChangesSublocationQuery** preference to modify which queries are used for each sublocation. The **Cm1MyChangesProvider** uses the preference value pairs to select the query which provides the data to the page. The preference definition contains the details of implementation.



Following are the default queries for each tab to display changes with a closure setting of **Open** and the logged-on user is the requester, analyst, or change specialist:

- **All tab**

```
Get all ChangeItemRevs where Closure=Open AND (Requestor = Logged-in User  
OR Analyst =  
Logged-in User OR Change Specialist1 = Logged-in User)
```

- **Submitted tab**

```
Get all ChangeItemRevs where Closure=Open AND (Requestor = Logged-in User  
OR Analyst =  
Logged-in User OR Change Specialist1 = Logged-in User) AND  
ProcessStageList != NULL
```

- **Saved tab**

```
Get all ChangeItemRevs where Closure=Open AND (Requestor = Logged-in User  
OR Analyst =  
Logged-in User OR Change Specialist1 = Logged-in User) AND  
ProcessStageList = NULL
```

- **Approved tab**

```
Get all ChangeItemRevs where Closure=Open AND (Requestor = Logged-in User  
OR Analyst =  
Logged-in User OR Change Specialist1 = Logged-in User) AND Disposition =  
Approved
```

- **Disapproved tab**

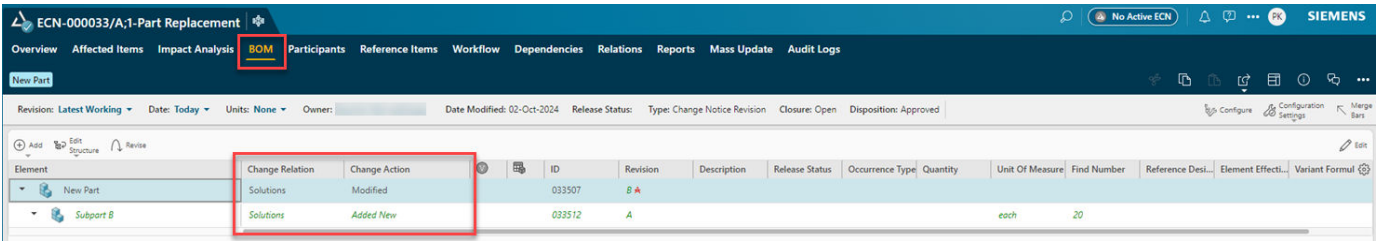
```
Get all ChangeItemRevs where Closure=Open AND (Requestor = Logged-in User  
OR Analyst =  
Logged-in User OR Change Specialist1 = Logged-in User) AND Disposition =  
Disapproved
```

Viewing the BOM tab for a change notice

To simplify managing multiple BOMs affected by a change notice, enable the **BOM** tab. This tab displays all impacted and solution items in an assembly, eliminating the need to navigate away from the Change Management interface. It provides essential BOM-related functionalities, including **Add**, **Edit**, **Revise**, **Revert**, **Remove**, and **Replace**.

Using the **BOM** tab, you can revise and modify parts in an assembly. It shows all affected items in one place, enabling BOM authoring and displaying redlines as in the Change Summary. The **Change**

Relation column indicates the relation of the object and the **Change Action** column indicates the actions performed on the object.



You must modify the **EnableMultiBOM** preference to true from false (the default) to enable the **BOM** tab.

Configure the Affected Items table

Use the **CM_affected_items_table_source** (string array) preference to configure objects to display in a single Affected Items table.

This preference specifies the relation and business object type (separated by a dot) to be shown in the consolidated **Affected Items** tab.

Values include:

- CMHasSolutionItem.ItemRevision
- CMHasSolutionItem.Fnd0AbstractOccRevision
- CMHasSolutionItem.MdIOModelElement
- CMHasSolutionItem.Cfg0AbsConfiguratorWSO
- CMHasSolutionItem.Bom0ConfigurableBomElement
- CMHasImpactedItem.ItemRevision
- CMHasImpactedItem.Fnd0AbstractOccRevision
- CMHasProblemItem.ItemRevision
- CMHasProblemItem.Fnd0AbstractOccRevision

Overview Affected Items Impact Analysis Participants Reference Items Workflow Dependencies Relations Reports Mass Update											
Show in Single Table: ☒											
This table contains Problems, Impacted and Solutions.											
Object	Change Relation	Requested Cha...	Lineage	Proposed ...	Release Status	Cancelled	Date Relea...	Owner	Disposition	Incorporation ...	
047675/A;1-Bike	Solutions							ed (ed)			
047675/A;1-Bike	Problems							ed (ed)			
047675/A;1-Bike	Impacted							ed (ed)	Use As Is	Unincorporated	
047679/C;1-Seat	Impacted		1					ed (ed)	Use As Is	Unincorporated	
047679/D;1-Seat - 1	Solutions		1					ed (ed)			
047680/B;1-Pedals	Impacted		2					ed (ed)	Use As Is	Unincorporated	
047680/C;1-Pedals - 4	Solutions		2					ed (ed)			

Customize Change Summary indicators

Add custom indicator columns to the Change Summary to track additional modifications such as attachments, vendor parts, or custom relations by modifying JavaScript files and preference settings.

Restrictions and limitations

Before you begin, note the following:

- Indicator columns are only available when the preference **CM_enable_changesummary_with_indicators** is set to true (Teamcenter 2506 and later).
- This configuration requires modifying JavaScript source files in the Teamcenter installation.

Prerequisites

- Before you begin, ensure that you have access to the following files:
 - `src/thinclient/changejs/propertyRendererTemplates.json`
 - `src/thinclient/changejs/src/propRendererTemplates/changeSummaryIndicatorRenderer.js`
- The preference **CM_enable_changesummary_with_indicators** is set to true.

Procedure

Add the indicator column to the preference

- Add a new value to the preference **ChangeSummaryWithIndicatorsTableColumns** using the following format:

Type:Indicator,Property:Indicator_IndicatorPropertyName,Width:PIXEL_WIDTH

Example: `Type:Indicator,Property:Indicator_HasVendorPart,Width:150`

- Create a corresponding preference **ChangeSummaryWithIndicatorsTableColumns_Indicator_IndicatorPropertyName** with the value formatted as:

- Relation:properties on relation** (separated by comma), OR
- Relation:properties on relation**

Example: Create preference

ChangeSummaryWithIndicatorsTableColumns_Indicator_HasVendorPart with value **CMHasVendorPart**.

Modify the JSON template file

- Locate and open the file `propertyRendererTemplates.json`.
- Locate the **changeSummary2IndicatorColumnHeaderRenderer** template.
- Add **Indicator_IndicatorPropertyName** to the columns array within this template.

Update the indicator cell renderer

9. Locate and open the file `changeSummaryIndicatorRenderer.js`.

This file contains two renderers for displaying icons in the change summary table:

- **indicatorColumnCellRenderer:**

Renders the indicator cell.

- **indicatorColumnHeaderRenderer:**

Renders the indicator column header.

10. Open the `indicatorColumnCellRenderer()` function.
11. Add the indicator to the `indicatorNames` list using `Indicator_IndicatorPropertyName` as the key.
12. Set the value as an object containing:
 - a. `indicatorName`: The name of the cell indicator icon.
 - b. `tabName`: The tabKey or name of the corresponding tab in the **Details of Changes** pane.

```

7
8  /**
9   * This function is used to render the indicator column cells.
10  * @param {Object} vmo Viewmodel object which contains props to get property display name.
11  * @param {Object} containerElem Container dom element in which indicator is attached.
12  * @param {Object} columnName - columnName of the current column which is rendered.
13  */
14  export let indicatorColumnCellRenderer = function( vmo, containerElem, columnName ) {
15    // Check if the column value is 'true'
16    if ( vmo.props[columnName]?.dbValues?.[0]?.toLowerCase() === 'true' ) {
17      // Define the indicator names
18      // For custom indicator it can be added in this list in following format.
19      const indicatorNames = {
20        Indicator_HasPropertyChange: {
21          indicatorName: 'indicatorProperties16',
22          tabName: 'tc_xrt_properties'
23        },
24        Indicator_HasSubstitute: {
25          indicatorName: 'indicatorSubstitute16',
26          tabName: 'tc_xrt_substitutes'
27        },
28        Indicator_HasAttachment: {
29          indicatorName: 'indicatorAttachment16',
30          tabName: 'tc_xrt_attachments'
31        },
32        Indicator_HasVendorPart: {
33          indicatorName: 'indicatorVendorPart16',
34          tabName: 'tc_xrt_vendorParts'
35        },
36        default: {
37          indicatorName: 'indicatorMissingImage16',
38          tabName: ''
39        }
40      };
41
42      if ( !vmo.visibleTabs ) {

```

Update the indicator column header renderer

14. Open the `indicatorColumnHeaderRenderer()` function.
15. Add the indicator to the `indicatorNames` list using `Indicator_IndicatorPropertyName` as the key.
16. Set the value as the header icon name.

```

/*
 * Change summary indicator column header renderer
 * we need to show icon instead of column name
 */
export let indicatorColumnHeaderRender = function( containerElement, columnField, tooltip, column ) {
  let headerContent = document.createElement( 'div' );
  headerContent.className = 'sw-row aw-change-columnHeaderTextWrapAuto';

  if ( column.name.includes( 'Indicator_' ) ) {
    const indicatorNames = {
      Indicator_HasPropertyChange: 'IndicatorHeaderProperties16',
      Indicator_HasSubstitute: 'IndicatorHasSubstitutes16',
      Indicator_HasAttachment: 'IndicatorAttachment16',
      Indicator_HasVendorPart: 'IndicatorHeaderVendorParts16',
      default: 'miscTableHeaderMissingInSource16'
    };

    let indicatorName = indicatorNames[column.name] || indicatorNames.default;
    let context = {
      indicatorName,
      tooltipDetails: {
        prop: column.displayName,
        isIndicator: true
      }
    };
  }
}

```

Results

After completing these steps:

- The custom indicator column appears in the Change Summary table when **CM_enable_changesummary_with_indicators** is set to true.
- Clicking the indicator icon opens the corresponding **Details of Changes** pane showing the tracked relation or property data.
- The indicator column header displays the custom icon instead of a text label.



For information about Change Summary column configuration preferences, see [Configure Change Summary columns](#).

Configure Change Summary columns

Administrators can configure which columns appear in the Change Summary table and define their default properties. The configuration system varies depending on which version of Change Summary is enabled in your Teamcenter environment.

Determine your Change Summary version

The **CM_enable_changesummary_with_indicators** preference controls which Change Summary version is active:

Value = **true**: Change Summary with indicators (only available from Tc2506 onwards)

Value = **false**: Change Summary without indicators

How column configuration works

1. Site-Level Configuration

Administrators configure the default columns using the appropriate site-level preference (**ChangeSummaryWithIndicatorsTableColumns** or **ChangeSummaryTableColumns**) for the active Change Summary version.

2. User-Specific Configuration

When users customize their column arrangement through the UI, the system saves their personal configuration to user-specific preferences (**ChangeSummaryWithIndicatorsColumnsShownPref** and **ChangeSummaryWithIndicatorsColumnsShownWidthPref** for Change Summary with indicators, and **ChangeSummaryColumnsShownPref** and **ChangeSummaryColumnsShownWidthPref** for Change Summary without indicators).

3. Tracked properties override

The **CM_bomline_tracked_properties** preference overrides the site-level column configuration and automatically adds columns for tracked BOM line properties (such as **bl_quantity**, **bl_sequence_no**) to the Change Summary by design.

Note

Users can reset their customized column configuration back to the site-level defaults using the **Reset** function in the **Arrange** panel. This clears their user-specific preferences and restores the original columns as defined in the site-level preference.

Configure site-level column preferences

Column value types

The site-level preferences (**ChangeSummaryWithIndicatorsTableColumns** or **ChangeSummaryTableColumns**) define the column names to be displayed. There are three types of values:

- **Type 1: Direct property mapping**

Format: Type:TYPE_NAME,Property:PROPERTY_NAME,Width:PIXEL_WIDTH

The system reads the property name for the column from **PROPERTY_NAME** and displays the name for the column based on the owning type, **TYPE_NAME**.

Example: Type:WorkspaceObject,Property:object_name,Width:200

- **Type 2: Key-based property mapping**

Format: Type:Key,Property:KEYNAME,Width:PIXEL_WIDTH

The system searches for the preference **ChangeSummaryTableColumns_KEYNAME** (or **ChangeSummaryWithIndicatorsTableColumns_KEYNAME** for Change Summary with indicators) to retrieve source property names for the column.

Example: Type:Key,Property:ID,Width:150

- **Type 3: Action column**

Format: Type:NoType,Property:action,Width:PIXEL_WIDTH

This entry displays the Action column in the Change Summary table.

Example: Type:NoType,Property:action,Width:150

Configure tracked BOM line properties

Use the **CM_bomline_tracked_properties** preference to define which BOM line properties are tracked in the Change Summary.

Note

By default, the system automatically adds columns to the Change Summary for properties specified in **CM_bomline_tracked_properties**, as these properties are tracked and displayed by design. This preference overrides the column configuration specified in the respective Change Summary master preference (**ChangeSummaryWithIndicatorsTableColumns** or **ChangeSummaryTableColumns**).

Additional information

- For information about customizing column display as a user, see [Customize your Change Summary or comparison view](#).
- For information about adding indicator columns to the Change Summary, see [Customize Change Summary indicators](#).

Configure revert behavior

The **CM_keep_solution_item_after_revert** site preference determines whether solution items remain in the change object after a revert action and controls which types of changes are reverted during change management operations. Administrators configure this preference to determine whether solution items stay in the change object or are removed after a revert action.

Settings and behavior

The **CM_keep_solution_item_after_revert** preference is a Boolean site-level preference with a default value of **true**. The preference value determines the following behavior:

When set to true (default):

- Keeps the solution item in the change object
- Reverts structural updates and property edits
- Preserves changes to attachments and vendor parts

When set to false:

- Removes the solution item from the change object
- Reverts all changes including structural updates, property edits, attachments, and vendor part modifications
- No redlines are shown if there is no lineage between revisions

Revert command behavior

The **CM_keep_solution_item_after_revert** preference also affects when the **Revert** command is visible and enabled. When set to true, the **Revert** command is disabled if the only change in a line is a revision change, as the solution item remains intact.

The system checks several prerequisites before enabling the **Revert** command. These include whether object modifications are performed as part of the change notice, whether objects are revised with proper lineage tracking, and whether the object revisions are kept in **Impacted items**.

Users receive appropriate error messages if prerequisites are not met, such as when attempting to revert objects without proper lineage or when selecting items that cannot be reverted due to redlined children.

User notifications

Confirmation messages vary based on the preference setting. When set to **true**, users see messages indicating that structural and property changes will be reverted while attachments and vendor parts are preserved. When set to **false**, users see messages indicating that all changes will be reverted and the solution item will be removed.

During multi-select operations, the system does not dynamically check redlines for each selected item. This means the confirmation message shows only the changes that can be reverted based on the preference setting, not the actual changes present in each selected item. Users should verify individual items before performing multi-select revert operations to ensure the expected changes are reverted.

For information about the confirmation messages users see during revert operations, see [Understanding revert confirmation messages](#).